

CAMBRIDGE TEXTBOOKS IN LINGUISTICS

Mood and Modality

Second edition

F. R. Palmer

Since the publication of F. R. Palmer's *Mood and Modality* in 1986, when the topic of 'modality' was fairly unfamiliar, there has been considerable interest in the subject as well as in grammatical typology in general. Modality is concerned with mood (subjunctive, etc) and with modal markers such as English modal verbs (can, may, must, etc.) and is treated as a single grammatical category found in most of the languages of the world. In this new edition, Palmer draws on a wealth of examples from a wide variety of languages to investigate the category of modality in detail. He discusses familiar features in a number of mainly European languages, and also looks at less familiar features including 'evidential' systems and the contrast of realis/irrealis, both to be found in unrelated languages.

F. R. PALMER is Emeritus Professor of Linguistic Science at the University of Reading and is a Fellow of the British Academy. He is the author of innumerable articles and many books, including *Semantics* (2nd edition, 1981), *Mood and Modality* (1986) and *Grammatical Roles and Relations* (1994), also published by Cambridge University Press. He has lectured all over the world including in the USA, Brazil, India, China, Japan, Australia and most of Europe.

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MOOD AND MODALITY

Second Edition

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F. R. PALMER

*Professor Emeritus, Department of Linguistic Science,
University of Reading*



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To
ANNA, ELIZABETH *and* RACHEL
(*not forgetting* RICHARD *and* CLAIRE)

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PREFACE

Since the publication of the first edition (Palmer 1986) there has been considerable interest in modality (as well as in grammatical typology in general). A symposium on Mood and Modality held in the University of New Mexico in 1992 was successful in bringing together over forty researchers, and resulted in the publication of eighteen papers (Bybee and Fleischmann 1995a). Yet, in contrast, a workshop on modality at the International Congress of Linguists only ten years before had attracted only four scholars. This symposium was followed in 1993 by a symposium on Modality in Germanic languages, which resulted in a further eight papers (Swan and Westvik 1997). Indeed, Bybee and Fleischmann (1995b: 1) suggest that the first symposium succeeded in establishing for modality the kind of status that had been established for tense and aspect by a symposium on those subjects ten years before (Hopper 1982). It should, however, be pointed out that the first edition of the present volume was published six years before the symposium in New Mexico and that a volume on modality in English appeared thirteen years before that (Palmer 1979).

It was recognized in the first volume that the most appropriate name for the relevant category is simply ‘modality’, and that ‘mood’ is more appropriate as the traditional name for indicative, subjunctive, etc., in both classical and modern European languages. Nevertheless, it was thought that, since the topic was fairly unfamiliar, *Mood and modality* might be the most informative title, and that was the title used in the 1993 symposium.

I have been assisted by a whole host of scholars who have provided examples or answered queries about their work. I fear that a list of names would inevitably contain omissions, and hope they will accept my sincere thanks and forgive me for not mentioning them individually.

NOTE ON THE TEXT

In addition to the usual conventions for the use of italics, quotation marks and asterisks, the following notation is used:

In the text:

<i>Initial capitals</i>	major typological categories
SMALL CAPITALS	verbs, including modals
‘single quotation marks’	terms by the authors quoted

In the inset examples

SMALL CAPITALS	grammatical categories and language forms where the grammatical category is not established
- hyphen	morphemic boundary in the language material and corresponding division in the gloss
+ plus sign	combined categories in the gloss represented by a single element in the language material

To a large extent, the transcriptions, glosses and translations used in the glosses are those of the original authors, except that

- (i) no distinction is made between affixes and clitics;
- (ii) where both a phonemic or phonetic and a morphophonemic representation (showing morpheme boundaries) are given by the author, either the morphophonemic version will be given or the morphophonemic form will be given and followed by the phonemic representation;
- (iii) the abbreviations used for the names of the categories are standardized for consistency – see list on next page.

ABBREVIATIONS

1	first person
2	second person
3	third person
ABS	absolute
ABL	ablative
ABSOL	absolute
ACC	accusative
ADM	admirative
AFF	affirmative
AG	agent
AOR	aorist
APP	apparent
ASS	assertion/assertive
ASSUM	assumed/assumptive
AUD	auditory
AUX	auxiliary
BEN	benefactive
CARIT	caritative
CATEG.ASS	categorical assertion
CAUS	causative
CERT	certain
CIRC	circumstantive
CL	classifier
CM	conjugation marker
CNTF	counterfactual
CNTR.ASS	counterassertion
CO.AG	co-agency
COMPL	complementizer
COMPLET	completive
COMPUL	compulsional

List of abbreviations

CONCESS	concessive
COND	conditional
CONJ	conjecture
COOP	cooperative
COP	copulative
COREF	coreferential
CUST	customary
DAT	dative
DEB	debitive
DEC	declarative
DED	deduced
DEF	definite
DEFOC	defocus
DEM	demonstrative
DES	desiderative
DIFF	different event
DIR	direct case
DS	different subject
DUB	dubitative
DUR	durative
EF.INF.	emphatic first-hand information
EMPH	emphatic
ERG	ergative
EV	evidential
EXCL	exclamative
EXPECT	expectational
FACT	factual
F.INF	first-hand information
FOC	focus
FREQ	frequentative
F.SG	feminine singular
FUT	future
FUT.INT	future intention
GEN	generic
GEN.COND	generic conditional
GEN.KNOW	general knowledge
HAB	habitual
HAB.P	habitual past
HORT	hortative
HR.EV	heard evidence

HSY	hearsay
HYPOTH	hypothetica;
IGNOR	ignorative
IMM.PAST	immediate past
IMP	imperative
IMPERS	impersonal
IMPF	imperfect
IMPFV	imperfective
INC	incompletive
INCH	inchoatived
INCL	inclusive
IND	indicative
INDEF	indefinite
INDIR	indirect
INF/INFER	inference/inferential
INFIN	infinitive
INFREQ	infrequentative
INT	interrogative
INTEN	intensifier
INTR	intransitive
IRR	irrealis
IRRELEV	irrelevance
JUSS	jussive
LING.EVID	linguistic evidence
LOC	locative
M.E.	multiple event
M.SG	masculine singular
MOD	modal
NEG	negative
NOM	nominative
NONFEM	non-feminine
NONFUT	non-future
NONVIS	non-visual
OBJ	object
OBL	obligative
OPT	optative
PART	participle
PAT	patient
PER	period
PERF	perfect

List of abbreviations

PERFV	perfective
PERM	permission/permissive
PERS.AG	personal agency
PERS.EXP	personal experience
PL	plural
PL.EXC	exclusive plural
PLUP	pluperfect
POSS	possessive
POT	potential
PRED	predicate
PREDICT	prediction
PREP	preposition
PRES	present
PROG	progressive
PROH	prohibitive
PUNCT	punctual
PURP	purpose, purposive
QUES, Q	question
QUOT	quotative
REAL	realis
RECIP	reciprocal
RED	reduplication
REFL	reflexive
RELEV	relevance
REL.FUT	relative future
REM	remote
REP	report, reported, reportive
REP.DEP	report in dependent clause
REP.IND	report in independent clause
RESP	responsive
RESP.SCEP	sceptical response
SAME	same event
SCEP	sceptical
SEC	second-hand
SEE.EV	seen evidence
SEMEL	semelfactive
SENS.EVID	sensory evidence
SEQ	sequential
SG	singular
SIM	simultaneous
XX	

SIMULAT	simulative
SM	subject marker
SPEC	speculated
SS	same subject
STAT	stative
SUB	subject
SUBJ	subjunctive
TOD.PAST	today's past
TOP	topic
UNCERT	uncertain
VIS	visual
YEST.PART	yesterday's past

I

Introduction

It has come to be recognized in recent years that modality is a valid cross-language grammatical category that can be the subject of a typological study. It is a category that is closely associated with tense and aspect in that all three categories are categories of the clause and are generally, but not always, marked within the verbal complex.

In notional terms all three are, in some way, concerned with the event or situation that is reported by the utterance (though for simplicity the term ‘event’ will, throughout this volume, be used to cover events, actions, situations, states, etc.). Tense, rather obviously, is concerned with the time of the event, while aspect is concerned with the nature of the event, particularly in terms of its ‘internal temporal constituency’ (Comrie 1976: 3). Modality is concerned with the status of the proposition that describes the event.

1.1 Basic concepts

1.1.1 *Realis and irrealis*

Modality differs from tense and aspect in that it does not refer directly to any characteristic of the event, but simply to the status of the proposition. One possible approach to its analysis is to make a binary distinction between ‘non-modal’ and ‘modal’ or ‘declarative’ and ‘non-declarative’, and to associate this distinction with the notional contrast of ‘factual’ and ‘non-factual’, or ‘real’ and ‘unreal’.

However, these terms are not really satisfactory (see 1.1.2), and in recent years ‘realis’ and ‘irrealis’ have been used for this distinction. They have the advantage that they are obviously technical, so that their use can avoid any possible connotations of the more familiar terms. For Mithun (1999: 173) the distinction is that ‘The realis portrays situations as actualized, as having occurred or actually occurring, knowable through direct perception. The irrealis portrays situations as purely within the realm of thought, knowable only through imagination.’

Doubts have been raised about the validity of the realis/irrealis distinction in a typological study (see 7.3), and it must be allowed that it is not necessary for the analysis of everything that is subsumed under modality. Nevertheless, the idea that notional features of realis and irrealis are grammaticalized as the typological categories of Realis and Irrealis is a useful one and will be followed in this book. (The terms are used initially as terms for the notional features that determine the grammatical status of propositions, but are also used, by extension, to refer to the typological categories ‘Realis’ and ‘Irrealis’ with initial capitals – see 1.7. They are also used, in some languages, for the related language-specific grammatical categories see 1.7 and 6.2.)

In all typological studies there is considerable variation in the ways in which languages deal with grammatical categories, and there is probably more variation with modality than with other categories. Indeed, there is variation, at a quite fundamental level, in the categories that are treated as Realis and Irrealis in different languages. One language may mark commands as Irrealis, another may mark them as Realis, while yet another may not treat them as part of a system of modality at all. There are similar variations in the treatment of futures, questions, denials, reports, etc.

A few examples will show both what kinds of feature are involved and the very different ways in which they are treated in different languages. First, English uses a modal verb to distinguish a judgment about a proposition from a categorical statement. Compare the first example, with no modal verb, with the other two:

Mary is at home
Mary may be at home
Mary must be at home

Secondly, Spanish distinguishes what is believed to be true from what is doubted, by the use of the indicative and subjunctive mood, as in Klein (1975: 356, 353):

Creo que aprende
I believe that learn+3SG+PRES+IND
‘I believe that he is learning’

Dudo que aprenda
I doubt that learn+3SG+PRES+SUBJ
‘I doubt that he’s learning’

Thirdly, forms that are actually given the grammatical labels of ‘realis’ and ‘irrealis’ in the Papuan language Amele (Roberts 1990: 371–5) are used to distinguish ‘remote past’, ‘today’s past’, ‘habitual past’ and ‘present’ from ‘future’, ‘imperative’, ‘hortative’, ‘counterfactual’ and ‘negative’. Examples of

realis ‘remote past’ and irrealis ‘future’, with the realis marker *-en* and the irrealis marker *-eb* on the verb in the subordinate clause, are:

ho bu-busal-en age qo-in
 pig SIM-run out-3SG+DS+REAL 3PL hit-3PL+REM.PAST
 ‘They killed the pig as it ran out’

ho bu-busal-eb age qo-qag-an
 pig SIM-run out-3SG+DS+IRR 3PL hit-3PL-FUT
 ‘They will kill the pig as it runs out’

1.1.2 The notion of assertion

The contrast between ‘factual’ and ‘non-factual’ or ‘real’ and ‘unreal’ is not clear enough to explain fully the distinction between Realis and Irrealis. Explanations in terms of ‘true’ and ‘untrue’ are even less satisfactory, and, in any case, it may be wise to avoid these terms in view of their logical connotations (see Bybee, Perkins, and Pagliuca 1994: 239).

There is, however, a body of literature that has proved very helpful. It has been argued that the use of the ‘indicative’ and the ‘subjunctive’, which are the traditional terms used in many European languages for the distinction between Realis and Irrealis, can be accounted for in terms of ‘assertion’ and ‘non-assertion’ (Bolinger 1968; Terrell and Hooper 1974; Hooper 1975; Klein 1975). The most succinct account is that of Lunn (1995: 430), who directly links the choice of the indicative to assertion and the choice of the subjunctive to non-assertion. Using examples of subordinate clauses in Spanish, she suggests that a proposition may be unworthy of assertion for three reasons:

- (i) the speaker has doubts about its veracity:
 Dudo que sea buena idea
 I doubt that be+3SG+PRES+SUBJ good idea
 ‘I doubt that’s a good idea’
- (ii) the proposition is unrealized:
 Necesito que me devuelvas ese libro
 I need that me return+2SG+PRES+SUBJ that book
 ‘I need you to return that book to me’
- (iii) the proposition is presupposed:
 Me alegra que sepas la verdad
 me it pleases that know+2SG+PRES+SUBJ the truth
 ‘I’m glad that you know the truth’

The important thing about this analysis is that it shows quite clearly that the choice of the Irrealis marker, the subjunctive, does not depend on the distinction between what is factual and what is not (and still less on what is

true and what is not true). It depends on the distinction between what is asserted and what is not asserted. This is particularly clear from the third example, which relates to what is presupposed, for here the proposition ('that you know the truth') is clearly factual. It is known to, presupposed by, both the speaker and the addressee. Indeed, Givón (1994: 304) comments that it would appear to be '*super realis*' since it is known to be true. It is not, therefore, factuality, certainty or truth that is at issue here. What is at issue is that nothing is being asserted, that there is no information value, because both speaker and hearer accept the proposition. It is for that reason that the proposition is treated as *Irrealis*, for propositions that are presupposed are not asserted.

There are other types of non-asserted proposition. These are discussed in 1.4.

1.2 Two basic distinctions

1.2.1 Mood and modal systems

Basically, there are two ways in which languages deal grammatically with an overall category of modality. These are to be distinguished in terms of (i) modal system and (ii) mood. Both may occur within a single language, e.g., in German, which has a modal system of modal verbs and mood (indicative and subjunctive), and in Central Pomo (see below, this section, and 6.3.2). In most languages, however, only one of these devices seems to occur or, at least, one is much more salient than the other. In some of the European languages the subjunctive appears to be falling into disuse, while in English it has virtually disappeared and, at the same time, a modal system of modal verbs has been created (see 4.3.1).

Typically with mood, all or most clauses are either *Realis* or *Irrealis*: the system is basically ('prototypically') binary. An obvious example is the contrast made in European languages between the indicative and the subjunctive, the indicative marking clauses as *Realis* and the subjunctive marking them as *Irrealis*. A similar distinction (though with rather a different set of notional-grammatical functions) is one that is mostly found in the description of some Native American languages and some languages of Papua New Guinea. In the study of these languages the distinction has been made in terms of (the grammatical categories) *realis* and *irrealis*, rather than in terms of indicative and subjunctive.

Examples of the indicative/subjunctive contrast from Spanish and of the *realis/irrealis* contrast from Amele were given in 1.1.1, but are repeated here:

Creo que aprende
 I believe that learn+3SG+PRES+IND
 'I believe that he is learning'

Dudo que aprenda
 I doubt that learn+3SG+PRES+SUBJ
 'I doubt that he's learning'

ho bu-busal-en age qo-in
 pig SIM-run out-3SG+DS+real 3PL hit-3PL+REM.PAST
 'They killed the pig as it ran out'

ho bu-busal-eb age qo-qag-an
 pig SIM-run out-3SG+DS+IRR 3PL hit-3PL-FUT
 'They will kill the pig as it runs out'

Although the contrast of Realis/Irrealis can be said to be basically or prototypically binary, there are exceptions, especially that (i) Imperative and Jussive are outside the indicative/subjunctive system of mood (see 5.4), and (ii) where there are realis and irrealis markers, some clauses may be unmarked for this distinction (see 6.5.2). However, even where there are these exceptions, there is usually still a pair of forms that essentially mark a binary contrast of notional realis and irrealis (see 7.2 for discussion).

Strictly, there is no typological difference between indicative/subjunctive and realis/irrealis. Both make express the distinction between the notional features of realis and irrealis and can be seen as instances of the typological categories of Realis and Irrealis. However, there are some differences in their distribution and syntactic functions, some of which are illustrated by the examples given above:

- (i) The subjunctive is found mainly in subordinate clauses (5.1, 5.5).
- (ii) Irrealis and realis often co-occur with other grammatical markers (6.1).
- (iii) Both are often notionally redundant, but in different ways – irrealis because of its co-occurrence with other markers, subjunctive because, in subordinate clauses, its occurrence is determined by the type of complementizer (7.1).
- (iv) Unlike the indicative/subjunctive system, realis/irrealis systems do not usually occur together with tense systems. In general past and present are marked as realis, future as irrealis.

For these reasons they will be dealt with in separate chapters, indicative/subjunctive in Chapter 5 and realis/irrealis in Chapter 6. It must be emphasized, however, that this is to some degree a decision based more on practical considerations than on a clear set of criteria for the distinction (see

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7.1). (For the problem of terminology and the decision to use the terms ‘realis’ and ‘irrealis’ for the notional features, the grammatical markers and the typological categories, see 1.7.)

Within modal systems different kinds of modality are distinguished within a single system of commuting terms. A clear example of such a system is that of the English modal verbs when they are used to make judgments about the factual status of a proposition as in:

Kate may be at home now
Kate must be at home now
Kate will be at home now

These can be interpreted and labelled as:

a possible conclusion	speculative
the only possible conclusion	deductive
a reasonable conclusion	assumptive

A rather different system is the evidential system of Central Pomo (Mithun 1999: 191), where, as well as a (rarely used) unmarked form, there are forms with markers for general knowledge, first-hand personal experience (usually visual), auditory evidence, hearsay and inference:

čʰéemul
rain fell
‘It rained’
čʰéemul-ʔma
rain fell-GEN.KNOW
‘It rained’ (that’s an established fact)
čʰéemul-ya
rain fell-VIS
‘It rained’ (I saw it)
čʰéemul-nme·
rain fell-AUD
‘It rained’ (I heard it)
čʰéemul-ʔdo
rain fell-HSY
‘It rained’ (I was told)
čʰéemul-ʔka
rain fell-INF
‘It rained’ (everything is wet)

(There are also two markers for personal experience of actions performed by the speaker and for actions that affected the speaker. These are discussed in 2.2.5.) For a similar system in Tuyuca see 2.2.1.

The distinction between Realis and Irrealis is an integral part of systems of

mood. It can generally usually be applied to modal systems as well, in that notionally the terms in a modal system are irrealis, and there is usually an unmarked realis form. In English, for example, there are the three types of (irrealis) modality that are marked by the modal verbs, but there is also a realis form, the ‘declarative’ that contains no modal verb:

Kate is at home now

Similarly, in Central Pomo it would seem that there is an unmarked realis form and five irrealis forms indicating different kinds of evidence for the proposition.

However there are two types of situation in which it is less easy to apply a simple Realis/Irrealis distinction. First, there are some languages with a modal system but no unmarked (realis) declarative, so that it would seem that either the system contains realis and irrealis forms, or there is no realis form in the language – see 2.7.1. Secondly, although Central Pomo has a modal system in which it is reasonable to see the Realis/Irrealis distinction, it also has a completely independent system of mood with realis and irrealis markers (see 6.3.2).

There is one final point. It is not always possible to draw a clear distinction between mood and modal system, since, in some languages, the overall system of modality has characteristics of both – see 6.5.3 and 6.5.4.

1.2.2 *Propositional and Event modality*

There is a clear contrast in the notional features involved in the following pairs of examples (the first two already quoted in 1.2.1 and the second two indicating permission and obligation):

Kate may be at home now

Kate must be at home now

Kate may come in now

Kate must come in now

The distinction is usually made in terms of ‘epistemic’ and ‘deontic’ modality, and illustrated by the use of paraphrases using ‘possible’ and ‘necessary’:

It is possible (possibly the case) that Kate is at home now

It is necessarily the case that Kate is at home now

It is possible for Kate to come in now

It is necessary for Kate to come in now

The important distinction between the two pairs is indicated by the words ‘that’ and ‘for’. This clearly suggests that the first pair are concerned with the speaker’s judgment of the proposition that Kate is at home, whereas the

second are concerned with the speaker's attitude towards a potential future event, that of Kate coming in. For that reason a basic distinction may be made between 'propositional modality' and 'event modality'. (This contrast is essentially the same as that of Jespersen (1924: 329–31), between 'containing an element of will' and 'containing no element of will'.)

This distinction does not apply only to epistemic and deontic modality, as illustrated here (where it is largely associated with the notion of possibility and necessity – see 4.1). Other types of modality are to be included, especially dynamic modality (Palmer 1979: 36–7; 1986: 102–3) within event modality and evidential modality within propositional modality. (Chung and Timberlake (1985: 244) use the term 'epistemological' for 'evidential', but 'epistemological' is too similar to 'epistemic' to be a useful term.) The terms 'epistemic', 'deontic' and 'dynamic' are taken from Von Wright (1951: 1–2).

In summary, epistemic modality and evidential modality are concerned with the speaker's attitude to the truth-value or factual status of the proposition (Propositional modality). By contrast, deontic and dynamic modality refer to events that are not actualized, events that have not taken place but are merely potential (Event modality).

1.3 The classification of modality in modal systems

1.3.1 Epistemic and Evidential

Epistemic and Evidential systems are the two main types of Propositional modality and were exemplified in 1.2.1 from English and Central Pomo. The essential difference between these two types is (as is implicit in the discussion) that with epistemic modality speakers express their judgments about the factual status of the proposition, whereas with evidential modality they indicate the evidence they have for its factual status. The English examples were of three kinds of (epistemic) judgment – speculative, deductive and assumptive. They may be regarded as examples of the typological categories Speculative, Deductive and Assumptive. The Central Pomo examples included five types of evidence: 'general knowledge', 'first-hand personal experience (usually visual)', 'auditory evidence', 'hearsay' and 'inference'. The first four of these exemplify the evidential categories of Reported(gen), Visual, Auditory and Reported(1), while the last ('inference') is the epistemic category Deductive.

One immediate observation is that the typological category Deductive appears in both systems. There is an obvious reason for this – deduction, as expressed by English *MUST* usually suggests that the judgment was based on evidence. Coates (1983: 41) comments, 'In its most normal usage, Epistemic

1.3 The classification of modality in modal systems

MUST conveys the speaker's confidence in the truth of what he is saying, based on a deduction from facts known to him (which may or may not be specified).' Thus *Kate must be at home* may be a judgment based on the observation that she is not in her office. In a similar way, the 'inference' form in Central Pomo indicates that there is indirect, usually visual, evidence for the proposition. Notionally these are very close and can both be treated as Deductive. Because both a deduction and evidence are involved, it is reasonable to treat the English form in a judgment system and the Central Pomo form in an evidential system. However, purely for the purpose of discussion and exemplification, Deductive will be discussed in the section on Epistemic (2.1).

A second observation is that evidentials may occur within what is primarily an epistemic (judgment) system. The modal verbs in German, for instance, which are similar to those of English, express the judgments Deductive and Speculative, but SOLLEN and WOLLEN are used as evidentials (Hammer 1983: 231, 232):

Er soll	steinreich sein
He SOLLEN+ 3SG+PRES+IND	very rich be
'He is said to be extremely rich'	
Er will	eine Mosquito abgeschossen haben
he WOLLEN+3SG+PRES+IND	a Mosquito shot down have
'He claims to have shot down a Mosquito' (plane)	

The first indicates what is reported, the second what is claimed by the person designated by the subject of the sentence. In both cases the speaker merely provides the evidence for the proposition.

Within Evidential modality there are two main types, which may be identified as Sensory and Reported, the former subsuming Visual, non-Visual and Auditory (at least) and the latter, three sub-types of report. It could be argued that these form distinct types of propositional modality, so that there are three types – Epistemic, Reported and Sensory. In practice, however, the binary distinction between Epistemic and Evidential modality will be maintained in this book.

1.3.2 Deontic and Dynamic

Deontic and Dynamic are the two main types of Event modality. In the simplest terms the difference between them is that with deontic modality the conditioning factors are external to the relevant individual, whereas with dynamic modality they are internal. Thus deontic modality relates to obligation or permission, emanating from an external source, whereas

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dynamic modality relates to ability or willingness, which comes from the individual concerned. The distinction can be seen in:

John may/can come in now	(permission)
John must come in now	(obligation)
John can speak French	(ability)
John will do it for you	(willingness)

The first two exemplify the typological categories of (Deontic) Permissive and Obligative. The second pair exemplify the categories of (Dynamic) Abilitive and Volitive. This is, however, an oversimplification in three respects.

First, although Deontic modality stems from some kind of external authority such as rules or the law, typically and frequently the authority is the actual speaker, who gives permission to, or lays an obligation on, the addressee. It is for this reason that Searle (1983: 166) speaks of ‘directives’, which he defines as ‘Where we try to get [others] to do things’.

Secondly, Ability, one of the categories of Dynamic modality, has to be interpreted rather more widely than in terms of the subjects’ physical and mental powers, to include circumstances that immediately affect them (but not, of course, deontic permission). This can be seen in the contrast between deontic and dynamic CAN, where deontic CAN indicates permission, while dynamic CAN may indicate not merely ability, but also the possibility in a more general sense, as shown by:

He can go now	(Deontic: I give permission)
He can run a mile in five minutes	(Dynamic: he has the ability)
He can escape	(Dynamic: the door’s not locked)

Thirdly, there is another category, that of Commissive, which is defined by Searle as ‘where we commit ourselves to do things’, and exemplified by English SHALL:

You shall have it tomorrow

Since the conditioning factor for the action lies with the speaker, not the subject, this is best seen as another type of Deontic modality.

It must be admitted, however, that Commissive and the two forms of Dynamic modality are much more clearly marked in English than in many other languages.

1.4 Other types of modality

The sub-types of modality discussed in the last section are mainly relevant to the analysis of modal systems. There are, however other grammatical

categories that are associated with modality, especially (but not solely) with respect to mood.

1.4.1 Presupposed

The treatment of presupposed propositions as irrealis was discussed in 1.1.2, with the example:

Me alegra que sepas la verdad
me it pleases that know+2SG+PRES+SUBJ the truth
'I'm glad that you know the truth'

The use of the subjunctive in subordinate clauses, as in this example, is, perhaps, the clearest indication of its use for presupposition, since it is clear that nothing is being asserted by the proposition in the subordinate clause. Yet examples of presupposed propositions being marked as irrealis can be found in main clauses too, e.g., the use of the subjunctive in Italian in a concessive clause, where the speaker admits to the truth of the proposition (see 5.2.5):

sia pure come dici tu ma io non vengo
be+3SG+PRES+SUBJ perhaps as you say but I not come
'It may be as you say, but I'm not coming'

It is also relevant that, as the translation shows, English uses a modal verb (may) here. More striking is the marking of the category 'admirative' as irrealis in Caddo (Oklahoma – Chafe 1995: 357):

hús-baʔa-sayi-k'awih-saʔ
ADM-1+BEN+IRR-name-know-PROG
'My goodness he knows my name'

This Chafe finds 'surprising', yet it is clear that the speaker is not asserting the proposition, but is indicating his surprise at what is obvious to the addressee (see 6.6.7).

'Presupposed' is very similar to what Kiparsky and Kiparsky (1971) call 'factive', with the argument that REGRET, RESENT and many other verbs and constructions are 'factive predicates.' However, the term can be misleading, as can be seen in the remark by Coates (1983: 235) that 'KNOW is the classic example of a factive predicator'. For although know clearly suggests that the speaker accepts the information as factual, presupposition is not involved – it does not suggest that the addressee equally accepts the factuality. The complements of KNOW are not Factive in the presuppositional sense.

1.4.2 Negative and Interrogative

Negatives and Interrogatives are classed together as being 'non-assertive' by Quirk, Greenbaum, Leech and Svartvik (1985: 83) for grammatical reasons

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in English – their association with certain ‘non-assertive’ forms such as *any* and *yet*. It is, therefore, not surprising that they sometimes occur within modal systems or are marked as irrealis where there is marking of mood.

A striking example is to be found in what is clearly an epistemic system in Imbabura, a variety of Quechua (Cole 1982: 164), where the same marker is used for both question and negation. Cole lists the markers as:

ma(rí)	emphatic first-hand information
mi	first-hand information
shi	conjecture
cha(rí)	doubt
chu	Yes–No question
chu	negation

The first two are probably to be seen as indicating evidence from the senses, while the second two are markers of judgments. The third pair shows that the same marker is used for Interrogative and Negative. Examples are:

kan-paj ushi-wan	Agatu-pi-mi
you-of daughter-with	Ageto-in-F.INF
‘I met your daughter in Ageto’	
kaya-shi	kan-paj churi shamu-nga
tomorrow-CONJ	you-of son come-3+FUT
‘I suppose your son will come tomorrow’	
Juzi-ka kitu-man	chaya-shka-chá
José-TOP	Quito-to arrive-PERF-DUB
‘Perhaps José has arrived in Quito’	
mayistru-chu	ka-ngui
teacher-QUES	be-2sg
‘Are you a teacher?’	
ñuka-ka	mana chay llama-ta shuwa-shka-ni-chu
I-TOP	not that sheep-ACC steal-PERF-2SG-NEG
‘I didn’t steal that sheep’	

Similarly, but in a system of mood (realis and irrealis), Interrogative and Negative are both marked as irrealis in Caddo (Chafe 1995: 334–5; see 6.6.3, 6.6.4):

kúy-t’ayi-bahw
neg-1+AG+IRR-see
‘I don’t see him’
sah?yi-bahw-nah
2+AG-+IRR-see-PERF
‘Have you seen him?’

It may be thought that interrogatives and negatives are never marked as Irrealis in European languages. It is true that there is no indication of them being treated as Irrealis in main clauses, but there is evidence of such treatment in subordinate clauses. Thus reported (indirect) questions are expressed with the subjunctive in Latin (see 5.2.4), while a negated verb of saying in Spanish and Italian is also usually followed by the subjunctive (see 5.2.3).

1.4.3 *Wishes, fears, etc.*

The status within modality of wishes and fears is a little more obscure, although it is clear that notionally they express attitudes towards propositions whose factual status is not known or propositions that relate to unrealized events. They are thus partly deontic, partly epistemic. There are plenty of examples of wishes and fears being expressed by the subjunctive in subordinate clauses in Latin and other languages (see 5.3.3), but the subjunctive is also used in main clauses, as in Latin and Classical Greek:

modo valeres	(Cic. Att. 9.22)
only be well+2SG+IMPERF+SUBJ	
'If only you were well'	
mé: sóus diaphthéire:i gámous	(Eur. Alc 315)
not your ruin+3SG+PRES+SUBJ marriage	
'I'm afraid she may ruin your marriage'	

Similarly in the West African language Fula (Arnott 1970: 299ff.), wish is one of the features signalled by what is identified as subjunctive:

njuutaa	balde
be.long+2SG+SUBJ in days	
'May you live long!'	

Expressions of wanting, desire and preference relate similarly to unrealized events and may be marked as irrealis, but generally only in subordinate clauses, as in Spanish:

quiero que estudias	más
I want that study+3SG+PRES+SUBJ more	
'I want you to study more'	

1.4.4 *Past tense as modal*

There is one further feature that affects modal systems – the use of past tense forms to express 'unreality', 'tentativeness', 'potentiality', etc. This often functions independently of mood, but may interact with it. This is discussed in Chapter 8, but two illustrations may be given.

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First, three of the English modal verbs have what are formally present and past forms:

WILL	<i>will</i>	<i>would</i>
CAN	<i>can</i>	<i>could</i>
MAY	<i>may</i>	<i>might</i>

All three past tense forms can be used with past time reference, but past time reference is not the most common function of the past tense forms. Rather they are used to express greater tentativeness, as in:

He'll be there now
He'd be there now
He may be there now
He might be there now
He can't be there now
He couldn't be there now

Secondly, in a large number of languages, a distinction is made between 'real' and 'unreal' conditionals, as can be illustrated by English:

If the children are here, John will be happy
If the children were here, John would be happy

If Mary comes, John will stay
If Mary came, John would stay

The second example of each pair is unreal. In the first of these second examples (which refers to a present situation, where the facts are known), there is a clear implication that the children are not here so that John will not be happy. In the second, which refers to the future, where the facts are not knowable, the implication is that it is unlikely that Mary will come. This distinction, as can be seen from the examples, is marked by tense, with present tense forms in both clauses of the real conditionals and past tense forms in those of the unreal ones.

This use of the past tense will be referred to as the 'modal-past'.

1.4.5 Complex systems

Ideally, there would be discrete systems of modal markers, such that each of the typologically relevant modal systems was in a one-to-one correspondence with a formal system in each language. In practice, the situation, as might be expected in a typological study, is far less simple and neat. There are three ways in which the systems are far from being orderly and simple.

First, two apparently different systems may use the same (or most of the same) set of markers. This is so for the epistemic and deontic/dynamic systems of English, which use the same modal verbs, especially MAY, CAN and

MUST, as was illustrated in 1.2.1. (Indeed many languages have the same forms for epistemic and deontic modality – see 4.2.1.) CAN, for instance, may be used as epistemic, deontic or dynamic:

He can't be in his office now (epistemic)
 He can come in now (deontic, permission)
 He can run a mile in four minutes (dynamic, ability)

However, there are some formal distinctions between the uses, e.g., that for positive epistemic modality MAY is used rather than CAN, that MAY is an alternative to CAN only for deontic, and that the past tense form *could* is used to refer to past time only when it is dynamic. These features are discussed in much more detail in 4.2.3.

Secondly, in a few cases, notionally identical modal categories may be treated as members of one system in one language, and of another system in another language. This is so with Deductive which, because it involves a judgment based upon evidence, may be a term in a judgment system or in an evidential system, as in English and Central Pomo respectively (see 1.3.1).

Thirdly, a single formal system often contains forms that belong to two or more modal systems. Thus in the German system of modal verbs not only do most of the modal verbs function as both epistemic and deontic, but also there are forms (SOLLEN and WOLLEN) that are clearly evidentials (see 1.3.1). The system in Imbabura (Cole 1982: 164, described in 1.4.2) has an even more varied set of terms.

Fourthly, there may not be a simple one-to-one relation between the terms of a formal system and some of the typologically relevant notional features. Thus in Lega (Bantu, Eastern Zaire – Botne 1997: 511–22, and personal communication), there are three particles, whose primary functions seem to be those of Reported, Speculative and Sensory. These are familiar categories, but there are additional features associated with two of them. The term for Reported often expresses doubt or disbelief:

nkumgwágá (bónɔ), ámbɔ bazongo bé kulyágá merende
 I hear that whites EV eat frog
 'I hear that Westerners eat frogs (though I find that unlikely)'

The term used for Sensory (Visual and Auditory) may also express both confidence of the kind associated with Deductive and reports that can be trusted:

ampó mbulá zésaloka
 EV rain rain
 'It is already raining' (I can see it)

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ampó ékukúrá momponge
EV she.is.pounding rice
'She's assuredly pounding rice' (I can hear it)

ampó Kisangá éndilé kw isòkò; kikápu kyágé také gáno
EV Kisanga went to market; basket is not here
'Kisanga surely went to the market; her basket is not here'

ampó Moké ákorwa
EV Moke tired
'Moke is tired' (Moke told me)

See 2.2.4 for more details.

Fifthly, the same formal system may contain not only modal categories, but others such as those of tense, e.g., in Ngiyambaa (Donaldson 1980: 159–62):

- (i) *Imperative*
ñindu bawuñ-ga yuwa-dha
you+NOM middle-LOC lie-IMP
'You lie in the middle!'
- (ii) *Past*
yuruñ-gu ñidjiyi
rain-ERG rain+PAST
'It rained'
- (iii) *Present*
yuruñgu ñidja-ɾa
rain-ERG rain-PRES
'It is raining'
- (iv) *Irrealis*
yuruñ-gu ñidja-l-aga
rain-ERG rain-CM-IRR
'It might/will rain'
- (v) *Purposive*
ñadhu bawuñ-ga yuwa-giri
I+NOM middle-LOC lie-PURP
'I must lie in the middle'
- yuruñ-gu ñidja-l-i
 rain-ERG rain-CM-PURP
 'It is bound to rain'

In fact, the situation in Ngiyambaa is even more complex than suggested above (Donaldson 1980: 251–65). There is a set of clitics that relate to features that are clearly modal and to others that are clearly not. Donaldson divides them into three types, 'Belief' clitics, 'Knowledge' clitics and 'Evidence' clitics. Examples are:

‘Belief’ clitics:

- (i) *Assertion*
waɣaɣy-ba:-na yana-nhi
NEG-ASS-3ABS walk-PAST
‘He didn’t walk’ (again)
- (ii) *Categorical assertion*
guni:m-baɣa-nu: balu-y-aga
mother+ABS-CATEG.ASS-2OBL die-CM-IRR
‘Your mother is bound to die’
- (iii) *Counter-assertion*
guyan-baga:-dhu gaɣa
shy+ABS-CNTR.ASS-1NOM be+PRES
‘But I’m shy!’
- (iv) *Hypothesis*
galir-ɲinda-gila ɲiyanu balu-y-aga
water-CARIT-HYPOTH we+PL+NOM die-CM-IRR
‘We’ll probably die for lack of water’

‘Knowledge’ clitics:

- (i) *Exclamative*
minja-wa:-ndu dha-yi
what+ASS-EXCL-2NOM eat-PAST
‘What did you eat?’ / ‘You ate what?’

guya-wa:-ndu dha-yi
fish+ABS-EXCL-2NOM eat-PAST
‘So you ate a fish!’ / ‘What? You ate a fish’
- (ii) *Ignorative*
minjaŋ-ga:-dhu dha-yi
what+ABS-IGNOR-1NOM eat-PAST
‘You ate something, I don’t know what’ / ‘I don’t know what you ate’

guya-ga:-ndu dha-yi
fish+ABS-IGNOR-2NOM eat-PAST
‘Did you eat a fish’ / ‘You ate a fish, I don’t know’

‘Evidence’ clitics:

- (i) *Sensory evidence*
ɲindu-gara girambiyyi
you+NOM-SENS.EVID sick+PAST
‘One can see you were sick’

gabuga:-gara-lu ɲamumiyyi
egg+ABS-SENS.EVID-3ERG lay+PAST
‘It’s laid an egg by the sound of it’

(ii) *Linguistic evidence*

ṅindu-dhan girambiyi
you+NOM-LING.EVID sick+PAST
'You are said to have been sick'

The evidence clitics are clearly markers of evidentials, but most of the others are, at best, marginally modal.

1.5 Typology

It is now well understood that grammatical typology cannot be undertaken on the basis of purely formal, grammatical marking. As Croft (1995: 88) says, 'In general, cross-linguistic identification cannot be accomplished on purely formal (structural) grounds for two reasons. First, variation across languages is too great . . . Second, formal definitions are internal to the structural system of a single language . . . For these reasons, typologists generally use definitions . . . that are "external" to the linguistic system, that is semantic, pragmatic or discourse-based definitions'. (The features involved in the definitions are referred to in this book as 'notional', thereby avoiding any debate about semantic vs pragmatic, etc.)

It may be useful to consider briefly the ways in which notional and formal criteria may be related.

First, a single category may be identified across languages, e.g. Speculative in English, Tamil, Ngiyimbaa and Quechua (2.1.2).

Secondly, a category may be distinguished formally in one language, but may be only one of several categories indicated by a single marker in another. Thus, Future is uniquely marked in Amele, but is only one of five categories indicated by an irrealis marker in Manam (both of them Papuan languages – see 6.1).

Thirdly, a single marker may indicate a number of different notional categories that are not formally distinguished in any language, but are nevertheless of typological interest. Thus Presupposed is indicated by subjunctive in Spanish (5.2.5). It is not uniquely distinguished elsewhere, but is of typological interest in that it accounts for irrealis marking in Caddo (6.6.7).

Finally, it is sometimes useful to compare a set of forms that is formally marked with a set whose formal status is doubtful in another language. Thus the modal verbs in English are a formally defined set (4.2.1), but it is useful to compare them with similar sets not only in German, where there are some formal markers, but also in the Romance languages where their formal status is very much in doubt (see 4.2.2).

1.6 Grammatical markers

The grammatical markers of modality are very varied, and it would be unhelpful to exemplify them in detail here. There is, however, a discussion of the grammatical status of modal verbs in 4.2 and of the markers of realis and irrealis in 6.3.1. However, a few comments may be made.

Basically there are three types of marker: (i) individual suffixes, clitics and particles, (ii) inflection and (iii) modal verb.

Some examples of individual items have already been given, e.g. those of Imbabura in 1.4.2. They are found as markers both of terms in modal systems and of mood.

Wholly inflectional forms that involve the form of the lexical verb itself as well as tense, mood and person-number are characteristic of indicative and subjunctive. There are, however, simultaneous markers of both mood and person-number in some languages that are described in terms of realis/irrealis, and there is at least one such language (Takelma) where the forms involve the lexical verb also (see 6.3.1).

The use of modal verbs seems to be confined to modal systems.

There are also some syntactic conditions on the occurrence of mood markers. In 6.3.2 there is one example of a Native American language and several examples of Papuan languages where realis and irrealis are marked only in 'linked' clauses.

Finally, in some languages there are large sets of particles or clitics. Thus Mithun (1986: 99) says that there are over fifty different evidential particles in Cayuga (Northern Iroquoian). There is no easy way of deciding to what extent such sets are relevant to a grammatical study of modality though it is likely that most of the members of the set will not be modal in any of the senses discussed here. Such large sets are usually best regarded as lexical rather than grammatical. But some of the items may be of typological interest.

1.7 Terminology

Some remarks about terminology and notation have already been made, but it may be useful to consider the issues briefly here.

In a typological study there are three distinct but related sets of concepts (three levels of discourse) – the typological categories that are seen as common to various different languages, the grammatical markers associated with them in individual languages and the notional features that justify the typological identification. It might seem wise to use different terminology and different notation for each of the three levels, but this is virtually impossible for several reasons, and would probably bring confusion rather

than clarity; it is actually both normal and helpful to use the same terms at all levels. For instance, grammar books talk about the ‘future tense’ and say that it indicates the (notional) future. Indeed, the names for grammatical terms are generally derived from the notional features associated with them. There are, admittedly, alternative names for some of the categories and Lyons (1977: 30), following Bar-Hillel (1970: 370), suggests that a distinction should be made between ‘declarative’, ‘interrogative’ and ‘imperative’ on the one hand and ‘statement’, ‘question’ and ‘command’ on the other. This advice is not consistently followed here for two reasons. First, there are no obvious pairs of terms for some of the categories, and, moreover, there is little consistency among writers. Secondly, provided ambiguity can be avoided, it is sometimes actually helpful to use the same terminology. It is more illuminating, for instance, to say that the Latin subjunctive with a negative is used for a negative imperative (and so compare it with the morphologically defined imperative) than to say that it is used for a prohibition.

One convention that will be used is the use of capitals to distinguish the typological categories, e.g., *Realis*, *Irrealis* (1.1.1), *Speculative*, *Deductive*, *Assumptive*, etc. (1.3.1). However, this convention will not always be used, for two reasons. First, the same terms are often used for the language-specific grammatical categories, e.g., imperative and future, which may be instances of Imperative and Future. In such cases it is simpler to use the form without initial capitals, unless it is important to stress the typological status. (One case where the distinction is important is with (grammatical) *realis* and *irrealis*, since indicative and subjunctive are also instances of (typological) *Realis* and *Irrealis*.) Secondly, it seems hardly worthwhile using capitals throughout for the terms to describe types of modality – event, propositional, epistemic, deontic and dynamic, which refer to the main typological classes, and never to language-specific grammatical categories. Initial capitals will be used only when they are first mentioned.

There is no simple way of distinguishing the notional and grammatical categories, except by using terms such as ‘question’ and ‘command’, where these are readily available. One possibility might have been the use of quotation marks for the notional categories, but quotation marks are needed to indicate those terms used by individual authors that differ from those used in this book. Instead, where there is a danger of confusion the glosses ‘notionally’ or ‘notional’ can be added to indicate notional categories.

There is also a problem with the facts that there is wide variation in the terminology used by different authors and that their terminology is often different from that used in this book. An extreme case is the use of the terms

‘report’, ‘quotative’ and ‘hearsay’, which are used by different authors in various ways (see 2.2.2). It is for that reason that quotes will be used for all such authors’ terms, if they are different from those used in this book. The authors’ terms will also be used in the glosses to the examples (where they are shown in abbreviated small capitals), but the abbreviations used for these categories will be standardized (a list is given in the preliminary pages).

In summary:

- (i) The terms for the grammatical categories will generally be the same as those for the typological categories, though an exception is made in the use of ‘indicative/subjunctive’ in some languages and ‘realis/irrealis’ in others to relate to the typological Realis/Irrealis.
- (ii) Typological categories may be identified by the use of initial capitals. There will be no variation in the terms used for them.
- (iii) Where necessary notional categories will be glossed with ‘notional(ly)’.
- (iv) Terms used by individual authors will be indicated by quotation marks. These terms will also be shown (in small capitals) in the glosses to the examples but in a standardized abbreviated form.

Something needs to be said too about the choice of terms used for the typological categories. The guiding principles here are to avoid confusion (such as that created by ‘report’, ‘quotative’ and ‘hearsay’), and to use terms that are reasonably familiar and not too off-putting to the reader. The issues that were considered (and the decisions that were made) are as follows:

- (i) There are some familiar grammatical terms, such as Imperative and Jussive, for which there are no sensible alternatives.
- (ii) There are other terms with both adjectival and nominal forms such as Interrogative/Interrogation and Negative/Negation. For consistency with (i) the adjectival form will be used.
- (iii) Most of the terms have the ending *-ive* but *Visual* is preferred to the etymologically correct, but wholly invented, *Visive*. Two other adjectival forms are Future and Conditional.
- (iv) There is no obvious term for what is presupposed – ‘Presuppositive’ is too unfamiliar and clumsy. The participial (*-ed* form) ‘Presupposed’ is preferred.
- (v) For the confusing ‘report’, ‘hearsay’ and ‘quotative’ the term ‘Reported’ (with *-ed*) is preferred to ‘Quotative’ (with *-ive*). There are, however, three sub-types of Reported and to avoid

the confusion that would be created by using the other terms, these will be referred to as ‘Reported (2)’, ‘Reported (3)’ and ‘Reported (Gen)’, to indicate second-hand information, third-hand information and what is generally said to be true (see 2.2.2).

- (vi) No adjectival forms are suggested for wishes and fears or for habitual past, though ‘Desiderative’ and possibly ‘Timitive’ might have been used for the first two.

1.8 Summary of basic categories

It may be useful to summarize the main typological categories and their classification, already mentioned.

Within modal systems these categories can, not surprisingly, be arranged neatly, but there are other categories that are not easily classified, most of them found mostly, but not entirely, with Realis/Irrealis mood.

Propositional modality

Epistemic

Speculative

Deductive

Assumptive

Evidential

Reported: Reported (2), Reported (3), Reported (Gen)

Sensory: Visual, non-Visual, Auditory

Event modality

Deontic

Permissive

Obligative

Commissive

Dynamic

Abilitive

Volitive

Other important categories, that may be marked as Irrealis but are mostly found with mood, are: Future, Negative, Interrogative, Imperative-Jussive, Presupposed, Conditional, Purposive and Resultative, Wishes (‘Desiderative’) and Fears (‘Timitive’?) and, less commonly, Habitual-Past.

1.9 Organization

This book is basically in two parts.

The first part, Chapters 2, 3 and 4, is concerned with modal systems. Chapter 2 deals with propositional modality, with main sections on epistemic and evidential modality, and Chapter 3 with event modality, with main

sections on deontic and dynamic modality, although in both chapters there are some sections dealing with other, but related, matters. Issues affecting both types of modality are considered in Chapter 4.

The second part, Chapters 5, 6 and 7, is concerned with mood. Chapter 5 deals with indicative and subjunctive and Chapter 6 with realis and irrealis, while issues affecting them both are considered in Chapter 7.

Chapter 8 deals with the issue of past tense being used as modal.

2

Modal systems: Propositional modality

As was suggested in 1.2.2, epistemic modality and evidential modality are concerned with the speaker's attitude to the truth-value or factual status of the proposition and may thus be described as 'propositional modality'. The basic difference between epistemic modality and evidential modality is that with epistemic modality speakers make judgments about the factual status of the proposition, whereas with evidential modality they indicate what is the evidence that they have for it.

However, as was noted in 1.3.1, there are two respects in which epistemic and evidential systems are, in practice, not always wholly distinct. First, the category Deductive, which involves both a judgment and evidence, may occur in both types of systems, and this is true also, though less commonly, of other categories (e.g., Assumptive in Tuyuca – see 2.1.3). It would not be helpful to deal with these twice, once in a section on epistemic modality and once in a section on evidential modality and, for that reason, they will be considered in the section on epistemic modality (2.1). The section on evidential modality (2.2) will deal with purely evidential features, most importantly with Reported and Sensory. Secondly, evidential categories sometimes occur within what are primarily epistemic (judgment) systems (e.g., the uses of *SOULEN* and *WOLEN* in German that were exemplified in 1.3.1), but these can still be considered in the section on evidentials (2.2.2).

Later sections deal with what seem to be other types of propositional modality.

2.1 Epistemic modality

2.1.1 *The three types*

There are three types of judgment that are common in languages, one that expresses uncertainty, one that indicates an inference from observable evidence, and one that indicates inference from what is generally known. These may be identified typologically as Speculative, Deductive and Assump-

tive, respectively. An alternative name for the first is ‘Dubitative’, but, since the forms do not generally indicate positive doubt (Bybee 1985: 179), ‘Speculative’ is to be preferred.

There seem to be few languages that have a system with all three markers, but English is an exception, using the three modal verbs *MAY*, *MUST* and *WILL*:

John may be in his office
John must be in his office
John’ll be in his office

The first indicates that the speaker is uncertain whether John is in his office. With the second, the speaker makes a firm judgment, on the basis of evidence, e.g. that the office lights are on, that he is not at home, etc. With the third, the judgment is based on what is generally known about John, e.g. that he always starts at eight, that he’s a workaholic, etc. (a reasonable conclusion).

MUST seems to draw a firmer conclusion than *will*, so that the three might be explained in terms of:

a possible conclusion
the only possible conclusion
a reasonable conclusion

There are, it may be argued, two not entirely compatible contrasts in the English system. The first involves the strength of the conclusion, and distinguishes between what ‘may’ be and what ‘must’ be, i.e., between what is epistemically possible and what is epistemically necessary (see 4.1.2). This distinguishes Speculative (*MAY*) and Deductive (*MUST*). The second distinguishes between an inference from observation and an inference from experience or general knowledge, i.e., between Deductive (*MUST*) and Assumptive (*WILL*). In many languages it seems that one of these contrasts is either the only contrast in an epistemic system or, at least, the more important contrast. Not surprisingly, the single contrast between inference from observation and inference from experience is mostly found in languages with evidential systems.

The common element in the two contrasting pairs is Deductive, but the notional features associated with it differ somewhat depending on which contrast is the more salient. For these reasons the two contrasting pairs will be considered in separate sections, first dealing with Speculative and Deductive, the second with Deductive and Assumptive.

However, although these three are the most common categories of propositional modality, there are different systems in some languages.

Ladakhi (Tibeto-Burman), for instance, has a marker for ‘inference’ (= Deductive) in a basic set of modal markers, but also a second set of markers that distinguish four types of inference. However, these involve evidential features and are discussed in 2.2.1.

2.1.2 *Speculative and deductive*

English then, has the contrast of Speculative and Deductive, as in:

John may be in his office
John must be in his office

There is one further point. These examples illustrate judgments about propositions relating to the present, but with MAY (and, less commonly, MUST) the proposition can refer to the future:

John may come tomorrow

They may also be used with HAVE plus the past participle for propositions relating to the past:

John may/must have been in his office

Many of the European languages have the same contrast, e.g., Danish:

Det kan være sandt
that may+3SG+PRES be true
‘That may be true’
Det må være sandt
that must+3SG+PRES be true
‘That must be true’

Danish is a Germanic language like English, but the same contrast is found in the Romance language, Italian:

Può essere nell’ufficio
can/may+3SG+PRES be in the office
‘He may be in the office’
Deve essere nell’ufficio
must+3SG+PRES be in the office
‘He must be in the office’

It is found also in Modern Greek, but through the use of impersonal verbs:

bori na ine sto yrafio tus
can+IMPERS that they are in the office their
‘They may be in their office’
prepi na ine sto yrafio tus
must+IMPERS that they are in the office their
‘They must be in their office’

This is not a feature of modern European languages alone. In Tamil (Dravidian, S. India and Sri Lanka – Thiagarayan 1981; Asher 1982) there are a number of potentially modal forms, but only two are in the form of a verbal suffix, *-laam* and *-um*. These Asher (1982: 16772) calls ‘debitive’ and ‘permission’, though they are used epistemically as Speculative and Deductive, as well as deontically (see 3.2.1). In the example below the term ‘concessive’ is used to indicate that ‘permission’ is used in the epistemic sense:

Kantacaami vantaalum vara-laam
Kandaswami come+CONCESS come-PERM
‘Kandaswami may perhaps come’

Gaṇēecan ippa Mannaarkuṭiyile irukka-ṇum
Ganesan now Mannargudi+LOC be-DEB
‘Ganesan must be in Mannargudi now’

The two types of judgment are often found in complex systems of the type that were considered in 1.4.5, e.g., in Ngiyambaa (Donaldson 1980: 159–62):

yuruṇ-gu ṇidja-l-aga
rain-ERG rain-CM-IRR
‘It might/will rain’

yuruṇ-gu ṇidja-l-i
rain-ERG rain-CM-PURP
‘It is bound to rain’

It was also noted in 1.4.2 that Imbabura has markers for ‘conjecture’ and ‘doubt’, but a clearer illustration is to be found in Inga, which, like Imbabura, is a variety of Quechua (Levinsohn 1975: 14–15, 22). The distinction is made between ‘action deduced’ and ‘action speculated’, the full system being:

mi	action witnessed – affirmative
chu	action witnessed – negative
si	action reported to the speaker
cha	action deduced by the speaker as having occurred
(char-	probability reinforced)
sica	action speculated as possible by the speaker

Levinsohn calls these markers ‘deictic aspect enclitics’, because, when not attached to the main verb, they indicate the ‘rheme’ – what is normally new information, but this is not relevant to the issue of modality. He gives examples of them, but with no indication of the modal status in the translations and without glosses for the enclitics:

- nis puñuncuna-mi (15)
 then they.slept-MI
 ‘Then they slept’ (witnessed affirmative)
- chica nuca mana yachani-chu (31)
 that I not I.know-CHU
 ‘I don’t know that myself’ (witnessed negative)
- chacapi-si yallinacú (22)
 on.bridge-*si* they.were.crossing
 ‘They were crossing on the bridge’ (reported)
- chihoraca mal-cha cado carca fiide del bautismo (23)
 at.that.time bad-CHA it was baptismal certificate
 ‘At that time my baptismal certificate must have been incomplete’
 (deduced)
- chipica diablo-char ca (19)
 there devil-CHAR it.was
 ‘A devil was presumably there’ (deduced, probability reinforced)
- yuyanacurca-sica chica cajta huamra (27)
 they.were.thinking-SICA that be child
 ‘I suppose they were thinking he was a child’ (speculated)

It will be noted that there is a further distinction here for Deductive in terms of probability – ‘deduced’ and ‘deduced, probability reinforced’. This is discussed in 2.1.5.

One final point may be made. It is no coincidence that Tamil ‘debitive’ and ‘permissive’ forms are used for both epistemic and deontic modality, for, as noted in 1.2.2, the same is true of English MAY and MUST. This can be explained in terms of epistemic and deontic possibility and necessity, which is discussed in 4.1.

2.1.3 Deductive and Assumptive

English has the contrast between Deductive and Assumptive, as shown in 2.1.1:

John must be in his office
 John’ll be in his office

The difference between WILL and MUST is illustrated in (see Palmer 1990: 58):

It’s nine o’clock – John will be in his office now
 Yes, the lights are on, so he must be there

As with MAY and MUST, WILL can be used to refer to the future (but is then difficult to distinguish from the ‘pure’ future use – see 4.3.2) and with HAVE to refer to the past.

The contrast is also found in a number of other languages, but in most of them Deductive and Assumptive occur together in systems that include evidential markers of report and sensation. One such language is Tuyuca (Colombia – Barnes 1984: 257), where the markers are labelled ‘apparent’ and ‘assumed’. The full paradigm is:

diiga apé-wi
 soccer play+3SG+PAST-VIS
 ‘He played soccer’
 (I saw him play)

diiga apé-ti
 soccer play+3SG+PAST-NONVIS
 ‘He played soccer’
 (I heard the game and him, but I didn’t see it or him)

diiga apé-yi
 soccer play+3SG+PAST-APP
 ‘He played soccer’
 (I have seen evidence that he played: his distinctive shoe print on the playing fields. But I did not see him play)

diiga apé-yigɪ
 soccer play+3SG+PAST-SEC
 ‘He played soccer’
 (I obtained the information from someone else)

diiga apé-hīyi
 soccer play+3SG+PAST-ASSUM
 ‘He played soccer’
 (It is reasonable to assume that he did)

These can reasonably be identified as instances of the typological categories Visual, Auditory, Deductive, Reported and Assumptive respectively.

As has been already argued in 1.3.1, it is probably wrong to treat this as a ‘mixed system’, one that contains markers of both judgments and evidentials. It would be more reasonable to say that Deductive and Assumptive can be seen as both judgments and evidentials in that the relevant judgments are based upon evidence – in the case of Tuyuca from visual evidence and from what is more generally known. As such these fit into systems that are primarily either epistemic or evidential.

A pair of markers that can equally be identified as Deductive and Assumptive is also to be found in Wintu (N. California, now almost extinct – Schlichter 1986: 51–3): the ‘inferential’ *-re* and what Schlichter calls the ‘expectational’ *-ʔe*. Examples of the ‘inferential’ (Deductive) are:

Modal systems: Propositional modality

heke ma'n hara'ki-re'-m
somewhere EXCL go+COMPLET-INF-DUB
'He must have gone somewhere' (I don't see him)

piya mayto'n deknə'sto'n piya ma'n biyaki-re'-m
those feet steps that EXCL be+COMPLET-INF-DUB
'Those tracks of steps! That must have been him'

Examples of the 'expectational' (Assumptive) are:

tima min-el? pira'-?el
cold die-EXPECT starve-EXPECT
'He might freeze to death, he might starve' (it's cold and he's alone,
helpless, sick)

?imto'n nuqa'-?l
berries ripe-EXPECT
'The berries must be ripe' (it's that time of year)

Schlichter translates the assumptive as 'might' and 'must', but he also comments that the speaker believes the statement to be true. In the case of the Deductive it is 'because of circumstantial sensory evidence', most often visual, and in the case of the Assumptive 'because of his experience with similar situations, regular patterns, or repeated circumstances common in human life'. These remarks together with the bracketed comments suggest that these are examples of Assumptive, which would have been more clearly indicated by a translation with WILL:

'He'll freeze to death, he'll starve'
'The berries will be ripe'

From the data given, it seems that the Assumptive in evidential systems differs little from WILL in English, but that there is a slight difference between the Deductive in such systems and English MUST. It appears that MUST is used only when there is emphasis upon the deduction, whereas the Deductive in evidential systems is used much more widely – whenever any degree of deduction is involved. Oswalt (1986: 38–9) says of the 'inferential' in Kashaya that it 'appears in many situations in which English would not normally use "must have"'. Examples are:

kalikak^h dima ši-qa-ċ-q^h
book holding make-cause-self-INF
'He has a picture taken of himself holding a book'

he?én šin-iwa ma mace'-t^hi-qa-m
how doing-QUES you guard-NEG-INF-RESP
'Why did you not guard her?'

He comments that 'the existence of the picture (which is seen) is the evidence

that one was taken (which act was not seen)' and that 'the lack of guarding was not seen but is inferred . . . because the prisoner is gone'.

It may also be noted that for one example in Makah, Jacobsen (1986: 12) does not use 'must' in the translation for an example of the marker for 'inference from physical evidence':

haʔukaʃ piˈdic
'I see you ate'

Similarly, Mithun (1986: 101–2), in the discussion of an inferential particle in Wyandot, Seneca, Cayuga and Mohawk, uses ‘must’ in the translation for only two out of eight examples.

There is some emphasis in some of the discussion on the fact that the speaker believes the proposition to be true. Thus Schlichter (1986: 51) says of the 'inferential' that 'the speaker believes his statement to be true' (see above), while Oswalt (1986: 34) says 'The Kashaya Inferential suffix implies no lack of certainty, merely lack of higher ranking evidence.' Whether Deductive in these languages is significantly different from English MUST is discussed in 2.7.1.

2.1.4 *MAY as concessive*

MAY is used as a concessive, i.e., in the sense of 'although', in e.g.,

He may be rich, but he's not very lucky
He may have been rich, but he wasn't very lucky

There is a similar use of MÖGEN in German (Hammer 1983: 261):

Er mag noch so gescheit sein, . . . aber . . .
 he MÖGEN +3SG+PRES+IND ever so intelligent be, . . . but . . .
 'He may be intelligent, but . . .'

With a concessive clause the speaker does not indicate doubt about the proposition, but rather accepts it as true, in order to contrast one state of affairs with another. The use of *MAY* is to be explained, therefore, not in terms of Speculative, but of Presupposed. In other languages, such as Latin, Italian or Spanish, the subjunctive would be used (see 1.4.1 and 5.2.5).

2.1.5 Modifications

Some languages have ways of indicating that the judgments are either weaker or stronger. English, for instance, uses the past tense forms of the modals, the ‘modal-past’ (see 1.4.4 and 8.2.1). These express a lower degree of commitment, judgments that are more ‘tentative’.

For MAY and WILL the past tense forms *might* and *would* are used:

Mary might be at the school by now
Mary would be at the school by now

MUST has no past tense forms, but the modal-past equivalent is *ought to* or, more commonly, *should* (even though formally this is the past tense form of SHALL):

Mary ought to / should be at school by now

But this carries with it some notion of conditionality – ‘provided things are as I expect them to be’, etc., and is not just a weaker or more tentative form of MUST (Palmer 1979: 49, cf. 100–2).

There is another past tense form – *could*, which is also used to imply some kind of inference, although the present tense form *can* is not used in a similar sense, as in:

Mary could be at the school by now

This might suggest a possible conclusion, but one based on knowing facts such as what time it now is or how long it takes to get there.

There is a similar use of past tense forms in German, but together with the subjunctive. The present form of *MÖGEN* is most typically used to express a weak epistemic judgment, and its past tense form in the subjunctive indicates greater tentativeness (Hammer 1983: 227ff.):

Man möchte meinen, dass . . . (229)
one MÖGEN+3SG+IMPF+SUBJ think that . . .
'One might think that . . .'

However, the same forms of KÖNNEN and DÜRFEN are all also used in similar senses:

Er könnte krank sein (227)
 he KÖNNEN+3SG+IMP+SUBJ ill be
 'He might be ill'

Er dürfte krank gewesen sein (228)
 he DÜRFEN+3SG+IMPF+SUBJ ill been be
 'He might well have been ill'

According to Hammer, ‘*möchte* often denotes possibility or probability or conveys a hesitant or polite doubt’, while ‘*dürfte* implies greater probability than *könnte*, but at the same time more tentative and polite’. (The use of similar forms in conditionals is discussed in detail in 8.2.)

A distinction can be made, then, in English and German between weaker and stronger judgments. There appears to be a similar contrast in Inga (Levinsohn 1975, see 2.1.2 above):

chihoraca mal-cha cado carca fiide del bautismo (23)
 at.that.time bad-CHA it was baptismal certificate
 ‘At that time my baptismal certificate must have been incomplete’
 (deduced)

chipica diablo-char ca (19)
 there devil-CHAR it was
 ‘A devil was presumably there’ (deduced, probability reinforced)

However, there is a difference. In Inga, it seems, the contrast is between a normal conclusion and a more positive one, though the translation in terms of ‘presumably’ is a poor indication of reinforced probability. In English and German the contrast is between a normal conclusion and a less positive one.

2.1.6 Past time reference

The present tense forms of MAY and WILL can be used to make a judgment in the present about events in the past by the addition of *have* plus the past participle:

Mary may/will/must have arrived by now

But it is not possible to use the past tense forms of these modals to indicate a past judgment by the speaker. It is not possible to say:

*Yesterday Mary might/would arrive

In other words the proposition can be in the past, but the modality (the judgment) cannot. (Admittedly, in literary language *Mary might/would be there by now* could occur as an independent sentence, but only to represent the thoughts of one of the characters, as in *John relaxed – Mary would be there by now*. This is essentially reported speech (see below) with the verb of reporting ‘understood’.)

There is a reason for this. Inferences or conclusions are essentially subjective and performative. They are actually made by the speaker, at the time of speaking. They cannot be made in the past, although it is perfectly possible to report them in the past with lexical verbs, e.g., *Mary thought that . . .*, *Mary concluded that . . .*, etc. Lyons (1977: 798) offers a theoretically possible example of objective epistemic modality, but it is contrived and he concedes that the distinction between subjective and objective epistemic modality ‘is, to say the least, uncertain’.

MUST has no past tense form, so that its morphology alone rules out **Mary must be there yesterday*. There is, however, an unexpected use of the past tense of HAVE TO to indicate that there was only one possible conclusion, and, usually, that it proved to be right.

I found the book at last – in the bookcase – it had to be there

This, however, can be explained – HAVE TO is not a modal verb, and so not subjective as MUST is. It can therefore be paraphrased as ‘It was epistemically necessary that . . .’, i.e., ‘It couldn’t have been otherwise than that . . .’

However, the past tense forms of the modals are used in reported speech with past tense verbs of reporting (and in that sense only may indicate past time – see Palmer 1990: 43).

He may/will be there
He said he might/would be there

Since MUST has no past tense form, it may be used in similar constructions, or replaced by *had to*:

He must be there
He said he must be there

2.1.7 Inference and confidence

Coates (1983: 41, 131, 177) says of the English epistemic modals:

In its most normal usage, Epistemic MUST conveys the speaker’s confidence in the truth of what he is saying, based on a deduction from facts known to him (which may or may not be specified).
MAY and MIGHT are the modals of Epistemic Possibility, expressing the speaker’s lack of confidence in the proposition expressed.
WILL expresses the speaker’s confidence in the truth of the proposition; unlike epistemic MUST, the speaker’s confidence is not based on a process of logical inference. Instead, it is based on common sense or repeated experience.

However, the repeated reference to ‘confidence’ may be a little misleading since there are two issues here, one a matter of ‘confidence’ or degree of commitment by the speaker, the other a matter of the strength of the conclusion. For degrees of confidence can be expressed by adverbs:

Perhaps she’s there
She’s probably there
She’s certainly there

These are not, or not wholly, equivalent to sentences with the modal verbs.

With MUST there is often some indication of the facts (often observable) on which the inference is based. An example from Coates (1983: 41) is:

His teeth were still chattering but his forehead, when I felt it, was hot and clammy. He said ‘I must have a temperature.’

A similar example is found in Palmer (1979: 44):

All the X-rays showed absolutely negative. There was nothing wrong, so it must just be tension, I suppose.

It is clear that it is the notion of deduction or inference from known facts that is the essential feature of *MUST*, not just the confidence of the speaker, which is expressed by the adverbs *certainly*, *definitely*, etc.

Similarly, *will* is not paraphrased by *probably*, but draws an inference, not so clearly from observable facts, but from generally known facts. Thus *Mary will be at the school by now* indicates a conclusion based on knowing what the time now is and what are Mary's movements.

By contrast, there seems to be little difference between *Mary may be at school* and *Perhaps Mary is at school*, but this may be because both are negative in the sense that the first suggests there is absence of good grounds for a conclusion, the second that there is absence of real confidence. There is, then, little or no difference, just as there is no difference between coffee without milk and coffee without cream.

There is, then, a difference between inference and confidence. With the former, speakers indicate that they are inferring from available information, while in the latter they indicate the degree of confidence they have in what they are saying. There is a clear difference between *MUST* and adverbs such as *certainly*, the modal expressing inference and the adverb confidence. The same is true of *WILL*, which is different from *probably*. There is less obvious difference between *MAY* and *perhaps*, but they can still be seen, in a negative sense, as instances of inference and confidence respectively.

However, markers of confidence may be added to the markers of inference. With the modal verbs, it is possible for the speaker's commitment to be modified by the addition of what Halliday (1970: 331) calls 'harmonic combinations', which reinforce the speaker's commitment, or by 'hedges', which reduce it. Coates (1983: 46, 138) notes for epistemic *MUST* the occurrence of harmonic *I'm sure*, *surely* and *certainly*, and of the hedges, *I think* (the most common), *I mean*, *I suppose*, *I fancy*, *I take it* and *I would guess*. For *MAY*, she suggests harmonic *perhaps*, *possibly*, and hedges *I suppose*, *I think*, *I don't know*, *I wouldn't know*, *I'm not sure*, *I mean* and *It seems to me*. These do not belong to a grammatical system and an analysis of them is beyond the scope of this work. (For discussion of similar issues see 2.2.7 and 2.7.2.)

2.2 Evidential modality

2.2.1 Evidential systems

Although there are formal systems with multiple terms that are primarily evidential, there are basically only two types of purely evidential categories, Reported and Sensory (evidence of the senses). Indeed, in Ngiyambaa there

are, according to Donaldson (1980: 275–6), just two ‘Evidence clitics’, those for ‘sensory evidence’ and ‘linguistic evidence’ as in:

ṇindu-gara girambiyi
you+NOM-SENS.EVID sick+PAST
‘One can see you were sick’

ṇindu-dhan girambiyi
you+NOM-LING.EVID sick+PAST
‘You are said to have been sick’

(‘Sensory evidence’ in Ngiyambaa may involve any of the five senses, see 2.2.3.)

However, many formal systems include other categories, including those discussed as judgments in 2.1. One clear example of such an extended evidential system (from Central Pomo) was given in 1.2.1. Another is that of Tuyuca (Brazil and Colombia – Barnes 1984: 257, cf. also Malone 1988 for a reanalysis of the data), where there are evidential categories labelled ‘visual’, ‘non-visual’, ‘apparent’, ‘second-hand’ and ‘assumed’:

dīiga apé-wi
soccer play+3SG+PAST-VIS
‘He played soccer’
(I saw him play)

dīiga apé-ti
soccer play+3SG+PAST-NONVIS
‘He played soccer’
(I heard the game and him, but I didn’t see it or him)

dīiga apé-yi
soccer play+3SG+PAST-APP
‘He played soccer’
(I have seen evidence that he played: his distinctive shoe print on the playing fields. But I did not see him play)

dīiga apé-yigt
soccer play+3SG+PAST-SEC
‘He played soccer’
(I obtained the information from someone else)

dīiga apé-hīyi
soccer play+3SG+PAST-ASSUM
‘He played soccer’
(It is reasonable to assume that he did)

The first two indicate evidence of the senses, the fourth evidence from what

the speaker has been told, while the third and fifth indicate judgments based on the evidence from what has been observed or is known.

A rather different system is suggested for Hidatsa (Siouan, USA) by G. H. Matthews (1965: 99–100). This is interesting because of the glosses provided, though there are serious problems about the interpretation (see 2.7.1). Matthews suggests that the final clause of each sentence ends in a mood morpheme; six such morphemes are glossed as follows:

Emphatic: ‘indicates that the speaker knows the sentence to be true; if a sentence that ends with the Emphatic is false, the speaker is considered a liar’

Period: ‘indicates that the speaker believes the sentence to be true; if it should turn out otherwise, it would mean that he was mistaken, but by no means a liar’

Quotative: ‘indicates that the speaker regards what he has said to be something that everyone knows’

Report: ‘indicates that the speaker was told the information given in the sentence by someone else, but has no evidence of its truth value’

Indefinite/Question: ‘both indicate that the speaker does not know whether or not the sentence is true. The Indefinite also means that the speaker thinks the listener does not know; whereas the Question indicates that the speaker thinks that the listener does know.’

The first three are illustrated by:

wacéo iikipi kurè héo ski
man pipe carried EMPH
‘The man (sure) did carry the pipe’

wacéo iikipi kurè héo c
man pipe carried PER
‘I suppose the man carried the pipe’

wacéo iikipi kurè héo wareac
man pipe carried QUOT
‘The man carried the pipe, they say’

(The markers for ‘Report’ and ‘Indefinite’ are *rahe* (see 2.2.2 for an example) and *toak* respectively, while ‘Question’ ‘has no actual phonemic shape’.)

Most of the extensive evidential systems that have been described are to be found in Native American languages, but Foley (1986: 165) reports that there are evidentials in some of the Papuan languages. He quotes Loeweke and

May (1980) for examples from Fasu, where the list includes ‘seen’, ‘heard’, ‘deduced from evidence’, ‘hearsay from a known source’, ‘hearsay from an unknown source’ and ‘supposition’:

a-pe-re
SE[EN]-come-[SE]EN
‘[I see] it coming’

pe-ra-rakae
come-CUST-HEARD
‘[I hear] it coming’

pe-sa-reapo
come-PAST-DED
‘[I’ve concluded] it’s coming’

pe-sa-pakae
come-PAST-HSY.UNKNOWN
‘[I’ve heard] it’s coming’

pe-sa-ripo
come-PAST-HSY.KNOWN
‘[I’ve heard] it’s coming’

pe-sa-pi
come-PAST-SUPPOSE
‘[I think] it’s coming’

Even more striking is the situation in Ladakhi (Tibeto-Burman – Koshal 1979), where there are no less than three systems relating to epistemic modality (as well as a set of deontic suffixes).

First, there is a system of six suffixes, of which four are evidential – ‘reportive’ (Reported), ‘observed’ (Visual), ‘experiential’ (not fully explained) and ‘inference’ (Deductive). Of the other two, one (‘possible’) indicates Speculative and the other is labelled ‘generic’:

pəlldən-ni spe-čha sill-ət
Paldan-ERG book-DIR read-REP
‘Paldan reads a book’ (a report)

kho-e lcəŋ-me cəd-duk
he-ERG tree-die cut-OBSERVED
‘He cuts the tree’ (direct observation)

kho che-ərək
he go-EXPERIENTIAL
‘He goes’ (speaker’s feelings)

kho-ə zur-mo sante duk ši-ok
he-DAT pain-DIR very be die-INF
‘He will die, because he is very sick’

kho-e thore ɲə əʦo thuk-cen
 he-DAT tomorrow my brother-DIR meet-POSSIBLE
 ‘He is likely to meet my brother tomorrow’

ñi-mə zəktəŋ sər-ne sərrənok
 sun-DIR daily east-ABL rise-GEN
 ‘The sun rises daily in the east’

Secondly, four different types of inference that involve some evidential features are indicated by suffixes together with *-thig-*. The four types are ‘from sounds or habitual occurrences’, ‘observation not correctly remembered’, ‘unobserved, partial or vague knowledge’ and ‘guessed, as, for instance, about events that occurred at a distance and hence cannot be seen clearly’:

dolmə yon-thig-rek
 Dolma come-INF-SOUND
 ‘Dolma is coming’ (hearing footsteps, voice, etc.)

khoe kəne pene ɲe khyer-thig-yot
 he me from money take-INF-OBSERVED
 ‘He might have taken money from me’

kho i-khəŋpe nəŋŋə duk-thig-son
 he this-house in live-INF-UNOBSERVED
 ‘He might have lived in this house’

əpumə rdemo yot-thig-duk
 that-girl beautiful be-INF-GUESSED
 ‘That girl might be beautiful’

Thirdly, there are four copulas used for simple statements, definite knowledge, experience or feeling, and what is seen or heard:

ɲə məg-mi yin
 I soldier-DIR be
 ‘I am a soldier’ (simple statement)

khon-ɲə pe-ne yot
 he-DAT money-DIR be
 ‘He has money’ (definite knowledge)

ɲə go-ə zur-mo rək
 I head-DAT pain-DIR be
 ‘I have a head-ache’ (experience)

pu-mo rdemo duk
 girl-DIR beautiful be
 ‘That girl is beautiful’ (seen)

2.2.2 *Reported*

Examples of Reported (with labels such as ‘second-hand’, ‘linguistic evidence’, ‘hearsay’) from Tuyuca and Ngiyambaa and Fasu were given in the last section.

Reported is also found in languages that have a system of mood, but only where the system is ‘joint’ (see 6.1), i.e., where the grammatical markers of Realis and Irrealis co-occur with grammatical markers of other categories. This co-occurrence suggests that these categories are notionally ‘realis’ or ‘irrealis’ (and themselves typologically Realis or Irrealis). Not surprisingly, Reported is generally so marked as Irrealis, as can be seen from the system in Hixkaryana (Carib, N. Brazil – Derbyshire 1979: 1435), where ‘uncertain’ is the mark of Irrealis and ‘hearsay’ the term used for Reported (see 6.5.4):

nomokyan ha-ti
he.come+NONPAST+UNCERT INTEN-HSY
'He's coming (they say)'

However, similar forms are to be found in languages that have what are predominantly judgment systems, e.g., German (Hammer 1983: 231, 232) and Danish (Davidsen-Nielsen 1990: 96):

Er soll steinreich sein
He SOLLEN+3SG+PRES+IND very.rich be
'He is said to be extremely rich'

Han skal være enormt rig
He SKULLE+PRES be enormously rich
'He is said to be enormously rich'

For some languages, it is necessary to recognize sub-categories of Reported. Willett (1988: 57, 96) suggests that there are three kinds of reported evidence:

- 1) *Second-hand evidence*: the speaker claims to have heard of the situation described from someone who was a direct witness.
- 2) *Third-hand evidence*: the speaker claims to have heard of the situation described, but not from a direct witness.
- 3) *Evidence from folklore*: the speaker claims that the situation described is part of established oral history.

It is not easy to find suitable names for these sub-categories. It might have been possible to use the terms 'Quotative', 'Hearsay' and 'Folklore', but there is great inconsistency in the literature in the use of these terms, and this might lead to confusion. Moreover 'folklore' is misleading in that the third sub-category usually refers to what is generally believed to be true, not

necessarily as part of folklore or tradition. For these reasons the sub-categories are to be labelled ‘Reported (2)’, ‘Reported (3)’ and ‘Reported (Gen)’. (‘Reported’ may be used to refer either to the undifferentiated category or to the overall category that includes the sub-categories.)

In many languages the same form is used for all three types of Reported. This seems to be the case with the examples provided by Mithun (1986: 102–3) for ‘quotative particles’ in the Northern Iroquoian languages Cayuga, Oneida, Mohawk and Seneca. Two of the examples are clearly from a story, while there is no indication in the others whether the evidence is ‘second-hand’ or ‘third-hand’. The same appears to be true of Tuyuca, where Barnes (1984: 261) talks of a speaker ‘reporting information that was relayed to him’, and gives an example of Reported (Gen), in this case part of a legend:

ãñã kĩre baka-yigĩ
‘A snake bit him’

Similarly the example from Hixkaryana given above is glossed as ‘they say’, but Derbyshire goes on to say that the same ‘hearsay’ particle has to be repeated continually through a story that is being related, and Givón (1982: 34–5) reports that in Sherpa (Tibeto-Burman) the ‘indirect/hearsay’ markers is used in passages from religious texts (see also 2.7.1).

Yet some languages have markers to distinguish the subtypes. Thus Fasu (2.2.1) has different markers for ‘hearsay from a known source’ and ‘hearsay from an unknown source’, i.e., Reported (2) and Reported (3):

pe-sa-pakae
come-PAST-HSY.UNKNOWN
‘[I’ve heard] it’s coming’

pe-sa-ripo
come-PAST-HSY.KNOWN
‘[I’ve heard] it’s coming’

Similarly for Hidatsa (Siouan, USA), G. H. Matthews (1965: 99–100) distinguishes between what he calls ‘quotative’ and ‘report’ (i.e., again, Reported (2) and Reported (3), respectively). These are glossed (see 2.2.1):

Quotative: ‘indicates that the speaker regards what he has said to be something that everyone knows’
Report: ‘indicates that the speaker was told the information given in the sentence by someone else, but has no evidence of its truth value’

Examples are:

wacéo iikipi kurè héo wareac
man pipe carried QUOT
'The man carried the pipe, they say'

wacéo wiira rakcí héo rahe
man pipe carried REP
'The man carried the pipe, they say'

In Central Pomo too (see 2.2.1 for examples), there is a distinction between 'general knowledge' and 'hearsay'. 'General knowledge' is to be identified as Reported (Gen), while 'hearsay' seems to be either Reported (2), or Reported (2) + Reported (3) (the example is glossed 'I was told').

A rather different distinction is to be found in German, which has yet another type of Reported (see 1.3.1). For while the modal verb *SOLLEN* indicates what others say, *WOLLEN* indicates what the person represented by the subject says (Hammer 1983: 231, 232):

Der Geschäftsführer sollte schon nach Hause gegangen sein
the manager SOLLEN+3SG+PAST already to house gone to.be
'The manager was said to have gone home already'

Er will eine Mosquito abgeschossen haben
he WOLLEN+3SG+PRES+IND a Mosquito shot.down have
'He claims to have shot down a Mosquito' (plane)

WOLLEN can also be used for non-verbal communication by the subject, as in:

Er wolle mich nicht erkennen
he WOLLEN+3SG+PAST me not to recognize
'He pretended not to recognize me'

There is one further point to note concerning German. As briefly mentioned above, the subjunctive is used to indicate what was said, i.e., Reported (2), as in (see 5.2.5):

Bei seiner Vernehmung berief sich H. auf Notwehr. Er
in his examination appealed H to self-defence. he
sei mit S. in Streit geraten und
be+3SG+PRES+SUBJ with S. in quarrel fallen and
habe sich von diesem bedroht gefühlt
have+3SG+PRES+SUBJ self by him threatened felt
'In the course of his cross-examination, H. pleaded self-defence. He had become involved in a quarrel with S. and had felt himself to be threatened by him'

The second sentence is not asserted by the speaker (or writer), but is shown to be what was said (by H.).

A further distinction between a report that is reliable and a report that is unreliable is to be found in Lega, but is discussed in 2.2.4.

2.2.3 *Sensory*

The situation with Sensory is like that of Reported in that, although some languages have a single category of Sensory, others have sub-categories, of which Visual (evidence from seeing) and Auditory (evidence from hearing) are the most obvious. In practice, however, there are generally three possibilities: (i) a single marker (Sensory), (ii) a marker for seeing and another for all of the other senses (which may be identified as Visual and non-Visual) and (iii) markers for seeing and hearing (Visual and Auditory). Only rarely is there a marker for the senses other than seeing and hearing (but see below).

Ngiyambaa (Donaldson 1980: 275–6) is of the first type – there is a clitic which is used for sensory evidence that may involve all five senses:

ɲindu-gara girambiɲi
 you+NOM-SENS.EVID sick+PAST
 ‘One can see you were sick’

 gabuga:-gara-lu ɲamumiɲi
 egg+ABS-SENS.EVID-3ERG lay+PAST
 ‘It’s laid an egg by the sound of it’
 (The chicken concerned was out of sight.)

 yura:bad-gara ɲidɟi guɽuga-nha
 rabbit+ABS-SENS.EVID here+CIRC be inside-PRES
 ɲama-ɽa-baɽa-dhu-na
 feel-PRES-CATEG.ASS-1NOM-3ABS
 ‘I can tell there’s a rabbit in here. I (can) feel it for sure’
 (The speaker had her hand in a burrow.)

 dhagun-gir-gara ɲina dhiŋga: ga-ɽa
 earth-nasty-with-SENS.EVID this+ABS meat+ABS.be+PRES
 ‘This meat tastes nasty with earth’
 (Said while attempting to eat it.)

 wara:y-gara-dhu-na bungiyamiɲi dhiŋga:-dhi:
 bad+ABS-SENS.EVID-1NOM-3ABS change.with.fire+PAST meat+ABS-1OBL
 ‘I have burnt my meat so it’s no good, to judge by the smell’
 (Said outside the house where the meat was cooking.)

The senses involved here are, quite obviously, those of sight, hearing, touch, taste and smell respectively.

The second type, in which there is one marker for sight (Visual) and another (non-Visual) for all the other senses is represented by Tuyuca (Barnes 1984: 260). Examples repeated from 2.2.1 are:

diiga apé-wi
 soccer play+3SG+PAST-VIS
 ‘He played soccer’
 (I saw him play)

diiga apé-ti
soccer play+3SG+PAST-NONVIS
'He played soccer'
(I heard the game and him, but I didn't see it or him)

In spite of the fact that the second example refers to hearing, Barnes states that this 'non-visual evidential may be used to report how someone, something or some event smelled, sounded, tasted or felt (smells, sounds, tastes or feels)'. Examples are:

yoáro susúhã-ta (*ta* = 3PL+PAST)
'They smelled (of liquor) a long way off'
mütúru bisiti (*tì* = 'other' (1, 2 or inanimate) +PAST)
'The motor roared'

It is also used to refer to the speaker's own emotion, pain or knowledge:

páaga pūnĩ-ga (*ga* = 'other'+PRES)
'My stomach hurts'
t sá-ga
'I like it'

The third type with separate markers for both Visual and Auditory are less common. Indeed Oswalt (1986: 43) says 'Kashaya, Southern Pomo and Central Pomo are apparently rare among the languages of the world in having an evidential for a specific non-visual sense, the Auditory.'

For Central Pomo, Mithun (1999: 181) offers a very neat paradigm that clearly marks Auditory as well as Visual (plus Reported and Deductive, as well as a simple Declarative, for which see 2.7.1). The full paradigm was given in 1.2.1. Examples (repeated) of Auditory and Visual are:

čbémul-ya
rain.fell-vis
'It rained' (I saw it)
čbémul-nme
rain.fell-AUD
'It rained' (I heard it)

For Kashaya, Oswalt notes a pair of markers for Visual and a marker for Auditory. The Visual markers are what Oswalt (1986: 36) calls 'factual' and 'visual', although the formal difference between them is that of imperfective and perfective respectively. These form a pair, and while both refer to what the speaker sees or has seen, the 'factual' may also be used, more widely, to refer to classes of actions that have been generally observed or are common knowledge:

qowa°q-ŵă (qowá'q^h)
 pack-FACT
 '(I see) he is packing'
 qowa°q-yă (qowahy)
 pack-VIS
 '(I just saw) he packed, I just saw him pack'

The Auditory markers occurs with marker for both perfective and imperfective:

mo-Ŵ°d-Ŵ°nnă (mo'dun)
 run-IMPV-AUD
 'I hear/heard someone running along'
 mo-m°ac-Ŵ°nnă (momá'cin)
 run-PERFV-AUD
 'I just heard someone run in'

However, Auditory marker are not limited to Native American languages. As noted in 2.2.1, both a Visual marker and an Auditory marker are to be found in the Papuan language Fasú (Foley 1986: 165; Loeweke and May 1980):

a-pe-re
 SE[EN]-come-[SE]EN
 '[I see] it coming'
 pe-ra-rakae
 come-CUST-HEARD
 '[I hear] it coming'

In some languages there is a marker that is used mainly, but not exclusively, for hearing. Gordon (1986b: 76–7) provides examples in Maricopa (Yuman) for sight and hearing:

m-iima-'yuu
 2-dance-SEE.EV
 'You danced' (I know because I saw)
 m-ashvar-a
 2-sing-HR.EV
 'You sang' (I know because I heard it)

However, she comments that the 'hearing evidential' is used for 'first-hand knowledge . . . gained not by having seen the event, but having otherwise sensed (usually heard) the state or event'. Moreover, it is most strongly associated with verbs strongly associated with sound, e.g. 'sing', and with verbs of saying as in:

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Pam-sh 'im nyip ny-mhank ii-'a
Pam-SUBJ say me 3/1-like say-HR.EV
'Pam told me that she likes me'

Similarly, for Makah (Nootka, Washington State), Jacobsen (1986: 9–10), has examples of an unmarked declarative and one with the marker for hearing (*q̣adʔi*):

wiki-caṣaw
'It's bad weather'
wiki-caṣ akq̣adʔi
'It sounds like bad weather'

Jacobsen, however, notes that the same marker can be used for personal feelings, and has, in recent times, been extended to refer to direct observation.

There are no attested examples of a specific marker for the senses other than seeing and hearing. However, in Kashaya, which unusually (see above) distinguishes Visual and Auditory, there is a 'catch-all class', which 'happens to include inference on evidence found apart from the act' (Oswalt 1986: 43). This is what Oswalt calls 'Inferential I', which expresses 'lack of high ranking evidence'. Examples are (Oswalt 1986: 38–9):

mu cohtoc-q̣ă
'He must have left, he has left'
cuhni' muʔta-q̣ă
'Bread has been cooked' (I can smell it)

In Maricopa, as noted above, the other senses can be expressed by the marker usually used for hearing.

The situation in Ladakhi (discussed with examples in 2.2.1) is complex. It would appear that there is an evidential for Visual in the basic set of evidentials, one for Auditory (or Auditory-Assumptive) in a set of suffixes used for inference and a copula related to Visual–Auditory. Examples, repeated from 2.2.1, are:

kho-e lcəŋ-me cəd-duk
he-ERG tree-die cut-OBSERVED
'He cuts the tree' (direct observation)
dolmə yoŋ-thig-rek
Dolma come-INF-SOUND
'Dolma is coming' (hearing footsteps, voice, etc.)
pu-mo rdemo duk
girl-DIR beautiful be
'That girl is beautiful' (seen)

In addition there is an ‘experiential’ (not fully explained) in the basis set plus a copula for experience and feelings:

kho che-ərək
 he go-EXPERIENTIAL
 ‘He goes’ (speaker’s feelings)
 ŋə go-ə zur-mo rək
 I head-DAT pain-DIR be
 ‘I have a head-ache’ (experience)

Finally it may be noted that even in English there is some connection between sensation and modality. The most common way of expressing what one sees, hears, smells, tastes or feels is with the modal verb CAN:

I can see the moon
 I can hear a funny noise
 I can smell something burning
 I can taste salt in this
 I can feel something hard here

All of these indicate that the speaker has the sensation, not that he has the ability to have it. The relevant point is that English does not normally present information about sensation with simple declarative statements, but chooses instead to use a modal form, just as Ngiyambaa indicates sensory evidence with an evidential clitic.

2.2.4 Direct and indirect evidence

There are some languages in which there is a single marker of modality that may indicate either what is said (Report) or what may be inferred to be so (Deductive). For Turkish, Aksu-Koç Ayhan and Slobin (1986: 159) exemplify a contrast between what they call ‘direct experience’ with the suffix *-dİ* and ‘indirect experience’ with the suffix *mİş*, giving as an example of the latter:

Ahmet gel-mİş
 Ahmet come-MİŞ
 ‘Ahmet came / must have come’

This may be interpreted as:

- (a) inference: the speaker sees Ahmet’s coat hanging in the front hall, but has not yet seen Ahmet.
- (b) hearsay: the speaker has been told that Ahmet has arrived, but has not yet seen Ahmet.

(Cf. also Lewis (1967: 101).)

However, the inferential interpretation is possible only where the inference is drawn from a state resulting from a previous process, such as the coat hanging on the wall resulting from Ahmet's arrival. As a result only the 'hearsay' interpretation, not the 'inferential' interpretation is, therefore, possible for:

yağmur yağ-acak-mış
rain rain-FUT-*miş*
'It is reported that it will rain'
*'It will probably rain'

In Sherpa too, there is a contrast between what Givón (1982: 32) calls 'direct' evidence and 'indirect-hearsay' evidence, illustrated by:

ti laḡ a ki-yin-no
he work DO-AUX-be+DIR
'He is working (and I have direct evidence to support this)'
ti laḡ a ki-yin-way
he work DO-AUX-be+INDIR
'He is working (I have indirect/hearsay evidence)'

There is something similar in Abkhaz (N. W. Caucasus – Hewitt 1979: 196), where there is a suffix on the verb in certain tense forms to 'indicate that the assertion is made as a result of inference or hear-say', e.g.:

də-r-s-xà-zaap'
him-they-kill-PERF-INFER/HSY
'Apparently they have already killed him'

However, Abkhaz has a number of constructions that express various modal notions.

Another possible example is an Iranian language of the USSR, Tajik. Rastorgueva (1963: 64) distinguishes between 'evident' and 'non-evident' (or 'narrative') forms, the latter being glossed as referring to 'an occurrence or action known to the speaker, not on the basis of personal experience, but from collateral sources, i.e. from someone else's words or from logical inference'. The non-evident verb forms are compounds formed from the perfect, including a form of the copula. An example given is:

aka fodi: *meomadaast*, rost-mi
'(they say that) Mr Shodi is coming. Is that true?'

There is no example of inference, but another example indicates non-immediate awareness, which can be seen as a (correct) inference in the past:

man avval u:ro nafjinoxtam, diḡkat karpa binam, Aḡmad *budaast*
'At first I didn't recognize him; then when I looked carefully, I saw it was Ahmed'

It is relevant to add that English uses *apparently* in both senses, as in the translation of the Abkhaz example.

Rather more striking is the situation in Lega (Bantu, Zaire – Botne 1997: 512–22, and personal communication), where, it seems, there is a clear contrast between indirect, unreliable evidence and direct, reliable evidence. There are three particles in this language with grammatical status in that they are invariant and occur only in clause-initial position.

One of these, labelled ‘possibility’, indicates ‘weak inference, often based on sensory input’, and so can be seen as a marker of the judgment Speculative (and is of no further interest here):

émbe mbula zékokó ka
SPEC rain fall
‘Maybe it will rain’

The other two are evidential. One (*ámbo*) is used for Reported, but may also be used to express doubt after verbs of saying and in questions, as in:

nkumgwágá (bónó), ámbo bazongo bé kulyágá merende
I hear that EV whites eat frog
‘I hear that Westerners eat frogs’ (though I find that unlikely)
ámbo Amisi éndileko Misisi?
QUOT Amisi went to Misisi
‘Is it really the case that Amisi went to Misisi?’

The other evidential (*ampó*) is used where the evidence is believed to be ‘particularly robust or cogent’. The evidence is of three main types:

- 1) Direct sensory evidence, particularly from sight or hearing;
- 2) Strong inferential evidence (Deductive);
- 3) Report that can be trusted.

Examples are (the first and last from personal communication):

ampó mbulá zésaloka
EV rain rain
‘It is already raining’ (I can see it)
ampó ékukúrá momponge
EV she is pounding rice
‘She’s assuredly pounding rice’ (I can hear it)
ampó Kisangá éndile kw isoko; kikápu kyágé také gáno
EV Kisanga went to market; basket is not here
‘Kisanga surely went to the market; her basket is not here’
ampó Moké ákorwa
EV Moke tired
‘Moke is tired’ (Moke told me)

(The last example could also be said if the speaker can see that Moke is tired.)

The essential difference between the two evidentials is that the first indicates evidence that is less than reliable, the second evidence that is wholly reliable. This contrast between two types of Reported is particularly striking.

2.2.5 Other evidential possibilities

There are other possibilities. In Makah (discussed above in relation to Auditory), there is an unmarked form that indicates visual or direct experience, but also a marker (*caqi*) for ‘uncertain visual’. There are, thus, two types of Visual:

čapač
‘It’s a canoe’
čapaccaqil
‘It looks like a canoe’

A different feature is found in Central Pomo (Mithun 1999: 181). In addition to the five evidentials illustrated in 1.2.1 and 2.2.3, it has markers for personal experience of actions performed by the speaker and for actions that affect the speaker (Mithun does not suggest labels for these):

da-čé-w-la
pulling-seize-PERF-PERS.AG
‘I caught it’ (I know because I did it)
da-čé-w-wiya
pulling-seize-PERF-PERS.AG
‘I got caught’ (I know because it happened to me)

The related language Kashaya has forms that Oswalt (1986: 34–6) calls ‘performative’, which are used to refer to actions performed by the speaker as in:

qowa°q-wela
‘I am packing (a suitcase)’ (imperfective)
qowa°q-mela
‘I just packed’ (perfective)

In addition, there is another marker, (*-yowǎ*), which Oswalt (39–40) calls ‘personal experience’, that is used in the initial stages of narrative in place of all the other markers except ‘quotative’ (= Reported):

mul-í-yow-e° hayu cáhno-w
then-ASS-PERS.EXP-NONFINAL dog sound-ABSOL
‘Then (I saw, heard, judged) the dog barked’

This is more than a general marker of Sensory, since it includes inference.

But it is equally not a general marker of propositional modality, since it excludes reports (Reported).

2.2.6 *Hierarchy*

Oswalt (1986: 43) says of the evidentials in Kashaya that they lie on a hierarchy:

performative > factual > visual > auditory > inferential > quotative
(i.e. 'Performative' > Declarative > Visual > Auditory > Deductive > Reported)

Here those that precede have priority over those that follow. Oswalt suggests that the order seems to be universal, even for English where evidential concepts are expressed by verbs. 'Thus, someone speaking of an act he himself is performing or has performed, would not normally attribute knowledge of that event to a lower type of evidence . . .' (than the 'Performative'). Similarly, 'a speaker of any language would not normally say "I heard someone approaching the door" when he can see the person approaching'.

Oswalt gives no specific examples from Kashaya, but there is much more exemplification and discussion from Tuyuca (Barnes 1984: 262–4). The hierarchy here is:

visual > non-visual > apparent > second-hand > assumed
(i.e. Visual > Non-visual > Deductive > Reported > Assumptive)

The lowest in the hierarchy is 'assumed' (Assumptive), which is used when the speaker has prior knowledge about the state of things or about habitual or general 'behaviour patterns', but 'only when no information about the state or event is being or has been received'. In that sense it is the negative member of the evidential system, used when no ready evidence is available.

In contrast, Visual is the preferred evidential. It is used whenever a speaker has seen, or is looking at, a state or event. Even if he has or has had another type of evidence, he will still use the Visual evidential rather than that appropriate to the other types. Thus, if a man hears his dog yelp when killed by a jaguar, he will still use the Visual evidential, not the non-Visual, if he later sees evidence of the kill. The visual evidence takes precedence over the auditory.

Barnes stresses the importance of giving visual information whenever possible. Thus if a person is asked, on leaving his mother's house, whether she is in, he will reply 'She was' with the Visual evidential, to indicate the past visual information, rather than 'She is', which would require the

Assumptive, since there is no other information available concerning her present whereabouts.

2.2.7 Evidentials and certainty

There was a discussion in 2.1.7 of the differences between markers of inference and markers of certainty, with only the former strictly belonging to a modal system. There are, however, formal systems that contain markers of both types, e.g., in Hanis Coos (Oregon – Frachtenberg 1922: 385–8; cf. Mithun 1999: 182):

kwa	‘it seems, as if, like, kind of’
yiku, k ^w	‘maybe, perhaps, I guess’
hakwał, kwał	‘it seems, as if’
qên	suspicion
qaiku	supposition
qaini	sudden recollection ‘Oh, I recollect’
natsi	‘I doubt’
heñin	hearsay, ‘I was told, it is said’
il	‘surely, certainly’
cku	knowledge by evidence, ‘it must have been that’

Most of these denote ‘degrees of certainty and knowledge’ and six of them look rather like types of Speculative, though these, together with the marker for ‘surely, certainly’, are more like indicators of degrees of confidence than of inference or deduction (see 2.1.7). Yet the system also contains what are clearly epistemic–evidential markers – Reported (‘hearsay’) and Deductive (‘knowledge by evidence’). The whole system is, therefore, of more interest to the issue of modality than, say, the particles in German (see 2.6.1).

2.3 Interrogative and Negative

Interrogative and Negative are sometimes, but not commonly, found as members of an epistemic modal system. Interrogative, for example is a term in a modal system in Menomini (Algonquian, USA), where the system according to Hockett (1968: 2378) is illustrated by:

pi’w	he comes, is coming, came
pi’wen	he is said to be coming, it is said that he came
pi’?	is he coming, did he come?
piasah	so he <i>is</i> coming after all! (despite our expectations to the contrary!)
piapah	but he was going to come! (and now it turns out that he is not!)

It should not, perhaps, be surprising that some languages use the same marker for both Negative and Interrogative, since these can be seen as ‘non-

assertive’ in, e.g., English (see 1.4.2). A clear example is Imbabura (Cole 1982: 164). The full system is given in 1.4.2, but the relevant examples (with *chu*) are:

mayistru-chu ka-ngui
 teacher-QUES be-2SG
 ‘Are you a teacher?’
 ñuka-ka mana chay llama-ta shuwa-shka-ni-chu
 I-TOP not that sheep-ACC steal-PERF-2SG-NEG
 ‘I didn’t steal that sheep’

Another example of the same form being used for both negation and interrogation is to be found in Tiwi (Australia – Osborne 1974: 43). In this language there is an unmarked ‘indicative’, but four marked moods – ‘imperative’, ‘subjunctive’, ‘compulsional’ and ‘incompletive’. The ‘subjunctive’ has three functions – for negation, for non-PAST interrogation and for what is possible but unlikely. The first two functions are illustrated by:

a.nuŋkwa ji-ma-kəðimi
 NEG he-SUBJ-do
 ‘He doesn’t do it’
 pu-ma-ta-wari
 they-SUBJ-FUT-fight
 ‘Are they going to fight?’

A different possibility is that there is no specific marker for Interrogative, but that questions are indicated by a marker that indicates lack of knowledge on the part of the speaker. This is illustrated in the system of ‘knowledge’ clitics of Ngiyambaa (Donaldson 1980). There are two such clitics, identified as ‘exclamative’ and ‘ignorative’. The ‘ignorative’ is used together with ‘counterfactual’ and ‘hypothesis’ to express what might have been and ‘perhaps’ respectively:

minjaŋ-gaɪ-ma-ndu dha-yi (253)
 What-IGNOR-CNTF-2NOM eat-PAST
 ‘You might have eaten I don’t know what’ (but you didn’t)
 guya-gila-gaɪ-lu dha-yi (257)
 fish+ABS-HYPOTH-IGNOR-3ABS eat-PAST
 ‘Perhaps he ate a fish’

However, it indicates a question when no other clitic occurs:

minjaŋ-gaɪ-ndu dha-yi (260, 262)
 what+ABS-IGNOR-2NOM eat-PAST
 ‘You ate something, I don’t know what’/‘I don’t know what you ate’

guya-ga:-ndu dha-yi (260, 263)
 fish+ABS-IGNOR-2NOM eat-PAST
 'Did you eat a fish?'/ 'You ate a fish, I don't know'

The 'exclamative' is used in a similar way to express both ignorance and surprise:

minja-wa:-ndu dha-yi (260, 262)
 what+ABS-EXCL-2NOM eat-PAST
 'What did you eat?'/ 'You ate *what*?'
 guya-wa:-ndu dha-yi (260, 262)
 fish+ABS-EXCL-2NOM eat-PAST
 'So you ate a fish!'/ 'What? You ate a fish!'

The indication of a question by a form that marks ignorance on the part of the speaker is shown, even more clearly, in the 'participant system' of Kogi, to be discussed in 2.6.2.

Rather differently, in Huichol (Mexico – Grimes 1964: 27) it appears that unmarked forms are often taken to be questions, whereas statements are specifically marked as 'assertive', the reverse of what is normal in most languages. Grimes says 'Any assertion can be agreed with or disagreed with, so that a plain assertion is the semantic equivalent of an English "Yes or No" question. In a conversational context, however, modeless assertions are most likely to be interpreted as questions.' Examples are:

pée-t áa
 ASS-direction go
 'He left'
 mázá tikuucúu
 deer asleep
 'Is the deer asleep?'

However, it should be pointed out that a tentative question is commonly signalled by an expression of ignorance in most languages. Thus in English the verb *WONDER* may be used to suggest a question, or possibly a request:

I wonder if he's arrived
 I wonder if you can help me

(But see Bolinger 1978 for the potential distinction of question and rumination marked by *if* and *whether*.) Conversely, an interrogative may be used to express doubt:

We're expecting him, but will he come?

Two other languages that exhibit Interrogative or both Interrogative and Negative within a modal system are Hixkaryana (Carib, N. Brazil) and Serrano (Uto-Aztecan, California), but in both languages there is also a

system of mood (realis and irrealis) and, for that reason, they are discussed in 6.5.4.

There are also languages that have more than one interrogative marker. Thus Tuyuca has a ‘first-hand’ and a ‘nonfirst-hand’ interrogative (discussed briefly in 2.5), while Khezha (Tibeto-Burman) has five types of marker for Yes–No questions and six types for *Wh*-questions (discussed in 2.6.1).

2.4 Two other possibilities

There is a ‘remote’ suffix in Kashaya (now rarely used), that indicates an activity in the remote past, often the personal experience of the speaker, or if less remote, a past that is irretrievable ‘because the participants are dead, or because the world has changed so much’ (Oswalt 1986: 40):

men ši-yi? ci? -thi-miy
 this do-PL+HAB-NEG-REM
 ‘They never used to do that in the old days’

It may be significant, however, that the example itself relates to a habitual activity, for a similar example (labelled ‘habitual’) is found with an irrealis marker (see 6.6.9) in Bargam (Papua – Roberts 1990: 384, quoting an unpublished manuscript by Hepner):

miles-eq leh-id teq anamren aholwaq-ad in
 return-SS+IRR go-DS+IRR then owner see-SS+SIM 3SG
 didaq tu-ugiaq
 food PERF-give+HAB.P+3SG
 ‘When (the pig) would return and then the owner would go and, on
 seeing it, used to give it food’

Roberts (1990: 283, fn. 13) notes that English also uses an (irrealis?) modal for expressing habitual past – *When we were children, we would often play at being grown ups*. It could be added that the use of a form of WILL suggests tendencies rather than just habitual action – that we tended to play rather than that we actually did play, and that it may be this that permits habitual past actions to be seen as irrealis. The relevance of this is, however questioned by Bybee *et al.* 1994: 239 (see 7.3).

Kashaya (Oswalt 1986: 42) also has a means of expressing ‘what proved to be true’, where what was believed in the past is in contrast with the present known situation. Here there are two suffixes, *-bi-*, which is labelled ‘Inferential II’ and *-wa-*, which is a marker of the verbal feature ‘absolutive’:

kbe hi?baya-?-bi-w
 my man-ASS-INF-ABSOL
 ‘It turned out to be my husband’

Slightly different are the forms noted by Hockett (1958: 237–8) for Menomini (Algonquian) – see 2.3 for the full paradigm:

piasah so he *is* coming after all! (despite our expectations to the contrary!)
piapah but he was going to come! (and now it turns out that he is not!)

In both languages there is a contrast between what was expected and what proved to be true. With the Menomini examples, moreover, it would seem that nothing is being asserted – possible translations might be ‘It is surprising that he is coming’ and ‘It is surprising that he isn’t coming.’ On this interpretation these could be handled in terms of presupposition (as in 1.4.1 and 6.6.7). The Kashaya example does not lend itself to a similar interpretation, but there is still the element of unexpectedness and it may be that this is more important than the assertion. These examples also have something in common with the use of English *had to* (2.1.6). All of them are concerned with the recognition of a present time fact, but relating it to what was known or assumed in the past.

2.5 Structure of propositional systems

The organization of this chapter is in terms of two types of propositional modality, Epistemic modality and Evidential modality, and of two main types of Evidential modality, Reported and Sensory. This is essentially the same as that of Willett (1988: 96) who arranges the ‘meanings’ of the ‘evidentials’ as:

- I. Direct Evidence
 - A. Visual
 - B. Auditory
 - C. Sensory
- II. Indirect Evidence
 - A. Reported
 - 1. Second-hand
 - 2. Third-hand
 - 3. From Folklore
 - B. Inferring
 - 1. From results
 - 2. From reasoning

Although the individual categories are usually clearly marked, the grouping of them into a structure such as that suggested by Willett appears to be largely based on notional criteria and this may seem to be a weakness of them (see, e.g., Croft 1955: 88). However, sometimes there is formal justification. An example of this can be seen in the analysis of the Tuyuca

evidentials, which have been discussed several times (see, e.g., 2.2.1). The five categories are Visual, non-Visual, Deductive, Reported and Assumptive and Barnes (1984: 267) suggests that the first three are ‘first-hand’ and the last two ‘nonfirst-hand’. However, Malone (1988: 123) argues that only the first two are first-hand, so that Deductive is nonfirst-hand. Malone’s argument is based on the fact that Tuyuca has two types of Interrogative markers that are to be distinguished as ‘firsthand’ and ‘nonfirsthand’, the choice being determined by the type of response expected. By this test, Deductive is firsthand like Visual and non-Visual, because it is a possible response to a first-hand Interrogative. (Another analysis in terms of first-hand and nonfirst-hand is to be found in Botne (1997: 523–5), following Chafe (1986: 263) and Frawley (1992: 412–15).)

Similarly, but more ingeniously, Steele (1997: 289–90) offers an analysis of Luiseño (Uto-Aztecan, S. California) that provides a formal basis for a structural analysis that makes the contrasts quotative/non-quotative, assertion/non-assertion and modal/non-modal. There is, in this language a ‘particle complex’, which, she argues, has four parts. The third part marks person and number and is of no relevance here. The first, second and fourth parts are marked as zero or with a particle (shown as *X* if other than *kun*), the possibilities being:

Non-quotative			
Non-modal assertion	0-0-3-0	OR	0-0-3-X
Non-assertion	X-0-3-0		
Modal assertion	X-0-3-X		
Quotative			
Assertion	0-kun-3-0	OR	0-kun-3-X
Non-assertion	X-kun-3-0	OR	X-kun-3-X

Examples are (with the particle complex analysed into its four parts):

noo n	takwayak	(0-0-n-0)
I	PART.COMPL sick	
‘I am sick’		
heyiquş	şum	(şu-0-m-0)
dig+PAST	PART.COMPL	
‘Were they digging?’		
heyi xumpo		(xu-0-m-po)
dig	PART.COMPL	
‘They should dig’		
nookunun	takwayak	(0-kun-n-0)
PART.COMPL	sick	
‘I’m sick, so I’m told’		

heyiquş şukunum (şu-kun-m-0)
 dig+PAST PART.COMPL
 'They were digging – is that what you said?'

It can be seen that the quotative/non-quotative distinction is made by *0/kun* in second place and assertion/non-assertion by *0/X* in first place. On this analysis it would seem that 'modal assertion' is a form of non-assertion (as would be expected – see 1.1.2) and that for non-quotative non-assertion only it is marked by *X* in fourth position. What is important, of course, is that all three distinctions are marked formally in the language and are not purely notional.

2.6 Discourse and participants

2.6.1 Discourse systems

Modals have an important part to play in discourse, as the participants express their opinions and attitudes and, in general, interact with one another. It is not wholly surprising, therefore, that there are systems which are more directly concerned with discourse relations.

A good example of a system that is more concerned with discourse than modality is that of Cashibo (Shell 1975: 178–93). The forms used for third person singular are:

ka	declarative
kaisa	report in independent clause
isa	report in dependent clause
kara	interrogative
ria	response
riapa	response where question implies scepticism
karaisna	interrogative report

Examples are:

Jorgenin ka ain lápiz q̣ãasiaša
 George DEC+3 his pencil break+PAST+3
 'George broke his pencil'
 Jorgenĩ kaĩsa ain lápiz q̣ãasiaša
 George REP.IND+3 his pencil break+PAST+3
 'George broke his pencil, it is said'
 ain lápiz isa q̣ãasiaša kişon ka
 his pencil REP.DEP+3 break+PAST+3 CLOSE OF QUOTE DEC+3
 Jorgenin ?i kaaşa
 George me say+PAST+3
 'George told me that he broke his pencil'

Jorge kara k^wan
 George INT+3 go+PRES+INT
 'Is George going?'
 (aš) ria k^wanín
 (he) RESP+3 go+PRES+3
 'Yes, he is going'
 (aš) riapa k^wanín
 (he) RESP.SCEP+3 go+PRES+3
 'Yes, he certainly is going!'
 an ka ñokáša ʔi karaisna k^wan kišon
 he DEC+3 ask+PAST+3 me INT.REP+3 go+PAST+1/2 CLOSE OF QUOTE
 'He asked me if I went'

Cashibo, then, has a very full discourse system, and, in particular, forms for making statements, asking questions, giving replies and giving emphatic replies (the first two being interpretable as Declaratives and Interrogatives).

There are some discourse features in the 'belief' clitics of Ngiyambaa (Donaldson 1980: 252–5), which were exemplified in 1.4.5 and are further discussed in 2.7.2. Donaldson defines them as:

<i>Assertion</i>	'used to draw the addressee's attention to a statement'
<i>Categorical assertion</i>	'the speaker presents the statement . . . as significant for its absolute truth'
<i>Counter-assertion</i>	'either contradicts a previous statement or is intended to counter some presupposition the speaker suspects his addressee of entertaining'
<i>Hypothesis</i>	'marks a statement as an unconfirmed hypothesis on the part of the speaker'

Examples, repeated from 1.4.5, are:

waɣa:y-ba:na yana-nhi
 neg-ASS-3ABS walk-PAST
 'He didn't walk (again)'
 guni:m-baɣa-nu: balu-y-aga
 mother+ABS-CATEG.ASS-2OBL die-CM-IRR
 'Your mother is bound to die'
 guyan-baga:-dhu gaɣa
 shy+ABS-CNTR.ASS-1NOM be+PRES
 'But I'm shy!'
 gali:-ɲinda-gila ɲiyanu balu-y-aga
 water-CARIT-HYPOTH we+PL+NOM die-CM-IRR
 'We'll probably die for lack of water'

'Counter-assertion' is obviously a matter of discourse. 'Categorical asser-

tion' also seems to be related to discourse in that it can be triggered by what someone else has said. In one example a place is being discussed and someone intervenes with:

ɲadhi-la:-baɾa-dhu	badhiyi
there+CIRC-EST-CATEG.ASS-1NOM	come+PAST
(EST=ESTablished reference)	
'That's exactly where I've (just) come from!'	

Another example is translated 'I've seen it already.'

It has not always been noted that, in an indirect way, English has a formal indication of discourse features, found in constructions associated with the NICE properties of the English auxiliary verbs. (This is an acronym suggested by Huddleston (1976: 333) and based on Palmer (1974: 15).) These features are, with examples:

Negative	I can't go
Inversion	Must I come?
'Code'	He can swim and so can she
Emphatic Affirmation	He <i>will</i> be there

Obviously, negation may be used to deny what someone has said, 'inversion' to form a question and 'emphatic affirmation' to reaffirm what may have been doubted. 'Code', in particular, is very relevant to discourse. It allows the use of modal verbs without a lexical verb, the lexical verb being 'understood' (recoverable) anaphorically from a previous lexical verb as in:

A. Can he do it?
B. Yes he can.

There could be a series of such modal verbs – *Well he may*, *But he must*, etc., so that anyone joining the conversation would not know what the topic was. Moreover, as was seen in 2.3, Interrogative and Negative are sometimes members of modal systems, and they too play an important role in discourse.

Many languages have sets of particles that may be used in discourse, e.g. the 'modal particles' of German (Curme (1905: 368 (1960: 350–2))). The most important ones (with Curme's glosses) are:

ja	'truly, why, don't you see, you know'
doch	'after all, though, just, truly, surely . . .'
denn	'evidently, as is well known, as I learn . . .'
schon	'never fear, no doubt, surely, as a matter of course'
wohl	'indeed, certainly'

However, these are not directly relevant to discourse and probably cannot be defined as members of a grammatical system.

More precisely definable is the set of sentence-final particles in Chinese which are glossed as follows by Li and Thompson (1981: 238ff.):

le	currently relevant state
ne	response to question
ba	solicit agreement
ou	friendly warning
a/ya	reduce forcefulness
ma	question

Some examples are:

- yíyàng de le (264)
 same it LE
 ‘It’s the same (you’re wrong in thinking that what you have is different)’
- tāmen yǒu sān tiáo niú ne (301)
 they exist three CL cattle NE
 ‘(Listen) they have three cows’
- wo he bàn bei ba (308)
 I drink half glass BA
 ‘I’ll drink half a glass, OK?’
- wo yào dǎ nǐ ou (308)
 I will hit you OU
 ‘Let me tell you, if you do this, I will hit you’
- nǐ lái a/ya cf. nǐ lái (315)
 you come A/YA you come
 ‘You come here’ ‘You come here!’
- nǐ hao ma (305)
 you well MA
 ‘Are you well? (= How are you?)’

Even more striking is the situation in Khezha (Tibeto-Burman – Kapfo, quoted by Bhat 1999: 80–1), where there are eight types of marker for Yes–No questions and another eight for *Wh*-questions, the choice being largely related to features of discourse.

For Yes–No questions the relevant issues are:

- (i) The speaker knows and expects confirmation;
- (ii) The speaker has reliable information and expects confirmation;
- (iii) The speaker indicates uncertainty (a negative marker);
- (iv) The speaker presumes the proposition to be correct, but is uncertain;
- (v) The speaker takes it for granted that the addressee will agree;
- (vi) The speaker is amazed at the proposition, asking for reaffirmation;
- (vii) The speaker has heard an unusual rumour and is asking for verification;

- (viii) The speaker is unconvinced, as the event is unusual, and wants reaffirmation.

Examples are:

nò merí-e nì-à-díre
 you Mary-ACC love-REAL-INT
 ‘You are in love with Mary, aren’t you?’

nò mhechə-à-ya/nì
 you work-REAL-INT
 ‘Are you working?’

àpfə wò-dà-mo
 father come-PERF-INT
 ‘I presume my father has come?’

àpfə wò-dà-níe
 father come-PERF-INT
 ‘Perhaps my father has (probably) come, hasn’t he?’

nò-nə méri-e nì-à-yo
 you-NOM Mary-ACC love-REAL-INT
 ‘Oh! You are in love with Mary? (I can’t believe it)’

ì-zò lésə-à-momí
 your-mother sick-REAL-INT
 ‘Is your mother ill?’

nò àwe-è nì-à-lè
 you me-ACC love-REAL-INT
 ‘Do you love me?’

(The suffix *-à* is glossed as ‘imperfective’ by Kapfo, but Bhat is almost certainly correct in treating it as *realis*, contrasting with *irrealis* future.)

2.6.2 *Participant systems*

In some languages there are systems indicating more direct reference to the participants in the discourse.

In Kogi (Chibchan, N. Colombia) there is, Hensarling (1982) tentatively suggests, an ‘evidential’ system indicating ‘who knows what about the situation being discussed’. She suggests a matrix for five particles, as shown in:

	<i>Speaker</i>	<i>Hearer</i>	<i>Gloss</i>
ni	+	+	remind
na	+	–	inform
shi	+	–	ask
skaN	–	–	doubt
ne	–	?	speculate

Examples are:

ni-gu-ku-á
 REMIND-do-I-NEAR.PAST
 ‘I did it just a while ago, as you know’

na-gu-ŋgú
 INFORM-do-INTERMEDIATE.PAST
 ‘I tell you he did it some time ago’

shi-ná
 ASK-be+PROXIMATE
 ‘Is that the way it is?’

skaŋ-gú
 DOUBT-do+PROXIMATE
 ‘Who knows if it did just now?’

năbbi nóŋgutse né haŋgna
 lion little SPECULATE think+PUNCTILIAR
 ‘“I wonder if it is a small lion”, he thought’

(né appears in sentence-final position here, but it can occur in the same place as the others.)

‘Remind’ relates what both speaker and addressee know, ‘Inform’ what the speaker knows but the addressee does not, ‘Ask’ what the addressee knows but the speaker does not, ‘Doubt’ what neither know and ‘Speculate’ what the speaker does not know (the addressee’s knowledge not being considered).

A more complex system is proposed by Lowe (1972) for Nambiquara (Brazil). He suggests that there is a two-dimensional system involving:

event verification:	individual, collective
speaker orientation:	observation, deduction, narration

The ‘speaker-orientation’ system is clearly an evidential system with the three familiar terms Visual, Deductive and Report:

observation	I report what I saw
deduction	I tell my deduction
narration	I was told

The ‘event verification’ system involves speaker and addressee – whether the speaker alone or both speaker and addressee saw the event, deduced that it occurred or were told about it. Lowe provides the following glosses:

individual, observation:	‘I report to you what I saw the actor doing’
individual, deduction:	‘I tell you my deduction of an action that must have occurred because of something I see or saw’

individual, narration:	'I was told by someone that a certain action occurred'
collective, observation:	'I report what both I and the addressee saw the actor doing'
collective, deduction:	'From what the speaker and the addressee saw, they deduce that a certain action must have taken place'
collective, narration:	'Both speaker and addressee were told that a certain event took place'

The individual verification third-person forms for the verb *wa³kon³* (with *hē³* glossed as 'past' and *ra²* as 'definite aspect' and two of the person + verification–orientation markers underlined> are given as:

<i>wa³kon³na¹hē³ra²</i>	'He worked'
<i>wa³kon³nū³hē³ra²</i>	'He must have worked'
<i>wa³kon³ta¹hē³ra²</i>	'I was told that he worked (past)'

The corresponding collective variation forms (with only final *wa²* glossed as 'indefinite mood') are given as:

<i>wa³kon³tait¹ti²tu³wa²</i>	'Both you and I saw that he worked'
<i>wa³kon³nait¹ti²tu³wa²</i>	'He worked, as deduced from what we saw'
<i>wa³kon³ta¹tē¹ʔti²tu³wa²</i>	'It was told us that he worked'

2.7 Declaratives

It is generally the case that, in languages with epistemic modal systems, there is a form that is unmarked for modality, and which, notionally, simply makes an unqualified assertion. It can thus also be seen as Realis, while the modal forms are Irrealis. This form is identified as the Declarative. Thus, for English, the Declarative is marked by the absence of a modal verb, although it will be marked for the other verbal categories, tense and aspect. (There is usually another unmarked form, the Imperative, but this is notionally not assertive but directive, and is best discussed together with deontic modality – see 3.4.)

Since it is unmarked for modality, the Declarative is not to be seen as in any sense 'stronger' than a modal form. It simply asserts without indicating the reasons for that assertion or the speaker's commitment to it. It is perfectly possible for a speaker to say *John is in his office* without being wholly sure or when there is strong evidence for it, i.e., in circumstances where *John may be in his office* or *John must be in his office* would also be appropriate.

It is perfectly possible to express modal notions without the use of modal systems. This can be achieved through the use of lexical verbs as in:

I think that Mary is in her office
 I saw John in his office this morning

Grammatically, the first of these is not an example of the modal judgment Speculative, and equally the second is not an example of the Visual evidential. They are both Declaratives, and the relevant notional interpretation is in terms of asserting that ‘I think’ and that ‘I saw’. (The situation is different and generally simpler in most of the mood systems discussed in Chapters 5–7, where there is generally a system of two, Realis and Irrealis. Grammatically neither is unmarked, both being of equal status within the system.)

2.7.1 *Declaratives and modal systems*

In many languages there is a Declarative form that is independent of any modal system. This is so of English where there is a contrast between a Declarative and epistemic modals:

Mary is at home
 Mary may/must/will be at home

In terms of the Realis/Irrealis distinction, the situation seems to be clear – the Declarative with no modal verb is Realis and the modal forms are Irrealis. However, this simple situation – unmarked Realis and a system of marked realis forms – is not typical of all languages with modal systems. There are several different situations.

First, in a few languages, the Declarative is a term in the same formal system as the modal categories. This is so in Ngiyambaa (New South Wales, Australia – Donaldson 1980: 159–62), as fully explained and exemplified in 1.4.5. Here past and present (Declaratives) occur in the same system as markers of Speculative (‘irrealis’) and Deductive (‘purposive’) as well as Imperative (discussed in 3.4).

Secondly, and much more problematic, is the situation in which there appears to be no formally and notionally unmarked Declarative because all the categories not only belong to a single formal system, but also are all notionally evidential. This is so in Tuyuca (see 2.2.1), where the ‘strongest’ evidential is Visual, and so the closest to Declarative, but it does not simply assert, it also indicates that there is visual evidence for the proposition. Examples, now repeated and glossed with the evidential labels used in this book, are:

Tuyuca
 dīga apé-wi
 soccer play+3SG+PAST-VIS
 ‘He played soccer’

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diiga apé-ti
soccer play+3SG+PAST-AUD
'He played soccer'

diiga apé-yi
soccer play+3SG+PAST-DED
'He played soccer'

diiga apé-yiḡ
soccer play+3SG+PAST-QUOT
'He played soccer'

diiga apé-hīyi
soccer play+3SG+PAST-ASS
'He played soccer'

It is instructive to compare an apparently very similar system in Central Pomo (repeated from 1.2.1) where there is both an unmarked form (though it is rarely used) and a set of marked evidential categories (though 'general knowledge' seems very close to being an instance of Declarative):

čbémul
rain fell
'It rained'

čbémul-ʔma
rain fell-GEN.KNOW
'It rained' (that's an established fact)

čbémul-ya
rain fell-VIS
'It rained' (I saw it)

čbémul-nme'
rain fell-AUD
'It rained' (I heard it)

čbémul-ʔdo
rain fell-HSY
'It rained' (I was told)

čbémul-ʔka
rain fell-INF
'It rained' (Everything is wet)

Central Pomo, then, has a Declarative and an evidential system, whereas Tuyuca has no Declarative but simply an evidential system.

There is, in fact, another language for which it has been claimed that there is no 'unmodalized declarative' (Lyons 1982: 110), this claim being based on the assertion by G. H. Matthews (1965: 98) that all main sentences in Hidatsa end in a 'Mood' (for details see 2.2.1). However, this is not strictly true, since

it is shown that the 'Period' marker does not occur with sentences that end with a past tense marker (G. H. Matthews 1965: 110, where it is stated that Past + Period is 'reduced' by a rule to Past). In fact, then, there is an unmarked Declarative with past tense.

Thirdly, even when the Declarative is formally distinct from the evidential system, sometimes it can be used only where there is direct or visual evidence. Thus in Serrano (Hill 1967: 18), which is discussed in 6.5.4 because it has mood as well as a modal system, the simple past is used only to report first-hand knowledge. The speaker, that is to say, identifies himself as a witness. Hill gives as an example:

'i:p bi' wahi' pinq
 here he+PAST coyote pass
 'The coyote passed here' (I saw him)

The importance of direct evidence is also shown in the account of Hixkaryana (see 2.2.2 and 6.5.4). Derbyshire (1979: 145) says that the distinction between 'hearsay' and 'eyewitness' is crucial and that 'in any connected discourse concerning events that have not been witnessed by the speaker', the 'hearsay' evidential occurs in almost every clause.

Perhaps even more striking is the contrast in Sherpa between direct evidence and 'indirect-hearsay' evidence as illustrated in 2.2.4. Givón (1982: 34–5). reports that a Lama narrating the *Life of Buddha* told the bulk of the story in the perfective/past using the hearsay/indirect evidence suffix, and that in the entire narrative only two direct evidence suffixes occurred, and that these were both in direct quoted speech. Givón comments that the story 'is undoubtedly considered the "truest" of all stories for a devout Tibetan Buddhist such as the story-teller himself, yet is told in the "hearsay/indirect evidence mode" because it was not witnessed by the speaker'.

Fourthly, as was noted in 2.1.3, in some languages, e.g. Kashaya and Makah, the Deductive is used much more widely than English MUST. It seems that, while MUST lays some emphasis on the fact that a judgment (based on evidence) is being made, Deductive in these languages is used for any statement for which there is circumstantial evidence. This relates to the question whether, in terms of the speaker's belief, the Deductive is stronger or weaker than the unmarked Declarative. On Kashaya, Oswalt (1986: 34) says 'the Inferential suffix implies no lack of certainty, merely lack of higher ranking evidence'. For Wintu, Schlichter (1986: 51) says that for Assumptive as well as Deductive the speaker believes the statement to be true 'because of circumstantial sensory evidence'. For Deductive this evidence is most often visual, and for Assumptive it is 'because of his experience with similar

situations, regular patterns, or repeated circumstances common in human life’.

However, it may be that the same is true of epistemic MUST and WILL in English. Someone who says *John must be in his office* or *John’ll be in his office* must, surely, believe that John is in his office. The issue of Declarative vs modal categories is not simply a matter of belief, but of the contrast between merely asserting and indicating either that a judgment has been made or that there is evidence for the proposition. The difference between English and languages such as Kashaya is that there is a greater tendency to use modal forms in Kashaya. Yet the fact that this language uses Deductive where English would use Declarative, does not necessarily imply that this Deductive is more indicative of the speaker’s belief than English MUST.

There is one final point. The Declarative may not be the unmarked form. Thus in Huichol (Mexico – Grimes 1964), an unmarked form is usually taken as a question, while the Declarative has the ‘assertive marker’ (see 2.3).

2.7.2 Assertion and ‘strong assertion’

In some languages there are, it seems, both a ‘weaker’ and a ‘stronger’ Declarative. Thus Imbabura (Cole 1982: 164) has forms for both ‘emphatic first-hand information’ and ‘first-hand information’ as in (the second repeated from 1.4.2):

ñuka-ta miku-naya-n-mari
I-ACC eat-DES-3-EF-INF
‘I want to eat’

kan-paj ushi-wan Agatu-pi-mi
you-of daughter-with Ageto-in-F-INF
‘I met your daughter in Ageto’

It was also noted in 1.4.5 and 2.6.1 that there are two ‘belief’ clitics in Ngiyambaa (Donaldson 1980: 252–5). These are defined as:

Assertion ‘used to draw the addressee’s attention to a statement. It is literally equivalent to a statement with “I assert that . . .”’
Categorical assertion ‘the speaker presents the statement . . . as significant for its absolute truth. Literally, [it] is equivalent to prefacing the statement with “I categorically assert that . . .”’

Examples (repeated) are:

wajary-bar-na yana-nhi (254)
NEG-ASS-3ABS walk-PAST
‘He didn’t walk’ (again)

wəŋa:y-ba:ɽ-na yana-nhi (254)
 NEG-CATEG-ASS-3ABS walk-PAST
 ‘He absolutely didn’t walk (again)’/‘He never walked again’

Similarly, for Hidatsa (see 2.2.1), G. H. Matthews (1965: 99–100) distinguishes between ‘Emphatic’ and ‘Period’, with the comments:

- *Emphatic*: ‘indicates that the speaker knows the sentence to be true; if a sentence that ends with the Emphatic is false, the speaker is considered a liar’
- *Period*: ‘indicates that the speaker believes the sentence to be true; if it should turn out otherwise, it would mean that he was mistaken, but by no means a liar’

Examples (repeated) are:

wacéo iikipi kurè héo ski
 man pipe carried EMPH
 ‘The man (sure) did carry the pipe’
 wacéo iikipi kurè héo c
 man pipe carried PER
 ‘I suppose the man carried the pipe’

The distinction, it is claimed, is one of knowledge vs belief, but the situation in all three languages is very similar.

If modal systems are treated in terms of Realis and Irrealis, it seems that in these three languages there is not only an Irrealis system, but also a two-term Realis system, in which two degrees of strength of assertion are distinguished.

3

Modal systems: Event modality

Deontic and dynamic modality refer to events that are not actualized, events that have not taken place but are merely potential, and may, therefore, be described as ‘event modality’. The basic difference between deontic and dynamic modality is that with dynamic modality the conditioning factors are external to the person indicated as the subject (that he is permitted, ordered, etc., to act), whereas with deontic modality they are internal (that he is able, willing, etc., to act). There are, however, some other points. First deontic modality is generally dependent on some kind of authority, often the speaker. Secondly, Commissive (where the speaker guarantees that the action will take place) may also be included under deontic modality. Thirdly, dynamic ability may sometimes be interpreted in terms of the general circumstances that make action possible or impossible (see 1.3.2) rather than the actual ability of the subject.

However, it must be admitted that a great deal of material for this chapter comes from English, for English appears to have a more extensive system of deontic modality than many other languages.

3.1 Formal systems

As already noted several times (see especially 1.4.5), the same forms are often used for epistemic and for deontic/dynamic systems. Equally the same form (e.g. CAN) may be either deontic or dynamic. Moreover, as also noted in 1.4.5, modal markers may occur in the same formal system as other grammatical markers, e.g., the ‘purposive’ marker in Ngiyambaa, which not only may be either epistemic or deontic, but also occurs in a formal system containing tense markers.

3.2 Deontic

The most common types of Deontic modality are the ‘directives’, ‘where we try to get others to do things’ (Searle (1983: 166)).

3.2.1 Directives

Two kinds of directive are expressed in English by MAY and MUST, the two modal verbs that also express epistemic Speculative and Deductive, though in colloquial speech CAN is preferred to MAY, but MAY still survives – see Palmer (1990: 71) – as in:

You may/can go now

You must go now

These can be identified as Permissive and Obligative.

There are similar pairs of verbs in other European languages, e.g. German:

Du magst herein kommen

You can/may+2SG+PRES in.here come

‘You may come in’

Du musst herein kommen

You must+2SG+PRES in.here come

‘You must come in’

So also in Italian, where, however, the third person form is used instead of the second for politeness:

Puó entrare

can/may+3SG+PRES come.in

‘You may come in’

Deve entrare

must+3SG+PRES come.in

‘You must come in’

Modern Greek also uses the same form as for the epistemic counterparts, but with one difference – that the Permissive form is not impersonal like epistemic Speculative, but agrees with the subject:

boris na fiyis

can+2SG+PRES that you.leave

‘You may leave’

prepi na fiyis

must+IMPERS that you.leave

‘You must leave’

Similarly, although French has two verbs, POUVOIR and DEVOIR, the form normally used for obligation is the impersonal *il faut*.

Rather more surprising, in Danish and North Frisian, the modal verb used for epistemic deduction is also used for both deontic permission and obligation, although other forms are available to avoid ambiguity. The relevant verb is *mätte* in Danish (Davidsen-Nielsen 1990: 84, 187, 194):

Modal systems: Event modality

Der må være flere andre grunde
There MÅTTE+PRES be several other reasons
'There must be several other reasons' (epistemic necessity)

Du må danse en dans til
you MÅTTE+PRES dance a dance more
'You may dance another dance' (deontic possibility)

Vi må bare tage chancen
we MÅTTE+PRES just take the chance
'We'll just have to take the chance' (deontic necessity)

There are pairs of forms in many other languages. Thus in Tamil (Dravidian – Asher 1982: 167–70) the suffixes *-laam* and *-um* are used, as with epistemic modality – see 2.1.2 – for permission and obligation (Debitive) respectively:

veṇum-ṇṇaakkaa, naalekki avan peeca-laam
want-COND tomorrow he speak-PERM
'If he wants, he can speak tomorrow'

avan aṇke pooka-ṇum
he there go-DEB
'He must go there'

Similarly in Lisu (Lolo-Burmese – Hope 1974: 122, 126), there is a set of intransitive verb stems that include forms for obligation and permission, though these belong to a wider system that includes dynamic modals – see 3.3.1:

ása nya ami khwa wà-à
Asa TOP field hoe obligatory-DEC
'It is obligatory for Asa to hoe fields'

ása nya ami khwa da-à
Asa TOP field hoe acceptable-DEC
'It is acceptable for Asa to hoe fields'

The modals of permission and obligation (Permissive and Obligative) can be interpreted (like epistemic Speculative and Deductive) in terms of possibility and necessity. This is discussed in detail in 4.1.2.

3.2.2 Commissive

Commissives are defined by Searle (1983: 166) as 'where we commit ourselves to do things'. They are signalled in English by the modal verb *SHALL*. They can usually be seen either as promises or as threats; the only difference between these seeming to be whether what the speaker undertakes to do is or is not welcome to the addressee:

John shall have the book tomorrow
 You shall do as you are told

Here the speaker commits himself to ensuring that the event takes place, by guaranteeing to arrange that John will receive the book and that the addressee will do what is demanded. (SHALL is also used together with WILL, in some formal dialects of English, for future time reference, but in this future use SHALL occurs only with first person subjects. SHALL used with 2nd or 3rd person subjects specifically signals a Commissive.)

Most languages do not have a specific grammatical form for Commissives, but it may also be noted that in Ngiyambaa (Donaldson 1980: 160, 161) the ‘irrealis’ marker is used epistemically for ‘might’ and ‘likely’, but deontically for authoritative ‘shall’ (see 1.4.5):

yuruŋu ɲidja-l-aga
 rain-ERG rain-CM-IRR
 ‘It might/will rain’

waŋa:y-ndu-gal dhagurma-gu yana-y-aga
 NEG-2NOM-PL cemetery-DAT go-CM-IRR
 ‘You shall none of you go to the cemetery’

It may be significant that the last example and other similar examples quoted by Donaldson are negative. This appears to be related to, but not identical with, the issues of negation with possibility and necessity discussed in 2.1.6 and 3.3.2. (Ngiyambaa has a different form for obligation, the ‘purposive’, which will be discussed in 3.5.)

3.2.3 Modifications

Just as the past tense forms of epistemic modals can be used to indicate more tentative, weaker, judgments, so past tense forms of some of the deontic modals are used to weaken the force of the modality. MUST has *ought to* and *should* (for any difference between these see Coates 1983: 7783), and MAY and CAN have *might* and *could*). Morphologically, *should* is formally the past tense form of SHALL, but notionally it functions as the modified form of MUST.

The position is clearest with *ought to* (*should*). There are two points. The first is that these verbs differ from MUST in that the speaker admits the possibility that the event may not take place. This is seen in:

He ought to /should come, but he won’t
 *He must come, but he won’t

The second example here is most unlikely, if not anomalous; if the speaker thinks that the obligation may not be fulfilled, *ought to/should* would be used.

Secondly, *ought to/should* (plus *have*) can refer to past events, whereas **MUST** cannot:

You ought to/should have come

Must have can be interpreted epistemically (2.1.6), but never deontically.

The explanation for these two points is that *ought to* and *should* are essentially conditional – referring to what would occur or would have occurred. The conditionality does not refer strictly to the modal, but to the event expressed in the proposition (Palmer 1979: 102; 1990: 125). Thus the following glosses are appropriate:

You ought to come

‘You have an obligation to come, and you would come if you fulfilled it’

You ought to have come

‘You had an obligation to come and you would have come if you had fulfilled it’

The situation with *might* is less simple. It is often used in questions as a more tentative, more polite, form for asking permission (Palmer 1990: 80):

Might I come in at the moment on this, Chairman?

But it is also used to make a quite positive suggestion, as in (Palmer 1990: 187):

You might try nagging the Abbey National

You might have told me

Here again, as with *ought to/should*, there is conditionality, referring to events that would take place or would have taken place under certain conditions. With *ought to/should* the conditions were the carrying out of an obligation; with *might* they seem to be those of behaving in a sensible or proper way. But this is a much stronger notion than mere permission, and it is clear that *might have* does not mean ‘You would have acted thus if you had had permission.’ Apparently then, *might* expresses a stronger kind of deontic modality than **MAY** – a positive suggestion rather than mere permission.

Degrees of obligation are, however, signalled in other languages in different ways. Thus in Albanian (Newmark, Hubbard and Prifti 1982: 102–3; Frawley 1992: 423) the ‘must’/‘should’ contrast is marked by different modal verbs:

kjo këmishë duhet larë

this shirt must wash

‘This shirt must be washed’

kjo këmishë do larë

this shirt want wash

‘This shirt needs washing’

3.2.4 Subjectivity

The deontic modals are often used to indicate permission and obligation emanating from the speaker, but it cannot be claimed that they are always subjective in this sense. The speaker may not be involved in, e.g.,

You can smoke in here

You must take your shoes off when you enter the temple

However, generally there is an implication that the speaker agrees with the permission or obligation. For obligation it is relevant that English has a form that is an alternative to *MUST*, and which generally indicates that the speaker takes no responsibility for the obligation. This is *HAVE TO*. There is a potential contrast between:

You must come and see me tomorrow

You have to come and see me tomorrow

The first could be no more than a suggestion or an invitation. The second suggests that there is some compelling reason independent of the speaker. If there is not, the addressee might take offence, regarding it as presumptuous of the speaker to say what he or she has to do.

Similarly, *BE SUPPOSED TO* is used instead of *should* or *ought to* where the speaker is not responsible:

You should go to London tomorrow

You are supposed to go to London tomorrow

It is difficult to judge to what extent directives, especially Obligative, are subjective in this sense. There is, in Tiwi (Australia – Osborne 1974: 44), a marker that is labelled ‘compulsional’ (see 2.3):

a-u-kəjimi

he-COMPUL-do

‘He has to do it’

a-u-ra-kəjimi

he-COMPUL-FUT-do

‘He will have to do it’

This, Bybee (1985: 167) suggests, on the basis of the translations, does not indicate obligation imposed by the speaker, but merely asserts that the agent has an obligation. However, Osborne says that this ‘compulsional’ form is ‘equivalent to “must” or “have to”’, which does not support Bybee’s suggestion. He also states that it occurs in the non-PAST or the present, which implies that it does not occur in the past. This might well suggest that it is, in fact, subjective (that the speaker imposes the obligation), as will be argued in the next section.

3.2.5 Past time reference

It was suggested in 2.1.7 that, because the epistemic modals are essentially subjective, they are not used in past tense forms to indicate a judgment in the past.

It is equally true, for English at least, that the past tense forms of the deontic modals are not used for past time reference. The past tense forms of MAY and CAN, *might* and *could*, are not used to indicate permission in the past, while MUST has no past tense forms. It is, therefore, not possible to say with a deontic sense:

*I might/could/must(ed) come yesterday

(*Could* is possible in a dynamic sense – 3.3.2.) Moreover, for an obligation in the past, which by definition cannot be imposed by the speaker at the present time, the past tense form of HAVE TO is available:

He had to be in London yesterday

As with the epistemic modals (2.1.7), past tense forms may be used in reported speech (with MUST unchanged or replaced by *had to*):

He may(can)/ must come on Tuesday

He said he might(could)/must/had to come on Tuesday

There is a slightly different situation with the past tense forms *should* and *ought to*, in that they do not strictly lay an obligation, but merely indicate what the speaker thinks is right. It is, therefore, possible to indicate such deontic judgments in relation to past events. English does this with *should have* and *ought to have*:

He should have/ought to have gone to London the next day

The restrictions on past tense are, however, essentially a feature of English. There is no similar restriction on, for instance, the comparable German modal MÜSSEN, which has past tense forms that may be used for past time reference:

Ich musste fleissig arbeiten

I müssen+3SG+PAST+IND hard work

'I had to work hard'

The same is true of the related verbs in the Romance languages.

3.3 Dynamic

3.3.1 Ability and willingness

There appear to be two types of dynamic modality, expressing ability and willingness (Abilitive and Volitive), which are expressed in English by CAN and WILL:

My destiny's in my control. I can make or break my life myself
 Why don't you go and see if Martin will let you stay?

Again it is to be noted that the verbs here are verbs also used for other types of modality. CAN is used both for epistemic modality (but only when negated) and for deontic modality. It can thus be seen as expressing a third type of modality, one that is internal to the subject of the sentence. Similarly the permissive forms of other European languages that were discussed in 3.2.1 are used to indicate ability. In many languages there is no formal distinction between permission and ability, but in English, however, the distinction is clear, in that MAY is not used to indicate ability (see 4.2.3). As mentioned in 1.3.2, however, dynamic CAN is used not only to refer to physical and mental powers, but also to include circumstances that might affect the person involved, as (repeated) in:

He can run a mile in under four minutes (ability)
 He can escape (there is nothing to stop him)

In Lisu (Lolo-Burmese – Hope 1974: 122–6) there are distinct forms to indicate two types of ability, one in the sense of ‘knowing how’, the other in the sense of physical ability:

ása nya ami khwa kwú-ǵ
 Asa TOP field hoe mentally.able-DEC
 ‘Asa is able (knows how) to hoe fields’

ása nya ami khwa kwhu-ǵ
 Asa TOP field hoe able-DEC
 ‘Asa is (physically) able to hoe fields’

ása nya ami khwa da-ǵ
 Asa TOP field hoe acceptable-DEC
 ‘It is acceptable for Asa to hoe fields’

There are three other types of dynamic possibility, indicating freedom from taboo, no hindrance and having sufficient courage:

ása nya ami khwa tyè-ǵ
 Asa TOP field hoe freedom.taboo-DEC
 ‘It is not taboo for Asa to hoe fields’

ása nya ami khwa bala-ǵ
 Asa TOP field hoe no.hindrance-DEC
 ‘Asa is free to hoe fields’

ása nya ami khwa pə-ǵ
 Asa TOP field hoe able.couragewise-DEC
 ‘Asa dares to hoe fields’

However, the Lisu verbs are members of a fairly large group, which includes

not only the deontic and dynamic forms, but also others with translations ‘It is normal for Asa to hoe the field’, ‘Asa’s hoeing the field has happened before’, ‘Asa is tired of hoeing the field.’

It is worth adding that there are European languages that distinguish these two kinds of ability, by using the verb ‘know’ for mental ability, e.g. French *savoir*:

Il	sait	nager
he	know+3SG+PRES+INDIC	to swim
‘He can swim’		

However, there are no formal grounds for treating *SAVOIR* as a modal verb (see 4.2.2). More strikingly, in Chinese the modal verb *hui* has the general meaning of ‘know how’. Hockett (1968: 62) comments that ‘If a person knows how to speak Swahili, he *hwèi* speak Swahili; if he smokes he *hwèi* smoke.’ He adds that it is used even with inanimate objects – ‘one can say that a high wind *hwèi* . . . blow down a tent or that an electron *hwèi* behave in accordance with the equations of wave mechanics’.

One further point may be made in connection with Lisu. There is a form to express ‘able courageously’, which can be translated by *dare* in English. *DARE* is a modal verb, though restricted to non-assertive environments (see 1.4.2). It can, then, be seen as a marker of another type of dynamic modality (Palmer 1990: 111–12).

WILL is used in English not only as an Assumptive (2.1.3) and to indicate future (4.3.2), but as a Volitive, to express willingness. Attested examples that clearly indicate such willingness rather than mere futurity are (Palmer 1990: 134):

Why don’t you go and see if Martin will let you stay?
She loves him and she won’t leave him.
Will you stand by the anchor?

There is one formal difference between future *WILL* and volitive *WILL* – only the latter is used in the protasis (*If*-clause) of a conditional sentence:

It’ll rain tomorrow	→	If it rains tomorrow
John’ll help you	→	If John’ll help you

Corresponding verbs in other European languages usually have much wider meanings, expressing wishing and wanting, and, in the case of German, *WOLLEN* also has the evidential meaning of what is said, claimed or pretended by the subject of the sentence (see 2.2.2).

There are examples in some languages of a form that indicates intention, e.g., the ‘intentives’ of Tonkawa (Texas – Hoijer 1931: 289–90) and Maidu (California – Shipley 1964: 46–52):

heul-a-ha'a 'I shall catch him'
 ʔjǣójʔtàs 'I'm going to go'

Ability can be interpreted in terms of dynamic possibility, as permission may be interpreted in terms of deontic possibility (see 3.2.1). It might be expected that there would be a similar use of **MUST** to express what is necessarily so in a similar way. Such a use of **MUST** is, it seems, fairly rare, but is, perhaps, found in (Palmer 1990: 130):

He's a man who must have money

Another example is (Perkins 1982: 260):

You must go poking your nose into everything

However, these two are slightly different. The first suggests an overwhelming desire, the second an overwhelming propensity. The second sense can also be expressed by *can't help* as in:

You can't help poking your nose into everything

3.3.2 *Past tense and time*

One respect in which these dynamic modals differ in English from the epistemic and deontic modals is that they can be used in the past tense to refer to past time:

When I was younger I could run much faster
 All he would accept was our thanks

However, neither of them can be used to refer to an event that was actually achieved as a result of the ability or willingness. Thus it is not possible to say:

*I ran fast and could catch the bus
 *I asked him and he would come

Yet there is no problem with the corresponding negative form or habitual forms:

I ran fast, but couldn't catch the bus
 I asked him but he wouldn't come
 I always ran fast and so could catch the bus
 I always asked him and he would come

There is, however, no similar restriction on the almost equivalent forms **BE ABLE TO** and **BE WILLING TO**

I ran fast and was able to catch the bus
 I asked him and he was willing to come

There is a very interesting explanation for this – that a modal, being irrealis, is inappropriate to refer to what is a realis situation – a completed or a past

action. But negation and habitual past can be regarded as irrealis, as will be illustrated in 6.6.4 and 6.6.9, and it may be for that reason that the past tense forms can be used when there is negation or reference to habitual action, for the situation is then irrealis. For more discussion see Palmer 1990: 92–7, 154–7.

Past tense forms, as with other modals, can be used as ‘modifications’ to express ability and willingness more tentatively, especially when making an offer as in:

I can/could do that for you
I will/would do that for you

However, there is often an implication of conditionality – ‘if you wanted me to’, ‘if you asked me’.

Could is also used, rather like *might*, for a suggestion:

You could try nagging the Abbey National
You could have told me

3.4 Imperative and jussive

Most languages have a specific form that can be identified as Imperative. In languages such as English where the modal systems are marked by modal verbs the imperative is quite independent of the modal systems, in English being indicated by the simple form of the verb, e.g. *Come here*.

Notionally, Imperative is closely associated with deontic modals. It is clearly directive and usually portrayed as indicating a command. In fact it is often thought to be the strongest of the directives, one that emanates from someone in authority, which, therefore, does not expect non-compliance. However, there are two points to be noted about its relationship to the modal verbs.

First, it can be used not only to give commands, but also simply to give permission or advice as in:

Come in!
Don’t worry about it

Thus, *Come in!* may be interpretable as either *You may come in* or *You must come in*. Secondly, Imperative is performative and subjective in that the speaker actually gives the ‘command’ in the act of speaking. For this reason, unlike the directives, it does not normally occur in a subordinate clause. Compare:

‘You must come’ I said that she must come
‘Come in!’ *I said that come in

(But see 5.4.1.)

Lyons (1977: 747) argues that the imperative can only be, strictly, second person, and never third person (or first person). This may, however, be no more than a terminological issue, since first and third person ‘imperatives’ are often simply called ‘jussives’. Bybee (1985: 171) suggests that where there is a full set of person-number forms the term ‘optative’ is used, but this is not entirely suitable in view of the fact that the term is used traditionally for the ‘optative’ mood in Classical Greek (8.2.2). (But see also 5.4.2 for data for which ‘optative’ might be more appropriate.) The term ‘Jussive’ (plus Imperative) is preferred here.

Imperative and Jussive may belong to a modal system. Thus in Afar (Cushitic, Ethiopia – Bliese 1981: 139–46), there is, in the same formal system, ‘imperative’, ‘jussive’, a ‘subjunctive’ and a ‘consultative’, as exemplified by:

imperative:	‘ab	‘do’
jussive:	‘nakay	‘let me drink’
subjunctive:	‘rabu	‘may I die’
consultative:	a‘boô	‘shall I do it?’

All of these could be regarded as deontic – as examples of event modality, relating to possible events in the future.

It was noted in 1.4.5 that in Ngiyambaa (Donaldson 1980: 159), Imperative occurs in the same system (of verbal inflection) as the Permissive and the Commissive (whose forms also have epistemic functions) – see 3.2.1, 3.2.2:

ŋindu bawuŋ-ga yuwa-dha
you+NOM middle-LOC lie-IMP
‘You lie in the middle!’

However, past and present Declaratives (see 2.7.1) also belong to the same formal system in Ngiyambaa. Imperative is not, then, formally, a member of a deontic system, or even, more strictly, of a purely modal system, but a member of a mixed system.

Similarly, in many of the Native American Indian languages, suffixes that mark Imperative and Jussive are often members of a larger system. There are sets of such suffixes in Tonkawa (Texas – Hoijer 1931: 83–94) and Maidu (Shipley 1964: 46–51), both quoted by Mithun (1999: 171), in which both Imperative and Jussive (‘hortative’, ‘exhortative’) appear. For Maidu, Shipley has the following tense-‘mode’ paradigms:

present-past indicative, future indicative, habitual past indicative,
past punctual indicative;

subjunctive;
monitive optative ('might'), intensive optative ('going to'),
hortatory optative ('let');
interrogative;
imperative I and imperative II.

Examples from the system for Tonkawa are:

Declarative	naxadjganaw-o-'o	'I married'
Assertive	do.na-na'a	'He lies'
Exclamatory	'awac'a'la hedoxa-gwa	'The meat is all gone'
Interrogative	yaxa-ga?	'Did you eat?'
Intensive	heul-a'ha'a	'I shall catch him'
Imperative	'andjo-u	'Wake up'
Potential	ya.dj-'a'-n'ec	'I might see him'
Exhortative	hama'am-do'x-a'dew-e'l	'Let him be burned up'

(The terms 'exclamatory' and 'exhortative' are offered by Mithun 1999: 171.)

There are two kinds of imperative in a number of North American languages (Mithun 1999: 171). In Maidu (North California – Shipley 1964: 51–2), 'Imperative I' 'is used when the action of the order is to be carried out in the presence of the speaker or when there is no interest in the place of the ordered action', while 'Imperative II' is used 'when the ordered action is to be carried out in the absence of the speaker':

Imperative I	sólpi 'Sing!'
Imperative II	mymýk púlkydi dákpaĵtipadà 'Stick it in his door!'

In Cheyenne (Algonquian, Montana, Leman 1980: 41) the difference is between commands to be carried out immediately and those to be carried out later:

méseestse	'Eat!'
méséheo?o	'Eat!' (later)

This is true also of Takelma (Sapir 1922: 94), and for a number of other languages (Bybee 1985: 171). Also, there are often two kinds of command in languages with mood systems, but differing in terms of politeness – see 5.4 and 6.7.2.

Maidu and Cheyenne also have jussives:

'Hortative'	mésęheha	'Let him eat!'
'Hortative'	?yk'ójtàs	'Let me go!'

3.5 'Purposive' in Australian languages

The category identified as 'purposive' in Australian languages expresses obligation (and epistemic necessity) in main clauses (see 1.4.5). An example from Ngiyambaa (Donaldson 1980: 162) is:

ɲadhu bawuŋ-ga yuwa-giri
I+NOM middle-LOC lie-PURP
'I must lie in the middle'

However, a 'purposive' may also be used in a main clause to suggest a result from an unknown cause, as in Dyirbal (Dixon 1972: 69):

balan ugumbil baŋgul yaŋangu balgali
CL+NOM woman+NOM CL+ERG man+ERG hit+PURP
'Something happened to enable or force the man to hit the woman'

In subordinate clauses the same marker is used for both purpose and result. Thus in Yidiny (Dixon 1977: 345–6), the purposive in a subordinate clause usually indicates purpose:

ɖada ɖuɖu:mbu gaɽbaɽba:alɲu ɲuɖu wawa:lɲa
child+ABS aunt+ERG hide+PAST not see+PURP
'Auntie hid the child so that it should not be seen'

But it can also be used to express 'a natural result':

ɲayu burawuŋal duga:l ɖinbiɖinbi:lɲa
I+SUB Burawuŋal+ABS grab+PRES struggle+RED+PURP
I grabbed the water sprite woman and as a result she kicked and struggled

The same suffix is also used in some Australian languages to express indirect commands, as in in Ngiyambaa (Donaldson 1980: 280):

ɲadhu-na ɲiyiyi girma-l ɲinu:
I+NOM-3ABS say+PAST wake-PURP you+OBL
'I told her to wake you'

Typologically this is of interest because these functions of the 'purposive' are very like those of the subjunctive in Latin, which can also be used for purpose, result and for indirect commands (see 5.3.2, 5.3.4, 5.4.2). Notionally these different uses can, perhaps, be seen as related in terms of showing the effect of some cause – an intended effect with purpose and with indirect commands, an actual effect with results. The similarity between the obligation and result interpretations in main clauses (not paralleled in Latin) can, perhaps, be shown by different possible translations – 'The man had to hit the woman' and 'It was inevitable that the man hit the woman.' There is a cause for the action, either the obligation imposed on the man or some unknown or unstated factor.

Similarly for Dyirbal, Dixon (1972: 689) speaks of ‘implicated verb complexes’ which are again marked with the purposive inflection and indicate either an intended action or a natural consequence (both in relation to a previous event):

ɲaɖa ɖiŋgaliɲu biligu
I+SUB run+PRES/PAST climb+PURP
‘I’m running (to a tree) to climb it’

bayi yaɖa wayniɖin yalu
CL+NOM man+NOM come.uphill+PRES/PAST towards.here
baŋgun dunduŋgu maɖali
CL+ERG bird+ERG point.out+PURP
‘The man came uphill towards here, resulting in the bird’s pointing out his presence’

Dixon (1977: 346) points out (for Yidiny) that the semantic distinction between purpose and result may be neutralized in the negative:

ɲayu guŋgagunga:ɾ gali:na garu ɲaɲaɲ
I+SUBJ north+RED go+PURP by and by I+OBJ
ɲamu:ray ɲudu ɲuma:lɲa
smell+ABS not smell+PURP
‘I must go by the north so that she will not smell me’

Moreover, there are examples in which it is not possible to decide whether the purpose or result sense is intended, or whether there is simply indeterminacy between the two possible meanings, as in Dyirbal (Dixon 1972: 68):

balan ɖugumbil baŋgul yaɖaŋgu balgan baɖigu
CL+NOM woman+NOM CL+ERG man+ERG hit+PRES/past fall+PURP
‘Man hits woman causing her to fall down’

3.6 An alternative analysis

Bybee (1995: 6; cf. 1985: 166) argues that ‘event modality should be divided into speaker-oriented and agent-oriented modality’. ‘Agent-oriented modality’ she says ‘encompasses all modal meanings that predicate conditions on an agent with regard to the completion of an action referred to in the main predicate, e.g., obligation, desire, ability, permission and root possibility.’ She also says, ‘markers of directives, such as imperatives, optatives or permissives, by which a speaker attempts to move an addressee to action, are called speaker-oriented’. In particular, it should be noticed that agent-oriented modals include deontic statements that describe obligations and permission.

There is a slightly strange dichotomy, for there is considerable difference between (i) permission and obligation, for which the conditions factors are

external to the agent and (ii) desire and ability for which the factors are generally internal to the agent. This is the basis of the deontic/dynamic distinction. Moreover, permission and obligation (especially permission) often stem from the speaker and are thus more speaker-oriented than agent-oriented. Indeed, it is rather difficult to understand what is the difference in Bybee's analysis between (agent-oriented) 'permission' and (speaker-oriented) 'permissive'. In view of this, the deontic/dynamic dichotomy seems more useful than that of agent-oriented and speaker-oriented.

4

Modal systems and modal verbs

The modal systems described in the last two chapters share a number of features, not only in the systems themselves, but also, for many languages, in the use of modal verbs and the association with possibility and necessity. A detailed discussion of these has been left to this chapter in order to avoid too much repetition and cross-referencing. These issues hardly affect evidentials, so that the discussion is almost entirely concerned with epistemic, deontic and dynamic modality.

4.1 Modal systems

4.1.1 *Formal identity of different systems*

Notionally, epistemic modality and deontic/dynamic modality might seem to have little in common. As suggested in 1.2.2, epistemic modality is concerned solely with the speaker's attitude to the truth value or factual status of the proposition (propositional modality), whereas deontic and dynamic modality refer to events that are not actualized, events that have not taken place but are merely potential (event modality). Yet in English (and many other languages) the same forms are used for both types. The following, for instance, can all be interpreted either epistemically or deontically:

He may come tomorrow
The book should be on the shelf
He must be in his office

Where the same verbs are used for the different types of modality, there are often slight differences in the forms, which will be summarized in 4.2.3. Such differences suggest that the types are grammatically as well as notionally different, but do not explain why the forms are basically the same. One example of a difference is to be found in Modern Greek, where an impersonal form is used for epistemic possibility, but a personal form of the verb (agreeing with the subject) is used for (deontic) permission or (dynamic) ability, as in:

ta peðjá bori na fiyun ávrio
 the children BORO+3SG+PRES+IMPFV that leave+3PL+PRES+PERFV tomorrow
 'The children may leave tomorrow'

ta peðjá borún na fiyun ávrio
 the children BORO+3PL+PRES+IMPFV that leave+3PL+PRES+PERFV tomorrow
 'The children may/can leave tomorrow'

In the first, the modal verb (*bori*) is singular and does not agree with the plural subject, and so is to be interpreted epistemically. In the second the verb (*borún*) is plural and agrees with the subject and so is to be interpreted deontically (permission) or dynamically (ability).

The use of the same form for both propositional and event modality is not a purely European phenomenon. The same is true of the Dravidian language Tamil (Asher 1982: 171), where two suffixes that are identified as 'permission' and 'debitive' are used for both epistemic and deontic modality:

Kantacaami vantaalum vara-laam
 Kandaswami come+CONCESS come-PERM
 'Kandaswami may perhaps come'

veenṇum-ṇṇaakkaa naaḷekki avan peeca-laam
 want-COND tomorrow he speak-PERM
 'If he wants, he can speak tomorrow'

Gaṇeekan ippa Mannaarkuṭiyile irukka-ṇum
 Ganesan now Mannargudi+LOC be-DEB
 'Ganesan must be in Mannargudi now'

avan aṇke pooka-ṇum
 he there go-DEB
 'He must go there'

There are other non-Indo-European languages in which a single form may be used either deontically or epistemically, as has already been partially noted (1.4.5, 3.2.2 and 3.5) for Ngiyambaa (Donaldson 1980: 160–2):

yuruṇ-gu ṇidja-l-i
 rain-ERG rain-CM-PURP
 'It is bound to rain'

ṇadhu bawuṇ-ga yuwa-giri
 I+NOM middle-LOC lie-PURP
 'I must lie in the middle'

yuruṇu ṇidja-l-aga
 rain-ERG rain-CM-IRR
 'It might/will rain'

Modal systems and modal verbs

waŋa:y-ndu-gal dhagurma-gu yana-y-aga
NEG-2NOM-PL cemetery-DAT go-CM-IRR
'You shall none of you go to the cemetery'

(But see 3.2.2 for comments on the second pair of examples.)

In some languages, there are pairs for 'may' only. Thus in Tütatulabal (Steele 1975: 207, quoting Voegelin) the 'permissive' suffix is used in the two senses, e.g.:

hatda:w-aha-bi
CROSS-PERM-SM
'You may cross it'

wiʔ-aha-dza
run-PERM-SM
'It might run'

In colloquial Cairene Arabic (Gary and Gamal-Eldin 1982: 98–9) the same form is used for both kinds of 'must', but there are different forms for 'may':

laazim tiʔallas bukra
must you+SG-finish tomorrow
'You must finish tomorrow'

laazim jikuun hinaak
must he.be there
'He must be there'

tiʔdari tifuuti min hina
you+SG.can/may you+SG.pass from here
'You can/may pass through here'

jimkin jikuun hinaak
probable/possible he.be there
'He may be there'

Bybee *et al.* (1994: 195) also provide an example from Abkhaz for 'must' and from Lao for 'may/can'. They also note that there is a similar epistemic–deontic pairing for a form expressing both weaker obligation and a weaker judgment comparable to English *should* (see 2.1.5 and 3.2.3) in Baluchi. The form (marked 'subjunctive') may be a matter of mood rather than of modal system, but still illustrates the relation between deontic and epistemic:

a ešā bybart
3SG 3PL take.away +3SG+SUBJ
'He ought to take them away'

ma bəkly adda kəssa bȳzanə
1PL perhaps there someone know+1PL+SUBJ1
'Perhaps we know someone there'

The examples here show the same forms being used for epistemic and deontic modality not only for both propositional and event modality, but also for both deontic and dynamic. Identical forms are also used for deontic and dynamic modality as in English:

He can come in now
He can run a mile in under four minutes

This is true of similar forms in other languages. e.g., of POTERE in Italian. It is also true of Lisu (discussed in 3.3.1):

ása nya ami khwa da-a
Asa TOP field hoe acceptable-DEC
'It is acceptable for Asa to hoe the field'
'Asa is (physically) able to hoe fields'

4.1.2 Possibility and necessity

There is, in fact, a simple explanation for why the same forms are used in the different types of modality. The explanation is in terms of possibility and necessity, which are, Lyons (1977: 787) says, 'the central notions of traditional modal logic'. Indeed, the terms 'epistemic', 'deontic' and 'dynamic' are taken from a pioneering work on modal logic by Von Wright (1951: 1–2, 28).

Thus epistemic Speculative and Deductive can be interpreted in terms of what is epistemically possible and what is epistemically necessary:

John may be in his office
= It is epistemically possible that John is in his office
John must be in his office
= It is epistemically necessary that John is in his office

Similarly, deontic Permissive and Obligative can be interpreted in terms of what is deontically possible and deontically necessary:

You may/can go now
= It is deontically possible for you to go now
You must go now
= It is deontically necessary for you to go now

There is one reservation. The word *necessary* itself is not used in an epistemic sense in ordinary language (as opposed to logical terminology). It would not be normal to say **It is necessary that John is in his office*, although it is possible to say, in semi-logical language, *It is necessarily the case that John is in his office*. There is no problem with the *possible* – *It is possible that John is in his office* is perfectly normal.

Moreover, the epistemic and deontic uses can be distinguished in terms of ‘possible/necessary that’ and ‘possible/necessary’:

It is possible that John is in his office
It is ‘necessary’ that John is in his office
It is possible for John to be in his office
It is necessary for John to be in his office

The first two are epistemic, the last two deontic (although the third might be interpreted as epistemic also).

The importance of possibility and necessity in the modal systems is also shown by the distribution of the verbs in terms of negation. However, the situation with the forms used for negation of the modal verbs differs considerably in the various languages of the world and will be examined in greater detail in the next two sections.

4.1.3 Possibility, necessity and negation in English

There are two different ways in which a modal expression may be negated, as can be seen in the modal forms for epistemic possibility in English:

Mary may be at school
Mary may not be at school
Mary can’t be at school

With the use of the notions of possibility and necessity, it is easy to explain the difference between the two types of negation. One is to be interpreted in terms of ‘possible not’, the other in terms of ‘not possible’. With ‘possible not’ it may be said that there is negation of the proposition, whereas with ‘not possible’ there is negation of the modality.

It is possible that Mary is not at school (‘may not’, proposition negated)
It is not possible that Mary is at school (‘can’t’, modality negated)

Formally, the distinction is made, it can be seen, by changing the verb for ‘not possible’ – using CAN in place of MAY.

A similar distinction can be made between ‘necessary not’ and ‘not necessary’. Yet there are no comparable forms of MUST that are used for the negation of epistemic necessity. The only forms available are:

Necessary	John must be in his office
Necessary not	John can’t be in his office
Not necessary	John may not be in his office

What is immediately noticeable about these negative forms is that they are the same as those used for epistemic possibility, but in the reverse order.

There is quite a simple explanation for this. It involves the logical relations between possibility and necessity, in that ‘not possible’ is logically equivalent

to ‘necessary not’ and, conversely, ‘not necessary’ is equivalent to ‘possible not’. In logical terms:

Not possible $\equiv$ Necessary not
 Not necessary $\equiv$ Possible not

So for expressing ‘necessary not’, English uses the form for ‘not possible’ (can’t), and for ‘not necessary’ it uses the form for ‘possible not’ (*may not*).

Thus the forms of MAY and CAN provide all the negatives for epistemic possibility and necessity. There is suppletion – the negative forms for epistemic necessity are provided, suppletively, by the forms for epistemic possibility. Lyons (1977: 802) remarks that ‘in English at least, possibility rather than necessity should be taken as primitive in the analysis of epistemic modality’.

There are, then, two characteristics of these negative forms, the use of a different verb (CAN for MAY) and suppletion. These two features are to be found with the deontic modals too, but in a rather different way. The possibility forms are illustrated by:

Possible	Mary may/can come tomorrow
Possible not	Mary needn’t come tomorrow
Not possible	Mary may not/can’t come tomorrow

Examples of the necessity forms are:

Necessary	Mary must come tomorrow
Necessary not	Mary mustn’t come tomorrow
Not necessary	Mary needn’t come tomorrow

It can be seen here that with deontic modality it is the necessity that is ‘primitive’ in Lyons’s sense. For it is with the necessity forms that a different verb is used (NEED in place of MUST) and for ‘Possible not’ the ‘Not necessary’ form *needn’t* is required.

There is no problem with dynamic modality. Normally only the modality is negated:

He can’t run a mile in four minutes	(is unable)
He won’t answer	(is unwilling)

Being able or willing not to do something is more likely to be expressed by *able not to* and *willing not to*.

There is, then, an important question: why is it that *may not* and *can’t* have different interpretations? And the same question may be asked of *mustn’t* and *needn’t*. Formally, they are clearly negations of modals, and, if the modality were formally associated with the modal, the expected interpretation of the first two would be ‘not possible’ and of the second two ‘not necessary’, with negation of the modality. The interpretation of *may not* as ‘not possible’ but

of *mustn't* as 'necessary not' would suggest that the grammar and the notional interpretation are not in a one-to-one relationship, and that, in a sense, one of the negative markers is in the 'wrong' place. (If it is argued that the negation belongs with the lexical verb, the problem remains, but with the negative in the 'wrong' place for different forms.)

It should be said, that, although there does not appear to be a consistent over-all pattern for negation in English, some scholars have looked for an explanation. Cormack and Smith (forthcoming) argue that the 'split' between negation of the modality and negation of the proposition mainly (but not entirely) corresponds to possibility/necessity, while Coates (1983: 237–9) argues that it depends on the deontic/epistemic distinction, with only deontic *mustn't* being exceptional. In fact, the contrasts between epistemic *may not* and *can't* and between deontic *mustn't* and *needn't* provide exceptions to both views. The first view fails to account for *may not* (possibility with negation of the proposition) and *needn't* (necessity with negation of the modality). The second is contradicted not only by the 'exceptional' *mustn't* (deontic with negation of the proposition – and the same is true of *shouldn't*), but also by *can't* (epistemic with negation of the modality).

It will be helpful to look at other languages.

4.1.4 Possibility, necessity and negation in other languages

It might be expected that there are languages in which the negation of all the relevant modals is regular in the sense that there is no suppletion in terms of the logical equivalences and no use of alternative verbs. In fact, in an investigation into over thirty languages and dialects (Palmer 1995, 1997), none was discovered, the closest being Modern Greek in which only the deontic 'not necessary' form was irregular. The forms were:

bori na ine sto yrafio tus
BORO+3SG+IMPFV that they.are in.the office theirs
'They may be in their office' (epistemic possibility)

ðen bori na ine sto yrafio tus
NEG BORO+3SG+IMPFV that they.are in.the office theirs
'They can't be in their office' (not possible)

bori na min ine sto yrafio tus
BORO+3SG+IMPFV that NEG they.are in.the office theirs
'They may not be in their office' (possible not)

prepi na ine sto yrafio tus
PREPI+3SG+IMPFV that they.are in.the office theirs
'They must be in their office' (epistemic necessity)

ðen prepi na ine sto yrafio tus
 NEG PREPI+3SG+IMPFV that they.are in.the office theirs
 'They may not be in their office' (not necessary)

prepi na min ine sto yrafio tus
 PREPI+3SG+IMPFV that NEG they.are in.the office theirs
 'They can't be in their office' (necessary not)

boris na fiyis
 BORO+2SG+IMPFV that you.leave
 'You may leave' (deontic possibility)

ðen boris na fiyis
 NEG BORO+2SG+IMPFV that you.leave
 'You can't/may not leave' (not possible)

boris na mi fiyis
 BORO+2SG+IMPFV that NEG you.leave
 'You needn't leave' (possible not)

prepi na fiyis
 PREPI+2SG+IMPFV that you.leave
 'You must leave' (deontic necessity)

prepi na mi fiyis
 PREPI+3SG+IMPFV that NEG you.leave
 'You mustn't leave' (necessary not)

Only the deontic 'not necessary' form is missing from this paradigm. To express that, there are two possible forms, both using different verbs:

ðen ine anangi na fiyis
 NEG is necessary that you.leave
 'You needn't leave' (not necessary)

ðen xriaxete na fiyis
 NEG it.is.needed that you.leave
 'You needn't leave' (not necessary)

Where there is irregularity in the sense being discussed here, there are several possibilities, of which three are the most important:

- (i) the use of a different verb
- (ii) logical suppletion
- (iii) the positioning ('misplacement') of the negative marker.

These will be considered in turn.

The use of a different verb has already been illustrated for English and Modern Greek. It is also found in German (as well as other languages):

Du magst herein kommen
 you MÖGEN+2SG+PRES+IND here come
 'You may/can come in'

Du darfst nicht herein kommen
you DÜRFEN+2SG+PRES+IND not here come
'You can't/may not come in'

Du kannst nicht herein kommen
you KÖNNEN+2SG+PRES+IND not here come
'You can't/may not come in'

Where there is logical suppletion, it is of two types. In the first type, the possibility forms are used suppletively for the necessity forms. This has already been illustrated for epistemic modals in English, but an unrelated language for which there is a similar situation is Assamese (personal communication from Jiyoti Tamuli). The distinction between 'possible not' and 'not possible' is made by the position of the negative marker:

ɔfisɔt tʰak-ibɔ pare
office be-INF he.can
'He may be in his office'

ɔfisɔt na tʰak-ibɔ pare
office not be-INF he.can
'He may not be in his office'

ɔfisɔt tʰak-ibɔ nɔare
office be-INF NEG+he.can
'He can't be in his office'

For the negatives of necessity, however, the possibility forms are used suppletively, as in English:

ɔfisɔt tʰak-ibɔ lage
office be-INF he.must
'He must be in his office'

ɔfisɔt tʰak-ibɔ nɔare
office be-INF NEG+he.can
'He can't be in his office'

ɔfisɔt na tʰak-ibɔ pare
office not be-INF he.can
'He may not be in his office'

With the second type of suppletion, it is the forms that negate the modal ('not possible' and 'not necessary') that are used suppletively for the forms that negate the proposition, 'possible not' and 'necessary not'. This can be illustrated from Danish epistemic modals, where there are two forms only (Davidsen-Nielsen 1990: 78–87):

Det kan ikke være sandt
that can not be true
'That can't be true'

Det behøver ikke være sandt
 that must not be true
 'That may not be true'

These need a little explanation. In Danish (and most Germanic languages) the negative normally occurs after the verb, so that it is the modal that is formally negated in the examples above, and the expected interpretation would be 'not possible' and 'not necessary', with negation of the modality. For negation of the proposition ('possible not' and 'necessary not') there is suppletion, the first example expressing 'necessary not' and the second 'possible not'. But, as can be seen, suppletion is not in terms of possibility forms being used for necessity forms, as in English, but of forms negating the modal being used for forms negating the proposition.

This feature is also to be found in the deontic modals of a formal variety of Welsh, where the two forms are:

(Ni) gewch chi ddim ddod yfory
 (not) get you not come tomorrow
 'You can't come tomorrow'
 ('d oes) ddim rhaid i chi ddod yfory
 (It is not) not necessary to you come tomorrow
 'You needn't come tomorrow'

These are literally 'not possible' and 'not necessary', but are also used to express 'necessary not' and 'possible not'.

The third possibility is that the negation is, or appears to be, misplaced in the sense the grammatical marking and the notional interpretation do not correspond. To make this point quite clear, the Modern Greek examples above may be considered again. Because they involve subordination with the conjunction *na*, it can be seen that the negative may occur either in the main clause or in the subordinate clause. Moreover, when the negative occurs in the main clause (which contains the modal) the interpretation is 'not possible' or 'not necessary', and, when it occurs in the subordinate clause (which contains the lexical verb) the interpretation is 'possible not' or 'necessary not'. This is what might be expected – negation of the modal verb indicates negation of the modality.

This is so also in Catalan, according to Picallo (1990: 287):

En Jordi pot no haver sortit
 the Jordi can not have+INFIN leave+PART
 'It is possible that Jordi hasn't left'
 En Jordi no ha pogut sortir
 the Jordi not has can+PART leave+INFIN
 'Jordi hasn't been able to leave'

However, here, as the translations show, there is the further point that negation of the main verb is always associated with propositional (epistemic) modality and negation of the modal verb with event (deontic/dynamic) modality.

To avoid any possibility of ambiguity here, two distinctions are needed, the grammatical distinction between the modal and lexical verb and the notional distinction between the modality and the proposition. In a regular system such as that of Modern Greek (for the most part), (grammatical) negation of the modal indicates (notional) negation of the modality and (grammatical) negation of the lexical verb indicates (notional) negation of the proposition.

There are many languages, however, in which there is irregularity in this sense. Most commonly it occurs where there is grammatical negation of the deontic necessity modal. Consider the following examples from Kinyarwanda and Arabic:

agomba kwinjira
he.must to.come.in
'He must come in'

ntagomba kwinjira
NEG+he must to.come.in
'He mustn't come in'

lazim jizi
must you.come.in
'You must come in'

ma:/mu lazim jizi
NEG must you come in
'You mustn't come in'

In both sets of examples, the modal verb is formally negated, in Kinyarwanda by its morphology, in Arabic by the fact that the negative precedes it. Yet, notionally, it is the proposition, not the modality, that is negated. The interpretation is 'necessary not', not 'not necessary'. There is, it may be said, 'misplacement' or, better, 'reversal' of the marking of the negative.

The same is true of French and Italian:

il faut partir
it is.necessary to.go
'We must leave'

il ne faut pas partir
it NEG is NEG necessary to go
'We mustn't go'

Deve venire
 dovere+3SG+PRES+IND to.come
 'You (he/she) must come'

Non deve venire
 not dovere+3SG+PRES+IND to.come
 'You (he/she) mustn't come'

(The Italian form is, in fact, ambiguous in that it may also have the regular interpretation, with no displacement, of 'You needn't come', but it is still the only normal way of expressing 'necessary not'.)

The situation in the Germanic languages is not completely certain since the negative can usually occur in only one position – between the modal and the lexical verb. Nevertheless, it is usually the case that the negative follows the verb that it (notionally) negates. The regular interpretation of a Germanic form with a negative would, therefore, be that it is the modality that is negated.

On this interpretation the German deontic necessity form is regular and that of Norwegian is irregular. Within German negated MÜSSEN expresses 'not necessary' and DÜRFEN is required suppletively for 'necessary not' (see above):

Du musst nicht herein kommen
 you MÜSSEN+2SG+PRES+IND not here come
 'You needn't come in'

Du darfst nicht herein kommen
 you DÜRFEN+2SG+PRES+IND not here come
 'You may not come in'

In Norwegian, by contrast the negation of the necessity verb indicates 'necessary not' and a non-modal form is used for 'not necessary':

Han må ikke forlate rommet
 he must not leave the.room
 'He mustn't leave the room'

Han trenger ikke forlate rommet
 he must not leave the.room
 'He needn't leave the room'

Grammatically the negative belongs with the modal. Notionally, it is the modality that is negated in German, but the proposition in Norwegian.

English is also irregular in this sense with:

You mustn't come

This is, perhaps, even more striking, because the negative is cliticised on the

modal, and might thus seem to suggest even more strongly (but wrongly) that it is the modality that is negated.

However, this irregular displacement of the negative is not confined to deontic possibility. Consider the two possible interpretations of English:

Mary may not come tomorrow

In the epistemic sense it is regular ('possible not'), but in the deontic sense (now very formal and rather unusual), it is irregular ('not possible' refusing permission).

This reversal of negative marking with deontic necessity is widespread in the languages of the world. It occurred in ten of the twenty languages investigated in Palmer 1995. (The languages investigated in Palmer 1996 were all Germanic, and reversal was much rarer.) It is a puzzle why this should be. There may, perhaps, be two reasons – that there is comparatively little need of the 'not necessary' form and that placing the negation early in the sentence emphasizes the prohibition. Thus 'necessary not' usurps the form more naturally associated with 'not necessary'.

For a detailed study of the relationship between modality and negation, see De Haan (1997).

4.1.5 The notional connection

The close relationship between epistemic and deontic modals has, of course, been noted before. Joos (1964: 195) comments 'within the modal system English does not distinguish between duty and logic'. But it has seldom been noted that there is no immediately obvious reason why the same forms should be used for expressing the speaker's degree of commitment to truth and for getting other people to do things. It is by no means obvious that permission is a 'related notion' to possibility, or requirement to certainty, as Steele, Alemajias and Wasow (1981: 21) comment without justification or explanation.

The most detailed and plausible explanation is found in Sweetser (1982 and 1990) who argues (1982: 492) that the 'epistemic world is understood in terms of the sociophysical world', an idea that is virtually identical with that of locative case theory (Anderson 1971). Sweetser suggests that while MAY is 'an absent potential barrier in the sociophysical', epistemic MAY is the 'parallel case in the world of reasoning' and that its meaning would be that 'there is no barrier to the speaker's process of reasoning from the available premises to the conclusion expressed'. This idea of 'no barrier' had also been anticipated by Ehrman (1966) who interprets it as *nihil obstat*. Similar arguments hold for MUST.

This would be no more than plausible speculation if it were not the case that Sweetser adduces evidence from other forms in English. She finds a similar ambiguity in verbs such as INSIST, SUGGEST, EXPECT:

I insist that you go to London
 I insist that you did go to London

 I suggest that you leave the room now
 I suggest that you left the room to avoid being seen

The following sentence is ambiguous between the two senses:

I expect him to be there

To these observations can be added the fact that verbs such as *ask*, *promise* and *swear* can be used epistemically or deontically:

I asked if he had come
 I asked him to come
 I promise you he's here
 I promise you he'll come
 I swear he's here
 I swear I'll give it to you

Sweetser even notes a somewhat similar ambiguity in causal conjunction:

He came because he heard me screaming
 He heard me screaming, so he came
 (You say he's deaf, but) he came, so he heard me screaming
 (You say he's deaf, but) he heard me screaming, because he came

The first two are concerned with the reasons for the action, the third and fourth with the reasons for the judgment. Similar arguments can be found for *therefore*, *since*, *although*, *despite* and *anyway*.

The same is true of other languages. For example, *INSISTIR* in Spanish is used in both senses, though the indicative in the subordinate clause indicates the epistemic use and the subjunctive indicates the deontic sense (Klein 1975: 356, who does not specifically note the epistemic/deontic distinction):

Insisto que aprende
 I.insist that learn+3SG+PRES+IND
 'I insist that he's learning'

 Insisto que aprenda
 I.insist that learn+3SG+PRES+SUBJ
 'I insist that he learn'

It may be added that it has been argued, quite convincingly, that epistemic uses have developed, diachronically, from dynamic/deontic uses (see Traugott 1989).

4.2 Modal verbs

Modal verbs are used in all four of the main types of modality that are found in modal systems – judgments, evidentials, deontic and dynamic (though to a very limited degree with evidentials).

4.2.1 Modal verbs in English

It is, perhaps, fortunate for a study of modal verbs in which there is considerable discussion of English that there is no doubt that English has a set of modal verbs that can be formally defined. They are MAY, CAN, MUST, OUGHT (TO), WILL and SHALL, and marginally, NEED and DARE (including *might*, *could*, *would* and *should*). The facts have been stated many times (e.g. Palmer 1979, 1990) and will only be briefly summarized here.

First, they are members of a larger set of auxiliary verbs, which exhibit what Huddleston (1976: 333) has called the NICE properties – their occurrence with negation, inversion, ‘code’ and emphatic affirmation (see Palmer 1987: 14–21), as in (see 2.6.1):

I can’t go
Must I come?
He can swim and so can she
He will be there

These properties the modal verbs share with the other auxiliary verbs BE and HAVE but, in addition, they have formal features of their own:

- (i) They do not co-occur: there is no **will can come*, **may shall be*, etc. (though in a few dialects there are some very restricted possibilities of co-occurrence such as *might could* – see Brown 1991 and comments by Cormack and Smith forthcoming).
- (ii) They have no *-s* forms for their third person singular: e.g. **He oughts to come*. The form *will’s* exists, but as a form of the lexical verb, as does *cans*, in entirely different senses. (DARE has forms both with and without *-s*: *He dares to come*, *Dare he come?* This suggests that it functions both as a lexical verb and as a modal.)
- (iii) They have no non-finite forms: e.g. no **to can* or **canning*; there is no **I hope to can come tomorrow*.
- (iv) They have no imperatives: **Can be here!* **Must come now!*
- (v) MUST has no morphologically past tense form, although the others do (*could*, *should*, *might*, etc.); of those forms, only *could* is used to refer to past time (though all may occur in reported speech).
- (vi) There are suppletive negative forms (4.1.3).

- (vii) There are formal differences between the modal verbs, in their epistemic and deontic senses, in terms of negation and tense (4.2.3).

4.2.2 Modal verbs in other languages

It is instructive to compare the English verbs with similar verbs in German and French. German certainly has six potential candidates for the title of modal verb: KÖNNEN, DÜRFEN, MÖGEN, MÜSSEN, SOLLEN and WOLLEN, obvious cognates of the English ones (though, unlike the English modals, they have infinitive forms and fewer restrictions on their use). These are used both epistemically and deontically, epistemic MAY being translated by either KÖNNEN or MÖGEN, and deontic MAY by either KÖNNEN or DÜRFEN, while both epistemic and deontic MUST are translated by MÜSSEN (see Hammer 1983: 223ff. for examples):

Er kann/mag	krank sein
he KÖNNEN/MÖGEN+3SG+PRES+IND	ill be
'He may be ill'	
Du kannst/darfst	den Bleistift behalten
thou KÖNNEN/DÜRFEN+2SG+PRES+IND	the pencil keep
'You can keep the pencil'	

These are less clearly identified as members of a grammatical system than the English modals, but they have some idiosyncrasies, notably:

- (i) There is no final *-t* in the third person singular of the present indicative: e.g. *kann, mag* (cf. *gibt* 'gives').
- (ii) In subordinate clauses the order of the elements of the verbal complex is changed if one of them is a modal verb in the infinitive occurring with the finite form of another auxiliary verb (as it cannot occur in English). The usual rule is that the finite form occurs last, but in these circumstances it occurs at the beginning of the verbal complex (Hammer 1983: 224):

Es war klar, dass er sich	würde	anstrengen müssen
It was clear, that he himself	be+3SG+IMPF+SUBJ	exert must+INF
'It was clear that he will have to exert himself'		

(Note that the word order is not *anstrengen müssen würde.) There are similar modal auxiliaries in the Scandinavian languages, but they 'differ in complex ways from language to language, making translation difficult' (Haugen 1976: 80).

There are similar verbs in the Romance languages – French *POUVOIR* and *DEVOIR*, Italian *POTERE* and *DOVERE*, Spanish *PODER* and *DEBER*. The situation

in spoken French and Spanish is complicated by the use of impersonal *il se peut que* and *puede que* for epistemic possibility and *il faut* for deontic necessity in French, but the Italian verbs are used in both epistemic and deontic senses:

può essere nell' ufficio
POTERE+3SG+PRES+IND be in.the office
'He may be in the office'

deve essere nell' ufficio
DOVERE+3SG+PRES+IND be in.the office
'He must be in the office'

può venire domani
PODERE+3SG+PRES+IND come tomorrow
'He may/can come tomorrow'

deve venire domani
DOVERE+3SG+PRES+IND come tomorrow
'He must come tomorrow'

There appear to be no clear grammatical indications that the modal verbs in the Romance languages are a clearly defined set. (For French see Ruwet (1967: 185ff.), though there is one marginal feature noted by Kayne 1975: 1–27.) However, the verbs of Italian and Spanish are of typological interest in terms of their functions with negation.

The use of auxiliary verbs to express modality is very characteristic of European languages, but not exclusive to them. Thus Mandarin Chinese (Li and Thompson 1981: 173–88) has a set of auxiliary verbs that can be formally defined, in that:

- (i) They must co-occur with a lexical verb (or, at least, one must be understood from the context).
- (ii) They have no aspect markers or intensifiers.
- (iii) They cannot precede the subject.
- (iv) They cannot have objects.

All of these are semantically modal; the list is:

yīnggāi, yīngdang, gāi	'ought to, should'
néng, nénggòu, huì, kěyì	'be able to'
néng, kěyì	'have permission'
gǎn	'dare'
kěn	'be willing to'
děi, bixū, biyào, bíděi	'must, ought to'
huì	'will, know how'

Although the verbs of Chinese have no inflections, these auxiliaries appear to be essentially verbal. There is also a set of auxiliary verbs in Cashibo (Peru – Shell 1975: 178–91) that are inflectional and marked for person–number (though the stems may combine), but which may, perhaps, belong more to a discourse system than a modal system (see 2.6.1).

4.2.3 Formal differences

Although the same modal verbs may be used in English for both epistemic and deontic modality, generally the distinction is quite clear, and it can be seen from the previous discussion that there are also some formal distinctions between the modals in their two uses:

- (i) Deontic MUST has negative *mustn't* and a suppletive *needn't*, but epistemic MUST has no morphologically related negative.
- (ii) *May not* negates the modality when deontic (no permission), but the proposition when epistemic ('It may be that it is not so').
- (iii) MAY and MUST followed by *have* are always epistemic, never deontic.
- (iv) MAY is replaceable by CAN only in the deontic use, though *can't* may be epistemic.

Furthermore, when it refers to the future, *MUST* is almost always deontic; the epistemic sense is provided by *BE BOUND TO* (see Palmer 1979: 45–6, and, for discussion, Coates 1983: 42–3, and Palmer 1983: 291). Moreover, *might* is closely related in its meaning to present tense *may* only in its epistemic sense.

In Modern Greek, similarly, although the same forms are used for deontic and epistemic modality, there are differences in the syntax. First, as was seen in 4.1.1 and 4.1.4, forms of the verb *BORO* are used for possibility, but in the epistemic sense an impersonal (third person singular) form is found, while in the deontic (and dynamic) there is a fully inflected form that agrees with the subject.

Moreover, in many languages, if the lexical verb is marked as past, e.g., as in English with *may have* and *must have* above, only an epistemic interpretation is possible. Examples from Modern Greek are:

o janis prepi na fiyi avrio
the John PREPI that leave+3SG+PRES+PERFV tomorrow
'John must leave tomorrow'

o janis prepi na efiye
the John PREPI that leave+3SG+PAST
'John must have left'

4.3 Modal systems and other categories

4.3.1 Mood

The decision to treat modal systems and mood in separate parts of this book rests largely on the fact that most languages can be characterized as having one or the other. However, some languages have both, in two possible ways.

First, in a few languages that have a mood system with realis and irrealis markers (Chapter 6), the categories associated with irrealis may form what is clearly a modal system. Examples are Hixkaryana and Serrano, in which the categories marked as irrealis clearly belong to a familiar evidential system. This is discussed in some detail in 6.5.4. Rather differently, Central Pomo has a system of evidentials that is independent of its mood system (see 2.2.1 and 6.3.2).

Secondly, the Romance languages have a system of mood, marked by indicative and subjunctive, but also have a set of modal verbs. However, as noted in 4.2.2, the modal verbs have not been fully grammaticalized, and in French and Italian, at least, the subjunctive appears to be losing ground – it is used much less in colloquial French and has largely disappeared from the dialects of Southern Italy. Moreover, in English, the appearance of the modal system (of modal verbs) has been accompanied by the disappearance of Anglo-Saxon mood (see Lightfoot 1979, Plank 1984, who differ in their views about the way in which this development took place). There are, then, reasons for suggesting that, in general, the two are not likely to co-exist, or that, if they do, one will, in time, replace the other.

4.3.2 Future

WILL and SHALL are formally modal verbs by the criteria proposed in 4.2.1, yet they are often used to refer to future time (though SHALL is now much less common, especially in American English, in this sense). Indeed, traditional grammar books treated them as markers of a ‘future tense’ in English. However, as first argued by Fries (1927), it is reasonable to argue that English does not have a future tense, this being an idea carried over from Latin grammar. There are several arguments for this, but one important one is that, quite simply, they do not often indicate pure futurity, but are usually associated with conditional futures. Indeed BE GOING TO is a better candidate for the ‘future tense’ in English. (See Palmer 1990: 160–1, and the very persuasive arguments of Huddleston 1995.)

It is not surprising that modal verbs should have future time reference. The future is not fully known and it is always no more than a reasonable

assumption that a future event will ensue. Indeed, Lyons (1977: 677, 816) says:

Futurity is never a purely temporal concept; it necessarily includes an element of prediction or some related notion.

What is conventionally used as a future tense . . . is rarely, if ever, used solely for making statements or predictions, or posing or asking factual questions, about the future. It is also used in a wider or narrower range of non-factive utterances, involving supposition, inference, wish, intention and desire.

It may be added that even languages that have future tenses that are not formally modal, but belong within the inflectional system, often use these tenses for similar purposes. They are used in an assumptive sense, like English WILL, in, e.g., French and Italian (Lyons (1968: 310; Lepschy and Lepschy 1977: 139):

Ça sera le facteur
that be+3SG+FUT+IND the postman
'That'll be the postman' (epistemic)

Suonano, sarà Ugo
they.are.ringing be+3SG+FUT+IND Ugo
'The bell's gone; it'll be Ugo'

In colloquial Spanish, moreover, the future tense is normally used not to express future time, but in this epistemic sense. Future time is signalled by the verb IR ('to go') plus the infinitive.

Similarly, although English uses SHALL with second and third person forms as a Commissive (and WILL can be interpreted in the same way, though not as unambiguously as SHALL), this is true also of the French future tense:

You shall have it tomorrow
You'll have it tomorrow
Vous l'aurez demain

The connection between future and modality can also be shown historically. There are also plenty of examples of future tenses that are historically derived from subjunctives; this is true of some Latin forms (see Handford 1947: 15). Other languages have future tenses that have their origins in a modal-type auxiliary. Thus in Modern Greek, *tha* is a reflex of THELO: 'I wish', and the marker of future in Swahili is derived from (*ku*)-*taka* '(to) wish', while the futures of modern Romance languages were derived from the infinitive plus HABEO 'I have'.

Finally, it will be seen in the next two chapters that future time is signalled

by mood, both by the subjunctive and by irrealis. This too illustrates the potentially modal characteristics of future time reference.

4.3.3 Negation

The most important relationship between modality and negation is the one that involves possibility and necessity. But there are a few other points to note.

First, negation (usually together with interrogation) is involved in both modal systems (2.3) and mood (5.2.3 and 6.6.4).

Secondly, some languages have what might be thought to be negative modal verbs. Thus Latin may use the verb *NOLO* ‘I refuse’ in negative commands (as well as *ne* plus the subjunctive – see 5.4.2), as in:

Nolite	facere
refuse+2PL+IMP	do
‘Don’t do it’	

Welsh has a verb *PEIDIO* ‘I cease’ that is used in a similar way:

peidiwch	â	gadael	iddo	fynd
cease+2PL+IMP	from	let	him	go
‘Don’t let him go’				

5 Indicative and subjunctive

It was noted in 1.2.1 that many languages have the grammatical category of mood, and that in European languages particularly, but not exclusively, this is dealt with in terms of the distinction between the indicative and the subjunctive. An example from Spanish is (Klein 1975: 356):

Insisto que aprende
I.insist that learn+3SG+PRES+IND
'I insist that he is learning'

Insisto que aprenda
I.insist that learn+3SG+PRES+SUBJ
'I insist that he learn'

It was also noted that other languages, particularly the languages of the Americas and of Papua New Guinea, have markers that are labelled 'realis' and 'irrealis'.

It was suggested that basically there is no typological difference between indicative/subjunctive and realis/irrealis, and that both are instances of mood (Realis/Irrealis). There are, however, considerable differences between the functions of what have been labelled 'subjunctive' and 'irrealis', and, therefore, for practical reasons they will be dealt with in separate chapters (5 and 6). A further chapter (7) will discuss the similarities and differences between them.

Most of the examples in this chapter will be taken from the classical languages, Latin and Greek and the Romance languages, especially Spanish, since there is a large body of literature concerning the subjunctive in that language. Some mention is made of German, which has considerable uses of the subjunctive in its literary form, although in other Germanic languages, e.g. Danish, the use of the subjunctive is almost dead. However, there are other languages that have been usefully and plausibly described in terms of indicative and subjunctive. In some of the Bantu languages of Africa, for instance, the basic structure of verbs consists of a sequence of markers for

subject, tense, verb stem and mood, indicative being marked by final *-a* and subjunctive by final *-e* (though no tense distinctions occur with subjunctive). Examples from Swahili (Steere 1943: 43, 57) are:

n-me-pend-a
I-PERF-love-IND
'I have loved'

ni-pend-e
I-love-SUBJ
'Let me love'

Examples from another Bantu language, Luvala (Zimbabwe and Zaire – Horton 1949) and from the non-Bantu, West African, language Fula (Arnott 1970: 299ff.) are to be found later in this chapter (5.1.1 and 5.4.2). There will also be discussion of some Semitic languages, particular the Ethiopian languages, but with these there is a problem concerning the status of the subjunctive (see 5.4.2).

5.1 Main and subordinate clauses

Jespersen (1924: 314) noted that one of the functions of the subjunctive is simply that of being subordinate, in that it is typically the mood used in subordinate clauses. It is, in fact, no coincidence that the term 'subjunctive' is a translation of the Classical Greek *hypotaktiké* which literally means subordinate. Indeed, in Latin, the subjunctive was increasingly used in subordinate sentences even where there seemed to be no notion of irrealis (see below, 5.5).

However, the subjunctive is also used in main clauses, and its uses there are rather more simply explained than its uses in subordinate clauses. For that reason, main clauses will be considered first.

5.1.1 Main clauses

Lakoff (1968: 172ff.) recognizes six different uses of the subjunctive in main clauses in Latin. Her terminology is a little idiosyncratic (or at least very different from that used in this book), and, for that reason they are characterized here in terms of the modal categories used in the present volume, with Lakoff's terms in brackets :

Jussive ('Imperative'):

Naviget! haec summa est, hic nostri nuntius esto
sail+3SG+PRES+SUBJ this point is, this of.us message let.it.be

Virg. (*Aen.* 4. 237)

'Let him sail, this is the point, let this be our message'

Volitive ('Optative'):

Ut illum di . . . perduint (Pl. *Aul.* 785)
 that him gods destroy+3PL+PRES+SUBJ
 'May the gods destroy him!'

Obligative ('Jussive'):

Sed maneam etiam, opinor (Pl. *Trin.* 1136)
 But remain+1SG+PRES+SUBJ still, I.think
 'But I should still stay, I think'

Obligative ('Deliberative'):

Quid agam iudices? (Cic. *Verr.* 5.2)
 what do+1SG+PRES+SUBJ jurymen
 'What am I to do, gentlemen of the jury?'

Speculative ('Potential'):

Iam absolutos censeas quom incedunt infectores (Pl. *Aul.* 520)
 now paid.off think+2SG+PRES+SUBJ when come.in dyers
 'You may think they are already paid off, when in come the dyers'

Presupposed ('Concessive'):

Sit fur, sit sacrilegus . . .
 be+3SG+PRES+SUBJ thief be+3SG+PRES+SUBJ temple.robber
 at est bonus imperator (Cic. *Verr.* 5.4)
 yet he.is good general
 'Though he is a thief, though he is a temple-robber . . . he is a good general'

These can be related to the three types of non-assertion discussed first in 1.1.2 and the distinction of epistemic versus deontic/dynamic modality (1.1.2 and 1.3.2). The first four are deontic (and the subjunctive is used because the proposition is unrealized). The last but one is epistemic (the speaker has doubts about the veracity of the proposition). The last one is an example of presupposition (nothing is being asserted).

Similar examples can be found in Italian (Lepschy and Lepschy 1977: 223–4):

entri pure
 enter+3SG+PRES+SUBJ if.you.please
 'Please come in'
 che venga anche lui
 that come+3SG+PRES+SUBJ also him
 'Let him come too'
 potessi venire anch' io
 can+1SG+PRES+SUBJ come also I
 'If only I could come too'

Indicative and subjunctive

sapessi che lusso
know+2SG+PRES+SUBJ that grand
'You should see how grand'

che sia finito
that be+3SG+PRES+SUBJ finished
'I wonder if it's finished'

sia pure come dici tu ma io non vengo
be+3SG+PRES+SUBJ perhaps as say you but I not come
'It may be as you say, but I'm not coming'

The first is usually regarded as the polite command – Imperative (see 3.4), but, unlike the Imperative proper, in which the verb is in the second person, it has a verb in the third person. In form it is like the second, which is Jussive. The third is a wish (Volitive) and the fourth another example of obligation (Obligative). All are deontic (and irrealis). The last but one is epistemic, expressing possibility or a degree of doubt (Speculative), while the last is probably best seen as concessive indicating something presupposed, though it too could also be interpreted as Speculative, expressing doubt, as in the previous example. Again they exemplify the three types of non-assertion.

There is something very similar in the West African language Fula (Arnott 1970: 299ff.). Not surprisingly, the term 'subjunctive' is used. There is a paradigm of the verb that has functions very similar to those of the Latin subjunctive (though it lacks tense distinctions).

Four types of directives are illustrated:

Injunction

ngaraa
come+2SG+SUBJ
'Come on!'

Report for instruction or permission

minasta-na?
come in+1SG+SUBJ-INT
'May I come in?'

Offer or request for permission or invitations with HAA

haa njahen
HAA go+1PL+SUBJ
'Let's go'

Obligation with SEY

sey ngurtodaa
SEY come.out+2SG+SUBJ
'You ought to come out'

These are examples of Imperative, Permissive, Jussive and Obligative respectively. There is also an example of a wish (Volitive):

Wish or prayer

njuutaa balde
be.long+2SG+SUBJ in.days
'May you live long!'

In all of these the proposition is unrealized. No example of epistemic usage (the speaker has doubts about the veracity of the proposition) was given, but there is an example that is perhaps to be explained in terms of presupposition – that nothing is being asserted:

Expostulation or rhetorical question

njoofodaa
sit+2SG+SUBJ
'What? You sit down!'

An example of the subjunctive in Swahili used as jussive (see 5.4.2) was given above. Examples of the same use in another Bantu language, Luvale (Horton 1949: 302–4) are:

tu-y-e
we-go-SUBJ
'Let's go!'

va-iz-e (veze) waxi
they-come-SUBJ quickly
'Let them come quickly'

5.1.2 Subordinate clauses

Examples of the use of the subjunctive in subordinate clauses are even more varied. They include not only those uses that were discussed in Chapters 2 and 3, but others such as future reference and wishes and fears. For that reason the contents of the next two sections will be distinguished in terms of propositional modality and event modality rather than in terms of epistemic, deontic and dynamic. There is then a section on the imperative, which often involves the subjunctive, and a section showing the use of the subjunctive for no other reason, apparently, than that the verb occurs in a subordinate clause.

As argued in 1.1.2, the distinction between the indicative and the subjunctive is associated with assertion and non-assertion, and one of the reasons for non-assertion is that the speaker has doubts about the veracity of the proposition. But there is some ambiguity about the word 'speaker'. In a main clause it clearly refers to the actual speaker, the originator of the utterance. It is this speaker who doubts the veracity of the proposition, as in:

Quizá viene
maybe come+3SG+PRES+SUBJ
'Maybe he's coming'

In a subordinate clause, however, the relevant ‘speaker’ is the reported speaker, indicated by the subject of the main verb, as can be seen in:

María duda que sea buena idea
 María doubts that be+3SG+PRES+SUBJ good idea
 ‘Mary doubts that’s a good idea’

(This is a modification of the example in 1.1.2, which was unhelpful because the actual speaker and the reported speaker were the same, both 'I'.) In general, the choice of the subjunctive in main clauses depends on the attitudes, beliefs, etc., of the actual speaker. In subordinate clauses, however, it depends on the attitudes, beliefs, etc. of the person represented by the subject of the main clause (with those attitudes, beliefs, etc., indicated by the main verb). This is even true of the use of the subjunctive for what is presupposed, as in:

Le alegra que sepas la verdad
him it.pleases that know+2SG+PRES+SUBJ the truth
'He's glad that you know the truth'

Generally, however, the actual speaker also accepts that the proposition is true. (This is debatable, but will not be discussed here.)

No examples of the uses of subjunctive in subordinate clauses are given here – they are to be found in the following sections, which exemplify both main and subordinate clauses.

5.2 Propositional modality

5.2.1 Speculative

An example of the subjunctive being used to express epistemic possibility (Speculative) in Latin was given in 5.1.1. Further examples from Italian and Spanish are:

che sia finito
that be+3SG+PRES+SUBJ finished
'I wonder if it's finished'

Quizá viene
maybe come+3SG+PRES+SUBJ
'Maybe he's coming'

These are examples of main clauses, although the Italian example is introduced by *che* which is usually a subordinating conjunction ('that').

An example from a very different language, Baluchi (Bybee *et al.* 1994: 195), was given in 4.1.1:

ma bækly adda kəssa bȳzanə
1PL perhaps there someone know+1SG+SUBJ
'Perhaps we know someone there'

The subjunctive may also be used in subordinate clauses after verbs of belief, but here there is usually a choice between indicative and subjunctive, the choice being determined by the speaker's judgment about the truth of the relevant proposition. Thus Butt and Benjamin (1988: 228, 227) state that CREER 'believe' and similar verbs such as PARECER 'appear', SUPONER 'suppose' and SOSPECHAR 'suspect' may also be followed by the subjunctive if there is hesitancy or what is stated is more hypothetical. Similarly Lepschy and Lepschy (1977: 227) state that either mood may be used with Italian CREDERE 'believe', PARERE 'appear' and SEMBRARE 'seem'. An example with CREDERE is:

Credo che tu abbia/hai ragione
 I.think that you have+3SG+PRES+SUBJ/IND right
 'I think you are right'

The subjunctive is also used in Italian after impersonal expressions that express possibility and probability such as *è probabile/improbabile che* 'it is likely, unlikely that' and *può darsi che, può essere che* 'it may be that' (Lepschy and Lepschy 1975: 225), and after similar expressions in Spanish.

Slightly differently, in literary German the subjunctive is used with verbs of belief, but only with the past tense (and also with verbs of saying – see 5.2.2):

Ich glaubte er wäre krank
 I thought he be+3SG+IMPF+SUBJ ill
 'I thought he was ill'

A different issue is the use of the subjunctive with negated verbs of belief. This is discussed in 5.2.3.

5.2.2 Reported

The subjunctive is often used to indicate what is reported. In German, it may be used in main clauses to indicate that it is what is said or was said and not part of the writer's or speaker's own statement:

Bei seiner Vernehmung berief sich H. auf Notwehr. Er
 in his examination appealed H to self-defence. he
 sei mit S. in Streit geraten und
 be+3SG+PRES+SUBJ with S. in quarrel fallen and
 habe sich von diesem bedroht gefühlt
 have+3SG+PRES+SUBJ self by him threatened felt
 'In the course of his cross examination, H. pleaded self-defence. He had become involved in a quarrel with S. and had felt himself to be threatened by him'

The second sentence here is what H. claimed, not part of the writer's own report.

Indicative and subjunctive

In subordinate clauses, the subjunctive is usually to be found in German with verbs of reporting (as well as verbs of belief – see 5.1.1) in the past tense (examples are from Hammer (1983: 265–71), and are mostly from a literary style):

Ich glaubte er wäre krank
I thought he be+3SG+IMPF+SUBJ ill
'I thought he was ill'

Er sagte er wäre krank
he said he be+3SG+IMPF+SUBJ ill
'He said he was ill'

However, in colloquial German, the indicative may be used:

Er glaubte, ich war krank
he thought I be+3SG+IMPF+IND ill
'He thought I was ill'

In a literary style, the subjunctive is often used even with a present tense verb of reporting, especially when, as in all of the above examples, the conjunction *dass* is not used. With *dass* the indicative is more likely:

Er sagt, er müsse nach Hause
he says he must+3SG+PRES+SUBJ to house
'He says he must go home'

Er sagt, dass er nach Hause muss
he says that he to house must+3SG+PRES+IND
'He says he must go home'

The indicative without *dass* may suggest that the speaker accepts the reported proposition as true, as in German:

Er sagte, er schwimmt gern
he said he swim+3SG+PRES+IND with.pleasure
'He says he likes swimming'

Italian does not generally use the subjunctive after a verb of reporting that is neither negated or interrogative (5.2.3, 5.2.4), but it does so after *si dice* 'one says', which indicates 'hearsay' (Reported (3)). Compare (Lepschy and Lepschy 1977: 226):

Ada dice che i soldati sono partiti
Ada says that the soldiers be+3PL+PRES+IND left
'Ada says that the soldiers have left'

Si dice che i soldati siano partiti
One says that the soldiers be+3PL+PRES+SUBJ left
'They say that the soldiers have left'

This is also true of German:

Man sagt, er sei gestorben
 One says he be+2SG+PRES+SUBJ died
 'They say he's died'

Latin does not use the subjunctive for reported statements (see 7.4.1). However, a relative clause within a subordinate clause of reported speech in Latin is placed in the subjunctive if the subordinate clause is part of what is reported:

Dicit se de Gallis . . . postulare triumphum quos
 he.says self from Gauls to.demand triumph whom
 acie vicerit (Liv. 36.40.3)
 by.battle defeat+3SG+PERF+SUBJ
 'He says he claims a triumph from the Gauls, whom he has defeated in battle'

If it is part of the information being provided by the speaker (the original speaker) the indicative is used:

Diogenes . . . dicere solebat Harpalum, qui temporibus illis
 Diogenes to.say used Harpalus who in.times those
 praedo felix habebatur, contra deos
 brigand happy be.held+3SG+IMP+IND against gods
 testimonium dicere (Cic. D. N. 3.34)
 witness to.speak
 'Diogenes used to say that Harpalus (who at that time was generally thought to be a fortunate brigand) was a witness against the gods'

There is an exactly parallel situation in German (Hammer 1983: 268):

Er sagte, er werde das Buch kaufen da sein
 he said he be+3SG+IMPF+SUBJ the book to.buy as his
 Onkel . . . es ihm empfohlen hätte
 Uncle . . . it to.him recommended have+3SG+IMPF+SUBJ
 'He said he would buy a book as his uncle . . . had recommended it to him'

Er sagte, er bewerbe sich um diese Stelle,
 he said he apply+3SG+PRES+SUBJ self to this job
 für die er gar nicht geeignet ist
 for which he at all not suited be+3SG+PRES+IND
 'He said he was applying for this job, for which he is not at all suitable'

The subjunctive is also used in Latin for a reported cause:

Aristides . . . nonne ob eam causam expulsus est patria
 Aristides NEG INT for that cause expelled is from.country
 quod praeter modum iustus esset (Cic. T. D. 5.36.105)
 because beyond mean just be+3SG+IMPF+SUBJ
 'Was not Aristides exiled because (it was said) he was excessively just?'

(For an example from Classical Greek, but with the optative, see 8.2.2.)

The subjunctive is similarly used in German, e.g. (Hammer 1983: 269):

Papa möchte auch gern selbst lenken, Mama will es
Papa would too with.pleasure self to drive, Mama wishes it
aber nicht weil es die Nerven angreife
however not, because it the nerves strain+3SG+PRES+SUBJ
'Daddy would like to drive, but Mummy doesn't want him to because
(she says) it is a strain on the nerves'

Only slightly different is the Spanish example offered by Lavandera (1978: 19):

Mientras que a vos no te falte nada, como
as.long as to you not you lack+3SG+PRES+SUBJ nothing as
vos decís . . .
you say
'As long as you don't need anything as you say . . .'

Here the comment *como vos decís* clearly indicates that the reason for inaction (presumably) was given by the addressee.

5.2.3 Negative

The most common association of Negative with subjunctive is in subordinate clauses where the superordinate clause is negated. This association is rare in main clauses, in spite of the fact that within event modality the subjunctive is often used with negative imperatives (5.4.2) and that there are instances of the association of negative with irrealis in main clauses (6.6.4). However, one example of subjunctive with negative in a main clause is to be found in Luvale (Bantu – Horton 1949: 302–4), where it is used with a particle translated as 'not yet':

kanda tu-mum-on-e
not.yet we-him-see-SUBJ
'We haven't seen him yet'

In subordinate clauses the subjunctive is regularly used after negated verbs of belief and report, but rather different issues seem to be involved and, for that reason, the two types will be dealt with separately.

With verbs of belief, if the verb is negated, the subjunctive is generally used in the subordinate clause in the Romance languages. Compare from Spanish (Klein 1975: 353):

Creo que aprende
I.believe that learn+3SG+PRES+IND
'I believe that he is learning'

No creo que aprenda
not I.think that learn+3SG+PRES+SUBJ
'I don't think that he is learning'

Examples of similar negated form in Italian (Hall 1964: 222) and French (Bloomfield 1933: 273) are:

Non credo	che sia	Corelli
not	I.think that be+3SG+PRES+SUBJ	Corelli
'I don't think that it's Corelli'		
Je ne	pense pas	qu'il vienne
I	not think that he	come+3SG+PRES+SUBJ
'I don't think he'll come'		

The subjunctive is also used with verbs that express doubt even though there is no formal negation, e.g., in Spanish, (Klein 1975: 356, 353):

Dudo que aprenda
I.doubt that learn+3SG+PRES+SUBJ
'I doubt that he's learning'

From this it might be suggested that the use of the subjunctive is not directly determined by the negative, but by the expression of doubt. On this view the subjunctive is used because 'not-think' is an expression of doubt, and verbs of doubting take the subjunctive, as shown by the previous Spanish example and its Italian counterpart:

Dubito che impari
I.doubt that learn+3SG+PRES+SUBJ
'I doubt that he's learning'

This view is supported by the fact that with a verb of belief that is not negated, the use of the subjunctive indicates the speaker has some doubt about the truth of the proposition, as was noted in 5.2.1.

There is, however, another way of looking at this. *I don't think he is stupid* can be interpreted as 'I think he's not stupid.' This has been treated in theoretical studies as 'negative raising' or 'negative transportation', which suggests that the negative originates in the subordinate clause, but is raised or transported to the main clause. An even better analysis is in terms of the scope of the negation. With the most obvious interpretation there is narrow scope in *I don't think he is stupid*, since only the subordinate clause is (notionally) negated. In contrast, on the less likely interpretation 'It's not the case that I think he's stupid', there would be wide scope, negating the whole sentence.

In this respect it is relevant that, with a negative imperative, the indicative, not the subjunctive, is used:

No crea usted que es tonto
not think+3SG+PRES+SUBJ you that be+3SG+PRES+IND stupid
'Don't think that he is stupid'

Here, the interpretation is not in terms of 'not-think', i.e., 'doubt', but in terms of a command not to think – 'Not (think(he is stupid))' rather than 'Think (not(he is stupid))'. In terms of scope, it appears that the contrast is between narrow scope with the subjunctive (as in the previous examples) and wide scope (as here) with the indicative.

The indicative is also to be found with negative questions:

¿No es verdad que ha dicho eso?
not is truth that have+3SG+PRES+IND said that
'Isn't it true that he said that?'

Butt and Benjamin (1988: 227) comment that negative questions are not semantically negative, and so the indicative is used, but in terms of scope it is clear that this does not mean 'Is it true that he didn't say that?' (with narrow scope).

Now, what is important for comparison of verbs of belief with verbs of saying is that the subjunctive is used in the former where there is narrow scope. This, as will now be seen, does not apply to verbs of saying.

With verbs of saying in Romance the verb 'say' is generally followed by the indicative unless it is negated. If it is negated either the indicative or the subjunctive may be used, as in Italian:

Dico che ha torto
I.say that have+3SG+PRES+IND wrong
'I say that he is wrong'

Non dico che lui abbia/ha torto
not I.say that he have+3SG+PRES+SUBJ/IND wrong
'I do not say that he is wrong'

The crucial point here is that there is now wide scope – the interpretation of the negated example is 'It is not the case that I say he is wrong' not 'I say that he is not wrong.' With verbs of saying, unlike the verbs of belief, the subjunctive occurs where there is wide scope of the negation.

The same situation (with the same argument) applies to 'know', as in Spanish:

Yo sabía que él estaba ahí
I knew that he be+3SG+PAST+IND here
'I knew that he was here'

Yo no sabía que él estaba/estuviera ahí
I not knew that he be+3SG+PAST+IND/SUBJ here
'I did not know that he was here'

The second does not mean ‘I knew that he was not here’, but ‘It is not the case that I knew that he was here’.

Since the explanation in terms of narrow scope (and ‘not-think equals doubt’) is not available for negated verbs of saying, a different reason for the use of the subjunctive with negated verbs of saying must be sought. In fact, there is a fairly obvious answer – that what is not said is not asserted. It will be seen in 6.6.4 that, in languages where the mood distinction is made in terms of Realis/Irrealis, Negative is similarly associated with Irrealis.

It may be noted, however, that 'deny', also requires the subjunctive in Spanish:

Niego que haya venido
I.deny that have+3SG+PRES+SUBJ come
'I deny that he has come'

For with 'deny' there is (notionally) narrow scope. The example above means 'I say that he has not come.' So both wide scope and narrow scope of the negative are associated with the subjunctive.

The situation in German is similar to that in the Romance languages. The subjunctive is found even with present tense verbs if they are negated:

Er sagt, er ist müde
He says he be+3SG+PRES+IND tired
'He says he is tired'

Er sagt nicht, er wäre müde
he says not he be+3SG+IMPF+SUBJ tired
'He does not say he is tired'

A rather different example of negative and subjunctive is to be found in causal clauses in Latin. These are introduced by a causal conjunction and are generally followed by the indicative:

Torquatus . . . filium suum quo is contra imperium in hostem
 Torquatus son his because he against rule in enemy
 pugnaverat, necari iussit (Sall. C. 52)
 fight+3SG+PLUP+IND be.killed he.ordered
 ‘Torquatus ordered his son to be executed, because he had fought
 against the enemy contrary to orders’

However, the subjunctive is used when the clause is negated, i.e., it is not the true reason:

Pugiles . . . ingemescunt non quod doleant . . . sed
 boxers groan not because be.in.pain+3PL+PRES+SUBJ but
 quia profundenda voce omne corpus intenditur (Cic. *TD*. 2.23.56)
 because with.bursting voice whole body be.stretched+3SG+PRES+IND

Indicative and subjunctive

'Prize-fighters groan not because they are in pain, but because their whole body is made more tense by the burst of sound'

(This may be compared with the subjunctive for a reported cause in 5.1.2.)

The same kind of feature is found in Spanish (Lavandera 1978: 21):

Yo no lo digo porque a mi me moleste
I not it say because to me me bother+3SG+PRES+SUBJ
'I don't say it because it bothers me' (i.e. 'not because . . .')

There is something similar in the more stereotyped expression:

No es que no me guste
not is that not me pleases+3SG+PRES+SUBJ
'It isn't that I don't like it'

Here a possible cause is being rejected.

Slightly differently, Spanish uses the subjunctive after *sin que* 'without', to indicate that the relevant action did not take place:

Desechó el cigarrillo sin que el profesor
he.threw.away the cigarette without that the master
lo viese
it see+3SG+IMPF+SUBJ
'He threw away the cigarette without the master seeing it'

Similarly in Italian (Lepschy and Lepschy 1977: 225):

È entrato senza che noi lo sentissimo
is entered without that we him hear+1PL+IMPF+SUBJ
'He came in without our hearing him'

5.2.4 Interrogative

There are two points concerning Interrogative.

First, interrogation and negation often function in very similar ways, so that together they can be characterized as 'non-assertive' (Quirk *et al.* 1985: 83). There are, therefore, examples of verbs of belief and verbs of saying where interrogation, like negation, induces the subjunctive as in the Spanish and Italian:

¿Cree Vd que venga?
believe you that come+3SG+PRES+SUBJ
'Do you think he will come?'

sai se sia vero?
you.know if be+3SG+PRES+SUBJ true
'Do you know if it's true?'

This is also true of German even with present tense verbs of saying (see 5.1.2):

Sagt er er wäre müde
Says he he be+3SG+IMPF+SUBJ tired
'Does he say he is tired?'

There is an interesting opposite situation with negative interrogatives in Spanish, which use the indicative. This was discussed in 5.2.3 with the example:

¿No es verdad que ha dicho eso?
not is truth that have+3SG+PRES+IND said that
'Isn't it true that he said that?'

Secondly, the subjunctive is used in reported (indirect) questions, as in Spanish and Italian, where the indicative is more informal:

mi chiese se fosse/era possibile
me he. asked if be+3SG+IMPF+SUBJ/IND possible
'He asked me if it would be possible'

The subjunctive is always required for indirect (reported) questions in Latin, though the indicative is used in direct questions:

Quid agis? → Rogo quid agas
What do+2SG+PRES+IND → I. ask what do+2SG+PRES+SUBJ
'What are you doing?' → 'I ask what you are doing'

Rogavi pervenisset-ne Agrigentum (Cic. Verr. 2.4.12.27)
I. asked arrived+3SG+PLUP+SUBJ-INT Agrigentum
'I asked if he had arrived at Agrigentum'

5.2.5 Presupposed

It was noted in 1.1.2 that the subjunctive is used in Spanish where the proposition (in a subordinate clause) is presupposed, with the example:

Me alegra que sepas la verdad
me it. pleases that know+2SG+PRES+SUBJ the truth
'I'm glad that you know the truth'

There is a similar situation in Italian (as well as other languages), as can be seen from the Spanish and Italian examples:

Sp. Lamento que aprenda
It. Mi dispiace che impari
I. regret that learn+3SG+PRES+SUBJ
'I regret that he learns/is learning'

Verbs like REGRET have what Kiparsky and Kiparsky (1971: 245–8) call 'factive complements' (see 1.4.1), in that the proposition in the complement is presupposed, i.e., accepted as true by the speaker (and, the speaker assumes, by the addressee). The essential point, as explained in 1.1.2, is that, since the

proposition is presupposed, nothing is being asserted and, when there is non-assertion, the subjunctive is used.

Other examples of the subjunctive in a subordinate clause from Spanish and Italian are:

me molesta	que te	quejes	tanto
me it.bothers	that you+REFL	complain+2SG+PRES+SUBJ	so
'It bothers me that you complain so much'			
mi sorprende	che tu	dica	questo
me it.surprises	that you say+2SG+PRES+SUBJ	that	
'It surprises me that you say that'			

According to Givón (1994: 304), the subjunctive is used in Spanish with verbs such as SENTIR in the sense of 'sympathize', SORPRENDERSE, 'be surprised', MOLESTARSE, and ALEGRARSE 'be glad'. There are similar verbs in Italian.

The subjunctive is similarly used with impersonal constructions that express emotion (Spanish and Italian):

fué	una lástima	que no	me lo dijeras	
it.was a	pity	that not me it	tell+2SG+PAST+SUBJ	
'It was a pity that you didn't tell me'				
è	peccato	che sia	già	partit
it.is	pity	that be+3SG+PRES+SUBJ	already	left
'It is a pity that he has already left'				

With some expressions, Givón states, there is a choice of indicative or subjunctive depending on the 'degree of emotional reaction', the subjunctive expressing greater emotion, e.g. HORRORIZARSE 'be shocked' and SER INCREÍBLE 'be incredible'. In Italian there is a stylistic choice according to Lepschy and Lepschy (1977: 225) with verbs and expressions such as RINCRESCERE, 'be sorry', STUPIRSI 'be surprised', ESSERE CONTENTO/SCONTENTO 'be pleased/displeased', ESSERE ARRABIATO 'be cross' etc.

The contrast between the use of the indicative for what is asserted and the subjunctive for what is presupposed is well illustrated by verbs that can be used in two senses. Compare the Spanish and Italian pairs (Klein 1975; Lepschy and Lepschy 1977: 228):

Siento que aprende
I feel that learn+3SG+PRES+IND
'I feel that he is learning'
Siento que aprenda
I.feel that learn+3SG+PRES+SUBJ
'I regret that he should learn'

5.2 Propositional modality

Si capisce che sono arrabiati
 One understands that be+3PL+PRES+IND cross
 'It's clear that they are cross'

Si capisce che siano arrabiati
 One understands that be+3PL+PRES+SUBJ cross
 'It's understandable that they should be cross'

In the first of each pair the speaker presents the proposition as new information and the indicative is used, while in the second the speaker expresses his attitude towards shared knowledge, and the subjunctive is used.

Powerful evidence that what is relevant is not the emotional attitude itself, but the fact that there is no new information, is shown by the use of the subjunctive after *el hecho que* 'the fact that' (Butt and Benjamin 1988: 221):

el hecho que España no tenga petróleo
 the fact that Spain not have+3SG+PRES+SUBJ oil
 explica las dificultades económicas
 explains the difficulties economic
 'The fact that Spain doesn't have any oil explains the economic difficulties'

Although there is specific reference to 'fact' here, it is clear that the proposition is not being asserted, but is presupposed. By contrast where the 'fact' (*el hecho*) is new information, the subjunctive is not used (Butt and Benjamin 1988: 229):

No lo hace por de hecho que no le gusta
 not it does for of fact that no her please+3SG+PRES+IND
 'She doesn't do it, because she doesn't like it'

Finally, but quite importantly, the subjunctive may be used in concessive clauses to grant or admit that something is true. These may be in a main clause in apposition to a following clause (see 5.1.1), or an (oblique) subordinate clause as in Latin:

Sit fur, sit acrilagus . . .
 be+3SG+PRES+SUBJ thief be+3SG+PRES+SUBJ temple robber
 at est bonus imperator (Cic. Verr. 5.4)
 yet he.is good general
 'Though he is a thief, though he is a temple-robber . . . he is a good general'

Quamvis sis molestus, numquam te esse
 although be+2SG+PRES+SUBJ troublesome never you to.be
 confitebor malum (Cic. T.D. 2.25.61)
 I.will.confess evil
 'Although you are troublesome, I will never admit you to be evil'

Similar examples from Italian and Spanish are:

sia pure come dici tu ma io non vengo
be+3SG+PRES+SUBJ perhaps as say you but I not come
'It may be as you say, but I'm not coming'
Aunque sea difícil, lo haré
although be+3SG+PRES+SUBJ difficult it I.will.do
'Although it is difficult, I will do it'

What is conceded is not asserted, but is presupposed, and the subjunctive is used.

5.2.6 *Future*

Futurity is often marked as irrealis in languages in which mood is described in terms of realis/irrealis (6.6.1). This is much less common with indicative/subjunctive.

In main clauses, the only instance is the use of the subjunctive to refer to the future in Homeric Greek:

ou gár po: toíous idon anéras, oudé
not for ever such see+1SG+AOR+IND men nor
ído:mai (Hom. *Il.* 1.262)
see+1SG+AOR+SUBJ
'I have never seen such men, nor shall I see'

However, in temporal clauses, there is a widespread use of the subjunctive to refer to hypothetical future events, with the indicative used for past events. Examples from Classical Greek and Spanish are (the first of each pair has the indicative, the second the subjunctive):

epeí dé eteleúte:se Dareíos . . . , Tissaphérne:s
when but die+3SG+AOR+IND Darius Tissaphernes
diabéllei tón Kúron (Xen. *An.* 1.1.3)
slanders the Cyrus
'When Darius died, Tissaphernes slanders Cyrus'

epeidán dé diaprákso:mai há déomai, hé:kso: (Xen. *An.* 2.3.29)
when but finish+1SG.AOR+SUBJ what I want I.will.come
'When I have finished what I want to do, I will come'

Cuando llegó en Inglaterra, vino a ver-me
when arrive+3SG+PAST+IND in England he.came to see.me
'When he arrived in England, he.came to see.me'

Cuando se termine la guerra, volveré a
when REFL finish+3SG+PRES+SUBJ the war I.will.return to
Inglaterra
England
'When the war ends, I will return to England'

There are similar examples in locative clauses:

Dondequiera que era, me escribía
 wherever that be+3SG+IMPF+IND to.me he.wrote
 'Wherever he was, he would write to me'

Dondequiera que yo esté, te escribiré
 wherever that I be+1SG+PRES+SUBJ to.you I.will.write
 'Wherever I am, I will write to you'

The subjunctive may also be used in Classical Greek in the apodosis (the *if*-clause) of a real conditional clause (see 8.3) that refers to the future (though the future indicative is also possible):

eán toúto poié:is, hamarté:seis
 if this do+2SG+PRES+SUBJ, err+2SG+FUT+IND
 'If you do this (should do this), you will be wrong'

5.2.7 Conditional

The subjunctive is used in conditional clauses in a number of languages. In Latin, for instance, the present subjunctive is used in:

Si hoc facias, erres
 if this do+2SG+PRES+SUBJ err+2SG+PRES+SUBJ
 'If you did/were to do this you would be wrong'

However, 'modal tense' (1.4.4, 8.2) is usually involved as well, and, for that reason, a detailed discussion of subjunctive in conditional sentences is left to 8.3.2.

5.2.8 Other types

There is a distinction in Spanish between:

El que asasinó a Smith está loco
 he that kill+3SG+PAST+IND to Smith is mad
 'The man who killed Smith is mad'

and

El que asasinara a Smith está loco
 he that kill+3SG+PAST+SUBJ to Smith is mad
 'The man who killed Smith is mad'

The precise interpretation of these and the status of the form *asasinara* is a matter of some debate (see Rivero 1975, 1977; Rojas 1977), but it seems that the first refers to a particular person, while the second has a 'whoever' sense that links the killing with the insanity. Paraphrases would be 'I conclude that a certain person (the one who, in fact, killed Smith) is insane' and 'I conclude that whoever killed Smith is, thereby, shown to be insane.'

The subjunctive is being used, it would appear, for indefiniteness. Such indefiniteness is treated as irrealis because it is not asserted that a certain person was the killer.

Slightly different is the use of the subjunctive in:

Busco un empleado que hable inglés
I.look.for an employee who speak+3SG+PRES+SUBJ English
'I'm looking for an employee who speaks English'

Busco a un empleado que habla inglés
I.look.for to an employee who speak+3SG+PRES+IND English
'I'm looking for an employee who speaks English'

The first means that I am looking for an employee (anyone) who can speak English; the second means that I am looking for a certain employee who can, in fact, speak English. (However, as can be seen, the distinction is also marked by the use of the preposition *a* before specific objects.) There are similar constructions in Italian (Lepschy and Lepschy 1977: 229):

Cerco una ragazza che sappia cinese
I.look.for a girl who know+3SG+PRES+SUBJ Chinese
'I am looking for a girl who speaks Chinese'

Cerco una ragazza che sa cinese
I.look.for a girl who know+3SG+PRES+IND Chinese
'I'm looking for a girl who speaks Chinese'

This too could be seen in terms of indefiniteness, but may also be compared with the use of the subjunctive for 'relative purpose' (5.3.2), and also with its use in complement clauses of wanting (5.3.3). Comparison with the latter suggests that the subjunctive is determined or governed by the main verb.

To make the point in a better way, it may be said that the whole relative clause is within the semantic scope of the verb, in that speaking English, etc., is part of what the speaker is looking for. The indicative, by contrast, shows that the relative clause is not within the scope of the verb. It may be recalled that, similarly, relative clauses in Latin that are within complement clauses of indirect speech are in the indicative if they are not part of what was said (see 5.1.2). This too can be seen as a matter of scope – the indicative shows that the relative clause is not part of what is being reported.

5.3 Event modality

Most of the types of event modality for which the subjunctive is used are very different from those expressed by modal verbs, apart from Directives.

5.3.1 Directives

The subjunctive is also used to express the deontic notions of weak obligation (expressed in English by *should*, the modal-past equivalent of *must* (2.1.5)):

Italian

Sapessi che lusso
know+2SG+PRES+SUBJ that grand
'You should see how grand'

Latin

Iniurias fortunae . . . defugiendo relinquas (Cic. *T.D.* 41)
wrongs of.fortune by.fleeing leave+2SG+PRES+SUBJ
'Flee from and leave behind you the blows of fortune'

Latin

Sed maneam etiam, opinor (Pl. *Trin.* 1136)
but remain+1SG+PRES+SUBJ still, I.think
'But I should still stay, I think'

An example from a very different language, Baluchi (Bybee *et al.* 1994: 195), was given in 4.1.1:

a ešā bybart
3SG 3PL take away+3SG+SUBJ
'He ought to take them away'

There is, however, a problem with some of these. Those with second person subjects are not always easily distinguished from what are treated as 'polite imperatives' (to be discussed in 5.3.2), and those with first and third person subjects are not easily distinguished from jussives ('first and third person imperatives'), which are discussed in 5.4.1. Examples from Spanish are:

Tome su libro
take+3SG+PRES+SUBJ your book
'Take your book'

Hablemos de otra cosa
talk+1PL+PRES+SUBJ about other thing
'Let's talk about something else'

Que entre
that enter+3SG+PRES+SUBJ
'Let him come in'

The imperfect and pluperfect subjunctive are used in Latin to refer to what should have been done in the past. Here they are clearly distinct from imperatives and jussives, which are essentially performative, and so relate to the present:

Indicative and subjunctive

sed tu dictis, Albane, maneres (Virg. *Aen.* 8. 643)
but thou to.words Alban remain+3SG+IMPF+SUBJ
'But thou, Alban, shouldst have kept thy word'
Adservasses hominem (Cic. *Verr.* 65)
keep+2SG+PLUP+SUBJ man
'You should have kept the man'

One other use that seems to belong here is the 'deliberative', with which speakers question what they should (or perhaps must) do:

Latin
Quid agam iudices? (Cic. *Verr.* 5.2)
what do+1SG+PRES+SUBJ jurymen
'What am I to do, gentlemen of the jury?'
Greek
ó:moi egó pá:i bó:? (Eur. *Hec.* 1056)
alas I where go+1SG+PRES+SUBJ
'Ah me! Where shall I go?'

These too can refer to the past:

haec cum viderem, quid agerem . . . ? (Cic. *Sest.* 19)
these when I.saw what do+1SG+IMPF+SUBJ
'When I saw this, what was I to do?'

There is not a great deal to be said about directives and subordinate clauses. In most European languages the infinitive is used after expressions of permission and obligation (see 7.4.1), e.g. after Latin *licet* 'it is allowed' and *oportet* 'it is necessary', though Modern Greek uses a conjunction with a finite verb (see 4.1.4 for examples). However, some languages use subjunctive with such constructions. This is so with the Western Nilotic languages (Bavin 1995: 112–13). Thus in Acholi the subjunctive is used after *wek* for permission, *myero* for obligation and *gwok* for prohibition. An example of *myero* is:

myero a-ngwec-i
must 1+SG-run-SUBJ
'I must run'

5.3.2 Purposive

Clauses that express purpose are marked with a conjunction plus the subjunctive in Latin and Classical Greek:

Haec acta res est ut ii nobiles
this done thing is that those nobles
restituerentur in civitatem (Cic. *Rosc. Am.* 51.1 49)
be.restored+3PL+IMPF+SUBJ in state
'This was done so that those nobles should be restored to the state'

tón gár kákon aei déi kolázein, hin'
 the for bad always it is necessary to.punish in.order.that
 ameino:n e:i (Plat. Leg. 944d)
 better be+3SG+PRES+SUBJ
 'For we must punish the bad man that he may be better'

The same kind of construction is used in Spanish and Italian:

Le presté el dinero para que pudiese
 him I.lent the money in order that can+3SG+IMPF+SUBJ
 comprar su billete
 buy his ticket
 'I lent him the money in order that he could buy his ticket'
 ti scrivo affinché tu capisca la situazione
 you I.write in order that you understand+2SG+PRES+SUBJ the situation
 'I am writing to you so that you understand the situation'

The usual way of expressing negative purpose is simply to use a negative in the subordinate clause, with the same conjunction. However, Latin generally uses *ne* in place of *ut* (and only rarely *ut ne*):

Dolabella, ne collum tonsori committeret, tondere
 Dolabella lest neck to.the.barber entrust+3SG+IMPF+SUBJ to.shave
 filias suas docuit (Cic. Tusc. 5.20.58)
 daughters his taught
 'So that he should not entrust his neck to a barber, Dolabella taught his daughters how to shave'

Similarly, Classical Greek usually has the negative *mé*: in the subordinate clause after a conjunction, but Homeric Greek may use *mé*: alone:

apóstiche mé: ti noé:se:i (Hom. Il. 1.522)
 depart lest something notice+3SG+AOR+SUBJ
 'Depart lest she notices anything'

Latin *ne* and Classical Greek *mé*: are here associated with the subjunctive, whereas their other negatives, *non* and *ouk*, are generally found with the indicative. It is tempting, for that reason, to treat these negative forms as irrealis and realis respectively. But the conditions for their use are complex, and, in both languages, there are exceptions to the general rule (see especially the use of *non* with the subjunctive for 'result' in 5.2.3). It is beyond the scope of this book to deal with them in detail, but attention is drawn to their use for fears (5.3.3) and negative imperatives and jussives (5.4.2).

Purpose may be expressed with conjunctions other than those that translate 'in order that', e.g. with Latin *dum* 'until':

Expecta . . . dum Atticum conveniam (Cic. Att. 7. 1.4)
 wait until Atticus meet+1SG+PRES+SUBJ
 'Wait until I meet Atticus'

Without the notion of purpose the indicative is used:

Dum anima est, spes esse dicitur (Cic. *Att.* 9.10.3)
while life be+3SG+PRES+IND hope to.be is.said
'It is said that while there is life there is hope'

Purpose can also be expressed with a relative clause plus the subjunctive (traditionally 'relative purpose'):

Scribebat tamen orationes, quas alii dicerent (Cic. *Brut.* 6.206)
he.wrote however speeches which others say+3PL+IMPF+SUBJ
'However, he wrote speeches for others to give'

Here *quas* has the function of *ut . . . eas* 'so that . . . them'. Idiomatic Latin requires this relative construction when there is a pronoun in the subordinate clause.

Slightly different, but interesting in this context, is a construction in Tigrinya (Palmer 1962: 38), where the relative is followed by the imperfective (which is the usual form in subordinate clauses):

qwal'a zihəqqäf, zisənkäl bəhuq may
child REL+it.be.nursed, REL+it.be.cooked dough, water
ziməsa' zahaqqwänsäba
REL+it.be.drawn REL+it.be.churned milk
'A child to be nursed, dough to be cooked, water to be drawn, milk to be churned'

Some of the examples treated in terms of 'indefiniteness' in 5.2.8 (those after 'look for') might also be treated in terms of purpose.

It is worth noting that purpose clauses have much in common with the complements of directives. Thus in Latin, *ut* and the negative *ne* with the subjunctive are used both to express 'in order that' and after verbs of ordering, requesting, etc. Compare:

Caesar singulis legionibus singulos legatos et quaestorem
Caesar to.single to.legions single legates and quaestor
praefecit ut eos testis quisque suae virtutis
put.in.charge that them witnesses each of.his virtue
haberet (Caes. *B. C.* 1.52)
have+3SG+IMPF+SUBJ

'Caesar put the legates and the quaestor each in charge of a legion so that everyone might have them as witnesses of his valour'

Rogat et orat Dolabellam ut de sua provincia
he.asks and he.begs Dolabella that from his province
decedat (Cic. *Verr.* 1.29.72)
withdraw+3SG+PRES+SUBJ
'He asks and implores Dolabella to leave his province'

So too, in Russian, the infinitive is used both with ‘want’ and for purpose when the subject of the two clauses is the same, while *chtóby* (‘that’ plus particle *-by*) with the past tense is used for both types when the subjects are different (and the past tense with the particle *-by* is often called the ‘subjunctive’ – see 8.5):

on khochet plavat’
 he wants swim+INFIN
 ‘He wants to swim’
 my prishli posetit’ bol’novo
 we have.come visit+INFIN patient
 ‘We’ve come to visit the patient’
 ja khochu, chto-by vy bol’she eli
 I want that-*by* you more eat+PAST+M.SG
 ‘I want you to eat more’
 chto-by nikto ne znal ob etom, nado molchát’
 that-*by* no-one not know+PAST+M.SG about this necessary be.silent
 ‘So that no-one should know about this, we must be silent’

There is a comparable feature in Ngiyambaa (Donaldson 1980: 2801), in that the marker for purpose clauses (the ‘purposive’, which is also used to mark Obligative – see 1.4.5) is also used in subordinate clauses with verbs expressing ‘knowing how’/‘remembering’ and ‘wanting’ (with the ‘caritative’ case marker also for the latter):

ŋadhu dhi:rba-nha guruŋa-giri
 I+NOM know-PRES swim-PURP
 ‘I know how to swim’
 bura:y wagayma-giri-ŋinda gaŋa
 child+ABS play-PURP-CARIT be-PRES
 ‘The child wants to play’

This is hardly surprising, since purpose clauses express what the subject wants or intends, and they are quite close in meaning to complement clauses with the intention, etc., expressed by lexical items, as in:

He did it hoping/intending that they would come
 He did it so that they would come

5.3.3 *Wishes, fears, etc.*

The subjunctive is used for wishes and fears. The term ‘Desiderative’ is available for the former (a possible alternative, ‘Volitive’, has been used in a different sense – see 3.3.1), but there is no recognized term for the latter, although the formation from Latin would be ‘Timitive’ (1.7).

Indicative and subjunctive

In main clauses, for wishes for the future, the present subjunctive is used in, e.g., Latin, Italian and Portuguese:

Ut illum di . . . perduint (Pl. *Aul.* 785)

that him gods destroy+3PL+PRES+SUBJ

'May the gods destroy him!'

Dio vi benedica

God you bless+2PL+PRES+SUBJ

'May God bless you'

Venha a dia

come+3SG+PRES+SUBJ the day

'May the day come!'

It is also used in Fula, as noted in 5.1.1:

njuutaa balde

be long+2SG+SUBJ in.days

'May you live long!'

Impossible wishes for the present and past are expressed in Latin and Italian by the imperfect subjunctive:

modo valeres (Cic. *Att.* 9.22)

only be well+2SG+IMPF+SUBJ

'If only you were well'

potessi venire anch'io

can+1SG+PRES+SUBJ come also I

'If only I could come'

Similarly, impossible wishes for the past are expressed by the pluperfect in, e.g., Latin:

utinam ne . . . tetigissent litora puppes (Catull. 64.171)

that not touch+3PL+PLUP+SUBJ shores ships

'Would that their ships had not touched the shore'

However, generally, in European languages, wishes have the same form as conditionals (this is true even of the languages exemplified above), and are therefore considered in detail in 8.4.

In subordinate clauses after verbs of wanting and hoping, Spanish normally requires the subjunctive if there is a different subject (Givón 1994: 286):

Quiero que venga

I.wish that come+3SG+PRES+SUBJ

'I wish that he would come'

Espero que venga pronto

I.hope that come+3SG+PRES+SUBJ soon

'I hope he comes soon'

So does Italian:

Voglio che venga
I.wish that come+3SG+PRES+SUBJ
'I wish that he would come'

Spero che Ugo ritorni sabato
I.hope that Ugo return+3SG+SUBJ Saturday
'I hope that Ugo comes back on Saturday'

Latin may have either the accusative and infinitive construction (see 7.4.1) or the subjunctive for wanting:

Ut mihi aedis aliquas conducatur volo (Pl. Merc. 560)
that to.me house some buy+3SG+PRES+SUBJ I.wish
'I want him to rent a house for me'

Nolo me in tempore hoc videat senex (Ter. And. 819)
I don't wish me in time this see+3SG+PRES+SUBJ old.man
'I don't want the old man to see me now'

But for 'hope' the accusative and infinitive is the normal construction in Latin (see 7.4.1).

Fears are usually expressed with a verb of fearing plus the subjunctive in a subordinate clause, as in Spanish:

Temo que haya muerto
I.fear that have+3SG+PRES+SUBJ died
'I fear that he has died'

Latin and Classical Greek require simply *ne* and *mé*: with subjunctive (these are the arguably 'irrealis' forms used for negative purpose – see 5.3.2):

Timeo ne laborem augeam (Cic. Leg. 1.4)
I.fear work increase+1SG+PRES+SUBJ
'I am afraid that I shall increase my work'

Dédoika gár mé: oud' hósion éi . . . apagoreúein
I.fear for lest not righteous be+3SG+PRES+SUBJ . . . to refuse
(Plat. Rep. 368B)

'For I am afraid it will not be righteous to refuse'

There is, however, an interesting usage in Classical Greek where an expression of fear can be indicated without a verb of fearing, simply by the subjunctive preceded by the negative *me*:. Often, however, this expresses little more than an unwelcome possibility, like English 'I'm afraid that' as in:

mé: dé: né:as hélo:si (Hom. Il. 16.128)
lest indeed ships take+3PL+AOR+SUBJ
'I'm afraid they'll take the ships'

Indicative and subjunctive

allá mé: ou tóut' é:i khalepón . . . thánaton
 but lest not this be+3SG+PRES+SUBJ difficult death
 ekphugeín (Plat. *Ap.* 3ga)
 to.avoid
 'I suspect it is not the avoidance of death that is the difficulty'

It may even express no more than 'perhaps'

allá mé: tóuto ou kaló:s ho:mologé:samen (Plat. *Men.* 89c)
 but lest this not well agree+1PL+AOR+SUBJ
 'Perhaps we have not fairly conceded this'
 mé: soús diaphthéire:i gámos (Eur. *Alc.* 315)
 not your ruin+3SG+PRES+SUBJ marriage
 'She may ruin your marriage'

There is something similar in To'aba'ita (Austronesian, Solomon Islands – Lichtenberk 1995: 294–8), although there is no mood marking. There is a form that expresses epistemic uncertainty, but also signals apprehension (Lichtenberk calls it 'apprehensional' and translates it as 'lest'). This too is used in main clauses, in subordinate clauses after a verb of fearing and also for negative purpose:

ada keka fanga sui na'a
 lest they+SEQ eat COMPLET PERF
 'They may have finished eating' (the speaker is worried about getting food)
 nau ku ma'u 'asia na'a ada laalae to'a baa ki keka
 I I+FACT fear very lest later people that PL they+SEQ
 lae mai keka thaungi kulu
 go hither they+SEQ kill us+INCL
 'I am scared the people might come and kill us'
 nau ku agwa 'i buira fau ada wane 'eri ka riki nau
 I I+FACT hide at behind rock lest man that he+SEQ see me
 'I hid behind a rock so that the man might not see me'

There are similar examples in Diyari (Australia – P. Austin 1981: 225).

It could be argued that fears and wishes are best treated as epistemic, since they indicate attitudes to propositions rather than unrealized events. Indeed, Givón (1994: 280) refers to hopes and fears as 'epistemic anxiety'. Wanting, however, seems to be different in that the emotion is more directed to the event, and so should be treated as deontic, a kind of directive. A comparison of 'wish' and 'want' may bear out this point in two ways. First, telling someone what is wanted is often a direction for action whereas an expression of wish is not so obviously so. Secondly, wishes, like

fears and hopes, can relate to the past as well as the present or future, whereas wanting cannot:

I wish John had come
I fear John came
I hope John came
??I want John to have come

(The last example may not be impossible, but is very unnatural.) These two features suggest that wanting should be treated as deontic, the others as epistemic.

However, it must be admitted that it is not always easy to distinguish wanting and wishing. Givón (1994: 283) offers two translations for the Spanish verb *QUERER* in:

Quiero que estudies más
I want/wish that study+2SG+PRES+SUBJ more
'I want you to study more'
'I wish you would study more'

The point of the question mark is that the infinitive complement suggests a 'stronger manipulation' than the Spanish, but Spanish has no obvious alternative.

5.3.4 Resultative

The construction used in Latin to express results (traditionally called ‘consecutive’ clauses) is almost identical with that used to express purpose. If the clause is positive, *ut* plus the subjunctive is used in both cases. If it is negative, a purpose clause has *ne* instead of *ut* (see 5.3.2), but a consecutive clause retains *ut* with a following negative *non*. Examples are:

Adeo turbati erant. . . ut quosdam consul manu ipse
so disturbed they.were that some consul by.hand self
reprehenderit (Liv. 34.14)
seize+3SG+PERF+SUBJ
'They were so disturbed that the consul himself seized them by the hand'

Vulneribus confectus ut am se sustinere non posset
by.wounds weakened that still self hold.up not was.able (Caes. *B.G.* 2.25)
'So weakened by his wounds that he could no longer hold himself up'

The use of the subjunctive here has puzzled scholars because results are essentially factual and the indicative might therefore be expected. Moreover, in this context the arguably ‘realis’ negative (5.3.2) occurs with the subjunctive. Moore (1934: 108) says ‘The subjunctive implies, not that the result is

unreal, but that it is *causally connected* and logically subordinate to the main clause.’ Alternatively, it can simply be argued that since the true function of the subjunctive is to mark subordination, it is precisely that that is exemplified here (see below). Hamp (1982: 118) comments ‘The generalisation of the subjunctive would be a Latin phenomenon.’ This may be strictly correct, but is a little misleading, for the similarity between purpose and result clauses is found elsewhere. Even in English, *so that* introduces both types, though the purpose clause may also contain *should*:

He worked hard so that he became rich
He worked hard so that he should become rich

The first sentence is ambiguous between the two senses.

Even more striking is similarity with the use of the purposive in the Australian languages. This was discussed in some detail in 3.5, but it may be useful to repeat two examples from Yidiny (Dixon 1977: 3456), one expressing purpose, the other a ‘natural result’:

ɖaɖa ɖuɖu:mbu ɡaɾbaɣaɾbaŋaɭu ŋuɖu wawa:lɲa
child+ABS aunt+ERG hide+PAST not see+PURP
‘Auntie hid the child so that it should not be seen

ŋayu burawuŋal duga:l ɖinbiɖinbi:lɲa
I+SUBJ Burawugal+ABS grab+PRES struggle+RED+PURP
‘I grabbed the water sprite woman and as a result she kicked and
struggled’

It can be seen on reflection that purpose and result clauses are alike in that one expresses an intended result, the other an actual result. It seems that there is a good case for believing that purpose and result are, in a variety of languages, closely related, or even indistinguishable, concepts. Even if Hamp (1982) is correct in arguing that the *ut* of purpose and the *ut* of result have different historical origins, it must be significant that they have now fallen together, except in the negative.

Yet one might have expected that the contrast between intended and actual results would be marked by the distinction of subjunctive and indicative, but the main distinction between purpose and result clauses is made by the two negatives *ne* and *non*. It is almost as if these, rather than subjunctive/indicative, are the distinguishing markers of ‘irrealis’ intended results and ‘realis’ actual results.

5.4 Imperative and Jussive

Imperative and Jussive were briefly discussed in 3.4, where they appear to be terms in modal systems. They also occur in languages that have

indicative/subjunctive mood, and, in this respect, the mood system is not strictly binary.

5.4.1 Morphology and syntax

Traditional grammars recognize second and third person ‘imperatives’ in Latin and Classical Greek (the latter being instances of Jussive – see 3.4). The ‘imperatives’ for Classical Greek are, for the verb ‘loose’, given as:

	<i>singular</i>	<i>dual</i>	<i>plural</i>
2	lú-e	lú-eton	lú-ete
3	lú-eto:	lú-eto:n	lúonto:n

Examples from Classical Greek are:

all’ ei dokéi,	pléo:men,	hormástho:	táchus
but if it.seems	sail+1PL+PRES+SUBJ	set.forth+3SG+PRES+JUSS	swift
			(Soph. <i>Phil.</i> 526)
‘If thou wilt, let us sail and let him set forth with speed’			
epímeinon,	Aré:ia	teúkhea	dúo:
wait+2SG+AOR+IMP	of.Ares	equipment	put on+1SG+PRES+SUBJ
			(Hom. <i>Il.</i> 6.340)
‘Wait, let me put on my war harness’			

(Both of these, however, also contain subjunctives with the function of first person jussives – see 5.4.2.)

There are no first person imperatives/jussives in Latin and Classical Greek, but they occur in other languages – an example from Amharic is given below. Imperatives may be marked for gender as well as number, e.g. in many Semitic languages that mark gender in their verbal paradigms. They are also marked for aspect, in, e.g., Classical Greek and Russian, but not for tense.

As noted in 3.4, imperatives and jussives do not normally occur in subordinate clauses or in questions, for the obvious reason that they are performative – the speaker is actually giving a command. There are, however, a few exceptions to this. There is one example of a second person imperative in a subordinate clause in Classical Greek, but, since it occurred in a comedy, it may have been a linguistic joke:

all’ oísth’	hó	drá:son	
but you.know	what	do+2SG+PRES+IMP	
			(Ar. <i>Av.</i> 54)
‘You know what do it!’			

Similarly, there is one example of a jussive in a question:

keístho:	nómos	humí:n	
lay.down+3SG+PRES+JUSS	law	for.us	
			(Plato <i>Legg</i> 801d)
‘Shall the law be laid down for us?’			

What appear to be jussives are commonly found in questions in Amharic (Ethiopian Semitic – Cohen 1936: 179–80):

ləwsada-w
take+3SG+JUSS-it
'Must he take it?'

mən ləwsad
what take+1SG+JUSS
'What shall I take?'

Yet, as the translations show, this rather suggests that these jussives are, perhaps, not to be seen as Jussives, but as deontic Obligatives, more like the subjunctive in European languages. There is more discussion of the relation between Jussive and the subjunctive in 5.4.2.

Similarly, it would not be expected that imperatives can refer to past time in the way that deontic modals (*must have*, etc.) can. Yet there is in Syrian Arabic (Cowell 1964: 36) a construction that looks extremely like a past tense imperative – the use of the perfect form of *KAN* 'to be' with the imperative, but with the meaning 'should have':

kənt kol lamma kənt fəl-be:t!
be+2SG+PERF eat+IMPF when be+2SG+PERF in.the-house
'You should have eaten when you were at home'

5.4.2 *Subjunctive*

The subjunctive is used instead of the imperative as a polite command, but in the third person, in both Italian and Spanish:

entri pure
enter+3SG+PRES+SUBJ if.you.please
'Please come in'

Tome su libro
take+3SG+PRES+SUBJ 3SG+poss book
'Take your book'

Yet the subjunctive can also be used with second person subjects, to express weak obligation, as noted in 5.3.1.

In Latin and Classical Greek the subjunctive is used in negative commands, once again with the negatives *ne* and *me*: (see 5.3.2), though they may also use the imperative:

Latin
Ne sis patruus mihi (Hor. *Ser.* 2.3.88)
not be+2SG+PRES+SUBJ uncle to.me
'Don't come the uncle with me'

Greek

mé: thé:sthe	nómon me:dena	(Dem. 3. 10)
not place+2SG+AOR+SUBJ law	none	
'Do not pass any law'		

Similarly the subjunctive is used in Spanish (but not Italian):

No hablas con él
not speak+2SG+PRES+SUBJ with him
'Don't talk to him'

For subordinate clauses, Latin uses the subjunctive with *ut* or (rarely) without the conjunction *ut* or, for the negative, *ne*:

Rogat et orat Dolabellam ut de sua provincia
he.asks and he.begs Dolabella that from his province
decedat
(Cic. *Verr.* 1.29.7)
withdraw+3SG+PRES+SUBJ
'He asks and implores Dolabella to leave his province'

Oppidanos hortatur, moenia defendant (Sall. 7.56)
townsmen he.urges walls defend+3PL+PRES+SUBJ
'He urges the townsmen to defend the walls'

suisque imperavit ne quod omnino telum in hostes
and.to.his.men ordered that.not any altogether weapon in enemies
reicerent (Caes. *B. G.* 1.46.2)
throw.back+3PL+IMPF+SUBJ
'and ordered his men not to throw back any missile at all at the enemy'

However, for a few verbs, notably IUBEŌ ‘I order’, the accusative and infinitive are used – see 7.4.1:

cum . . . eos . . . suum adventum expectare
since them his arrival to.wait.for+PRES+INFIN
iussisset (Caes. *B. G.* 1.27)
he.had.ordered
'since he had ordered them to wait for his arrival'

There is a very similar use of the subjunctive in both Italian and Spanish:

Gli hanno ordinato che tacesse
to.him they.have ordered that be.quiet+PAST+SUBJ
'They ordered him to be quiet'

Le mandaron que les-siguiera
him they.ordered that them-follow+3SG+PRES+SUBJ
'They told him that he should follow them'

However, in both languages an infinitive may also be used:

Indicative and subjunctive

Gli hanno ordinato di tacere
to.him they.have ordered PREP be.quiet+INFIN
'They ordered him to be quiet'

Le mandaron seguir-les
him they.ordered follow+INFIN-them
'They ordered him to follow them'

This use of the infinitive construction as an alternative to the Subjunctive construction will be discussed in detail in 7.4.1.

The subjunctive is similarly used as a (notional) jussive. Examples of first person use in Classical Greek and Spanish are:

all' ei dokéi, pléo:men, (Soph. *Phil.* 526)
but if it.seems sail+1PL+PRES+SUBJ
'If thou wilt, let us sail'

Levantemo-nos
raise+1PL+PRES+SUBJ-1PL+REFL
'Let's get up'

Examples of third person use from Italian and Spanish are:

Che venga anche lui
that come+3SG+PRES+SUBJ also him
'Let him come too'

Que entre
that enter+3SG+PRES+SUBJ
'Let him come in'

Use of the subjunctive as jussive in non-European languages has already been illustrated for Hausa and the Bantu languages Swahili and Luvale. Examples repeated from the introductory section of 5 and from 5.1.1 are:

haa njahen
HAA go+1PL+SUBJ
'Let's go'

ni-pend-e
I-love-SUBJ
'Let me love'

va-iz-e (veze) waxi
they-come-SUBJ quickly
'Let them come quickly'

There was also some discussion of the jussive in Semitic languages in 5.4.1. In fact, the forms that can be treated as Jussive for both formal and notional reasons are sometimes more like subjunctives, and are even treated by some scholars as subjunctives. In particular, they occur in subordinate clauses.

Thus, for Syrian Arabic, Cowell (1964: 345ff.) uses the term ‘subjunctive’ to refer to a form that is similar to the Amharic Jussive, which may function as an ‘optative’ (i.e., Jussive) as in:

n-rūh ʕas-sinama
1PL-go+SUBJ to.the cinema
‘Let’s go to the cinema’

This ‘subjunctive’ is used in subordinate clauses with ‘overt expressions of exhortation, suggestion, wish, fear, intention, etc.’. In addition, it is found with ‘must’, ‘may’, ‘be able’, ‘know how to’, ‘forget to’ and many others, not all of them describable as modal. Examples are

ʔana bəddi ʔərzaʕ ʕal-be:t
I I.want I.go+SUBJ to.the-house
‘I want to go home’

la:zəm ʔufi bwaʕdi
must I.keep+SUBJ to.promise.my
‘I must keep my promise’

There is another respect in which this jussive (or ‘subjunctive’) is like the European subjunctive – it is used as the negative of the imperative in (Cowell 1964: 345):

ruh
go+2SG+IMP
‘Go!’

la truh
not go+2SG+JUSS
‘Don’t go!’

However, there is no uniformity with the form that is most generally used in subordinate clauses. In Tigre (Ethiopian Semitic) the jussive is used for purpose (Leslau 1945: 200):

ʔəttu ʔəgəl ləʔraʕ
to.him in.order.that complain+3SG+JUSS
‘in order to complain to him.’

In the closely related language, Tigrinya, it is the imperfect that is used (Leslau 1941: 142):

məʔənti kət-bälləʕ
in.order.that-eat+3F.SG+IMPF
‘so that she could eat’

Amharic normally uses the imperfect in indirect commands (Cohen 1936: 304):

ənd-imatu azzaza
that-they come+IMPF he.ordered
'He ordered them to come'

However, if the superordinate verb itself is in the imperative the jussive may be used (Cohen 1936: 357):

yəzazwaccaw yəmtu
order.them they.come+JUSS
'Order them to come'

(This is certainly not direct speech, which would require the imperative 'Come'.)

This is, perhaps, the best place to note the ‘optative’ mood in Nahuatl (Aztec – Andrews 1975: 52, 384), which has marking for number and person and has both present and past tense. This is used for wishes, relating to both present and past. However, Andrews adds, ‘There is no special imperative mood for the purpose of giving commands. Command sentences and exhortation sentences are simply wish sentences.’ It might, perhaps, be suggested that this is more like a subjunctive and that it provides an example of a language in which expressions that are treated elsewhere as jussives and imperatives (as well as desideratives – 5.3.3) are indicated entirely by the subjunctive.

5.5 Subjunctive as subordinator

It was suggested earlier (5.1) that the subjunctive is typically associated with subordination. Usually there is some notionally irrealis feature involved as well, but sometimes it seems simply to be a marker of subordination. This became increasingly true of Latin, as shown by an example from Late Latin:

Pugnatum . . . incerto Marte, donec proelium nox
 it.was.fought with.uncertain Mars until battle night
 dirimeret (Tac. H. 4.35)
 break off+3SG+IMPF+SUBJ
 ‘The fight went on indecisively until night broke it off’

There are other examples where the use of the subjunctive may seem to lack any motivation in terms of irrealis. For instance, it is sometimes used in Latin simply to distinguish between different kinds of subordinate clause. The conjunction *cum*, for instance, can be used to mean ‘when’ or (causal) ‘since’, as indeed can English *as*:

As I was going to the shop, I saw John
As I was going to the shop, I couldn't wait

Latin uses the indicative and subjunctive to distinguish these two senses:

5.5 Subjunctive as subordinator

Cum haec leges, consules habebimus (Cic. Att. 5.12.2)
 when these read+2SG+FUT+IND consuls we.shall.have
 ‘When you read this, we shall have consuls’

Quae cum ita sint, Catilina, perge quo
 which since thus be+3PL+PRES+SUBJ Catiline proceed whither
 coepisti (Cic. Cat. 1.5)
 you.began
 ‘Since this is so, Catiline, pursue the course you have begun’

However, here it could be argued that ‘since’ indicates shared knowledge, and that this is another example of presupposition being marked by an irrealis mood.

There is also a contrast between the moods in:

Ante quam pro L.Murena dicere instituo, pro
 before that for L.Murena speak begin+1SG+PRES+IND for
 me ipso pauca dicam (Cic. Mur. 1)
 me self few let.me.say
 ‘Before I begin to speak for L.Murena, let me say a little on my own behalf’

Ante quam veniat in Pontum, litteras ad
 before that come+3SG+PRES+SUBJ in Pontus letter to
 Cn. Pompeium mittet (Cic. Agr. 2.20.53)
 Cn. Pompeius he.will.send
 ‘Before he arrives in Pontus, he will send a letter to Cn. Pompeius’

However, the problem is not the use of the subjunctive in the second, but the use of the indicative in the first (both refer to the future – see 5.2.6). It may be that it is the immediacy of the event that warrants the indicative.

Similarly for Fula, Arnott (1970: 3056) indicates that the subjunctive is used not only with verbs of (i) enjoining, (ii) wishing, (iii) fearing, taking care, (iv) requesting, (v) permitting or agreeing, (vi) causing or arranging, but also for other types of clause. Thus (Arnott 1970: 310–11), it is required with *haa* ‘until’, *d’ooke* ‘before’, etc., even though the clauses are notionally realis:

be-ngaḍay ka remuki haa be-timmina
 ‘They continue farming until they finish’
 d’ooke be-njottoo, ’o-’yami gorko ’on . . .
 ‘Before they arrived, he asked the man . . .’

Similarly, in a very different language, Mangarayi (Australia – Merlan 1982: 178, 184), there is a prefix that is labelled ‘irrealis’, which is used for epistemic possibility, but which is also often the only marker of subordination:

a-pani-yug

IRR-talk-AUX

'He might talk'

gawa-j

muyg jan? ya-ma-n

bury-PAST-PUNCT dog die IRR-AUX-PAST-PUNCT

'He buried the dog when it died / that died'

5.6 Other possibilities

So far the contrast has been between one set of indicative forms and one set of subjunctive forms. But it is perfectly possible to allow that there might be a further system within either of the two moods.

In Swahili (and, indeed, other Bantu languages) the subjunctive is not marked for tense-aspect as the indicative is. There is a single set of forms for subjunctive, but other sets for present continuous, past, future, consecutive, conditional, etc. Swahili does not appear to have any modality systems, such as those of English or the evidential systems. It seems that there is here the converse of what is found with modal systems in that there is one marker of irrealis, but a system of markers of realis.

There is what might seem to be the opposite situation in Maasai (East Africa – Tucker and Mpaayei 1955: 61ff., 96ff.), in that, for subordination at least, there are several types of subjunctive, although the choice of marker is not free but determined by the subordinating verb. These different forms are labelled 'subjunctive (a)', 'subjunctive (b)', 'infinitive (a)', 'infinitive (b)' and the 'N-tense' and there are (at least) five classes of predicator:

- (i) 'Let' takes either of the two subjunctives.
- (ii) Verbs of wanting, liking, disliking, etc. (including 'ought'), take the N-tense.
- (iii) Infinitive (a) is used mostly after verbs of motion, but also after verbs of liking, 'help', 'do again', 'do in the near future', 'start', 'repeat', 'do early in the morning'.
- (iv) Infinitive (b) is used with 'get to do', 'know how to', 'finish', 'be able to', 'dare'.
- (v) Either infinitive (b) or the present tense is used after 'refuse', 'be afraid', 'forget', 'lack'.

6

Realis and Irrealis

It was noted in 1.2.1 that, in many languages, particularly the Native American languages and those of Papua New Guinea, mood is described in terms of the grammatical markers of realis and irrealis. Although the distinction is basically the same as that between indicative and subjunctive, both being markers of the typological categories Realis and Irrealis, there are sufficient differences to warrant discussion of them in a separate chapter. Yet there is not always a clear distinction between them – see 7.1 for discussion.

6.1 Joint and non-joint marking

There are basically two ways in which realis and irrealis markers function. In some languages their main function is to co-occur with other grammatical categories. In others they mainly occur in isolation and are themselves the only markers of specific notional categories.

For instance, in Amele (Papuan – Roberts 1994: 372) an irrealis marker is required whenever a future marker is present in the sentence:

ho bu-basal-en age qo-qag-an
pig SIM-run.out-3SG+DS+IRR 3PL hit-3PL-FUT
'They will kill the pig as it runs out'

By contrast in another Papuan language, Muyuw (Bugenhagen 1994: 18, quoting a personal communication), the irrealis marker is itself the indication of future:

yey b-a-n Lae nubweg
I IRR-1SG-go Lae tomorrow
'I will go to Lae tomorrow'

Strictly, these might seem not to be comparable in that, in the Amele example, there is a syntactic relationship between two grammatical markers, irrealis and future, whereas in the Muyuw example there is simply a specific grammatical marker, irrealis, that indicates futurity. Yet from a typological point of view, they are similar, in that the common notional feature of

futurity is associated in both languages with the grammatical category of irrealis.

These different functions of realis and irrealis will be distinguished in terms of being ‘joint’ and ‘non-joint’ markers, the joint markers being those that co-occur (obligatorily) with other grammatical markers.

However, there are two important reservations. First, the distinction is not clear-cut, in that both features often appear in a single language, although the mood system of any one language usually contains markers that are predominantly either joint or non-joint. It will, therefore, sometimes be convenient to refer to the systems themselves as either ‘joint’ or ‘non-joint’. Secondly, non-joint systems are often not strictly binary, in that constructions may not only be marked as realis or irrealis, but may also be unmarked (see 6.5.2). More problematic for the realis/irrealis distinction is the fact that, in some languages, realis and irrealis markers occur together with other markers in even wider systems, which, *prima facie*, look more like modal systems than instances of mood (but see 6.5.3).

For Caddo, spoken in Oklahoma, Chafe (1995: 351–9) reports that pronominal prefixes of the verb may be distinguished as either realis or irrealis and that the choice is determined by sets of grammatical markers that occur before these pronominal prefixes. Irrealis prefixes are used in conjunction with grammatical markers indicating negation, prohibition, obligation, conditional, and also in conjunction with ‘simulative’, ‘infrequentative’ and ‘admirative’ prefixes. (In addition to the ‘simple negative’ prefix, there is also a ‘temporal negative’ expressing ‘not for a long time’, and, in addition to the conditional, a ‘generic conditional’ and a ‘negative conditional’):

kúy-t’a-yibahw
NEG-1+AG+IRR-see
‘I don’t see him’

kaš-sah?-yibahw
PROH-2+AG+IRR-see
‘Don’t look at it’

kas-sa-náy?aw
OBL-3+AG+IRR-sing
‘He should/is obliged to sing’

hí-t’a-yibahw
COND-1+AG+IRR-see
‘If I see it’

dúy-t’a-yibahw
SIMULAT-1+AG+IRR-see
‘As if I saw it’

wás-t'a-yibahw
 INFREQ-1+AG+IRR-see
 'I seldom see it'
 hús-ba-ʔasa-yik'awih-saʔ
 ADM-1+BEN+IRR-name-know-PROG
 'My goodness he knows my name'

Realis prefixes, it is implied, although Chafe gives no examples, occur with expressions that refer to the present or past states and events. They also occur with two markers of future ('future' and 'future intention'):

ci-yibahw-ʔaʔ
 1+AG+REAL-see-FUT
 'I'll look at it'
 ci-yibahw-čaʔ
 1+AG+REAL-see-FUT.INT
 'I'm going to look at it'

However, both realis and irrealis each have one function in which they occur alone, with no other grammatical marker, and are thus non-joint. A non-joint irrealis prefix indicates a (yes–no) interrogative:

sahʔ-yibahw-nah
 2+AG+IRR-see-PERF
 'Have you seen him?'

A non-joint realis prefix indicates an imperative:

yahʔ-yibahw
 2+AG+REAL-see
 'Look at it'

The Caddo system is, then, predominantly, but not entirely, joint.

A non-joint system is found in Manam (Papuan – Lichtenberk 1983, discussed in Bugenhagen 1994: 9–11). The system here is binary – Lichtenberk (1983: 181) states 'The realis–irrealis opposition is crucial to the Manam verbal system in the sense that every finite verb must be specified (by means of a subject/mood prefix) for one of the two moods.' Realis is used for (i) past events, (ii) present events and (iii) habitual events:

u-nóʔu
 1SG+REAL-jump
 'I jumped'
 úra i-pura-púra
 rain 3SG+REAL-come-RED
 'It is raining'

ʔi-zen-zén
 1PL.EXC. REAL-chew betel-RED
 'We (habitually) chew betel-nuts'

Irrealis is used for (i) future events, (ii) commands, exhortations and warnings ('lest'), (iii) counterfactual events and (iv) sequences of customary or habitual activities:

úsi né-gu mi-ásaʔ-i
 loincloth POSS-1SG 1SG+IRR-wash-3SG+OBJ
 'I will wash my loincloth'

go-moanáʔo
 2SG+IRR-eat
 'Eat'

ŋáu u-rére nóra boʔaná-be go-púra
 1SG+IRR 1SG-want yesterday SIM-FOC 2SG-come
 'I wish you had come yesterday'

úma ga-ʔoáriŋ-i ʔái ga-pólo-ø
 garden 1PL.EXC+IRR-clear-3SG+OBJ tree 1PL.EXC+IRR-fell-3PL+OBJ
 'We clear (the bush for) a (new) garden, we fell the trees'

There is one important point to be noted here. When a marker is joint, i.e., occurs with another grammatical marker, it is redundant in that it adds nothing to the notional features associated with that other marker, apart from characterizing them as realis or irrealis. By contrast, non-joint markers are the sole indicators of the notional features and so are not redundant. However, it is seldom the case that all the terms in a system are joint (and redundant) or non-joint (and non-redundant). Moreover, occasionally either realis or irrealis may occur with the same grammatical marker (e.g., with Future in Central Pomo see – 6.5.1), with a difference of meaning, and so non-redundantly, although joint. There is a debate concerning redundancy, which will be discussed in 7.3.

6.2 Terminology

Although they are transparent, it is, perhaps, a little unfortunate that the terms 'realis' and 'irrealis' have been adopted as grammatical terms in place of the traditional terms 'indicative' and 'subjunctive'. Bybee *et al.* (1994: 236) report that the earliest use of the terms in the relevant sense is not to be found in their corpus before 1970, the earliest instance being in the analysis of the Australian language Maung by Capell and Hinch (1970).

It might have seemed wise, in the interests of consistency, to use only one set of terms, either 'indicative' and 'subjunctive' or 'realis' and 'irrealis' for both sets of data, preferably the former. However, this is probably imprac-

tical now that both traditions are firmly established, and, moreover, there is some benefit in dealing with the two types of phenomena separately (with their different terminologies, and in separate chapters).

The result of this, however, is that there is, unfortunately, a problem with the terminology in that 'realis' and 'irrealis' can be used in at least three senses.

First, the terms may be used to refer to the notional features, rather than 'factual' and 'non-factual', or 'real' and 'unreal' (see 1.1.1). It may be noted that Givón (1994) uses these terms in his analysis of the Spanish subjunctive. Secondly, the terms are used by linguists as labels for the formal grammatical categories. Thirdly, if indicative/subjunctive and realis/irrealis are seen as the grammatical markers of mood, it would be appropriate to recognize 'Realis' and 'Irrealis' as the relevant typological (cross-language) categories.

As argued in 1.7, there is no problem about the terminology to be used to distinguish a typologically valid category from the notional features associated with it, since typological categories can be indicated by the use of initial capitals. Thus the typological grammatical categories of Realis and Irrealis are associated with the notions of realis and irrealis. Confusion between the terms as names of the language-specific categories and as labels for the notional features is possible, but where necessary can be avoided by talking of 'notionally realis' and 'notionally irrealis'.

There is another point, which was briefly discussed in 1.5. Typologically the categories in joint and non-joint systems discussed above in 6.1 are very similar, and it is convenient to use the same terminology for them. Thus, although 'future' is not distinguished grammatically from 'command' and 'counterfactual' in Manam (6.1), it is still reasonable to treat it as an instance of the typological category Future. Commonly, however, writers express the facts by saying that categories such as future 'are' realis or irrealis, either because of their co-occurrence with realis or irrealis markers or because they are signalled by the realis or irrealis markers. This clearly mixes the grammatical and notional levels, but may be allowed because it is a fairly obvious and instructive way of combining the facts of both joint and non-joint systems.

There is one further point. The term 'irrealis' is used in the description of some languages as the name of one of the terms in a modal system, where it would be more appropriate to use a term such as 'dubitative'. An example can be found in the account of Ngiyambaa (Donaldson 1980: 160, 162) where 'irrealis' occurs in the same system as 'purposive' which is used both deontically and epistemically, and in its epistemic use is essentially Deductive (see 1.4.5, 2.1.2 and 4.1):

Realis and Irrealis

Irrealis

yuruŋ-gu ŋidja-l-aga
rain-ERG rain-CM-IRR
'It might/will rain'

Purposive

ŋadhu bawuŋ-ga yuwa-giri
I+NOM middle-LOC lie-PURP
'I must lie in the middle'

yuruŋ-gu ŋidja-l-i
rain-ERG rain-CM-PURP
'It is bound to rain'

This is not relevant for the treatment of irrealis as mood. The term is used here to refer to a term in a modal system (rather than in a system of realis/irrealis mood).

6.3 Morphology and syntax

6.3.1 Morphology

Often realis and irrealis markers are single isolated forms, whether words, clitics or affixes. This is so in Kiowa (Oklahoma – Watkins 1984: 171) in which there is a single suffix *ʔɔ*:

hàgyà à-bá-ʔɔ.
perhaps 1SG-go-IRR
'Maybe I'll go'

Similarly in Maricopa (see 6.5.3) the marker is *-ha*:

ny-aay-ha
1/2-give-IRR
'I will give it to you'

There is a similar situation with the non-joint markers in the Papuan languages. According to Bugenhagen (1994: 36) the markers in eight such languages are verb prefix for subject and mood, pronoun for subject and mood, verb prefix (for mood only), preverbal particle and sentence-initial particle.

In some languages, however, there are markers that are cumulative exponents of several grammatical categories. Thus in Caddo (Oklahoma – Chafe 1995: 352–3), there are prefixes that are simultaneously markers of (i) person, first, second, third and 'defocussing', (ii) the grammatical relations (see Palmer 1994) agent, patient and beneficiary, as well as (iii) realis and irrealis. Part of the whole system is:

Realis prefixes			
	agent	patient	beneficiary
First person	ci-	ku-	ku-
Second person	yahʔ-	si-	si-
Irrealis prefixes			
First person	ʔa-/ʔi-	ba-	ba-
Second person	sahʔ-	saʔa-	saʔu-

There are, in fact, forty-eight possibilities. There is some syncretism, however, such as that shown by the formal identity of the first three pairs of patient and beneficiary above. This is, perhaps, striking in that cumulative exponence of this kind is somewhat reminiscent of the subjunctive (see 7.1).

Even more like indicative/subjunctive is the system described for Takelma (Southern Oregon) by Sapir (1922: 94), and discussed by Mithun (1999: 5). Here there are six ‘tense-modes’ – ‘aorist’, ‘future’, ‘potential’, ‘inferential’, ‘present imperative’ and ‘future imperative’. Each verb has two stem forms *yowo-* and *yu-* for ‘run’, *baxam-* and *baxma-* for ‘come’, etc. The first stem, ‘a derived or amplified form of the base’, is used for the aorist, which is the most frequently used of the tense modes. The second stem, ‘the generally more fundamental verb-stem’, is used for the ‘less important tense-modes’. Sapir’s examples are, unfortunately, far from transparent, e.g.:

Aorist	ʔlomōm	‘He killed him’
Imperative	dōʔm	‘Kill him’

Now the aorist is used to refer to past events, the present and the immediate future, and is clearly realis, while the others are all potentially irrealis. The realis/irrealis distinction is marked, then, in the choice of verb stem. This kind of redundant marking is not at all uncommon in inflected languages. In Latin, for instance, there are different stems for the non-perfect and perfect tenses and moods, e.g. *am-* for non-perfect and *amav-* for perfect with the verb ‘love’, and, more strikingly, *fer-* and *tul-* non-perfect and perfect with ‘carry’, even though the person–number paradigms are also different. (For further discussion, see 7.1.)

A further possibility is that there is double marking, as in Alsea, once spoken in Oregon (Buckley 1988, quoting an unpublished manuscript, Frachtenberg 1918). Here the distinction is marked both by complementizers – realis *mis* and irrealis *sis* – and, according to Mithun (1999: 175), by affixes that are cumulative exponents of Realis/Irrealis and Inchoative/Completive (page references to Buckley):

temʔhu	mis-axa	wiʔl-x	(12)
and then REAL+COMPL-back come-REAL+COMPLET			
‘And then, after she came back’			

mis-iłx mųhu' łáq'-s-t-əx (19)

REAL+COMPL-3PL+SUB at last cross-?STAT-REAL+COMPLET
'When they finally got across'

sips tqaiáld-i (21)

IRR+COMPL+2PL+SUB want-IRR+COMPLET
'If you desire it'

í'mst tém-in tqaiált-əx sis
thus and-1SG+SUB want-REAL+COMPLET IRREAL+COMPL
kexk-ái'-m (13)

assemble-IRR+INCH-INTR

'For that reason, I want [the people] to reassemble'

There are two other possibilities. First, there are languages in which there are realis and irrealis markers and, in addition, constructions that are unmarked (discussed in 6.5.2). Secondly, Realis may be unmarked (as it usually is where there are modal systems of the type presented in Chapters 2 and 3). Possible examples of this are Tolkapaya Yavapai and Mojave, which are discussed in 6.3.3 and 6.6.5.

6.3.2 *Linked clauses*

In the Caddo examples presented by Chafe (6.1), the joint markers occurred not only within a single clause, but also within a single word. One repeated example (for Negative and Irrealis) is:

kúy-t'a-yibahw
NEG-1+AG+IRR-see
'I don't see him'

However, there are both Native American languages and Papuan languages in which these joint markers are associated with separate clauses of a complex sentence.

One such language is Central Pomo (Pomoan, N. California – Mithun 1999: 176–7, cf. Mithun 1995: 368–73), which also has a modal system (see 1.2.1, 2.2.3). Here the realis/irrealis distinction is made in linked clauses (of a single type, but where the links would be translated as 'and', 'while', 'then', etc.). In the first clause the verb has markers that are cumulative exponents of (i) realis/irrealis, (ii) 'same'/'different', which shows whether the linked clauses refer to the same or to different events, (iii) for realis only, 'simultaneous'/'sequential' to distinguish between simultaneous and sequential events. The co-occurring grammatical category that determines the choice of realis and irrealis is in the final clause. Thus, in the examples below, the categories of imperative and conditional occur in the second clause and

these determine that the verb in the first clause is marked as irrealis (same+IRREALIS and different+IRREALIS respectively):

q^há č^hni'-ʔel dó-č-*hi* mú'tya-l
 water bread-the make-SEMEL-SAME+IRR 3PL-PAT
 qa'-wá-č'-ka-m
 biting-go-IMPF+PL-CAUS-IMP
 'Make the water bread *and* invite them to eat it'
 me'n mí-*hla*, mú'l ʔa' ʔ^há'ʔ le ʔa'
 so say-DIFF+IRR that 1+AG sit-COND 1+AG
 'If she said that, I'd stay longer'

The next examples show markers of present and past in the second clause and the marking of realis (same+simultaneous+realis and different+sequential+realis) in the first clause:

ʔa' Edna-to čá-l yó-'h-du-*n* hínʔil-ʔel
 1+AG Edna-PAT house-to go-PERF-IMPF-SAME+SIM+REAL Indian-the
 ča'nó-'d-an-ya mú'tu
 talk-IMPF-IMPF-PERS.EXP 3+PAT
 'I go to Edna's house and talk Indian to her'
 ʔo' meʔ ʔ-né'-ya-w-li míya' mé
 1+PAT such by.gravity-set-DEFOC-PERF-DIFF+SEQ+REAL 3+POSS father
 dá-'ʔ-du-w č^hó-w
 like-REFL-IMPF-PERF not-PERF
 'I was nominated and his father didn't like it' ('When I was nominated, his father didn't like it')

There is a similar situation in the Papuan language Amele, as described by Roberts (1990: 371–5). However, one important difference is that, although realis/irrealis marking occurs with linked clauses ('clause chaining'), it does so only where the clauses have different subjects. Again, the verb of the first ('medial' in Roberts's terminology) clause is marked as realis/irrealis (cumulatively by a marker for realis/irrealis, number and person and 'different subject'), while the verb of the final clause is marked for the co-occurring grammatical category (cumulatively with number and person). The following examples illustrate realis marking (together with person and number and different subject) on the verb of the first clause and markers of past and present (plus person and number) on the verb of the final clause:

ho bu-busal-en age qo-in
 pig SIM-run.out-3SG+DS+REAL 3PL hit-3PL+REM.PAST
 'They killed the pig as it ran out'
 ho bu-busal-en age qo-igi-na
 pig SIM-run.out-3SG+DS+REAL 3PL hit-3PL-PRES
 'They are killing the pig as it runs out'

Similarly, other examples show irrealis marking in the medial clause and future and imperative in the final clause:

ho bu-busal-eb age qo-qag-an
pig SIM-run.out-3SG+DS+IRR 3PL hit-3PL-FUT
'They will kill the pig as it runs out'

ho bu-busal-eb age qo-ig-a
pig SIM-run.out-3SG+DS+IRR 3PL hit-3PL-IMP
'Kill the pig as it runs out'

These examples show how mood is marked on the first of the linked clauses (the 'medial' clause), but is dependent on the grammatical marker in the second (the 'final' clause). (For a full list of categories involved, see 6.4.)

The translations of the linked ('chained') clauses suggest that they include both oblique subordinate clauses (temporal, purpose, conditional, etc.) and coordinate clauses with 'and', 'but', etc. In Amele, at least (but probably in other languages), these two types of clause can be distinguished syntactically in that a subordinate medial clause can be moved – either embedded in the final clause or transposed after it (Roberts 1994: 13–15). This can be seen from:

ho qo-qag-an nu dana age h-oiga-a
pig hit-3+PL-FUT PURP man 3PL come-3PL-TOD.PAST
'The men came to kill the pig'

The subordinate *ho qo-qag-an nu* 'to kill the pig' can be transposed:

dana age ho qo-qag-an nu h-oiga-a
man 3PL pig hit-3+PL-FUT PURP come-3PL-TOD.PAST
'The men came to kill the pig'

dana age h-oiga-a ho qo-qag-an nu
man 3PL come-3PL-TOD.PAST pig hit-3+PL-FUT PURP
'The men came to kill the pig'

By contrast no transposition is possible with the coordinate:

ho busale-i-a qa dana age qo-iga-a
pig run.out-3SG-TOD.PAST but man 3PL hit-3PL-TOD.PAST
'The pig ran out, but the men killed it'

6.3.3 Complement clauses

The last section dealt with a special kind of complex sentence, in which two clauses are involved in the determination of mood marking, but realis/irrealis marking is also to be found in more familiar types of subordinate clause, both oblique clauses and complement clauses (in the same way as indicative/subjunctive). Thus in Mojave (Yuman, Arizona – Munro 1976: 54–5) the

irrealis suffix appears on ‘subordinate verbs to mark unreal or hypothetical situations: desires, conditionals, obligations, counterfactuals and occasional futures’. Examples of an oblique and a complement clause are:

huwalʔəpay ya-ʔ-aʔa:v-θ ʔ-ʔahot-e
 Walapai x-1-understand-IRR 1-good-FUT
 ‘If I learn Walapai, I’ll be all right’
 humi’č ʔatay iyu’-h-a’r-pč
 children more have-IRR-want-TENSE
 ‘She wants to have more children’

A similar pair from Alsea, once spoken in Oregon (Buckley 1988 and Frachtenberg 1920) was given in 6.3.1:

sips tqaiáld-i (24)
 IRR+COMPL+2PL+SUBJ want-IRR+COMPLET
 ‘If you desire it’
 í.mst tém-in tqaialt-əx sis
 thus and-1SG+SUB WANT-REAL+COMPLET REAL+COMPL
 kexk-ái’-m (42)
 assemble-IRR+INCH-INTR
 ‘For that reason, I want [the people] to reassemble’

There are plenty of examples of similar complement clauses in Papuan languages. In a table showing the functions of irrealis in these languages, Bugenhagen (1994: 36) indicates that, of the six languages with extensive mood marking, all use irrealis for ‘want’ and ‘lest’, while five use it for ‘purpose’, ‘ability’ and ‘obligation’. Examples of ‘want’ from Manam and Sursurunga are:

tamá-gu i-rére go-púra záma
 father-1SG 3SG+REAL 2SG+IRR-come tomorrow
 ‘My father wants you to come tomorrow’
 iau nem ngo i-na han ur Ukarumpa
 1SG+REAL want that 1SG-IRR go to Ukarumpa
 ‘I want to go to Ukarumpa’

Here, clearly, ‘want’ is expressed with the use of an irrealis complement clause. The same is true of ability and weak obligation, as well as wanting, in Mangseng (although with ability the irrealis marker is optional):

i pavurvu (aro) i ro vai
 3SG able (IRR) 3SG go.up coconut
 ‘He is able to climb coconut trees’
 i siveŋa ako aro i oma i
 3SG good that IRR 3SG do 3SG
 ‘He should do it’

tho n-ve aro te oriet pua ponange
1SG say-thus IRR we hunt crocodiles today
'I want us to hunt crocodiles today'

However, the status of the forms is less clear in Muyuw (Bugenhagen 1994: 19, quoting a personal communication), where examples of 'want', 'ability' and 'obligation' are:

sivina-g b-a-n lae
need/desire-1SG IRR-1SG-go Lae
'I want to go to Lae'

kadiloka b-i-vag
ABILITY IRR-3SG-do
'He can do it'

awoum b-i-vag
PROH IRR-3SG-do
'He must not do it'

There is no problem with 'want', but a possible analysis of the other two examples is not in terms of complements of the markers of ability and prohibition, but in terms of joint marking of irrealis and grammatical markers for ability and prohibition.

Complement clauses can also be found in conjunction with mood-marked linked ('chained') clauses. For Amele, Roberts (1990: 373–4) lists five types of complement subordinate clause that also require irrealis marking – intensive, desiderative, habitual desire, abilitative, purpose. The subordinate clauses are marked in several different ways – with 'relative future', imperative and infinitive:

ho bu-basal-eb age qo-qaq-a bili tawe-ig-abe
pig SIM-run.out-3SG+DS+IRR 3PL hit-3PL-REL.FUT be stand-3PL-TOD.PAST
'They stood about to kill the pig as it ran out'

ho bu-basal-eb age qo-ig-a ade-ø-na
pig sim-run.out-3SG+DS+IRR 3PL hit-3PL-IMP 3PL+OBJ-3SG-PRES
'They want to kill the pig as it runs out'

ho bu-basal-eb age qo-oc nu ihoc
pig SIM-run.out-3SG+DS+IRR 3PL hit-INFIN for able
'They are able to kill the pig as it runs out'

It is important to understand that these subordinate complement clauses themselves are not marked as irrealis in the way that in other languages complement clauses are marked with subjunctive. Rather, it is the markers of subordination in these clauses (relative future, imperative and infinitive) that determine that the previous ('medial') clause is marked as irrealis. (Notice too that the markers in the main clause are irrelevant in this respect – in the

first two they are future and present, but the ‘medial’ clause is marked as irrealis.)

Generally the use of irrealis in a subordinate clause is redundant in that it is determined by the construction and allows no contrasts of meaning. An interesting exception is to be found in Mojave (Yuman, Arizona – Munro 1974: 54–5). The distinction between (unmarked) realis and irrealis is used to distinguish wanting, an attitude towards a notionally irrealis proposition, and liking, an attitude towards a notionally realis proposition:

?-isay-θ ?-a:r-mot-m
 l-fat-IRR I-want-NEG-TENSE
 ‘I don’t want to get fat’
 ?-isay-k ?-a:r-mot-č
 l-fat-ss I-want-NEG-STAT
 ‘I don’t like being fat’

6.4 Joint systems

It is, perhaps, rather surprising that the two Native American joint systems that have been discussed in detail, Caddo (6.1) and Central Pomo (6.3.2) are very different in two important ways. First, as shown in 6.3.2, the syntax is very different. The co-occurring forms in Caddo are within a single clause (and in all the examples given, within a single word), whereas in Central Pomo there are linked clauses with the mood marker being in one single clause, and the co-occurring grammatical marker in another. In this respect, Central Pomo is rather like some of the Papuan languages, as described by Roberts (1994: 36). Secondly, there is a difference in their treatment of some of the grammatical categories. Those illustrated for Caddo are:

Realis: future, future intention, imperative
 Irrealis: negation, prohibition, obligation, conditional,
 simulative, infrequentative, admirative, temporal negative,
 generic conditional and negative conditional.

The categories illustrated for Central Pomo were:

Realis: imperfective, perfective
 Irrealis: conditional, imperative, future (basically).

(Negative and Interrogative do not affect mood in Central Pomo, while future is usually, but not always, marked as irrealis. Notice that future and imperative are realis in Caddo – see the relevant sections of 6.6.1 and 6.7.1.)

Takelma (6.3.1) is, perhaps, to be regarded as another Native American language with a joint system. If so, it would seem that realis is unmarked

since the (realis) ‘aorist’ has the basic stem of the verb. It has a joint system in that a set of categories occurs with the derived (irrealis) stem. The categories would then be:

Realis: present/past (‘aorist’)

Irrealis: future, potential, inferential, present imperative and future imperative

These three Native American Indian languages differ considerably from each other, but in many of the Papuan languages the realis/irrealis system is more homogeneous with fairly predictable sets of obvious realis and irrealis categories. Examples from Amele were given in 6.3.2. For that language, Roberts (1990: 275) lists the grammatical categories that are associated with realis and irrealis as:

Realis: habitual past, remote past, yesterday’s past, today’s past, present tense

Irrealis: future, imperative, hortative, prohibitive, counterfactual/prescriptive, apprehensive.

He provides a similar set of examples from Nobonob, Anjam and Bargam (all from unpublished manuscripts). A pair of sentences from Nobonob (Roberts 1990: 280) is:

ah ag e he-egeg danab lag lag qag-pig
woman 3PL food do-3PL+DS+SIM+REAL man 3PL house tie-3PL+PAST
‘As the women cooked the food, the men roofed the house’

ah ag e he-bepeg danab lag lag qag-kulag
woman 3PL food do-3PL+DS+SIM+IRR man 3PL house tie-3PL+FUT
‘As the women cook the food, the men will roof the house’

He notes that these three languages are unrelated, though all from the same area (the Madang Province of Papua New Guinea), but he also provides examples from languages in other, far removed, areas – Wojokeso (West 1983), Angaataha (Huisman 1973) and Gahuka (Deibler 1976). Roberts (1990: 382–3, quoting an unpublished manuscript by D. Rucker) also gives examples from Anjam, where present, immediate past and remote past are marked as realis, while future, imperative and counterfactual are marked as irrealis. Examples of remote past and future are:

e tabir yans-eqn-a-m Rut
1SG dishes work-SIM+REAL-REM.PAST-1SG+REAL+DS Ruth
alaŋ-oqn-e-j
play-CNTF-REM.PAST-3SG
‘While I washed the dishes, Ruth played’

a wan-oqn-i-m nan̄gi b-q-ab
 3SG work-SIM+IRR-FUT-3SG+IRR+DS 3PL come-FUT-3PL
 'He will be working when they come'

However, there is an obvious difference here – that the medial clauses as well as the final clauses contain the grammatical markers that determine the realis/irrealis distinction (remote past and future).

It is quite apparent that there is a close correspondence between the notional and formal categories. Only propositions indicating past and present events are treated as realis, for these propositions are asserted; propositions indicating future events and all unrealized, potential, events are treated as irrealis, for these are not asserted. The unrealized, potential, events are those indicated both by imperative, hortative, prohibitive, counterfactual/prescriptive, apprehensive in main clauses and by intensive, desiderative, habitual desire, abilitative, purpose in subordinate clauses.

There is one other curious point. Amele has a number of sentence-final particles. Some of these occur only with irrealis final verbs (those that determine irrealis in the medial verb), but others can occur with realis final verbs, and, in spite of being notionally irrealis, do not affect the mood of the medial verb. Thus with 'maybe' or the particle for question, the medial verb still has realis marking if the final verb is realis (e.g., present tense):

ho bu-basal-en age qo-gi-na fa
 pig SIM-run.out-3SG+DS+REAL 3PL hit-3PL-PRES DUB
 'Maybe they are killing the pig as it runs out'
 ho bu-basal-en age qo-gi-na fo
 pig SIM-run.out-3SG+DS+REAL 3PL hit-3PL-PRES QUES
 'Are they are killing the pig as it runs out?'

Other such particles are translated as 'nevertheless', 'let them', 'really' and 'always'. Roberts (1990: 379–88) reports that studies by other scholars, mostly in unpublished manuscripts, show a similar situation in Nobonob, Anjam, Bargam and Wojokeso (West 1983) and probably several other languages.

In the Papuan languages, Future is always one of the categories marked by irrealis in joint systems (and in non-joint – see 6.5.1). There is some variation with the other potentially irrealis categories. A table presented by Roberts (1994: 31) shows that where there is marking of irrealis on the medial verb with chained constructions, the marker indicates Future only in two languages; Future and Imperative in five; Future, Imperative and Counterfactual in four; and Future, Imperative, Counterfactual and Habitual Past in one. (Where the only irrealis category is Future, there is, of course, no way of proving that it is a marker of mood rather than tense.)

6.5 Non-joint systems

6.5.1 Binary systems

Although it is clear that realis and irrealis occur non-jointly to indicate, quite directly, contrasts such as present and past vs future, etc. (see 6.1, 6.6.1), there are few, if any, languages where there is a simple binary contrast of realis and irrealis. This is hardly surprising, for such a binary contrast would allow for a great deal of ambiguity. Most languages with non-joint marking have other devices to make further distinctions. In fact, most languages with extensive non-joint marking do not have binary systems of mood at all, as will be clear from 6.5.2, 6.5.3 and 6.5.4.

However, one language that appears to have a purely binary contrast is Manam (Papuan – Lichtenberk 1983: 182–91, discussed in Bugenhagen 1994: 9–11), which was exemplified in 6.1. Lichtenberk, it will be recalled, states that ‘every finite verb must be specified . . . for one of the two moods’. He suggests that realis is used for (i) past, (ii) present and (iii) habitual events, while irrealis is used for (i) future events, (ii) commands, exhortations, warnings (‘lest’), (iii) counterfactual events and (iv) customary or habitual activities.

Yet irrealis frequently occurs in conjunction with particles to mark further distinctions. Thus, while in isolation it will indicate a certain or near future (see 6.1), together with the form *mása* ‘perhaps’ it indicates an uncertain future:

tanépwa mása bulé?a ŋa-éma?-i
chief perhaps 3SG+IRR-make-3SG+OBJ
‘The chief will give a feast’

Together with the ‘prospective’ form *?ána* it is said to correspond to English ‘going to’:

?úsi né-gu mi-ása?-i ?ána
loincloth POSS-1SG 1SG+IRR-wash-3SG PROSP
‘I am going to wash my loincloth’

Moreover, when the prospective form is preceded by *abe* ‘already’, it expresses imminence, and when preceded by a form meaning ‘very (very) near’ it expresses ‘the fact that an event almost took (has taken) place, but did (has) not’. There are other forms to indicate ‘lest’ and ability/permission, and the irrealis form is also used in subordinate clauses after ‘want’.

There is a potential problem here in that it could be argued that, since these forms regularly occur with irrealis, this is a joint system rather than a non-joint system (where the irrealis marking would be redundant). However, it is clear that the basic distinction is made contrastively, with no other

markers – between realis past and present and irrealis future. The system is, therefore, basically non-joint and it is more appropriate to see these co-occurring particles as having the function of ‘fine-tuning’ the basic distinction. Notionally and typologically they are very different from the grammatical categories that co-occur with irrealis in joint systems.

6.5.2 *Realis, irrealis and unmarked*

A typical characteristic of mood is the binary system of Realis and Irrealis. However, in some languages, although there is a contrast stated in terms of realis and irrealis, there are also forms that are unmarked for mood.

This is inevitably so in the Papuan language Amele, because mood is marked only on linked constructions with different subjects, as illustrated in 6.3.2. Where there are the same subjects, the construction is unmarked for mood (Roberts 1994: 11):

age cabi na beli-me-ig ceta ceh-eig-an
 3PL garden to.go-SS-3PL yams plant-3PL-YEST.PAST
 ‘They went to the garden and planted yams (yesterday)’

There is no problem with this; it is enough to say that the realis/irrealis contrast is restricted to certain constructions, and that in such constructions there are still only the two possibilities.

In some languages there is a choice, in some constructions, between unmarked and irrealis. Thus in Alambalak (Papua New Guinea – Roberts 1990: 390–1, quoting Bruce 1984), both the imperative and the hortative (i.e., the jussive) are used without a mood marker, but may also be used with irrealis. Examples of the imperative are (see 6.7.1 for further details):

nuat wa-ya-n-t
 sago.patty IMP-eat-2SG-3F.SG
 ‘Eat the sago patty!’
 (nikë) wa-roh-twa-kë
 (2PL) IMP-sit-FUT+IRR+IMP/HORT-2PL
 ‘You all be seated’

Similarly, in Muyuw (Bugenhagen 1994: 18, quoting a personal communication), present and past are either unmarked or marked as realis, while future is marked as irrealis:

nov i-weiy buluk
 yesterday 3SG-kill pig
 ‘Yesterday he killed the pig’
 buluk nov bo n-ei-weiy
 pig yesterday EMPH REAL-3SG-kill
 ‘I know he really killed a pig yesterday’

yey b-a-n Lae nubweig
I IRR-1SG-go Lae tomorrow
'I will go to Lae tomorrow'

(But it is not stated whether the presence of the emphatic marker determines the realis marking, so that it is not entirely clear whether the choice of unmarked or realis is determined or not.)

Slightly different is the situation in which there are three markers, but where two of them might appear to be realis and irrealis. For Dani, another Papuan language, Foley (1986: 163) suggests that there is a three-term contrast – which he calls 'real', 'likely' and 'potential'. His examples are taken from Bromley (1981: 28, 47), but with some reinterpretation:

wat-h-i
kill-REAL-1SG+AG
'I killed him'

was-ø-ik
kill-likely-1SG+AG
'I will kill him'

wa?-l-e
kill-POT-1SG+AG
'I may kill him'

There is a similar situation in the Northern Iroquoian languages (New York, Ontario, Quebec – Chafe 1995: 359–60). There is a three-way division in Seneca:

wa?-ke-kę-?
FACT-1AG-see-PUNCT
'I see/saw it' (direct perception or memory)

ę-wa?-ke-kę -?
FUT-1AG-see-PUNCT
'I'll see it' (prediction)

aa-ke-kę -?
OPT-1AG-see-PUNCT
'I should/might see it' (obligation or possibility)

Chafe interprets these in terms of a reality continuum with 'factual' at one end, 'optative' at the other and 'future' as intermediate.

However, it could be argued that the systems of Dani and Seneca are modal systems rather than mood. In addition to the fact that there is a three-term system, the use of 'optative' as both epistemic and deontic is typical of modal systems (see 4.1.1) and the gloss on 'factual' is reminiscent of an evidential term. An argument against this is that modal systems do not usually include Future.

Rather differently, in Alamblak (Roberts 1990: 390) there are two negative forms, each associated with a different irrealis marker (they are distinguished as ‘certain’ and ‘uncertain’ negatives and irrealis):

fiñji noh-r-fě-r
 NEG.CERT die+IRR.CERT-IMM.PAST-3SG+MASC
 ‘He did not die’
 afě noh-rhwa-t-r
 NEG.UNCERT die-FUT+IRR.UNCERT-3SG
 ‘He will not die’

However, as can be seen, the choice of the irrealis marker is determined by the negative. The ‘certain’ irrealis is also used alone for counterfactuals:

hik-r-fě-an-n
 follow+IRR.CERT-IMM.PAST-1SG-2SG
 ‘I would have followed you’

6.5.3 *Wider systems*

It was noted in 6.2 that occasionally the term ‘irrealis’ is used for a term in a modal system, where a term such as ‘dubitative’ would be more appropriate. However, in some languages both ‘irrealis’ and ‘realis’ have been applied to terms that belong to a wider system, which, on formal grounds, should be analysed in terms of a modal system rather than (typically binary) mood. There is, however, some justification for the terms and for considering them here – that the notional features associated with them (and with irrealis in particular) are similar to those associated with irrealis in the languages that have been discussed.

Thus, in Kiowa (Oklahoma – Watkins 1984: 170–2) there is a marker of ‘future’ which is treated by Mithun (1999: 173) as irrealis (but Watkins’s ‘future’ is used in the glosses in the examples below). It clearly indicates future:

ém-cán-t’ò·-nò hègó yá-mò·khól-dò-t’ò·
 2SG-arrive-IRR-and+DIFF now 2SG/1SG/PL-ready-be-FUT
 ‘I’ll have them ready when you come’
 hàgyà à-bá·-t’ò·
 perhaps 1SG-go-FUT
 ‘Maybe I will go’

It is also used in several (notionally) irrealis functions – what might have occurred but didn’t, conditional and obligation:

mágyá à-bá·-t’ò·
 might.but.not 1SG-go-FUT
 ‘I thought I might go, but didn’t’

à-t^h-ṣ-tó' gó à-tól
 2SG+SG.OBJ-find-FUT and+SAME 2SG+SG.OBJ-send+IMP
 'If you find him send him here'
 k'yákômdà mâ'-ò'-dò kyôdé á-t^hây-dò'-tó'
 life indeed-good-because longtime 1PL+SG.OBJ-with-hold-FUT
 'Because life is so enjoyable, we ought to hang on as long as possible'

There is also an unmarked 'indicative' (= realis?), and markers for imperative and 'hearsay' (= Reported (3)?). The imperative is illustrated in the second example above. An example of 'hearsay' is:

béthò' èm-kò'dó-òlthò -k^hop-òmdè-hél
 unknowing 2SG-very-head-hurt-become-HSY
 'I didn't know you had a headache'

There is also a marker for negative imperative, though this occurs (jointly) with 'future' (irrealis):

pòy té' mèn-tél-tó'
 PROH all 2SG+DUAL-tell-FUT
 'Don't tell them everything'

This 'future' (irrealis) in Kiowa has a number of uses that are associated with irrealis in other languages, but clearly it does not belong to a binary system or even a trinary system (of realis, irrealis and unmarked).

A more dubious example is that of Maricopa (Yuman, Arizona – Gordon 1986a: 27, 109). Here an 'irrealis' suffix is used (without any other grammatical marker) to indicate future, possible and contrary-to-fact:

ny-aay-ha
 1/2-give-IRR
 'I will give it to you'
 haat nyi-ttpooy-nt-ha
 dog+PL OBJ+PL-kill+PL.ACTION-too-IRR
 'It might kill dogs too'
 aanylyviim m-vaa-kis ?nym-yuu-ha
 yesterday 2-come-COND 2/1-see-IRR
 'If you had come yesterday, you would have seen me'

Similarly, a realis marker indicates present or past (Gordon 1986a: 24, 25):

hot-m
 good-REAL
 'It is good'
 aham-m
 hit-REAL
 'He hit him'

However, Maricopa also has markers for ‘completed actions’ (perfective), ‘incompletive’, ‘desiderative’ and two evidentials, Visual and Non-visual:

?-yuu-ksh
 1-see-PERFV
 ‘I saw it’

 nyaa ?-imaa-INC
 I 1-dance-INC
 ‘I can dance’

 m-we-lya
 2-do-DES
 ‘Please do it’

 imaa-yuu
 dance-VIS
 ‘He danced’ (I saw it)

 m-ashvar-?a
 2-sing-NONVIS
 ‘You sang’ (I heard you)

There is also an imperative, but this occurs jointly with realis:

k-truy-m
 IMP-kill-REAL
 ‘Kill it’

Another language with an ‘irrealis’ form with several functions is Mao Naga (Tibeto-Burman – Giridhar 1994: 67–9). Here an irrealis marker is used for reference to the future, but also for thwarted intentions and for obligation:

ai izo ocū vuta le
 I today home go IRR
 ‘I will go home today’

 Alemo-no oro hrü le-Ti-e
 Alemo-ERG pig buy IRR-RELEV-PRED
 ‘Alemo wanted to buy a pig, but couldn’t’

 pfo-no idu ru-oTi-le
 he-ERG yesterday write-IRRELEV-IRR
 ‘He must have written yesterday’

However, ‘irrealis’ occurs, as can be seen, with markers of ‘relevance’ and ‘irrelevance’ and it can be replaced by ‘inference’:

pfo-no idu ru-oTi-ahi
 he-ERG yesterday write-IRRELEV-INF
 ‘He must have written yesterday’

Strictly, all of these languages are languages with modal systems, but they all have an ‘irrealis’ marker that indicates features that are associated with the grammatical category of irrealis in other languages.

6.5.4 Complex systems

There is a complex situation in Hixkaryana (Carib, N. Brazil – Derbyshire 1979: 143–5), in that there appears to be both an evidential modal system of the kind discussed in Chapter 2 and a system of realis/irrealis mood. Mood appears to be indicated by a non-past and a non-past ‘uncertain’ form of the verb, which can, perhaps be seen as realis and irrealis (though the system also involves an ‘intensifier’ particle).

When occurring alone, the ‘non-past uncertain’ indicates a question:

nomokya
he.come+NONPAST
‘He is coming’

nomokya
he.come+NONPAST.UNCERT
‘Will he come?’

With the intensifier *ha-*, however, these express ‘may’ and ‘must’ respectively:

nomokya ha
he.come+NONPAST INT
‘He must certainly come’

nomokyan ha
he.come+NONPAST.UNCERT INTEN
‘He may come’

With other particles, all preceded by the intensifier *ha-*, the non-past also expresses:

‘hearsay’ with *-ti*
nomokyan ha-i
he.come+NONPAST.UNCERT INT-HSY
‘He’s coming’ (they say)

‘uncertainty’ with *-na*
nomokyan ha-na
he.come+NONPAST.UNCERT INT-UNCERT
‘Maybe he’ll come’

‘deduction’ with *-mi*
nomokyan ha-mi
he.come+NONPAST.UNCERT INT-DED
‘He is evidently coming’ (on hearing the sound of an outboard motor)

‘positive doubt, scepticism’ with *-mpe*
 nomokyatxow ha-mpe
 they.come+NONPAST.UNCERT INT-SCEP
 ‘They are coming! I don’t believe it’

The non-past may also be used with the intensifier plus another particle for ‘certainty’, ‘prediction’ or ‘warning’:

nomokyaha ha-mpi ni
 he.come+NONPAST INT-PREDICT
 ‘He’s coming – be warned’

These are all categories found in modal systems.

Even more complex is the situation in Serrano (Uto-Aztecán, California – Hill 1967: 21, and personal communication). There is a set of evidential particles which ‘specify the validity of the statement’. They are:

ha	inferential	(889)
k ^w ə’ə	potential ‘can’	
k ^w ənə	quotative	
may	‘may’	
na’a	volitative	
pata	intensive	
qáy	‘not’	
ta	dubitative	

The ‘dubitative’ seems to be a marker of irrealis. In isolation it too indicates a question (although this is also signalled by intonation):

k^wa’i ta-m č kih^wu:či
 eat DUB-PL you fish+ACC
 ‘Are you (pl.) eating fish?’
 hai:ŋk^wa ta-bi mi
 to.where DUB-he+PAST go
 ‘Where did he go?’

It co-occurs with future and deductive (‘inferential’):

’i:p t wahi’ pinkiv
 here DUB coyote pass+FUT
 ‘The coyote will pass here’
 ‘ama’ t X ma:mč
 he DUB INF hear
 ‘He must hear it’

However the ‘potential’ can also be an indicator of a question (Hill, personal communication), and the ‘quotative’ may also be used alone with reference to the future:

k^wiʔ-č pi yi:ʔi
POT-you you+them dry
'Can you dry them?'

pimia' kwini-č qučib
with him QUOT-we dwell+FUT
'We would live with him (so we were told)'

Hixkaryana and Serrano could, then, also have been considered in Chapter 2 rather than here, since they appear to have an evidential modal system as well as mood. However, both have Interrogative as a term in the system and Hixkaryana has Negative as well. These at least are more typical of mood than of modal systems.

6.6 Propositional modality

6.6.1 *Past, present and future*

There is very little to be said about reference to simple past and present. These (perhaps to be identified as 'Non-future') are always marked as *realis*. Some languages have more than one marker for past, e.g., Amele with both 'remote past', and 'today's past' both of which are *realis*.

It is important to realize, however, that it is not the case that reference to the present or past time will always imply *realis* marking. There is nothing strange about the fact that, for instance, habitual past is marked as *irrealis* in Bargam (see 6.6.9), and other categories discussed in this section, such as Speculative, Negative, Interrogative, etc., may imply *irrealis* marking, even though there is reference to past or present time. It is only simple statements (assertions) about present or past actions that are always *realis*.

By contrast, Future is the category most commonly associated with *irrealis* in both joint and non-joint systems. Indeed, it was shown in 6.4 and 6.5.1 that in the Papuan languages of both types Future is always marked by *irrealis*. Examples already quoted from (joint) Amele (Roberts 1990: 372) and (non-joint) Muyuw (Bugenhagen 1994: 18) are:

ho bu-basal-en age qo-qag-an
pig SIM-run.out-3SG+DS+IRR 3PL hit-3PL-FUT
'They will kill the pig as it runs out'

yey b-a-n Lae nubweig
I IRR-1SG-go Lae tomorrow
'I will go to Lae tomorrow'

The same is true of most of the Native American languages. In Kiowa (see 6.5.3), Future is one of the categories indicated by *irrealis* alone. In Takelma

(6.3.1) the future paradigm is in the irrealis set. In Serrano (see 6.5.4), too, Future is marked as irrealis ('dubitative'):

'i:p t wahi' pinkiv
 here DUB coyote pass+FUT
 'The coyote will pass here'

However, the situation is quite different in Caddo. In that language there are two markers of future time, and both are marked as realis, not, as expected, irrealis (repeated from 6.1):

ciyi-bahw-ʔaʔ
 1+AG+REAL-see-FUT
 'I'll look at it'
 ciyi-bahw-čah
 1+AG+REAL-see-FUT INT
 'I'm going to look at it'

There is also a different situation in Central Pomo (Mithun 1995: 370). Here the future occurs most commonly with irrealis:

té'nta-lil wá-'n-hi ʔá' qó-be-w-ʔkʰe
 TOWN-to go-IMPF-SAME+IRR 1+AG toward-carry-PERF-FUT
 'I'll go to town and bring it back'

However, the realis marker may be used instead to indicate greater expectation of the event taking place (Mithun 1995: 379):

ʔá' čhó-w-da má ʔbá'-n-č̣-w-ʔkʰe
 1+AG not-PERF-DIFF+SIM+REAL 2+AG suffer-DUR-REFL-PERF-FUT
 'When I am no longer here, you will suffer'

It is natural to look for an explanation for the difference between Central Pomo and Caddo. There is a discussion of this in Chafe (1995: 358), but it includes the imperative (see 6.7.1), which is also marked as realis. He comments that since so many other categories in Caddo are 'functionally motivated members of the Irrealis category', the occurrence of irrealis with Imperative is 'an inconsistency'. He goes on to say that the use of realis with Future is 'more problematic . . . especially in view of the fact that future is so fundamental to the expression of irreality in the Papuan languages'. He suggests two answers. The first is that 'reality is not a binary but a gradient dimension in which imperatives and futures express ideas that are judged to be more in accord with reality than, say, yes-no questions or negations. Speakers may have a relatively stronger expectation that commands will be obeyed or that predicted events will take place.' He suggests that in the Northern Iroquoian languages Future is intermediate between realis and irrealis. The second is a diachronic explanation – that imperatives and

futures were established in Caddo before the realis/irrealis distinction was formed (from interrogative verbs). But such a diachronic explanation would be only half the story – it still needs to be explained why interrogatives and futures were not absorbed into such a ‘functionally motivated system’. Both answers may be true.

There are other languages in which there is a close association of future with irrealis. In Burmese, the ‘non-future’ marker appears to be used to refer to present and past events, while the ‘future’ marker is used for future events (Okell 1969: 425, 426, 355, who, however, merely refers to them as ‘verb-sentence markers’):

săneinei-taiñ mye? hpta?-te
Saturday-every grass cut-NONFUT
‘He cuts the grass every Saturday’
da-cauñmoú mă-la-ta
that-because.of not-come-NONFUT
‘So that’s why they didn’t come’
măne?hpañ sá-me
tomorrow begin-FUT
‘We shall begin tomorrow’

However, Comrie (1985: 51) points out that the contrast is not a temporal one. The ‘future’ marker may be used to refer to judgments in the present and past (Okell 1969: 355):

hmañ-leiñ-me
be.true-undoubtedly-FUT
‘That may be true’
măcithi sà-hpù-me htiñ-te
tamarind eat-ever-FUT think-NONFUT
‘I think he must have eaten tamarinds before’

Similarly, Comrie (1985: 51) argues that the putative future tenses of Dyirbal (Australian – Dixon 1972: 55) are essentially modal, but only because the ‘future’ (‘irrealis’) is used for present habitual. Yet, by contrast, present habitual is marked by the ‘non-future’ in Burmese. There are, moreover, many examples of the use of what are generally regarded as future tenses to refer to probability (Assumptive), e.g., the morphologically defined future tenses of Italian, French or Spanish. English WILL, moreover, has the notional functions of assumptive, future and habitual (see 2.1.3, 4.3.2 and 6.6.9).

6.6.2 Speculative and Deductive

Surprisingly perhaps, epistemic categories such as Speculative and Deductive (and also Reported) that are regular members of the epistemic modal system

are not listed as categories associated with irrealis in the tables for the functions of irrealis in Papuan languages presented by Bugenhagen (1994: 36) and Roberts (1994: 31).

In the Native American languages, they are sometimes, but rarely, associated with irrealis. There are three marginal cases that were noted. First, in Takelma (see 6.3.1), the ‘inferential’ is used for both Ability and Speculative and the ‘potential’ is used for both Deductive and Reported. Secondly, in Kiowa (Oklahoma – Watkins 1984: 171), Speculative is indicated by irrealis plus ‘perhaps’ (though the example given refers to the future also, and Future is marked by irrealis):

hàgyà à-bá'tó
perhaps 1SG+GO+IRR
‘Maybe I’ll go’

Thirdly, Speculative is sometimes found in non-binary systems (6.5.2), where an interpretation in terms of realis, irrealis and ‘intermediate’ is possible, the irrealis being interpreted as Speculative. This is the interpretation given by Chafe (1995: 339) for Seneca (examples repeated from 6.5.2):

waʔ-ke-kɛ-ʔ
FACT-1+AG-see-PUNCT
‘I see/saw it’

e, -ke-kɛ-ʔ
FUT-1+AG-see-PUNCT
‘I’ll see it’

aa-ke-kɛ-ʔ
OPT-1+AG-see-PUNCT
‘I should/might see it’

A similar trio from Dani was also noted in 6.5.2 (Foley 1986: 163):

wat-h-i
kill-REAL-1SG+AG
‘I killed him’

was-ø-ik
kill-likely-1SG+AG
‘I will kill him’

waʔ-l-e
kill-POT-1SG+AG
‘I may kill him’

In Burmese, however (Okell 1969: 355; Comrie 1985: 51), the irrealis (‘future’) marker is also used for Speculative:

hmañ-leĩñ-me
be.true-undoubtedly-FUT
'That may well be true'

măcithì sà-hpù-me htiñ-te
tamarind eat-ever-IRR think-FUT
'I think he must have eaten tamarinds before'

In general, however, Speculative, Deductive and Reported do not appear to be marked by mood.

Mention should be made of Hixkaryana and Serrano (6.5.4) but these appear to have both a modal system and a system of mood. The modal system in Hixkaryana clearly includes Speculative and Deductive, and that of Serrano has Deductive (and both have Reported also – see 6.6.6).

6.6.3 Interrogative

In some languages the irrealis marker is used, without any other grammatical marker, to indicate a question. (Some of the examples have already been given but are repeated here.)

This is true of Caddo (6.1):

sah?yi-bahw-nah
2+AG+IRR-see-PERF
'Have you seen him?'

It is also true of Hixkaryana and Serrano (which have both a modal system and a system of mood – see 6.5.4). An example from Hixkaryana is:

nomokyano
he.come+NONPAST.UNCERT
'Will he come?'

The situation is more complex in Serrano (Hill 1967: 21) in that the irrealis ('dubitative') is obligatory in questions – even though questions are also marked by intonation (shown as '?' in the examples) – unless the potential is present:

k^wa'ĩ ta-m ɕ ? kih^wu:ɕi
eat DUB-PL you QUES fish+ACC
'Are you (pl.) eating fish?' (20)

There is, however, one exception. The 'dubitative' marker is not required if the 'potential' is already present (Hill, personal communication):

k^wi-ɕ pi yi:ʔi?
POT-you you+them dry
'Can you dry them?'

Unusually, then, it is the potential alone that may occur without any other marker for questions.

In Central Pomo, however, (Mithun 1995: 373, 381) although an interrogation marker is required it does not affect mood. Thus, interrogative will occur with irrealis when there is also a future marker, but with realis when there is also a marker of past time, the situation being the same as if there were no interrogative present, mood being determined solely by the (irrealis) future and (realis) perfective:

ʔi-wa ma ša-čó-t-ʔkʰe ya-l ca-l
 be-INT 2+AG swinging-whip-M.E-FUT 1PL+PAT house-to
 dé-m-ma-hi?
 lead+PL-M.E.-COOP-SAME+IRR
 ‘Are you going to whip us when you take us home?’
 tʰaná da-sé-č̌-ba-wa ma ʔéː cʰʔól-č̌i-w
 hand pulling-wash-REFL-SAME+SEQ+REAL-INT 2+AG hair comb-REFL-PERF
 ‘Did you wash your hands and comb your hair?’

Mithun (1995: 380–1) offers an explanation for the difference between the marking of Interrogative in Central Pomo (no effect) and Caddo (irrealis). She suggests that it is a matter of scope. In Central Pomo basic propositions are categorized as realis or irrealis and then may be questioned:

QUESTION (± REALIS (PROPOSITION))

In Caddo, the scope relations are reversed, the marking of realis/irrealis being sensitive to questions:

–REALIS (QUESTION(PROPOSITION))

6.6.4 Negative

Interrogative and Negative are often considered together because they appear to function in similar ways (see 2.3), as they do in English where they are both subsumed under ‘non-assertion’ – cf. Quirk *et al.* (1985: 83–4). There are similarities between them in respect of realis and irrealis marking.

In Caddo (Chafe 1995: 354, 355) negatives, like interrogatives, co-occur with irrealis:

kúy-tʰayi-bahw
 NEG-1+AG+IRR-see
 ‘I don’t see him’

This is also true of Mesa Grande Diegueño (Yuman, S. California – Langdon 1970: 159):

ʔənyaː puy ʔəxap-x-vu əwaːp-x umaːw
 I there I.go.in-IRR-SPEC they.want.it-IRR they+NEG
 ‘They didn’t want me to go there’

In Central Pomo, however, the negative is like the interrogative in that it does not affect mood. Thus, where there are markers of both negative and conditional, the mood marker will be irrealis, and where there are markers of both negative and past, the mood marker will be realis, just as they would if no negative were present (Mithun 1995: 374–5):

ma me'n ʔi-w čhó-w-hla ma bé-da
 2+AG such do-PERF NEG-PERF-DIFF+IRR 2+AG this-at
 ma' baséʔ ʔhabáʔč̣i-w ph-wí-w-ʔke
 things bad lie-*INCH*-PERF VIS-perceive-PERF-FUT
 'If you don't do that, you're going to see bad things happening round here'

ranch-ʔel q̣dí yhé-t-ač̣ čhó-w
 ranch-the good do-M.E-IMP+PL NEG-PERF
 ʔi-n ya-l qó-'l mča-w
 be-SAME+SIM+REAL 1+PL-PAT out-to throw-PL+PERF
 dá-'ʔ-c'i-w
 want-REFL-IMP+PL-PERF
 'Because they didn't keep up the rancheria, they wanted to throw us out'

There is a similar variation in the Papuan languages. Roberts (1990: 378) specifically remarks that, although negation could have modal status (no doubt thinking of the situation in the Native American languages), 'it does not interact with the medial simultaneous realis–irrealis verb' in Amele. However, he also notes that in Alamblak irrealis is associated with negation, so that negative past (immediate past and remote past) as well as future is marked as irrealis:

fiñji noh-r-fě-r
 NEG.CERT die-IRR-IMM.PAST-3M.SG
 'He did not die'

afě noh-rhwa-t-r
 NEG.UNCERT die-FUT-IRR.UNCERT-3SG
 'He will not die'

(As noted in 6.5.2, there are two negative markers in Alamblak, each associated with a different marker of irrealis.)

Similarly, Bugenhagen (1994: 19) reports that for just one of the eight languages that he presents, Muyuw, negation is either unmarked for mood or marked as irrealis (but never as realis) with both past and present as in:

nag i-n wa-ven
 NEG 3SG-go to-village
 'He is not going/did not go to the village'

nag b-ei-n wa-ven
 NEG IRR-3SG-go to-village
 'He is not going/did not go to the village'

Mithun (1995 381–2) provides the same explanation for the variation in the realis/irrealis status of Negative as she does for Interrogative. It is a matter of scope:

Central Pomo	NEGATIVE ($\pm$ REALIS (PROPOSITION))
Caddo	–REALIS (NEGATIVE(PROPOSITION))

6.6.5 Negative-like categories

There are several instances of the use of irrealis with categories or expressions that are notionally to some degree negative.

Thus the infrequentative co-occurs with irrealis in Caddo (Chafe 1995: 357):

wás-t'a-yibahw
 INFREQ-1+AG+IRR-see
 'I seldom see it'

This is best seen as a type of negation, since this sentence implies that generally I do not see it. (*Seldom* is a 'semi-negative' in English – see Palmer 1987: 22.)

There are also several examples of the use of irrealis to refer to events that, in some way or other, were not fulfilled or to states that no longer hold, rather like the 'contrary to expectation' forms discussed in 2.4. Thus in Kiowa (Oklahoma – Watkins 1994: 171) irrealis may indicate that an action might have occurred, but did not (repeated from 6.5.3):

mágyá à-bá'-t'ɔ'
 might but not 1SG-go-IRR
 'I thought I might go, but didn't'

Somewhat similar is the situation in Tolkapaya Yavapai (Yuman, California – Hardy and Gordon 1980: 189–92). There is an infix, *th*, which is labelled 'modal', but which appears to be more like an irrealis marker in spite of the fact that there is another particle that is labelled 'irrealis'. This 'modal' infix has a number of functions, which may be grouped into three types:

- (i) It is used for unreal conditions and (unreal) wishes:
 m-vaa-th-m ny-'u-h yi-tha
 2-came-MOD-DS 1/2-see-IRR AUX-MOD
 'If you had come I would have seen you'
 ma-ch'-yu-th-k wal'-yii-k 'yum
 1-SUB-be-MOD-SS I-wish-ss I-be
 'I wish I were you'

- (ii) It is used after ‘try’:

‘wi ’-yoov-a-k ’-wi-th-k ’-yum
money l/3-make-IRR-SS l-do-MOD-SS l-be
‘I’m trying to make money’

- (iii) It is used for ‘what used to be (but is no longer)’:

ma-ch m-se-ch m-yu-th-k m-yum
you-SUB 2-fat-SUB 2-be-MOD-SS 2-be
‘You used to be fat’

‘-ima-th-k ’-tlahv-k ’-yum
l-dance-MOD-SS l-tired-SS l-be
‘I was dancing, but now I’m tired’

The significant point is that in (iii) the form expresses what was previously so but is no longer. (This may be relevant to the discussion of habitual past in 6.6.9. For habitual past also suggests that the actions of the past are no longer being continued, so that this may be part of the reason why habitual past can be marked as irrealis.)

Slightly differently, there are two irrealis markers in Nakanai (Papuan – Johnston 1980: 64, discussed by Bugenhagen 1994: 24–5). One, ‘non-imminent irrealis’ *ge*, is used to express future, doubt and possibility, the typical categories associated with irrealis. The other, ‘imminent irrealis’ *ga*, ‘encodes the notion of imminent or frustrated action’ according to Johnston, though Bugenhagen suggests that ‘almost’ would be a better translation, as in:

eau ga tuga so-io, ouka
lsg IRR walk to-there, not
‘I was about to proceed, but didn’t’

eau ga la-lea
lsg IRR RED-sick
‘I’m getting sick’

(For examples of ‘non-imminent irrealis’, see 6.7.2.)

A similar example of frustrated action was given for Mao Naga in 6.5.3 (repeated here):

Alemo-no oro hrü le-Ti-e
Alemo-ERG pig buy IRR-RELEV-PRED
‘Alemo wanted to buy a pig, but couldn’t’

6.6.6 *Reported*

Reported occurs as one of the categories in many evidential systems (see 2.2.2). It might, therefore, be expected to be one of the categories marked as irrealis with mood, but the only example noted was from one of the two

languages discussed in 6.5.4, which appear to have both mood and a modal system. This is Hixkaryana (Carib, N. Brazil – Derbyshire 1979: 143–5), where ‘hearsay’ occurs with ‘uncertain’, which is the marker of irrealis:

nomokyan ha-ti
he.come+NONPAST+UNCERT INT-HSY
‘He’s coming (they say)’

The other language discussed in 6.5.4 was Serrano (Uto-Aztecan, California – Hill 1967: 89), but there ‘quotative’ (= Reported (3)?) can occur alone, without any marker of irrealis:

pɪmia’ k^wɪnɪ-č quçib
with.him QUOT-we dwell+FUT
‘We would live with him (so we were told)’

6.6.7 *Presupposed*

Chafe (1995: 357) offers from Caddo (Oklahoma) an example of the irrealis marker being used with the Admirative prefix *hús*:

hús-baʔa-sayi-k’awih-saʔ
ADM-1+BEN+IRR-name-know-PROG
‘My goodness, he knows my name’

Chafe says ‘This last usage is perhaps surprising’, since ‘the event or state is real enough’, and suggests that the fact that it is contrary to expectation is responsible for the use of the Irrealis, ‘as if the speaker were saying “It’s unreal that he knows my name”’. But it is no more surprising than the use of the subjunctive in Spanish and Italian for what is presupposed (5.2.5), as would be clear if the translation were ‘I’m amazed that he knows my name.’ The explanation that there is no new information is equally valid for this use of the irrealis in Caddo.

6.6.8 *Conditional*

Conditionals are marked as irrealis in Caddo (Chafe 1995: 356):

hí-t’a-yibahw
COND-1+AG+IRR-see
‘If I see it’

So too are other markers that are notionally similar – generic conditional and simulative:

nas-t’a-yi-bahw
gen COND-1-AG+IRR-see
‘Whenever I see it’

dúy-t'ayi-bahw
SIMULAT-1+AG+IRR-see
'As if I saw it'

Similarly in Central Pomo (Mithun 1995: 370), both real and unreal conditionals (see 8.1.) are marked as irrealis, though only the latter has the conditional marker:

wa-q-hi ?e ló'-h-du-w?k^{he}
go-level-SAME+IRR COP help-IMPV-IMPV-PERFV-FUT
'If I go, I'll be helping out'
me'n mí-hla mu'l ?a' ?čhá' ?le ?a'
so say-DIFF+IRR that 1+AG sit+COND 1+AG
'If she said that, I'd stay longer'

(This 'conditional' is also used to express obligation – see 6.7.3.)

There is an interesting situation concerning conditionals in the languages considered by Bugenhagen (1994). He distinguishes what he calls 'counterfactuals' and 'hypotheticals'. These are what have been called 'unreal' and 'real' conditionals (see 8.1), although it is obvious that Bugenhagen's choice of terms avoids the danger of confusion with realis and irrealis. Yet there seems to be some genuine interplay between these two different types of 'unreality' in this sense and 'irreality', in that Papuan languages deal with mood in conditionals in different ways. Generally both clauses of conditions of both types are marked as irrealis, as in Manam:

?áti té?e-o ŋa-pára-ra Bogiá n-lá?a
boat one-3SG IRR-arrive-assume Bogia 1SG+IRR
'If a boat should come, I would go to Bogia'
'If a boat had come, I should have gone to Bogia'

Yet in Sursurunga, quite paradoxically, the protasis (*if*-clause) of counterfactual (unreal) conditionals is marked as realis:

ngo á-k-te han balbal us i ráin na
if 3SG+REAL-DEF-EMPH go again blow SUBJ rain 3SG+IRR
han kopkom kuluk á namnam
go grow good SUBJ food
'If it had kept on raining regularly, the crops would have grown well'

Only in one of the languages quoted by Bugenhagen (1994: 33), Sinangoro, is the distinction between hypotheticals and counterfactuals marked quite simply by realis and irrealis in both clauses. Only the hypothetical, with realis markers, is exemplified:

yema yasi be iaɣoma-ni nai, ba iaɣo-ni
if canoe 3SG+REM+REAL come-IMPF time 1SG+REM+REAL go-IMPF
'If the canoe comes, I will go'

6.6.9 Habitual Past

A little surprising, perhaps, is the co-occurrence of habitual past with irrealis in Bargam (Papua – Roberts 1990: 384, quoting an unpublished manuscript), as was mentioned in 2.4:

miles-eq leh-id teq anamren aholwaq-ad in
 return-SS+IRR go-DS+IRR then owner see-SS+SIM 3SG
 didaq tu-ugiaq
 food PERF-give+HAB.P+3SG
 ‘When (the pig) would return and then the owner would go and, on
 seeing it, used to give it food’

However, it is possible to see why this should be treated as irrealis. Although past time reference is usually treated as realis, what is relevant here is that the habitual past does not relate to specific actions in the past, but to a tendency to act. Indeed, Givón (1994: 323) suggests the habitual is a ‘hybrid modality’, sharing some features of realis (higher assertive certainty) and some of irrealis (‘lack of specific temporal reference; lack of specific evidence; . . .’).

Moreover, there are similar examples in other languages. As noted in 2.4, there is an evidential suffix in Kashaya, that indicates ‘Remote’ – an action (habitual?) in the remote past (Oswalt 1986: 40):

men ṣ̌i-yiʔciʔ-tḥi-miy
 this DO-PL+HAB-NEG-REM
 ‘They never used to do that in the old days’

It may also be relevant that Tolkapaya Yavapai has a marker that looks like irrealis to indicate what was previously true but no longer is. This was discussed in 6.6.5.

It may also be recalled that in Manam the prefix for habitual events is realis, although the prefix for ‘sequences of customary or habitual activities’ is irrealis (6.5.1) and that what might be regarded as irrealis in Dyirbal (6.6.1) may refer to either future or habitual events. Moreover, as Chung and Timberlake (1985: 221) point out, English uses a modal verb to express habitual action:

We would go for a walk most weekends

However, this view has been challenged (Bybee *et al.* 1994: 236–40; Bybee and Fleischman 1995b: 9–10), and there will be a great deal more discussion in 7.3.

6.7 Event modality

6.7.1 Imperative and jussive

As noted in 3.4 (see also 5.4), two types of commands are often distinguished grammatically. Those that are directed at the addressee or addressees alone

are usually treated as imperatives; in that sense there are only second person imperatives. All others are treated as jussives, though the term ‘hortative’ is also used. These are usually introduced in English by *Let* and are either first or third person.

In Central Pomo, imperatives and jussives occur with irrealis, as is shown in the examples (6.3.2):

q^há č^hni-ʔel dó-č-hi mú^hʔuya-l
water bread-the make-SEMEI-SAME+IRR 3PL-PAT
qa'-wá-č^h-ka-m
biting-go-IMPV+PL-CAUS-IMP
‘Make the water bread *and* invite them to eat it’
ya-ka háy š-dí-č-ma-hi
1+PL+AG-INF wood drag-carry-INCH-COOP-SAME+IRR
q^ha-kay š-dí-č-ma-w-ʔk^he
water-too dragging-carry-INCH-COOP-PERFV-FUT
‘Let’s get some wood and haul in some water’

In Takelma (see 6.3.1) there are two imperatives, a present imperative for commands to be carried out immediately and a future imperative for action at a later time (for other examples see 3.4), and both are marked as irrealis.

By contrast in Maricopa (Yuman, Arizona – Gordon 1986a: 25) the imperative occurs with realis:

k-tpuy-m
IMP-kill-REAL
‘Kill it!’

The same is true of Caddo (Chafe 1986: 358)

yah?-yibahw
1+AG+REAL-see
‘Look at it’

Chafe sees the marking of Imperative as realis as ‘an inconsistency’ and offers two possible explanations. This was discussed in 6.6.1.

It is instructive to note that ‘prohibitives’ (negative imperatives) are marked as irrealis in both Caddo and Central Pomo, but for different reasons. In Caddo, although the imperative is marked as realis (see 6.1), there is a prohibitive marker that occurs with irrealis:

kaš-sah?-yibahw
PROH-2+AG+IRR-see
‘Don’t look at it’

In Central Pomo the imperative is marked as irrealis and is still irrealis when the negative also occurs:

dá'wi ʔč^h-č-hi khyá swé-lanʔkhe-ṭhín ʔe ma
 on road stop-SAME-IRR game play-NEG it.is you
 'Don't stop and play on the way home'

In Caddo, then, the negative imperative is irrealis because of the negative, while in C. Pomo it is irrealis because of the imperative. Again, Mithun (1995: 383) offers an explanation in terms of scope (see 6.6.3, 6.6.4):

Central Pomo	NEGATIVE (–REALIS (IMPERATIVE))
Caddo	–REALIS (NEGATIVE (IMPERATIVE))

However, in Kiowa (Oklahoma – Watkins 1994: 172), only the 'prohibitive' is marked as irrealis, the imperative having no mood marker (repeated from 6.5.3):

pòy té mèn-tél-tó'
 PROH all 2SG+DUAL-tell-IRR
 'Don't tell them everything'

à-th-ṣ-tó' gó à-tôl
 2SG/SG-find-IRR and+SAME 2SG/SG-send-IMP
 'If you find him, send him here'

By contrast, in Sursurunga (Papuan – Bugenhagen 1994: 16) there is irrealis marking in imperatives, but not in 'prohibitives':

u-na rumrum i mama-m mai kaka-m
 2SG-IRR respect OBJ mother-2SG and father-2SG
 'Respect your father and mother'

6.7.2 'Strong' and 'polite' commands

A number of languages make a distinction between 'strong' and 'polite' commands (imperatives and jussives) by means of realis/irrealis marking. Thus, in Alambalak (Roberts 1990: 390–1, quoting Bruce 1984: 137, 140), both the imperative and the hortative (i.e., the jussive) are used without an irrealis marker. Examples, repeated from 6.5.2, are:

nuat wa-ya-n-t
 sago.patty IMP-eat-2SG-3F.SG+OBJ
 'Eat the sago patty!'

nuat a-ya-nēm-t
 sago.patty HORT-eat-1PL-3F.SG+OBJ
 'Let's eat the sago patty'

However, both the imperative and the 'hortative' may also be used with a marker that is interpreted as exhibiting the cumulative exponents of future, irrealis and 'hortative'/imperative:

(nikë) wa-roh-twa-kë
(2PL) IMP-sit-FUT+IRR+IMP/HORT-2PL
'Would you be seated'

(rër) a-roh-twa-r
(3SG) HORT-sit-FUT+IRR+IMP/HORT-3M.SG
'May he be seated'

Here, then, marking with irrealis creates a politer command (with both imperative and jussive).

Similarly, in Nakanai (Johnston 1980: 62) both imperatives and 'prohibitions' (negative imperatives) may be expressed both with no mood marker and with the 'non-imminent' irrealis marker *ge* (see 6.6.5). The translations suggest that the unmarked sentences express a stronger imperative:

tola egite so-ma
call 3PL to-here
'Call them here'

amutou ge lolo-a
2PL IRR hear-3SG
'You are to listen'

umala kokue-a
don't hit-3SG
'Don't hit him'

eme umala ge kokue-a
2SG don't IRR hit-3SG
'You must not hit him'

Slightly differently, in Jamul Diego (Yuman – Miller 1990: 119) there are two types of command, neither of them with an imperative marker. The 'strong' form for politeness has an irrealis suffix, while the other has a special second person prefix and no irrealis suffix:

nya-m-mápa-pu m-rar-x-s
INDEF-2-NOM+want-DEM 2-do-IRR-EMPH
'Do whatever you want'

k-naw
2-run
'Run'

These examples would seem to have something in common with the situation in European languages, where polite commands are expressed by the subjunctive. This has been illustrated for Italian and Spanish (see 5.4.2), where even greater politeness is achieved by using the third person form, as well as the subjunctive:

Italian
 entri pure
 enter+3SG+PRES+SUBJ if.you.please
 ‘Please come in’

Spanish
 Tome su libro
 take+3SG+PRES+SUBJ your book
 ‘Take your book’

In Amele (Roberts 1990: 384) both strong and polite commands are marked as irrealis, but in rather different ways. There is a clear contrast in the final verb in that with a strong command the final verb is marked with the imperative, while with a polite command the final verb is simply marked as irrealis (with no imperative marker):

h-og-a
 come-2SG-IMP
 ‘Come!’

ho-ho-m
 SIM-come-2SG+DS+IRR
 ‘Would you come’

However, even with the strong imperative (itself not marked as irrealis) the medial verb in a linked clause is marked as irrealis, as in (repeated from 6.3.2):

ho bu-basal-en age qo-ig-a
 pig SIM-run.out-3SG+DS+IRR 3PL hit-3PL-IMP
 ‘Kill the pig as it runs out’

There is a similar situation in another Papuan language, Bargam (Roberts 1990: 384, quoting an unpublished manuscript):

ni bol
 2SG come+IMP
 ‘Come!’

ni bol-eq
 2SG come-SS+IRR
 ‘Would you come’

6.7.3 *Obligative*

In Caddo, there is an obligative marker that occurs with irrealis (repeated from 6.1):

kas-sa-náy? aw
 OBL-3+AG+IRR-sing
 ‘He should/is obliged to sing’

There is also a marker for weak obligation in Central Pomo (Mithun 1999: 177), but this is also a conditional marker (see 6.6.8):

cá-w-h̥təw ʔé'y-yo-hi ʔáwhal da-čé'-ʔle
house-LOC-from away-go-SAME+IRR work handling-catch-COND
'He should go home and get a job'

In Kiowa (Oklahoma – Watkins 1984: 172), irrealis alone may indicate weak obligation, though it is also used for future and for what might have occurred but didn't (repeated from 6.5.3):

k'yákômdà mâ'-q̣.-dò kyóde á-thây-dò'-tót̃
life indeed-good-because longtime 1PL+SG.OBJ-with-hold-FUT
'Because life is so enjoyable, we ought to hang on as long as possible'

In the Papuan languages, it appears that obligation is usually expressed by a marker of obligation with a complement (see 6.3.3).

6.7.4 Abilitive

In Takelma (see 6.6.2) there is a paradigm referred to as 'potential', which is also for dynamic ability and for general possibility.

In the Papuan languages, ability, like obligation, appears usually to be expressed by a form indicating ability plus a complement, but an example from Muyuw (Bugenhagen 1994: 19) might be interpreted as a grammatical marker used jointly with irrealis (see 6.3.3):

kadiloka b-i-vag
ABILITY IRR-3SG-do
'He can do it'

7 Subjunctive and irrealis

This chapter will deal with a number of issues relating to both subjunctive and irrealis that have not been addressed, or not fully addressed, in the preceding two.

7.1 Similarities and differences

The decision to use different terminology for indicative/subjunctive and to deal with them in separate chapters rests upon a number of considerations.

It is partly a result of different traditions. The terms ‘indicative’ and ‘subjunctive’ are the traditional terms used in the description of the classical and modern languages of Europe, though, not unreasonably, they have been used by many scholars for other languages. According to Bybee *et al.* (1994: 236) the earliest use of the terms ‘realis’ and ‘irrealis’ noted in an extensive corpus is to be found in the description of the Australian language Maung in Capell and Hinch (1970). These terms have been preferred by scholars working in the Native American languages and in the languages of the Pacific, particularly those of Papua New Guinea. However, since it is unlikely that the scholars working in these languages were unaware of the classical tradition, it is probable that this decision was influenced by perceived differences in the nature of the data being investigated.

One possible difference that might be suggested is that mood in the European languages is a morphosyntactic category closely integrated with person, number, tense and voice. The four categories are not independently marked, the form being simultaneously the marker of all the grammatical categories, as well as of the lexical item. Each form is what Matthews (1991: 174) calls a ‘cumulative exponent’ of person, number, tense, mood and voice. Such cumulative exponents of grammatical categories (and usually, but not always, of the lexical item), are of course, the essential characteristic of languages that are called ‘inflectional’.

Mood in the languages where it is characterized as ‘realis’ and ‘irrealis’ is often marked by single words or individual affixes and clitics, as was

illustrated in 6.3.1. However, cumulative exponents are also to be found, as was shown in that section for Caddo (Oklahoma – Chafe 1995: 352–3). Here it was noted that there are prefixes that are cumulative exponents of (i) person – first, second, third and ‘defocussing’, (ii) markers of the grammatical relations (see Palmer 1994), agent, patient and beneficiary, as well as (iii) realis and irrealis. A few examples are (repeated from 6.3.1):

	Agent	Patient	Beneficiary
Realis prefixes			
1st person	ci-	ku-	ku-
2nd person	yah?-	si-	si-
Irrealis prefixes			
1st person	ʔa-/ʔi-	ba-	ba-
2nd person	sah?-	saʔa-	saʔu-

There is even some syncretism, a characteristic of inflection, such as that shown by the formal identity of the first three pairs of patient and beneficiary above. Two other languages with similar features, Takelma and Alsea, were also discussed in 6.3.1, where it was noted that Takelma has a morphological system very like that of Latin.

One similarity is that both subjunctive and irrealis markers are often redundant, in that the notional irrealis feature is already marked elsewhere in the sentence. Yet there is also a major difference. The subjunctive is generally redundant only in subordinate clauses, where the subordinating verb clearly indicates the notional feature, as in the Italian:

Gli hanno ordinato che tacesse
 to.him they.have ordered that be.quiet+PAST+SUBJ
 ‘They ordered him to be quiet’

After a verb of ordering with the conjunction *che*, the use of the subjunctive is obligatory. By contrast, an irrealis marker commonly occurs in main clauses, where it co-occurs with a grammatical marker that is notionally irrealis, as with the marker of obligation in Caddo (Oklahoma – Chafe 1995: 356):

kas-sa-náyʔaw
 OBL-3+AG+IRR-sing
 ‘He should/is obliged to sing’

Syntactically, the structures in which there is redundancy are very different.

At first sight, the notional features associated with subjunctive and with irrealis may appear different, but only if their functions in main clauses are considered. For the notions associated with the subjunctive in main clauses do not include question, denial or futurity (except rarely), while these are

commonly associated with irrealis. However, in subordinate clauses these notional features are often associated with subjunctive, often redundantly – see 5.2.3, 5.2.4, 5.2.6.

Perhaps one of the most striking features is that both irrealis and subjunctive are used to indicate what is presupposed – known to both speaker and hearer (though again with the issue of subordination). Thus the subjunctive is used in the subordinate clause of the Spanish examples:

- Sp. Lamento que aprenda
 It. Mi dispiace che impari
 I regret that learn+3SG+PRES+SUBJ
 ‘I regret that he learns/is learning’

Irrealis similarly occurs to indicate what is presupposed with the ‘admirative’ in Caddo (Oklahoma – Chafe 1995: 357 (see 6.6.7)):

- hús-baʔa-sayi-k’awih-saʔ
 ADM-1+BEN+IRR-name-know-PROG
 ‘My goodness he knows my name’

7.2 A binary system

It has been suggested several times in this book that, basically (prototypically), the Realis/Irrealis distinction is binary as illustrated by the contrasts provided by indicative/subjunctive and realis/irrealis. However, there are obvious examples that seem to be counterexamples and these must now be considered.

The issue of a binary Realis/Irrealis contrast is not crucial to modal systems, though it can often be established, and is relevant in that it links modal systems to mood in the overall category of modality. For instance, there is a contrast in English between an unmarked Declarative and the constructions with modal verbs. Similarly in many of the languages with evidential systems there is an unmarked form and a number of marked terms that form the modal system. In these languages, it can be suggested that there is the contrast between Realis and Irrealis. Yet it is not crucial to the analysis, as shown by the fact that in Tuyuca there appears to be no unmarked form, in stark contrast to the situation in Central Pomo (see 2.7.1).

With the indicative/subjunctive contrast there are two problems. The first is that there are also forms for jussive and imperative. However, jussives and imperatives are not as fully inflected as indicative and subjunctive in that they are never (hardly ever?) marked for tense. The second problem is that Classical Greek has an optative mood in addition to the other two. However, it can be argued that the distinction between subjunctive and optative is one of tense (‘modal-tense’) rather than mood (see 8.2).

The situation is less clear with languages that have realis/irrealis marking. Yet, there is a binary contrast in most of the languages with joint marking, e.g., Caddo (6.1). Particularly interesting is the situation in Amele, where it is true that realis/irrealis marking does not extend over the whole language, but is restricted to linked clauses with different subjects (6.3.2). But what is important is that, where the marking is required, it is binary. A little more problematic is the situation where there are unmarked forms as well as those marked as realis and irrealis (6.5.2), but, again, where there is marking it is binary. Yet it must be admitted that, in some languages, the forms labelled 'realis' and 'irrealis' are merely two terms in a wider system (6.5.3). Yet even there it can be argued that the notional features are essentially those associated with Realis and Irrealis elsewhere, so that there is, within the wider system, a Realis/Irrealis contrast. Alternatively, such systems can be treated in terms of modal systems rather than mood.

7.3 The typological status of Realis/Irrealis

A fundamental question is whether the distinction between Realis and Irrealis as exemplified here in terms of mood (though it is applicable to modal systems as well) is a coherent and homogeneous category that has typological validity. This is disputed by Bybee *et al.* 1994, (see also Bybee and Fleischman 1995b: 9–10, and Bybee 1998). There appear to be four main arguments.

- (i) The distinction is rarely realized in a language as a simple binary morphological distinction.
- (ii) Irrealis and Subjunctive markers are often semantically redundant in that the meaning is carried by some other element in the context.
- (iii) There is great variation in the notional features marked by them, which makes it difficult to circumscribe a focal meaning for them.
- (iv) Some of the notional features appear to be wholly inappropriate.

The first argument that 'realis/irrealis is rarely realized in a language as a binary morphological distinction' (Bybee *et al.* 1994: 237–8) is an interpretation of the statement by Foley, in his book on Papuan languages (1986: 158) that the 'basic distinction . . . is a binary one . . . but few languages express it in just this way'. Foley goes on to show that many languages make 'a number of distinctions along the continuum from real to unreal'. Nevertheless there is a significant number of languages in which the binary distinction is found, and many of them have been well illustrated in the

preceding two chapters. Roberts (1994: 7–8) provides a large list of languages in which a binary distinction is marked, and more are to be found in Bugenhagen (1994).

The second argument, like the first, is rather an overstatement. It is perfectly true that both subjunctive and irrealis markers are often redundant. Bybee *et al.* (1994: 10) say ‘it is difficult to determine in many instances whether the modal meaning of an utterance is contributed specifically by the so-called Irrealis marker or by some other element (lexical or morphosyntactic), in which case the Irrealis marker is functionally redundant’. Yet there are plenty of examples in the two preceding chapters where this is not so. Thus the subjunctive in main clauses generally contrasts with the indicative (see 5.1.1). Even in subordinate clauses there is often a choice between them, e.g. with verbs of belief in Italian and Spanish, depending on the degree of hesitancy about the proposition, or the degree to which the proposition is hypothetical (see 5.2.1). Similarly, Mithun (1995: 379) has a very clear and interesting example from S. Pomo to show that the irrealis marker is not redundant, since ‘realis’ and ‘irrealis’ futures can be distinguished. With realis the future indicates that the event is portrayed as being certain to happen, whereas the irrealis marker suggests some uncertainty. Examples (repeated from 6.6.1) are:

té·nta-lil wá·n·hi	ʔá·	qó-be-w-ʔkʰe
town-to go-IMPF-SAME+IRR	1+AG	toward-carry-PERF-FUT
‘I’ll go to town and bring it back’		
ʔá·	čhó-w·da	má ʔbá·-n·č̣i-w-ʔkʰe
1+AG	not-PERF-DIFF+SIM+REAL	2+AG suffer-DUR-REFL-PERF-FUT
‘When I am no longer here, you will suffer’		

Even more importantly, there can be redundancy only where there is joint marking (6.4). Where the marking is non-joint it is clearly not redundant (6.5), and may distinguish past–present from future time

The facts seem then to weaken or refute the first two arguments. The third is a little more difficult to deal with, since it must be admitted that there is great variation among languages in what is treated grammatically as *Realis* or *Irrealis*, especially in the languages in which mood is treated in terms of *realis/irrealis*. Thus Chung and Timberlake (1985: 241) say that ‘languages differ significantly as to which events are evaluated as actual (and expressed morphologically by the *realis* mood) vs non-actual (and expressed by the *irrealis* mood)’. Bybee *et al.* (1994) state that there is considerable misalignment between ‘the *irrealis* as defined and the actual distribution of forms in languages’. Indeed, they say that they did not find a single language in which

a single grammatical marker 'could be adequately described as marking off all the irrealis territory'. However, as Givón (1994: 323) points out, the degree of uniformity apparently demanded by Bybee *et al.* represents an extreme approach, and there is considerable cross-language variation with other generally recognized categories. He goes on to illustrate the complexity involved with passive. It may be necessary, then, to admit that the notional features associated with Realis/Irrealis are more complex and varied than those of, say, tense or aspect, but that should not preclude the recognition of mood as a typologically valid category.

Bybee *et al.* are particularly concerned with the fact that, in Maung (Australia) as analysed by Capell and Hinch (1970: 67), negative imperative is realis and positive imperative is irrealis. This may be a little surprising, but, notionally, both are potentially irrealis. There is probably a historical reason for the present situation, as there must be for the fact that the words for 'sun' and 'moon' are masculine and feminine respectively in French, but feminine and masculine in German, and there are plenty of similar examples – see Palmer (1984: 34–40). Most cases of variation, however, can be explained, in the sense that there is some motivation for treating them as either Realis or Irrealis. This is fairly obvious with statements about the future – they can be seen as either realis, as assertions, differing only from statements about present or past in terms of time, or they can be seen as irrealis, because, unlike the present and past, the future is unknown. The variation with Interrogative and Negative can similarly be explained – they can be seen as questions about, or denials of, asserted propositions or simply as non-assertions (see 6.6.3 and 6.6.4).

It should also be noted that Bybee *et al.* (1994: 237) actually say 'If one sits and thinks about what negative, possibility, hypothetical, and imperative have in common, one could well come up with the idea that they all describe situations that are unreal or non-actual in some sense.' Moreover, they also appear to accept the relevance of assertion. This seems to be an admission that there could well be a notional basis for the distinction.

The fourth argument is based on the fact that the habitual past is treated as irrealis in Bargam (see 6.6.9). Bybee *et al.* (1994: 238) say 'if this binary distinction differs so much across languages that a past aspect, which is usually considered one of the prototypical cases of realis (Foley 1986: 158ff.), can be considered irrealis in some languages, this binary distinction is not cross-linguistically valid'. This is not as strong an argument as might be supposed, for past is often marked as irrealis if there is negation, interrogation, conditionality, etc. It is only reference to simple acts in the past that is always marked as realis. Moreover, as suggested in 6.6.9, the past habitual

does not refer to actual events but simply to a tendency. As such it is not entirely surprising (as Givón 1994: 322 says) that it is treated as irrealis in some languages. It is not the fact of being past that may account for the irrealis marking, but the fact that it is habitual. Indeed, Givón (1994: 323) suggests the habitual is a ‘hybrid modality’, sharing some features of realis (higher assertive certainty) and some of irrealis (‘lack of specific temporal reference; lack of specific evidence; . . .’). (See 6.6.9.)

A further point is that there is at least one other example of irrealis marking of habitual past, the evidential suffix in Kashaya (2.4), which has an evidential modal system (2.2.3). To this might be added the use of *would* in English, which was cited by Chung and Timberlake (1986: 221) in relation to ‘non-actual mood’, but not in connection with the debate about Bargam. The reply of Bybee *et al.* (1994: 239) is that ‘The past tense of *willan* “to want” in Old English was already in use for habitual past . . . centuries before the future uses of *will* or the conditional use of *would* developed. Taking these two unrelated uses of *would* and trying to construct some common semantics does not improve our understanding of linguistic categories.’ But this is not as powerful an argument as they appear to think. For, in the first place, the Old English usage is obviously to be explained in terms of ‘wanting’ being interpreted as ‘tending’ to act, precisely the explanation given for the Bargam example. Secondly, since the modern form has lost so much of the semantics of the Old English, it may well be asked why the habitual sense survived when the modal auxiliary was formed. A plausible answer is that it did so because it was semantically consonant with the uses of the new auxiliary. When thinking about semantic change, it is also important to ask why some things have not changed. The uses of *WILL* are not as unrelated as Bybee *et al.* suggest. In particular, they share the potentially irrealis meaning of tending to act, which may, quite easily, be interpreted as either conditional future (conditional upon other events, see Palmer (1990: 98)) or as habitual. Roberts (1994: 23) also points out that the ‘future tense’ in Dyirbal (Dixon 1972: 55) is similarly used for habitual actions and is thus better viewed in terms of modality than tense. The argument about *would*, therefore, is not disproved, nor is the interpretation of the Bargam habitual past, nor, at an even higher level, is the view that mood in terms of realis and irrealis is a valid typological category.

7.4 Alternative subordinate constructions

It was seen in Chapter 5 that the subjunctive is often used to express irreality in subordinate clauses. There are, however, alternatives, and the contrast of these with the subjunctive can be interesting.

7.4.1 Non-finite clauses

The use of the accusative and infinitive construction as an alternative to a construction with subjunctive has been mentioned three times in Chapter 5 – in the sections on report (5.2.2), directives (5.3.1) and wishes, fears, etc. (5.3.3).

Latin uses the ‘accusative and infinitive’ construction for reported statements:

Dicit se de Gallis . . . postulare triumphum (Liv. 36.40.3)
he.says self from Gauls to.demand triumph
‘He says he claims a triumph from the Gauls’

Classical Greek uses a similar construction, but also has a construction with a finite complement (but with the indicative, not the subjunctive):

oíesthe khalkidéas tein hállada so:sein (Dem 9.74)
you.know Chalcidean+PL+ACC the Greece save+FUT+INF
‘You know that the Chalcidians will save Greece’
élegon hóti Kú:ros . . . téthne:ke, (Xen. An. 2.1.3)
they.said that Cyrus die+3SG+PLUP+INDIC
‘They said that Cyrus had died’

There is a contrast with verbs of wishing and hoping. In Spanish and Italian both occur with a conjunction plus the subjunctive (5.3.3), but in Latin an accusative construction is possible for wish, and is the only construction for hope:

sperabam tuum adventum appropinquare (Cic. Fam. 4.6.3)
I.hoped your arrival come-near+PRES+INF
‘I hoped your arrival would be soon’

These differences may reflect the fact that hopes are more likely to be realized than wishes – and so less likely to be marked as Irrealis.

For reported commands, Latin generally uses a finite complement with the subjunctive (see 5.4.2), but with *IUBEO* ‘I order’ it uses the non-finite construction:

cum . . . eos. . . suum adventum expectare (Caes. B. G. 1.27)
since them his arrival wait.for+PRES+INF
iussisset
he.had.ordered
‘since he had ordered them to wait for his arrival’

There is thus a contrast between the two verbs *IUBEO* and *IMPERO*, both translated as ‘I order’ (and there is a further difference in that the notional object is in the accusative with the first and in the dative with the second):

Eum iubet venire
 he+SG+ACC he.orders come+PRES+INFIN
 'He orders him to come'
 Ei ut veniat imperat
 he+SG+DAT that come+3SG+PRES+SUBJ he.orders
 'He orders him to come'

In Latin the choice of construction is determined by the superordinate verb, but in Italian there is freedom of choice with many verbs, as in:

Gli hanno ordinato di tacere
 to.him they.have ordered PREP be.quiet+INFIN
 Gli hanno ordinato che tacesse
 to.him they.have ordered that be.quiet+PAST+SUBJ
 'They ordered him to be quiet'

Among the verbs that take either construction are *ABBAIARE* 'bark' and *RINGHIARE* 'growl'!

It is not at all clear whether there is a notional difference between the two Latin verbs, or whether there is a difference between the two Italian constructions. However, Givón (1994: 281–3) offers an explanation for the fact that in Spanish some verbs can take either the finite or the non-finite constructions, while others take only the finite construction. He makes two suggestions.

First, he compares what he calls the deontic 'manipulation verbs' and claims that the infinitive invariably signals stronger manipulation than the subjunctive. Examples are:

Le mandaron seguir-les
 him they.ordered follow+INFIN-them
 'They ordered him to follow them'
 le mandaron que les-siguiera
 him they.ordered that them-follow+3SG+PRES+SUBJ
 'They told him that he should follow them'
 te prohibo cantar
 you I.forbid sing+INFIN
 'I forbid you to sing'
 te prohibo que cantes
 you I.forbid that sing+2SG+PRES+SUBJ
 'I forbid you to sing' (*'I forbid you that you sing')

The second suggestion is that there is a scale of manipulation and that there will be a changeover to obligatory use of the subjunctive (a 'cut-off point') at some place on the scale. Thus while both constructions are used with the verbs in the examples above, only the subjunctive is allowed with:

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le dijeron que les-siguiera
him they.told that them-follow+3SG+PRES+SUBJ
'They told him that he should follow them'

NOT

*le dijeron seguir-les
him they told follow+INFIN-them

Telling, he suggests, is much weaker manipulation than ordering. The deontic verbs that he illustrates are:

Stronger, with either construction: MANDAR 'order', PROHIBIR 'prohibit',
OBLIGAR 'make'

Weaker, with subjunctive only: DECIR 'tell', PEDIR 'ask', QUERER 'want'

He also suggests (Givón 1994: 283; cf. Givón 1971: 75) that, in respect of the stronger manipulation signalled by the infinitive, there is a similar situation in a Bantu language, Bemba. He compares three constructions, the non-finite construction with an object of the main verb, the finite construction with an object and the finite construction with no object as in:

n-à-mu-ebele uku-ya
I-REM-him-order INFIN-go
'I told him to leave'

n-à-mu-ebele (ukuti) a-y-e
I-REM-him-order (that) he-go-SUBJ
'I told him that he should leave'

n-à-ebele ukuti a-y-e
I-REM-order that he-go-SUBJ
'I said that he should leave'

This is very similar to the Spanish sentences with MANDAR and DECIR, but represented by a single verb, with three grades of manipulative force. He also considers another verb of ordering with higher force and again finds three degrees of manipulative force, but with a higher cut-off point.

n-à-mu-koonkomeshya uku-ya
I-REM-him-order INFIN-go
'I forced him to leave' (He left)

n-à-mu-koonkoomeshya (ukuti) a-y-e
I-REM-him-order (that) he-go-SUBJ
'I ordered him to leave' (He may or not have left)

n-à-koonkoomeshya ukuti a-y-e
I-REM-order that he-go-SUBJ
'I ordered that he should leave'

Here the non-finite construction has the sense of giving an order that is carried out (and there is no translational equivalent in English), but there are again three degrees of force.

Givón also points out that in English there are different possibilities with ORDER and with TELL:

- They ordered him to follow them
- Not *They ordered him that he (should) follow them
- They told him to follow them
- They told him that he should follow them

Of course, this is different from Spanish. In English, ORDER does not allow the 'subjunctive' (*should* – see 7.6), because, Givón says, it is too strong a manipulative verb, while TELL allows both. This might suggest that the difference between Spanish and English is a matter of the scale of manipulation. The higher verb in Spanish allows both constructions and the lower verb allows only the infinitive, while in English the higher verb has only the infinitive and the lower verb allows both constructions, as shown in:

<i>English</i>		<i>Spanish</i>	
order	Infin.	mandar	Infin., Subj.
forbid	Infin.	prohibir	Infin., Subj.
make	Infin.	obligar	Infin., Subj.
tell	Infin., Subj.	decir	Subj.
ask	Infin., Subj.	pedir	Subj.

However, the situation in English is rather more complicated than this would suggest. In all the examples above, the noun phrase following the verb is its object semantically and syntactically. One orders, forbids, makes, tells, asks somebody to do something. But there are verbs where the only object is the entire subordinate clause. Thus with *I ordered John to come*, the likely meaning is that I ordered John, i.e., that I gave the order to John, but with *I wanted John to come*, I did not, in any sense, want John – what I wanted was 'John-to-come'. Now there are many verbs like this in English; they take either the infinitive or the subjunctive construction. Surprisingly, perhaps, ORDER is among them, as can be clearly seen where a passive is used in the subordinate clause:

- He ordered the flag to be raised
- He ordered that the flag be raised/should be raised

It is obvious that orders were not given to the flag. However, this is rather curious, since PREFER and REQUIRE allow both constructions, WANT and EXPECT allow only the infinitive construction, while SUGGEST and PROPOSE permit only the subjunctive, and TELL does not fit in here at all:

I prefer/require the flag to be raised
I prefer/require that the flag (should) be raised
I want/expect the flag to be raised
*I want/expect that the flag (should) be raised
I suggest that the flag (should) be raised
*I suggest the flag to be raised
I propose that the flag (should) be raised
*I propose the flag to be raised
*I told the flag to be raised
*I told that the flag (should) be raised

These cannot be directly compared with Spanish sentences, because, in this type of construction, Spanish permits only the finite form with subjunctive. Yet Givón's scale of manipulation may still be relevant in that all the stronger verbs allow the infinitive and only the weaker verbs, SUGGEST and PROPOSE, do not.

Yet there is a problem with TELL. Why does it not fit in here at all? Why is it not like ORDER? The most obvious reason is that it always requires an animate object when it has a complement. One always has to tell someone. Another reason may be that it is mainly used for reported statements rather than reported commands. This is clear enough in:

I told them that I was coming

But this may also be true of:

They told him that he should follow them

Should here is not the 'subjunctive' *should* within an indirect command, but the modal verb used for obligation within an indirect statement. The evidence for that is that the simple form as 'subjunctive' cannot be used in this construction:

!He told them that they follow him

This cannot mean 'He told them to follow him', but only 'He told them that they (in fact) follow him.'

However, in Italian, the verb DIRE 'tell/say' is one of the many verbs that take either construction – it has none of the restrictions of DECIR in Spanish and TELL in English.

The choice between the two constructions is not, then, clearly motivated by notional features, though there is a tendency for the subjunctive to signal a greater degree of irrealis.

7.4.2 Direct speech

Another alternative to the subjunctive in a subordinate clause is the use of direct speech, the reporting of the actual words of the original utterance.

Direct speech is often used for reporting statements (where the use of the subjunctive is not very common – see the opening paragraph of the last section), but it can also be used for reporting questions and commands (where the use of the subjunctive is fairly common):

He said ‘I’m coming’
I asked ‘Are you coming?’
She said ‘Come!’

Direct speech is almost certainly to be found in all languages, either as an optional alternative to indirect speech, or as the only way of reporting. Some examples of direct speech (all for reporting commands) may be taken from Kobon (New Guinea – Davies 1981: 2), Yidiny (Australia – Dixon 1977: 524) and Syrian Arabic (Cowell 1964: 450):

nipe ip hag-öp ne ñel ud-ag-ø QUOTE g-öp
3SG 1SG say-PERF+3SG 2SG firewood take-NEG-2SG a do-PERF+3SG
‘He said to me “Do not take the firewood”’

damariŋgu buɟi:n bana: buɟi diga mamba
Damari+ERG tell+PAST water+LOC put+imp pour.water.on+IMP sour+ABS
‘Damari told him “Put it in the water, pour water on it, it’s bitter”’

marra w-marrtēn ʔəlt-əllo lā təʃab bəʃtari
time and-time I.told-to.him not play+IMP in.the.street
‘Time and time I’ve told him “Don’t play in the street”’

Rather more interestingly, there are languages that employ direct speech constructions much more widely than in English. Thus in spoken Amharic (Semitic, Ethiopia – Cohen 1936: 363), an expression of fear can be represented with the verb ‘fear’ plus the verb ‘say’ followed by a direct quotation, even though no actual words were spoken as in:

mangadum aagaññaum bəyye əfarallaüh
the.road he.will.not.find.it I.saying I.fear
‘I am afraid he won’t find the road’

It is even possible to express ‘refusal’ by an inanimate object in the same way:

səga albasləmm ala
meat.I.will.not.cook said
‘The meat refused to cook’

It has been claimed for a number of languages that they have no indirect speech, but it may be more accurate to say that indirect speech is rarely used in some of the languages. For instance, it appears that generally the Australian languages employ only direct speech, but consider this example from Ngiyambaa (Australia – Donaldson 1980: 280):

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ɲadhu-na ɲiyiyi girma-li ɲinu:
I+NOM-3ABS say+PAST wake-PURP you+OBL
'I told her to wake you'

Now the actual words would have been 'Wake him', so that this is clearly not direct speech. In fact, 'deictic shift' – changing pronouns, as here (and also other deictics such as time markers in some languages, e.g. English), is an essential feature of indirect speech. For English compare two sentences in which four deictics have been changed – pronouns, tense and temporal pronouns:

He said 'I'm coming tomorrow'
He said he was coming the next day

7.5 Speaker commitment

There is a further point to be made about indirect speech. Some languages have devices to indicate whether the speaker believes the reported utterance to be true. In English, there may be no deictic shift involving change of tense, if a reported statement is accepted as true by the secondary speaker:

I said 'I'm coming on Tuesday'
I said that I was coming on Tuesday
I said that I'm coming on Tuesday

The third example suggests that it is still the speaker's intention to come. A very clear example is given by Jespersen (1909–49: iv, 156):

The ancients thought that the sun moved round the earth; they did not
know that it is the earth that moves round the sun

The point here, quite clearly, is that the first proposition, with *moved*, is not accepted by the speaker, while the second, with *moves*, is accepted as true by him.

This can, obviously, be regarded as notionally a modal feature, and, in German, grammatical mood is involved, in that although – as noted in 5.2.2 – with past tense verbs, literary German uses the subjunctive in indirect speech, the indicative can be used to indicate confidence by the speaker:

Er sagte, er schwimmt gern
he said he swim+3SG+PRES+IND with.pleasure
'He says he likes swimming'

There is a contrast in:

Ich glaubte dass er wäre krank
I thought that he be+3SG+IMPF+SUBJ ill
'I thought he was ill'

Ich glaubte dass er krank war
 I thought that he ill be+3SG+IMPF+IND
 'I thought he was ill'

But the choice of mood depends also on stylistic and sociolinguistic factors (see Hammer 1983: 265–8). Notice that there is no change of tense as there is in English, only a change of mood.

Similarly, Classical Greek uses the optative mood with past tense verbs of reporting, but may also use the indicative, with the original (direct speech) tense:

e:peíle:s' hótí . . . badioíme:n (Ar. *Pl.* 88)
 I.threatened that go+1SG+FUT+OPT
 'I threatened that I would go'
 e:ngélthe autó:i hótí Mégara aphéste:ke (Thuc. 1.114)
 it.was.reported to.him that Megara revolt+3SG+PERF+IND
 'News came to him that Megara had revolted' (*not* 'has revolted')

There are some examples with both constructions in the same sentence, with, presumably, some indication of speaker commitment, as in:

élegon hótí Kú:ros mén téthne:ke,
 they.said that Cyrus on.the.one.hand die+3SG+PLUP+IND
 Ariaíos dé pepheugó:s . . . eíe: (Xen. *An.* 2.1.3)
 Ariaeus on.the.other having.fled be+3SG+PRES+OPT
 'They said that Cyrus had died, but that Ariaeus had fled'

In some languages, e.g. Japanese (Kuno 1973: 261), deictic shift does not affect tense – the reported speech retains the tense of the actual utterance. However, Japanese has three different conjunctions, *no*, *koto* and *to* (Kuno 1973: 213–22). It appears that the choice of *no* or *koto* rather than *to* depends on the degree of the speaker's commitment to the truth of the proposition (Suzuki 1994, in a reexamination of Kuno's arguments). According to Kuno (1973: 216–17), there is a choice of conjunctions in the following:

John wa Mary ga sinda to sinzinakatta
 John Mary died that not-believed
 'John did not believe that Mary was dead' (She might or might not have been)

John wa Mary ga sinda koto o sinzinakatta
 John Mary dead that not-believed
 'John did not believe that Mary was dead' (She was)

For this reason 'to forget' cannot occur with *to*, but only with *koto* (or *no*):

John wa Mary ga tunbo de aru koto/no (*to) o wasureteita
 John Mary deaf is that had.forgotten
 'John had forgotten that Mary was deaf'

In contrast, a ‘wrong notion’ can occur only with *to* (Suzuki 1994: 526):

John wa Mary ga shinda to (*koto/no) o gokaishita
John Mary died that formed.a.wrong.notion
‘John formed the wrong notion that Mary had died’

Similarly, in Kinyarwanda (Bantu, Rwanda – Givón 1982: 26–32, quoting Givón and Kimenyi 1974), there are three conjunctions *ko*, *ngo* and *kongo*, each expressing different kinds of commitment by the (actual) speaker:

ya-vuze ko a-zaa-za
he+PAST-say that he-FUT-come
‘He said that he’d come’ (and I have no comment)

ya-vuze ngo a-zaa-za
he+PAST-say that he-FUT-come
‘He said that he’d come’ (but I have direct evidence which makes me doubt it)

ya-vuze kongo a-zaa-za
he+PAST-say that he-FUT-come
‘He said that he’d come’ (but I have indirect/hearsay evidence which makes me doubt it)

Factive, non-assertive, complementizers such as ‘regret’, ‘forget’ require *ko*, the marker of the strongest degree of commitment.

Yet another language to make a similar distinction is the Jacaltengo dialect of Jacalteco (Mayan, Guatemala – Craig 1977: 268). There is a contrast between *chubil* (‘corresponds to a factive “that”’) and *tato* (‘introduces an expected, supposed or believed fact’). Both can be used with a verb of saying to indicate the reliability or unreliability of what is said:

xal naj tato chuluj naj presidente
said he that will.come the president
‘He said that the president is going to come’

xal naj alcal chubil chuluj naj presidente
said the alcalde that will.come the president
‘The alcalde said that the president is going to come’

The first indicates an unreliable source of information, the second a reliable source.

In some languages, the speaker’s doubt about the truth of what is reported may be indicated by a particle, e.g. (according to Frajzyngier (1985: 244–5)) Polish, Czech and Mapun (Chadic, Nigeria). Polish and Mapun examples are:

Powiedział, że niby jest chory
he.said that BE+3SG DOUBT sick
‘He said that he is sick, but I am not sure that it is true’

wur sat mun ni din paa yol muan
 3SG say 1PL that 3SG+COREF DOUBT go trip
 'He told us that he went on a trip, but maybe he didn't go'

In Polish, but not Mapun, the particle *may* be used where the subject of the main verb is first person, to indicate that the reported statement is a lie (by the speaker):

Powiedziałem, że niby jestem chory
 I said that be+3SG DOUBT sick
 'I said that I am sick, but I am not'

Slightly differently, in Lega (Bantu, E. Zaire – Botne 1997: 512–14) a particle that is commonly used as Reported *may* be used to express doubt about a statement (see 2.2.4):

nkumgwágá (bónɔ), ámbɔ bazongo bé kulyágá merende
 I.hear that whites QUOT eat frog
 'I hear that Westerners eat frogs (though I find that unlikely)'

It expresses disagreement when the statement is about the speaker:

bábolá bóno ámbɔ nne nekobákesá
 they.tell that QUOT I tricked
 'They say that I tricked them (though I take exception to it)'

7.6 The subjunctive in English

There are three constructions that have been thought to be subjunctive in English.

One is the use of *were* in *If I were rich*, but this is very marginal, partly because it is restricted to conditional sentences, but also because, strictly, there is nothing unusual about *were*. With all other verbs there is just one past tense form – *came, liked, walked*, etc., and this is also true of *were* in conditional sentences: *I were, you were, they were*, etc. In that sense *were* is quite regular. What is irregular is the use of *was* in the singular for all other past tense forms, for no other verb has different forms for singular and plural.

A second construction is with the simple form in sentences such as *I demand that he come tomorrow*. This use is becoming rare in English, especially in British English, though it seems to be alive in written American English, at least, although grammarians bewail its demise. But it is not a distinct form, like the subjunctive in other languages: it is merely the simple form of the verb, which has several functions in English, particularly those of the infinitive and the imperative. Little is gained by calling it 'subjunctive' rather than acknowledging that it represents one function of the simple, uninflected, form.

A third possibility is the use of *should* in, for instance, *It's odd that you should say that*. This, obviously, is the use of a modal verb, and as such comparable with the subjunctive. However, formally, this is the past tense of a modal verb, not a distinct paradigm like the subjunctive of this chapter. (For a similar reason there is, strictly, no future tense in English – see Huddleston (1995).)

One striking similarity between *should* and the subjunctive is its use in constructions where the proposition is presupposed, e.g. in the Italian example from 5.2.5 and one of its English translations:

mi sorprende che tu dica questo
me it.surprises that you say+2SG+PRES+IND that
'It surprises me that you say that'
'It surprises me that you should say that'

It should be added that *should* can refer to the present or the past. The second English sentence above is equivalent to either of the two:

'It surprises me that you say that'
'It surprises me that you said that'

Quirk *et al.*, (1985: 1014) refer to this use as 'putative', and suggest that it refers to putative situations that are recognized as possibly existing or coming into existence. They compare:

I'm surprised that he should feel lonely
I'm surprised that he feels lonely

They go on to suggest that the first questions the loneliness, while the second accepts it as true. Similarly, they suggest that there is a non-factual basis when the meaning is close to that of a conditional clause:

It's a pity that they should be so obstinate
It's a pity if they are so obstinate

Another use of *should* is in conditional sentences, such as:

If John should come, Bill will leave
Cf. If John comes, Bill will leave

The use of *should* here expresses some uncertainty. (This is not an example of an unreal conditional, the topic of 8.3. It is quite different from *If John came, Bill would leave*. See 8.3.4.)

It is not very important whether or not these uses of *should* are called 'subjunctive'. They have some parallel with subjunctives in other languages, but, on the other hand, the term 'subjunctive' is generally used to indicate an inflectional category. *Should* is formally merely the past tense form of a modal verb.

8

Past tense as modal

It was mentioned in 1.4.4 that past tense often has a modal function. There is, in fact a considerable body of literature that discusses this use of past tense forms to express ‘unreality’.

8.1 ‘Real’ and ‘unreal’

‘Unreality’ in this sense has much in common notionally with irrealis, in that it indicates some degree of lack of confidence by the speaker, but is best treated as a different feature for several reasons. First, it is marked differently – by tense. Secondly, it co-occurs with markers of irrealis, in both mood and modality systems; in fact, ‘real’ and ‘unreal’ in this sense mark a further distinction, another parameter within a wider field of modality. Not surprisingly, its functions are often quite different from those of irrealis.

8.2 Modal-tense

‘Unreality’ is not a satisfactory term, since it is too like ‘irrealis’ and, unfortunately, the terms ‘realis’ and ‘irrealis’ have been used in traditional grammars to refer to ‘unreal conditionals’ (below, 8.3). Instead the terms that will be used are ‘modal-past’ and, where relevant, ‘modal-present’ and ‘modal-tense’.

8.2.1 *Modal verbs*

There was a discussion in 2.1.5 and 3.2.3 of the use of past tense forms of the modals to provide weaker, more tentative, forms of the judgments and directives. These are, obviously, what are now being called ‘modal-past’.

Examples of the past tense forms of MAY and WILL as tentative epistemic judgments (Speculative and Assumptive) are:

He might be there (Cf. He may be there)

He would be on holiday now (Cf. He will be on holiday now)

MUST has no past tense form, but *ought to* and *should* appear to fill the gap:

He ought to/should be there by now (Cf. He must be there by now)

Formally *should* is the past tense form of SHALL, but it expresses tentative epistemic necessity (Deductive) and so is notionally one of the modal-past forms of MUST.

With deontic modals there is one-to-one correspondence only with WILL:

I'd do that for you (Cf. I'll do that for you)

Yet while *may* and *can* give permission, the corresponding past tense forms *might* and *could* are normally used to make suggestions:

You might/could ask your father

Ought to and *should* are again available for MUST:

You ought to/should come tomorrow (Cf. You must come tomorrow).

Commissive SHALL has no (notionally) corresponding modal-past form.

Because of these anomalies many scholars have argued *might*, *could* and *should* are not the past tense forms of MAY, CAN and SHALL. But the relationship in terms of tense can be demonstrated in two ways. First, by the 'sequence of tense' rule, present tense forms change to past tense after a past tense verb of reporting. This applies to the modals as well as to full verbs:

'I am coming'
He said that he was coming
'He may be there'
She said that he might be there
'You may come in'
He said that I might come in
'I'll do it for you'
She said that she'd do it for me

Secondly, these past tense forms are used with exactly the same relationship to the present tense forms in unreal conditionals, the subject of 8.3.

This use of tense is not confined to modal verbs. It is also to be found in polite, tentative, requests and questions, such as:

I wanted to ask you a question
Did you want to speak to me?

However, as Bybee (1995: 506) points out, WANT is notionally modal. This tense usage is not found with other types of verb.

8.2.2 Greek optative

In addition to the subjunctive, Classical Greek has an optative mood. One of its functions is in the expression of wishes:

ó: paí génoio patrós eutukhésteros (Soph. Aj. 550)
 O child become+2SG+AOR+OPT of.father luckier
 'My child, mayst thou be luckier than thy father'

It is also used in Homeric Greek for suggestions, rather like English *might*:

táut' eípois Akhiléi
 these say+2SG+AOR+OPT to.Achilles
 'You might tell Achilles this'

Since there are, apparently, three moods in Classical Greek, this might seem to be an argument against the binary nature of realis and irrealis, but, in fact, the optative can be seen as a modal-past form of the subjunctive.

Morphologically, there is evidence for this. To begin with, the subjunctive has similarities to the present tense, while the optative has similarities to the imperfect, as can be seen in the following forms (first person, middle voice):

- Present indicative lúomai
- Subjunctive lúoimai
- Past indicative elúome'n
- Optative lúoimēn

More importantly, a distinction is traditionally made between 'primary' and 'historic' tenses, which, with slight complications, is essentially the distinction between present and past tense, and this is of syntactic relevance (see below). The present and imperfect tenses are primary and historic respectively, but they differ only in the indicative – there is only one subjunctive and one optative paradigm associated with the pair of them. The same is true of the perfect and the pluperfect, which can be seen as differing from the present and perfect in aspect, not tense, with the pluperfect being the past of the perfect. (The traditional terminology is misleading. The present and imperfect would be better seen as the present and past of the imperfect, and the perfect and pluperfect as the present and past of the perfect.) With these tenses, then it could well be argued that the subjunctive and optative also differ in tense, the subjunctive being present and the optative being past:

<i>Imperfect</i>			
<i>Indicative</i>		<i>Subjunctive</i>	
Present	Past	Present	Past
'present indicative'	'imperfect indicative'	'present subjunctive'	'present optative'
<i>Perfect</i>			
<i>Indicative</i>		<i>Subjunctive</i>	
Present	Past	Present	Past
'perfect indicative'	'pluperfect indicative'	'perfect subjunctive'	'perfect optative'

The suggestion that Classical Greek has perfect and imperfect aspects, within which there are tense distinctions, is not new, but it has not been generally recognized that formally there are two other categories that belong to the same paradigm as imperfect and perfect. These are the aorist and the future, so that it could be argued that there are four, not two, aspects (though ‘aspect’ might not be the most suitable term). The reason for arguing that there are four ‘aspects’ is that each has its own infinitive and participles (not differentiated for tense), plus the fact that they also have subjunctives and optatives.

There are, however, some differences with the aorist and the future. The aorist does not have morphologically distinct present and past forms, yet it has both a subjunctive and an optative. However, syntactically, the aorist indicative can be either primary or historic, and so could be said to have a primary (present) subjunctive and a historic (past) optative:

<i>Aorist</i>			
<i>Indicative</i>		<i>Subjunctive</i>	
Present	Past	Present.	Past
‘aorist’ (‘primary’)	‘aorist’ (‘historic’)	‘aorist subjunctive’	‘aorist optative’

The future has only an optative, so that the pattern would seem to be:

<i>Future</i>			
<i>Indicative</i>		<i>Subjunctive</i>	
Present	Past	Present	Past
‘future indicative’	<i>none</i>	<i>none</i>	‘future optative’

However, the absence of a subjunctive may be thought to be due to the close association of the future with the subjunctive (and the fact that both are notionally irrealis).

The syntactic argument for Primary/Historic and so for the optative as modal-past is that the optative is required instead of the subjunctive in some types of subordinate clause if the superordinate verb is Historic. This is clearly so with purpose clauses, as illustrated by:

tón kakón deí	kolázein hín’	ameínon
the bad	it.is.necessary to.punish in.order.that	better
e:1		(Plat. <i>Legg.</i> 944D)
be+3SG+PRES+SUBJ		
‘We must punish the bad man to make him better’		
épempsa hó:s púthoito		
I.sent	that learn+3SG+AOR+OPT	
‘I sent him that he might learn’		

In the first example the main verb (*deí*) is present and primary, and the

subordinate verb is in the subjunctive. In the second example the main verb is aorist and historic, and the subordinate verb is in the optative.

The optative is also used to report a present tense form in the original utterance:

élegon hóti pantós áksia légoi seúthe:s (X. An. 8.3.13)
 they.said that of.all worthy say+3SG+PRES+OPT seuthes
 'They said that Seuthes's words were all-important'

The original words would have contained a present indicative (*légei*, 'says'). English similarly would use a past tense after a past tense verb of reporting, as the translation shows, where *were* reports an original *are*.

Similarly the optative is used for a reported cause, after a historic main verb (here an aorist participle), where Latin would use the subjunctive (see 5.2.2):

oístha epainésanta autón tón Agamémnona ho:s basileúste te
 you.know praise+AOR+PART he the Agamemnon as king both
 eíe: ágathos, kraterós t' aikhmeité:s (Xen. Symp. 4.6)
 be+3SG+PRES+OPT good, mighty and warrior
 'You know he praised Agamemnon because (he said) he was both a
 good king and a powerful warrior'

There is no need, therefore, to recognize optative as another mood, different from both indicative and subjunctive. It can be seen as a modal-past subjunctive.

8.3 Real and unreal conditions

The most striking and most widespread use of past tense for unreality is in conditional sentences. In a large number of languages a clear distinction can be made between real and unreal conditions, the latter being used where the speaker indicates some kind of negative attitude towards the condition expressed in the protasis (the 'if-clause'). In some languages this is marked with tense alone, in others by a combination of tense and mood.

8.3.1 Marking by tense

In English (see, e.g., Palmer 1990: 169) there is a contrast between:

If John comes, Bill will leave
 If John came, Bill would leave

These are examples of a real and an unreal conditional sentence. Notionally the difference is that, with the unreal conditional, the speaker indicates some doubt about the likelihood of the event indicated in the protasis (John's coming), whereas, with the real conditional, the possibility is simply left open. Formally, the difference between them lies in the tense marking in both

clauses, the real conditional having present tense verbs, the unreal conditional having past tense verbs in both protasis and apodosis (the main clause). To convert the real into the unreal, *comes* is replaced by *came* and *will* by *would*.

There is a corresponding unreal conditional in the past, which is:

If John had come, Bill would have left

Here past is marked twice, once for past time, once for unreality, the markers being past tense *had*, *would* and HAVE, since HAVE functions in English as both a marker of perfect aspect and of past time (for details see Palmer 1990: 170).

Conditionals of this type, which may be called ‘predictive’ since they predict, in terms of cause and effect, the occurrence of one event on condition of another, always require a modal verb in the apodosis (both real and unreal), e.g.:

If John comes, Bill can leave

If John came, Bill would leave

If John had come, Bill might have left

Non-predictive conditionals are such as *If John came, Mary left*, which merely indicate that if the protasis is true, so is the apodosis, without necessarily implying that there is cause and effect. There are no restrictions on the tense or modality of such sentences.

Examples of this use of tense for unreality can be found in other languages. Thus in Classical Greek the imperfect indicative is used in both clauses for present unreal conditional, and the apodosis also contains the particle *án*:

ei tóuto epoíeis, hé:mártanes án
if this do+2SG+IMPF+INDIC err+2SG+IMPF+INDIC án
‘If you were doing this, you would be wrong’

For past unreal conditions, it uses the aorist, instead of the imperfect:

ei tóuto epoíe:sas, hé:martes án
if this do+2SG+AOR+INDIC err+2SG+AOR+INDIC án
‘If you had done this you would have been wrong’

However, where there is clear future reference, real conditionals use the subjunctive (or the future) in the protasis and the future in the apodosis, and unreal conditionals use the optative in both:

ei tóuto epoíe:is, hamarté:seis
if this do+2SG+PRES+SUBJ err+2SG+PRES+FUT
‘If you do this, you will be wrong’

ei tóuto epoioíe:s, hamartánois án
if this do+2SG+PRES+OPT err+2SG+PRES+OPT án
‘If you did this, you would be wrong’

Although mood is required here, the difference between real and unreal is marked by subjunctive and optative. Essentially, then, in the light of the arguments in 8.2.2, the distinction is still one of tense.

There is a similar use of tense to mark unreality in the Ethiopian Semitic languages Tigre and Tigrinya, spoken in Eritrea (personal research). Tigre has a past tense (identified with the Semitic ‘imperfect’), plus the particle *wa* in both clauses:

London wa-gəsko, wa-mətko
 London COND-go+1SG+PERF COND-die+1SG+PERF
 ‘If I went to London, I would die’

Tigrinya has the present (imperfect) in the protasis, but a past (perfect) in the apodosis:

’əntä təharm-o, mə-mote
 if hit+2SG+IMPF-him COND-die+3SG+PERF
 ‘If you hit him, he would die’

In a neighbouring Cushitic language, Bilin (Palmer 1957), unreality is also marked by past tense but in a very different way. In this language the verbs have a large number of paradigms (inflected for person, gender and number), but these paradigms can be classed in terms of certain features of vowel quality and tonal patterns into two types, A and B. With the verb class illustrated below, the vowel immediately after the stem is open and without high tone in A, but front close and with high tone in B:

A		B	
jəbāk ^w	he buys	jəbík ^w	he bought
jəbāx ^w	which he buys	jəbéx ^w	which he bought
jəbāt	(says) that he buys	jəbét	(says) that he bought

As these examples show, many of the paradigms are paired, B referring to past time, A to present time. There are two forms for conditionals (used in both protasis and apodosis). These are quite different in form, but the paradigm used for real conditions is of Type A and the paradigm used for unreal conditions is of Type B (compare the vowels after the stem and the tones):

jəbän	‘will buy/if he buys’
jəbínädik	‘would buy/if he bought’

8.3.2 *Marking by tense and mood*

Often unreal conditions are marked by both mood and past tense. For unreal conditionals in the present, Latin uses the imperfect subjunctive in both clauses:

Past tense as modal

Si hoc faceres, errares
if this do+2SG+IMPF+SUBJ err+2SG+IMPF+SUBJ
'If you did/were doing this you would be wrong'

However, Latin, like Classical Greek, has a different set of forms for unreal conditionals with clear future reference – it has present subjunctive in both clauses:

Si hoc facias, erres
if this do+2SG+PRES+SUBJ err+2SG+PRES+SUBJ
'If you did/were to do this you would be wrong'

German is a little more complex. It has the imperfect subjunctive in the protasis but either the imperfect subjunctive or the 'conditional tense' in the apodosis (Hammer 1983: 253). However the 'conditional' tense is, in fact, formed by the imperfect subjunctive of the modal *WERDEN*, which often marks the future. Examples are:

Wenn ich ihm heute schriebe, bekäme
if I to.him today write+1SG+IMPF+SUBJ get+3SG+IMPF+SUBJ
er den Brief morgen
he the letter tomorrow

Wenn ich ihm heute schriebe, würde
if I to.him today write+1SG+IMPF+SUBJ be+3SG+IMPF+SUBJ
er den Brief morgen bekommen
he the letter tomorrow to.get

'If I wrote to him today, he would get the letter tomorrow'

Spanish also uses 'conditional tenses' and has either the imperfect subjunctive or the conditional in the protasis, but the conditional or the conditional subjunctive in the apodosis:

Si yo tuviese/tuviera bastante dinero,
if I have+1SG+IMPF+SUBJ/have+1SG+COND enough money,
comprara/compraría otro automóvil
buy+1SG+IMPF+SUBJ/buy+1SG COND+SUBS other car
'If I had enough money, I would buy another car'

However, Spanish, unlike English and German, does not employ a modal verb, but rather its (notionally very similar) future tense, and combines this with past.

For unreal conditionals in the past, Latin simply uses a more remote past marker, the pluperfect instead of the imperfect, but still with the subjunctive:

si hoc fecisses, errasses
if this do+2SG+PLUP+SUBJ err+2SG+PLUP+SUBJ
'If you had done that you would have been wrong'

German uses its ‘have’ form (HABEN), like English, to add the extra past tense (yielding what are, predictably, called the ‘pluperfect subjunctive’ and the ‘conditional perfect’):

Wenn ich ihm heute geschrieben hätte,
 if I to.him today written have+1SG+IMPF+SUBJ
 hätte er den Brief morgen bekommen
 have+SG+IMPF+SUBJ he the letter tomorrow got

Wenn ich ihm heute geschrieben hätte,
 if I to.him today written have+1SG+IMPF+SUBJ
 würde er den Brief morgen bekommen haben
 be+SG+IMPF+SUBJ he the letter tomorrow got to.have

‘If I had written to him today, he would have got the letter tomorrow’

In a similar way, past unreal conditionals in Spanish simply involve the addition of HABER (*hubiese* and *habría*).

8.3.3 Interaction of modality and unreality

Not surprisingly, perhaps, there is some interaction between modality and (conditional) unreality. This is most striking in Latin, but is most easily explained by first considering some English examples.

It has been argued that there are differences in the uses of *could have* in:

He could have jumped six feet, if he’d trained hard
 He could have jumped six feet, if he’d wanted to

The first example says that training hard would have resulted in the ability to jump six feet, whereas the second example says that wanting to would have resulted in jumping six feet, not in the ability to do so, for that ability was already there. In other words, *could have* indicates ‘I was able to, and would have’ not ‘I would have been able to’. Austin (1956: 163 [1961: 164]) suggests that *I could have* can be either a ‘past indicative’ or a ‘past conditional’ and may be equivalent either to Latin *potui* (‘I was in a position to’) or *potuissem* (‘I should have been in a position to’).

A clear example of the pluperfect subjunctive is:

quod certe si essem interfectus, accidere
 which certainly if be+1SG+IMPF+SUBJ killed to.happen
 non potuisset (Cic. Sest. 22.49)
 not can+3SG+PLUP+SUBJ
 ‘Which certainly could not have happened if I had been killed’

In contrast the perfect indicative is used in:

Past tense as modal

deleri totus exercitus potuit, si fugientis
be.destroyed whole army can+3SG+PERF+IND if those.fleeing
persecuti victores essent (Liv. 32, 12.6)
pursue+3PL+PLUP+SUBJ victors (discontinuous with persecuti)
'The whole army could have been destroyed, if the victors had pursued
those who were fleeing'

The sense is that it was possible for the whole army to be destroyed (that the victors were in the position of being able to destroy it), not that it would have been possible. Handford (1947: 132) sees the choice in terms of what is 'logically preferable'.

This phenomenon is found with other expressions that are notionally modal, such as LICITUM ESSE 'be allowed', MALLE 'prefer', DEBERE 'ought', OCCASIO ESSE 'be a chance' (Handford 1947: 131). An example of the last is:

quem si interficere voluisset, quantae quotiens
whom if to.kill wish+3SG+PLUP+SUBJ how.many how.often
occasiones fuerunt (Cic. Mil. 38)
occasions be+3PL+perf+IND
'If he had wished to kill him, how many chances, how often, there were!'

The same feature is found with present unreal conditionals, as in English with *could*:

He could jump six feet, if he trained hard
He could jump six feet, if he wanted to

So in Latin with LICITUM ESSE, with the present indicative instead of the imperfect subjunctive:

licitum est si velles (Pl. Trin. 566)
allowed be+3SG+PRES+IND if wish+2SG+IMPF+SUBJ
'You could if you wished'

In other examples there are different combinations of tense, but all with the subjunctive in the protasis and the indicative in the apodosis:

idque si nunc memorare hic velim . . . vere
and.that if now to.remember this wish+1SG+PRES+SUBJ truly
possum (Ter. Hec. 471)
can+1SG+PRES+IND
'If I now chose to recall this, I could do so with truth'
si ita sententia esset . . . tibi servire
if thus opinion be+3SG+IMPF+SUBJ to.you to.serve
malui (Pl. Miles 1356)
prefer+1SG+perf+IND
'If it were your view, I would have preferred to serve you'

eum contumeliis onerasti quem patris loco, si ulla in te
 him with.insults you.loaded whom of.father in.place if any in you
 pietas esset, colere debebas (Cic. *Phil.* 2.38.99)
 affection be+3SG+IMPF+SUBJ to.honour ought+2SG+IMPF+IND
 ‘You loaded with insult the man whom, if you had any affection in you,
 you ought to have honoured as a father’

However, with some examples the ‘logical’ explanation is not available. Handford (1947: 132) comments that (with POSSE at least) ‘a Latin author’s choice of mood depended largely on personal taste, the whim of the moment, metre or clausula’. He quotes:

non potuit . . . fieri sapiens, nisi
 not can+3SG+PERF+IND become wise unless
 natus esset
 born be+3SG+IMPF+SUBJ
 ‘He could not have become wise, if he had never been born’

There is one passage in which there is a contrast between the pluperfect subjunctive and the perfect indicative (in the apodosis) in two consecutive sentences. The first is:

in qua quid facere potuissem, nisi tum consul
 in which what to.do can+1SG+PLUP+SUBJ unless then consul
 fuissem?
 be+1SG+PLUP+SUBJ
 ‘What could I have done in that (crisis) if I had not then been consul?’

The second is:

Consul autem esse qui potui,
 consul however to.be who can+1SG+PERF+IND
 nisi eum vitae cursum tenuissem? (Cic. *Sest.* 22.49)
 unless that of.life course hold+1SG+PLUP+SUBJ
 ‘But how could I have been consul, if I had not held to that manner of life?’

Certainly the ‘logical’ explanation does not explain the second sentence, for the manner of life was the cause of the ability to become consul. However, Cicero was, in fact, able to become consul, and it may be that this is the reason for the choice of the indicative.

Giulio Lepschy (personal communication) has pointed out that there is something similar (but not quite the same) in Italian. The imperfect indicative *potevo* (= Latin *potui*) may be preferred to the ‘past conditional’ (formed by the conditional form of the verb ‘to have’ plus the past participle) if there is an implication that the ability was already in place, as in:

Potevi dir-me-lo!
 can+2SG+IMPF+IND tell-me-it
 'You might have told me!'

(The 'past conditional' *Avresti potuto dirmelo* is less natural.) The same is true of the protasis of past unreal conditional. Thus the imperfect indicative is possible in the first example, but almost impossible in the second:

potevo farlo se mi aiutavano
 can+1SG+IMPF+INDIC do-it if me you.had.helped
 'I could have done it if you had helped me'
 avrei potuto farlo se fossi stato più vecchio
 can+1SG+PAST.COND do-it if I.had been more old
 'I could have done it if I had been older'

The essential point here is that, with the first, there is an implication that I had the ability and merely needed help to fulfil it, but with the second it is clear that I did not have the ability, but would have had it if (impossibly) I had been older.

However, the Latin examples show that there is not always a 'logical' explanation, although the use of the indicative is more likely when there is an implication that the modal feature was already-in-place ability. The explanation may be simpler – that, since modal expressions are irrealis, it may seem redundant to mark them as unreal as well. There is, in fact, some support for this in English, in that, as Bybee (1995: 506) points out, it is possible for notionally modal expressions to be used in the apodosis of unreal conditionals, instead of a modal verb as in:

If I saw Judy	{	I wanted to I was gonna I was supposed to I intended to	}	tell her the news
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(But Bybee admits that some of these are not 'highly grammaticized'.)

8.3.4 Further possibilities

Not every language makes the distinction between real and unreal conditionals. According to Akatsuka (1985: 627), this is the case with many East Asian languages, such as Japanese, Chinese, Korean, Mongolian, Semai and Thai. An example from Japanese is:

musoko ga ikite i-tara ii noni naa!
 son SUB alive be-if good though EXCL
 'If my son is alive, I'll be so happy'
 'If my son were alive, I'd be so happy'

Some languages make the distinction between real and unreal conditionals, but make further distinctions as well. Thus, in Southern Agau (Ethiopian Cushitic – Hetzron 1969: 25), there are specific conditional paradigms (marking person, number and gender) for both the protasis and the apodosis, but tense (imperfect and perfect) is marked in the protasis and reality/unreality in the apodosis. There is, however, a further distinction between ‘indefinite’ and ‘definite’ that is also marked in the form used in the apodosis, and which indicates further degrees of speaker confidence.

Hetzron illustrates six ‘main combinations’. For real conditions, the conditional imperfect is used in the protasis and the regular imperfect, either definite or indefinite, in the apodosis:

'an desúni	ónɕye yégcé
I study+1SG+COND.IMPF	work find+1SG+IMPF.INDEF
‘If I study I shall find work’	
'an desúni	ónɕye yégcáyácé
I study+1SG+COND.IMPF	work find+1SG+IMPF.DEF
‘If I study I shall find work’	

For unreal conditionals either the conditional imperfect or the conditional perfect is used in the protasis and the conditional indefinite or the conditional definite in the apodosis, giving four possibilities:

'an desúni	yizágá	deséšiy ^{wà}
I study+1SG+COND.IMPF	my.relative study+3SG+COND.INDEF	
‘If I study my relative would (probably) study’		
'an desúni	yizágá	desáwšiy ^{wà}
I study+1SG+COND.IMPF	my.relative study+3SG+COND.DEF	
‘If I study my relative would (certainly) study’		
'an desášúni	yizágá	deséšiy ^{wà}
I study+1SG+COND.PERF	my.relative study+3SG+COND.INDEF	
‘If I had studied, my relative would (probably) have studied’		
'an desášúni	yizágá	desáwšiy ^{wà}
I study+1SG+COND.PERF	my.relative study+3SG+COND.DEF	
‘If I had studied, my relative would (certainly) have studied’		

Less striking, there are two further possibilities with conditional sentences in English, the use of *should* (see 7.6) and *were to*. The following two sentences may be compared with the first two given in 8.3.1 (in brackets here):

If John should come, Bill will leave
 If John were to come, Bill would leave
 (If John comes, Bill will leave)
 (If John came, Bill would leave)

The choice of *will* in the first and of *would* in the second clearly shows that these are real and unreal conditionals respectively. They appear to differ from the bracketed sentences only in suggesting slightly greater likelihood of the event referred to in the apodosis (John's coming). This, in combination with the real/unreal contrast, gives four degrees of the expression of likelihood.

There is a similar situation in Swahili (Bantu – Saloné 1983), where what Saloné calls the ‘imperfective’ is the usual marker of a real conditional. This is an infix that occurs in the place of tense-aspect markers, as in:

Mtumwa a-ki-taka ku-ondoka-na minyaro . . .
 slave SUB-IMPV-F want INFIN-leave-RECIP shackles
 i-na-m-lazimu a-fany-e mapambano
 SUB-PRES-OBJ necessary SUB-do-SUBJ struggle
 ‘If a slave wants to rid himself of his shackles, he must struggle.’

However, the form *ikiwa* or *iwapo* followed by a verb with full tense–aspect marking may be used to suggest that the condition is unlikely to be fulfilled:

ikiwa walawezi wote wa-na-fikiria hivyo ni-ta-jiuzili la
if settlers all SUB-PRES-think this.way 1SG-FUT-resign but
yuko kalika kundi la-wachache
he in group.of.few
'If all the settlers think that way, I shall resign, but he is in the minority.'

However, it must be noted that *ikiwa* contains the infix *-ki-* and is literally 'if it is' (and *iwapo* is a similar formation with a locative marker). It may not be surprising that an interpretation in terms of 'If it is the case that . . .' suggests some degree of doubt, usually towards a statement made by someone else.

8.4 Wishes

Wishes also use past tense forms, often along exactly the same lines as the protases of unreal conditions. Thus Classical Greek uses the optative, the imperfect and the aorist for future, present and past wishes respectively (compare the conditionals in 8.3.1). They are introduced by *ei gár*, and it is no coincidence that *ei* is the conjunction ('if') of conditionals:

ei gár genoíme:n téknon, antí sou
 oh that become+1SG+AOR+OPT son, instead.of you
 nekrós (Eur. *Hipp.* 1410)
 corpse
 ‘O that I might be a corpse, my child, instead of you!’
 ei gár tosaúte:n dúnamin eíkhon (Eur. *Alc.* 1072)
 oh that such strength have+1SG+IMPF+INDIC
 ‘Had I such strength’

ei gár m' hupó gé:n . . . hé:ken (Aesch. *P. V.* 152)
 oh that me below earth send+3SG+AOR+INDIC
 'Would he had sent me under the earth'

Similarly, Latin uses present, imperfect and pluperfect subjunctive, often with *utinam*:

Atque utinam ipse Varro incumbat in
 but that self Varro apply.self+3SG+PRES+SUBJ in
 causam (Cic. *Att.* 3. 15)
 cause
 'But if only Varro would apply himself to the cause'
 Modo valeres! (Cic. *Att.* 11.23)
 only be.well+2SG+IMPF+SUBJ
 'If only you were well'
 Utinam ne . . . tetigissent litora puppes (Catull. 64. 171)
 that not touch+3PL+PLUP+SUBJ shores ships
 'Would that the ships had not touched the shore'

English, like Classical Greek, sometimes uses an expression that contains the conditional conjunction – *if only*. More commonly it uses the lexical verb WISH. As with conditionals, past tense forms are used, but for wishes for the future, *would* or *could* are normally used:

If only/I wish John would/could speak tomorrow
 If only/I wish John had spoken yesterday

The use of *would* or *could* in future conditionals suggests that there is some willing (or able) agent, for wishes usually imply that there is some such agency.

For wishes about the present, a past tense form is used, as for conditionals – *I wish John liked curry*. This also explains the less likely wish for the future *If only I wish John spoke tomorrow*, for this is related not to *John will speak tomorrow*, but to *John speaks tomorrow*, with the meaning that it is arranged or planned for John to speak.

There are various ways in which other languages express wishes. Thus, for Fula, Arnott (1970: 299) offers:

njuutaa balde
 be.long+2SG+SUBJ in.days
 'May you live long!'

Particles are used for the same purpose in Serrano (Hill 1967: 88), and in Huichol together with a 'desiderative' suffix (Grimes 1964: 60–1).

8.5 Russian 'subjunctive'

There are other ways in which the past tense is associated with the subjunctive. In Russian, for instance, what is often called 'subjunctive'

consists in fact of the particle *-by* plus the past tense. This is used in several constructions where the subjunctive would be used in other languages. It is used, for instance, after the verbs ‘desire’, ‘demand’, ‘insist’ (whereas the accusative/dative and infinitive construction is used with ‘order’, and ‘suggest’ takes either construction):

Ja zhelaju/trebuju/nastaivaju chto-by ona ushla
I desire/demand/insist that-*by* she go+F.SG+PAST
‘I desire/demand/insist that she should go’

On poprosil/prikazal emu/emu uiti
he asked/ordered him/to.him go+INFIN
‘He asked/ordered him to go’

On predlozhil chto-by my ushli
he suggested that-*by* we go+PL+PAST
‘He suggested we should go’

On predlozhil nam uiti
he suggested to.us go+INF
‘He suggested we should go’

It is also used after the negative verbs of thinking, etc., where a Romance language would use the subjunctive (see 5.2.2):

Ja ne dumaju chto-by on byl glup
I not think that-*by* he be+M.SG+PAST stupid
‘I don’t think he’s stupid’

Compare the Italian:

Non credo che sia Corelli
no I.think that be+3SG+PRES+SUBJ Corelli
‘I don’t think that it’s Corelli’

Russian also uses this ‘subjunctive’ in purpose clauses, another type often associated with the subjunctive proper (see 5.3.2):

chtó-by niktó ne znal ob étom, nado
that+*by* no.one not know+3M.SG+PAST about this necessary
molchát
be.silent
‘So that no one may know, we must keep silent’

This Russian ‘subjunctive’ seems to show, once again, the connection between past tense and unreality.

8.6 Explanation

In 8.3 there were examples from a number of languages of past tense forms being used to indicate unreal conditions. It is easy enough to find further examples. Steele (1975: 200) mentions Garo (Tibeto-Burman), Chipewyan

(N. America, Athabaskan) and old Marathi (India, Indo-Iranian). James (1982: 376) adds Old Irish to the Western European languages exemplified, and also mentions the Bantu languages Tonga and Haya and further North American languages, Cree (Algonquian) and Nitinaht (Wakashan). Steele also argues, perhaps a little speculatively, that the protolanguage of Uto-Aztecan (N. America) had a morpheme reconstructed as *ta*, which combined the notions irrealis and past; the evidence for this is found, in languages within the family, in morphemes which can be regarded as reflexes of *ta*.

Naturally, this modal function of past tense can occur only in languages that have a tense system that differentiates past from present or non-past. It does not occur in languages such as those of Papua New Guinea (e.g., Amele – see 6.4) that deal with time relations within the category of Realis/Irrealis mood. Bhat (1999: 144) sees the use of past tense for unreal conditionals as a characteristic of ‘tense prominent’ languages, as distinguished from those that are ‘aspect prominent’ or ‘mood prominent’. However, although it is obvious that it does not occur in ‘mood prominent’ languages, it must be said that with ‘aspect prominent’ languages the categories used in conditionals appear to be restricted to perfective and imperfective, the categories that have a close association with past and present time. Bhat gives an example from Gujarati:

tuN wəkhətsər awe	to	apNe bəhar jəiej
you on.time	come+PERFV	then we out might.go
‘If you come on time, we might go out’		
e saurN kam kərlo	hOt to	prəphesər
he good work do+IMPFV	then	professor become
‘If he were doing good work, he would (have) become a professor’		

It would seem that, strictly, the relationship is with markers associated with past time rather than with grammatical past tense.

An obvious and important question is ‘Why is past tense (or, more strictly, forms associated with past time) so widely used to express this type of modality?’ Why is the future not used instead, since the future often seems to be a kind of modality (Lyons 1977: 816) and is often treated as irrealis?

The relation between past and unreality has often been noted, but the explanations seem to be largely circular. Joos (1964: 121–2), for instance, suggests that the essential common feature is remoteness, in time or reality. Similarly, James (1982: 396) speaks of ‘remoteness from reality’ and Langacker (1978: 855) uses the label ‘distal’. But this may be no more than giving a single label to two quite different meanings, and so may be uninformative.

Steele (1975: 217), argues for the ‘semantic primitive of disassociative’,

past time being disassociated from present time and unreality from reality. Of particular relevance is her interpretation of the use of past tense for polite requests, as in:

Would you pass the salt?
cf. Will you pass the salt?

She suggests that the past tense abstracts the speaker from the request. But why? A possible answer is that the use of the past leaves open the possibility that the willingness does not extend into the present, though it may imply that it does. Thus *I want to speak to you* demands attention, whereas *I wanted to speak to you* indicates only past desire and leaves it open for the hearer to make an excuse such as 'I haven't the time right now . . .' She also notes Hale's (1969: 22) suggestions that intentions in the past are usually unachieved and that unachieved intentions are 'a specific sort of irrealis'.

An argument along these lines is presented in much more detail and more convincingly by Bybee (1995: 506–8). She suggests that, with a modal verb (i.e. a notionally modal verb) such as WANT, there is a rather tenuous relation between the agent and the main predicate. *I wanted to* can be used 'both in contexts in which the desire was carried out and in contexts in which the desire was not carried out'. Moreover, as stative verbs the past forms of such 'modals' 'assert that a state existed before the moment of speech, but they do not say whether that state exists in the present or not'. There are two areas of vagueness: '(i) whether or not the predicate event was completed; and (ii) whether or not the modality remains in effect'. With a modal in past time, there is the possibility that some conditions on the completion of the main event were not met, as with:

I wanted to help you (but didn't)

She then considers the 'polite' use of past tense in:

I wanted to ask you a question

This, she suggests implies that there might be conditions that are unmet, including the question of whether the addressee wants to be asked a question. She could have added that it also leaves open the implication that the modality may no longer remain in effect. This seems even more important – for by suggesting that it is not necessarily the case that I still want to ask the question, I greatly weaken my insistence on the question being answered.

This explanation works for the dynamic modals CAN and WILL in sentences such as:

I could do that for you
I would do that for you

Here the use of present tense forms *can* and *will* might be taken as an instruction rather than an offer. For *can* may be used in (a mother to a child):

You can stay in your room until you behave

Similarly *will* may be used to instruct in:

You will do as I say

By contrast the past tense forms, for the reasons given above, can be interpreted as mere offers or suggestions.

It is worth recalling, at this point (see 3.3.2), that we cannot normally say:

*I ran fast and could catch the bus

*I asked him and he would come

For a past tense modal *may*, as argued above, imply that the event was not completed. It is, therefore, inappropriate when there is a clear implication that it was completed.

A somewhat similar explanation can be given for the use of *might* and *could* for suggestions, and of epistemic *might*. Yet it will not account in full for all the past tense modal forms. It does not, for instance, explain the use of *should* as the ‘unreal’ form of **MUST** rather than **SHALL**.

Moreover, this explanation does not seem to be applicable to the use of past tense verbs in unreal conditionals, especially where there is no modal verb, as in Classical Greek and Tigre (8.2.1). Bybee attempts to extend her analysis to *If she saw Judy, she would tell her the news*. She says ‘The past in both clauses . . . signals that the conditional relation held in past time. The modality in the *then*-clause signals that the relation is still in effect. (Cf. *If Mike saw her he told her the news*.)’. This is not entirely convincing, for although it explains why a past tense form might be interpreted as unreal, it does not explain why a past tense form should be used in the first place, since it is not actually true that the conditional relation held in past time.

Finally, and, unfortunately, negatively, it must be said that nothing in this section seems to account for the use of past tense in the Russian ‘subjunctive’. Not everything in language is capable of rational explanation!

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